Biochemistry



Section J
Carbohydrate metabolism

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13854808762

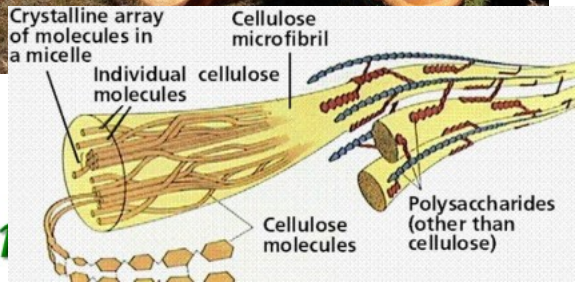


糖果



淀粉







几丁质

Biochem

J1 Monosaccharides and disaccharides

J2 Oligosaccharides and polysaccharides

J3 Glycolysis

(EMP: Embden-Meyerhof pathway)

L1 The Citric Acid Cycle

(TCA: tricarboxylic acid cycle)

J5 Pentose phosphate pathway

J4 Gluconeogenesis

J6 Metabolism of glycogen, starch and sucrose



J1 Monosaccharides and disaccharides

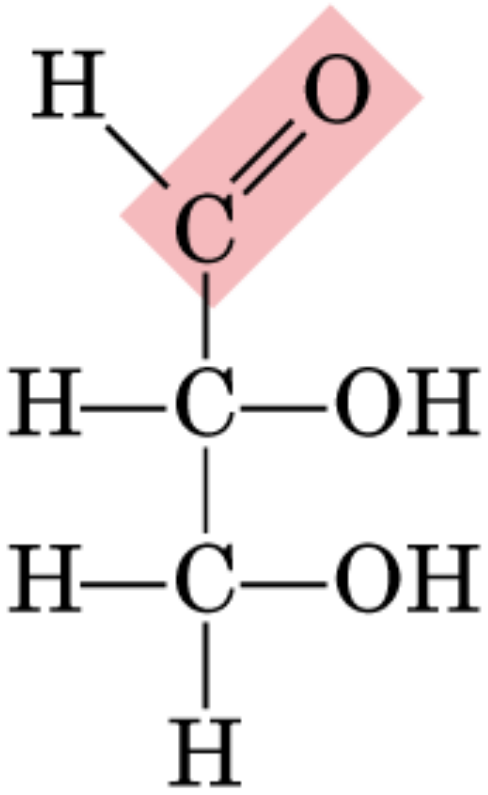


1. Monosaccharide

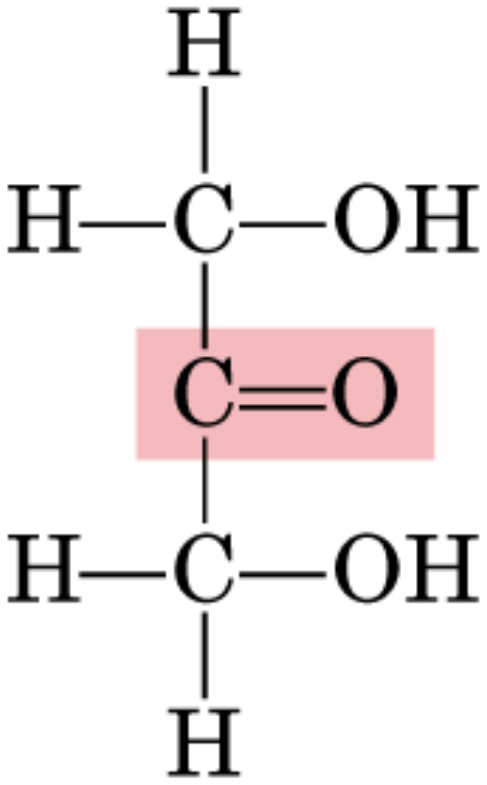
Aldoses (醛糖） and ketoses （酮糖）

A monosaccharide has the general formula $(CH_2O)_n$ and contains either an aldehyde group (an aldose) or a ketone group (a ketose).





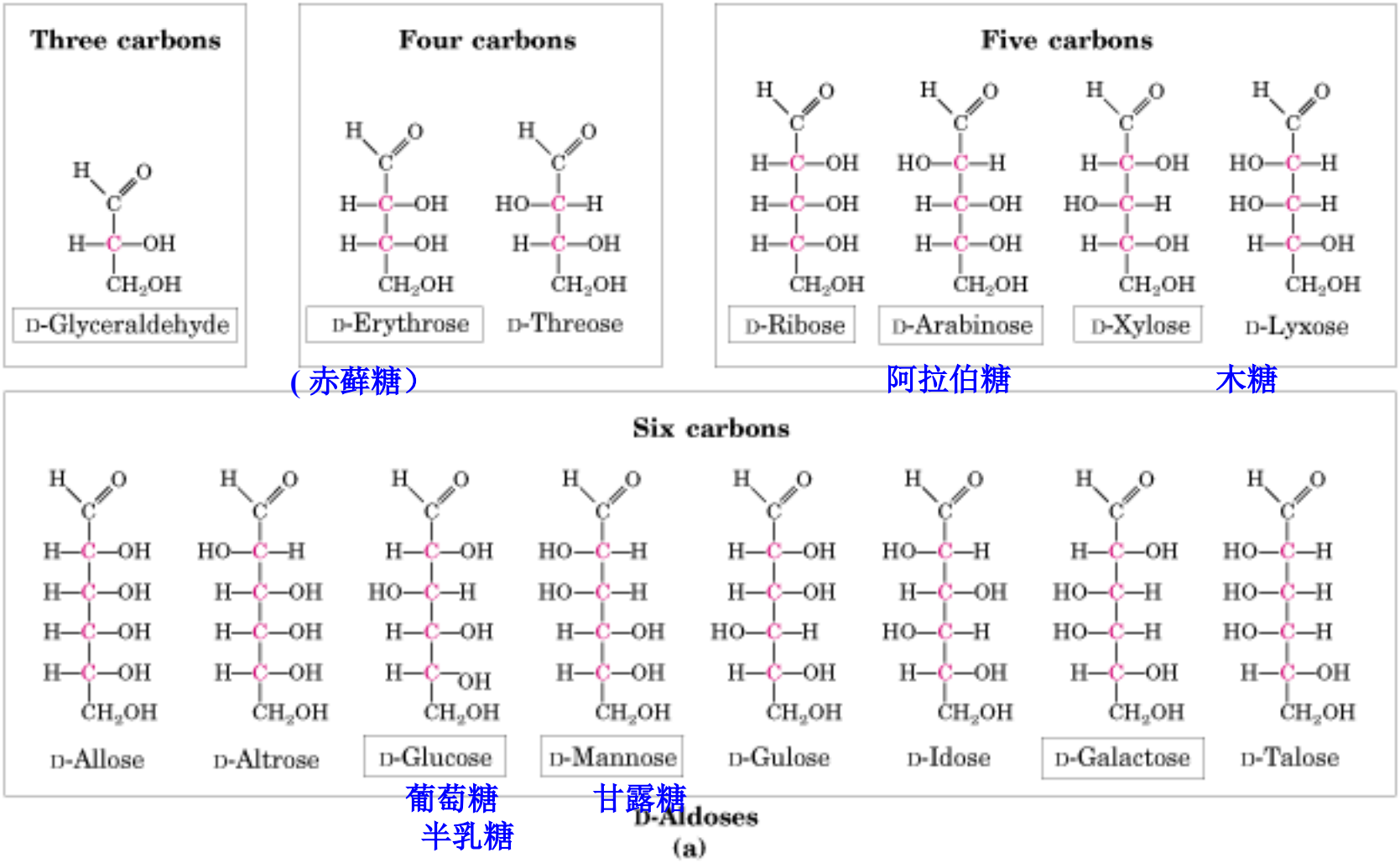
Glyceraldehyde, an aldotriose



Dihydroxyacetone, a ketotriose

(a)

Aldoses



（赤藓糖）

阿拉伯糖

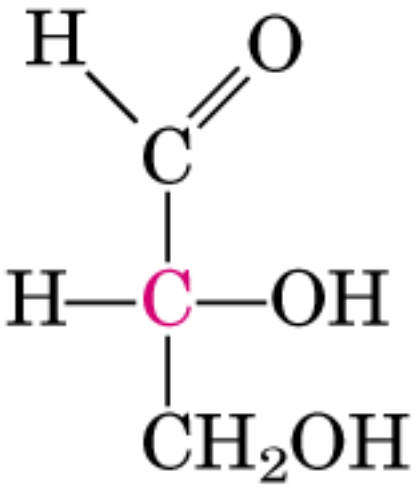
木糖

葡萄糖

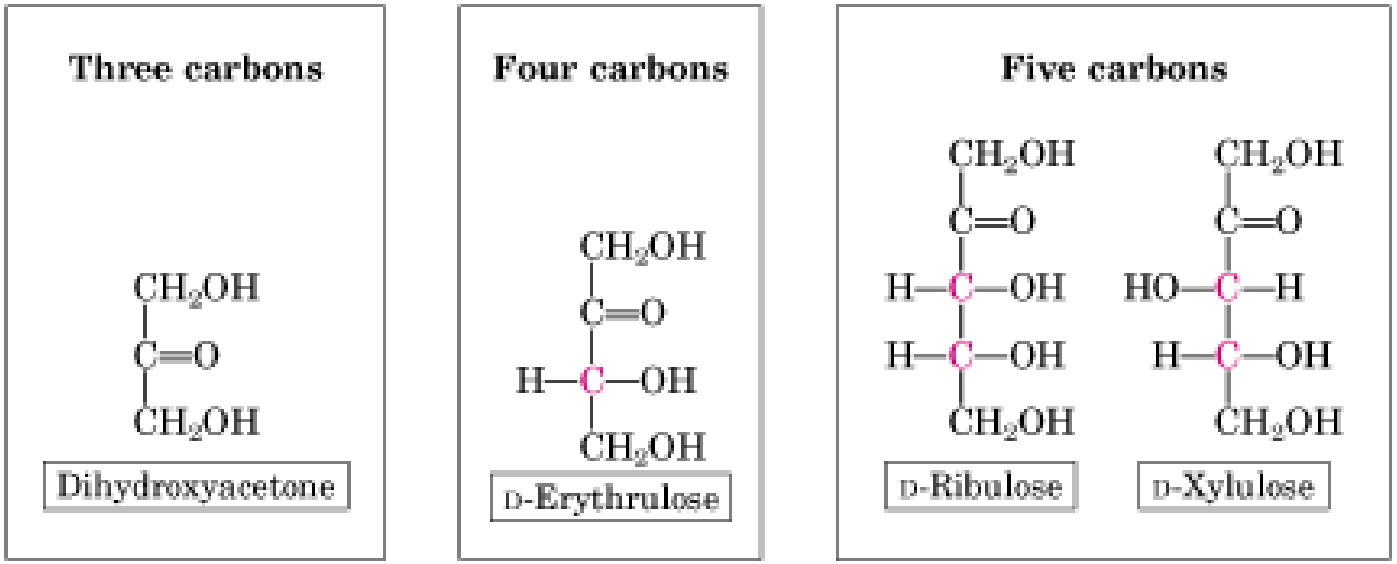
半乳糖

甘露糖

Three carbons



D-Glyceraldehyde

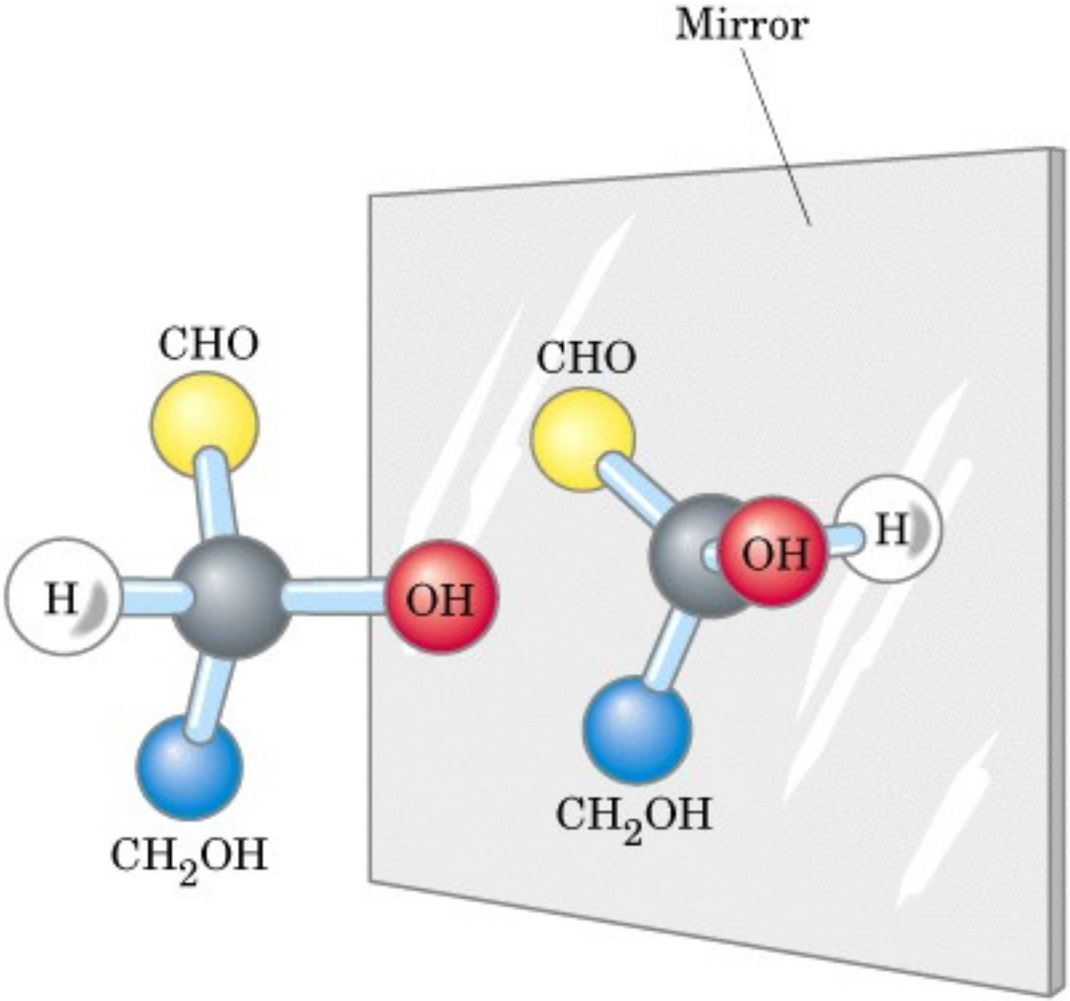

Three carbons
Dihydroxyacetone
Four carbons
D-Erythrulose
Five carbons
D-Ribulose
D-Xylulose
二羟丙酮
赤藓酮糖
核酮糖
木酮糖
Six carbons
D-Psicose
D-Fructose
D-Sorbose
D-Tagatose
果糖
D-Ketoses
(b)

Ketoses

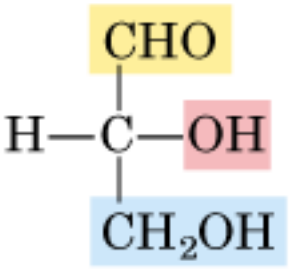
1.1.Stereoisomers

The D and L stereoisomers of sugars refer to the configuration of the asymmetric carbon atom **furthest** from the aldehyde or ketone group.

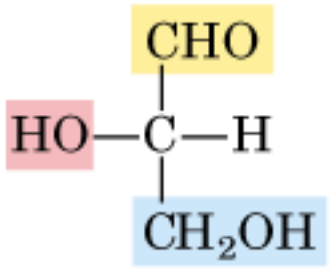




Ball-and-stick models

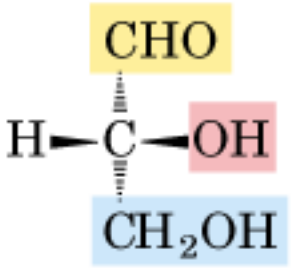


D-Glyceraldehyde

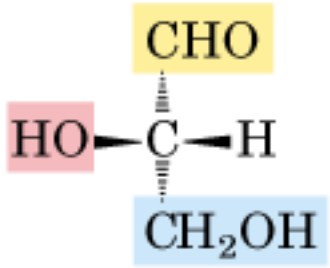


L-Glyceraldehyde

Fischer projection formulas



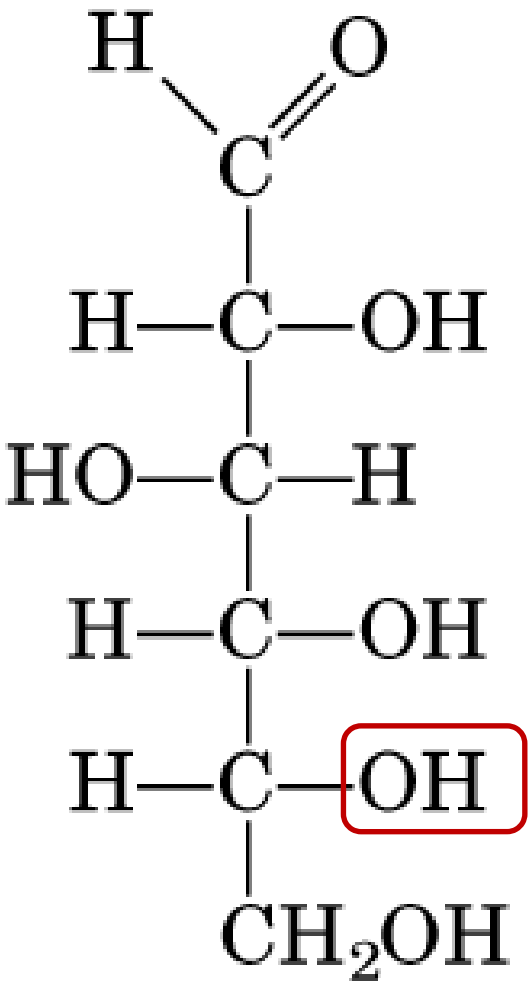
D-Glyceraldehyde



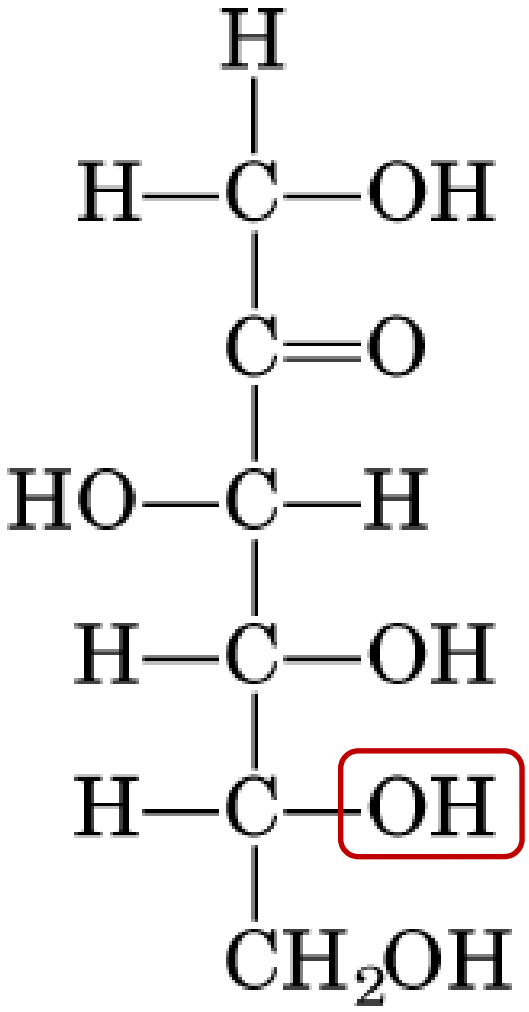
L-Glyceraldehyde

Perspective formulas

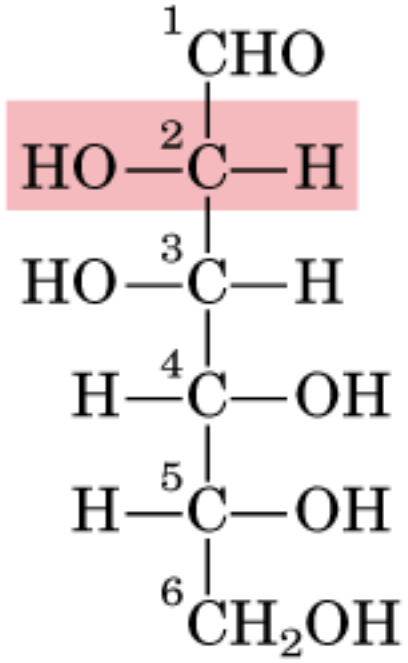




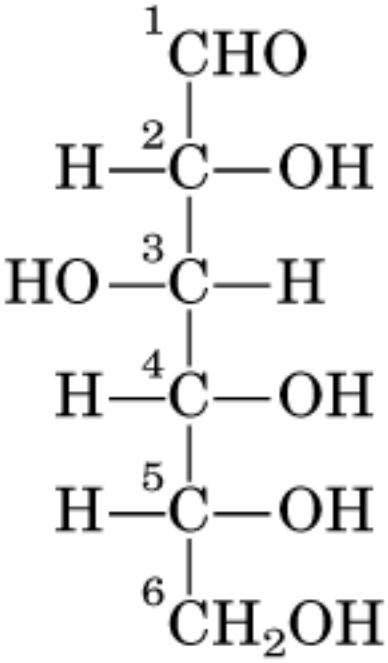
D-Glucose,
an aldohexose



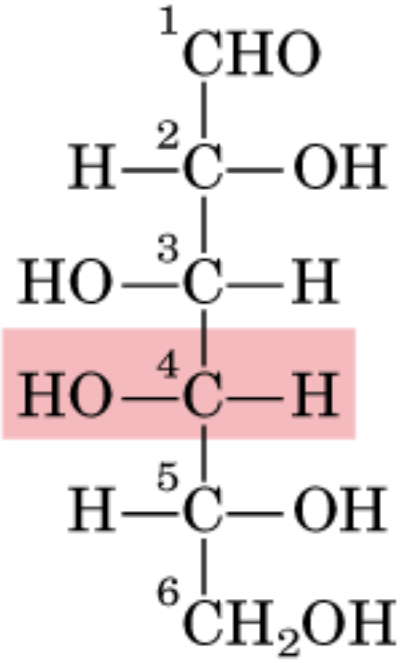
D-Fructose,
a ketohexose



D-Mannose
(epimer at C-2)



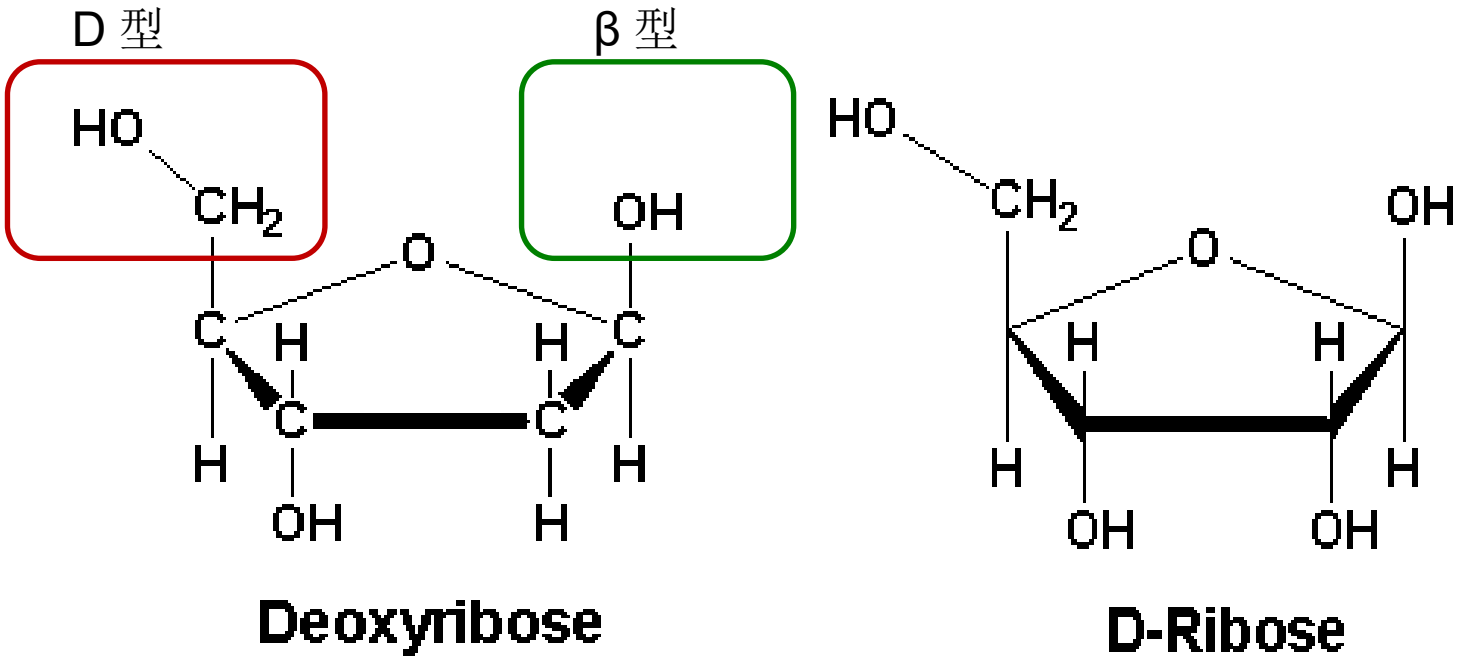
D-Glucose



D-Galactose
(epimer at C-4)

Two sugars that differ only in the configuration around one carbon atom are called **epimer.**

Deoxyribose and Ribose

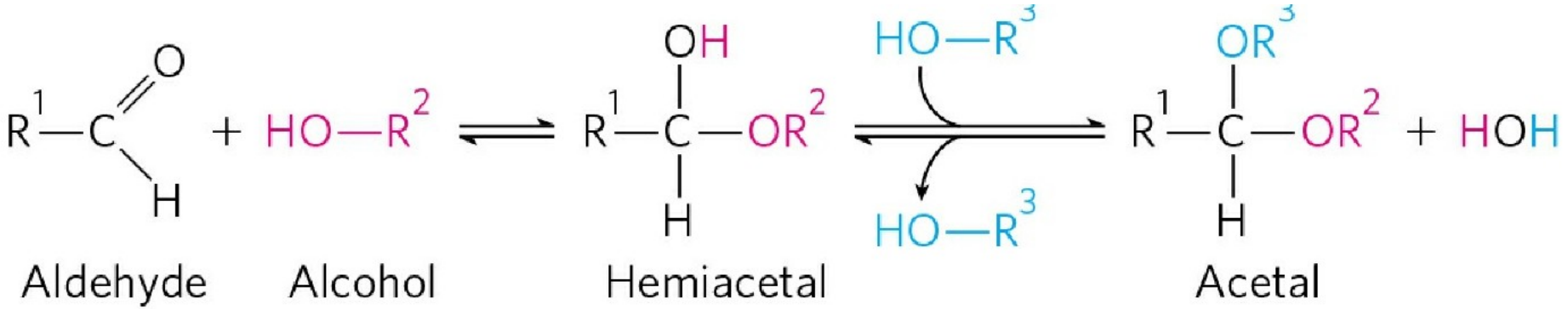


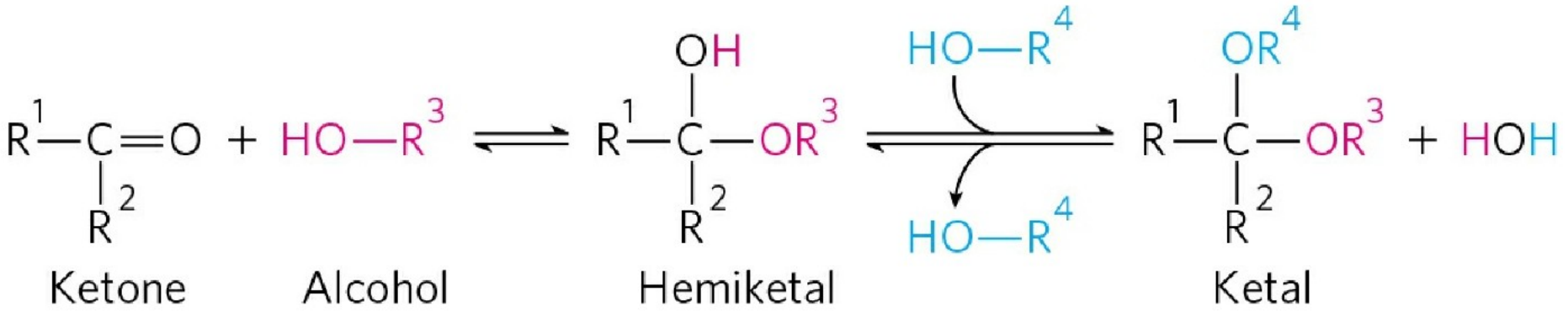
Isomeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called anomers, and the carbonyl carbon atom is called the anomeric carbon.

1.2 Ring structure

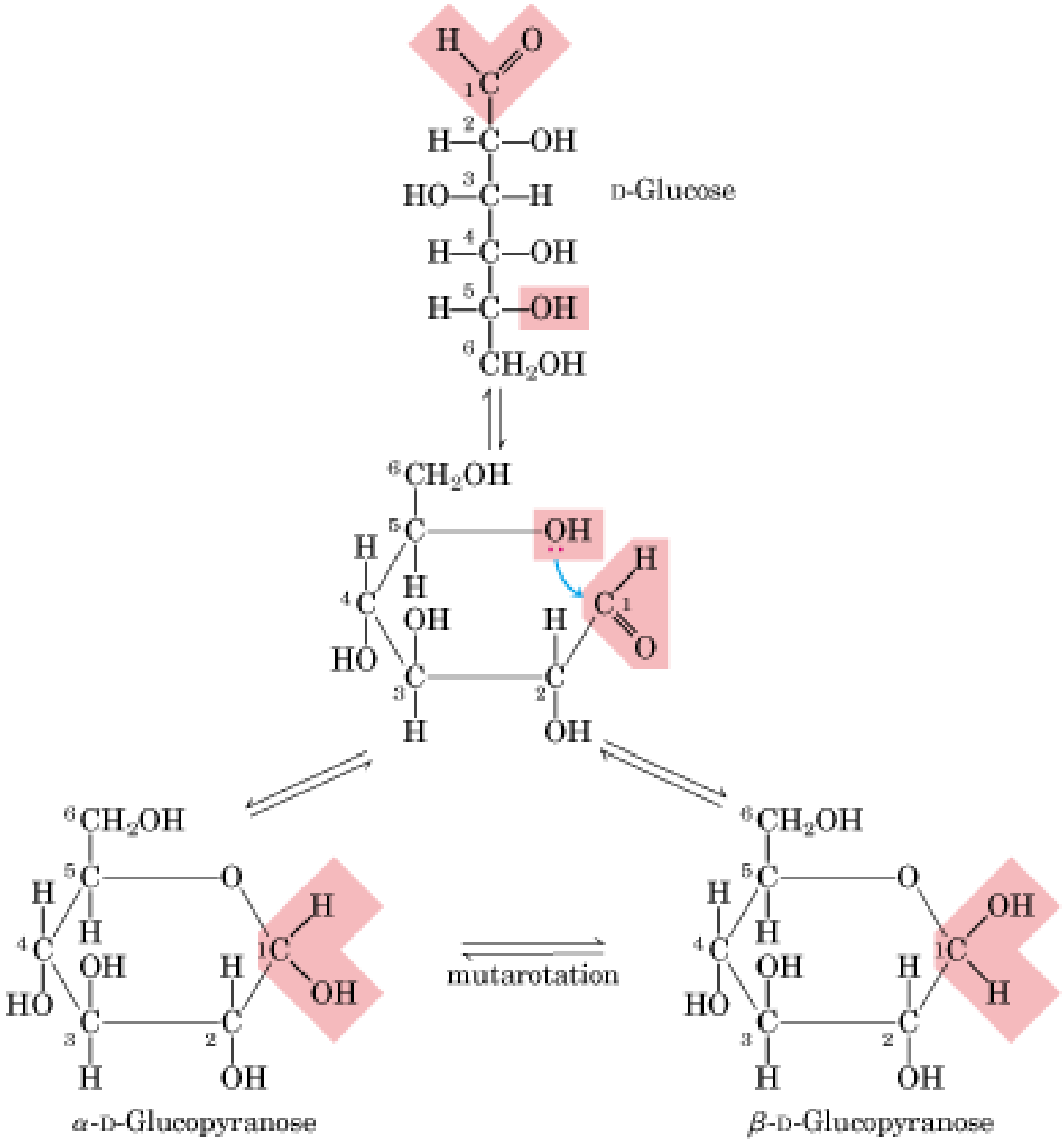
Aldotetroses and larger sugars can cyclized by reaction of the aldehyde or ketone group with a hydroxyl group on another carbon atom of the sugar.







Formation of hemiacetals and hemiketals

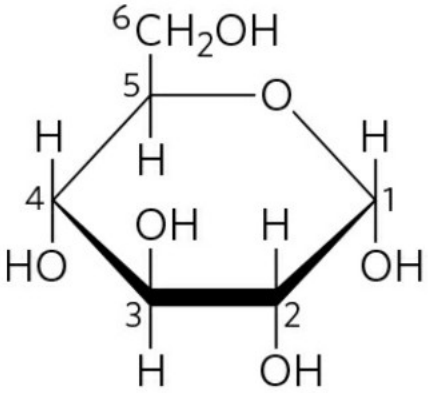


吡喃（型）葡萄糖

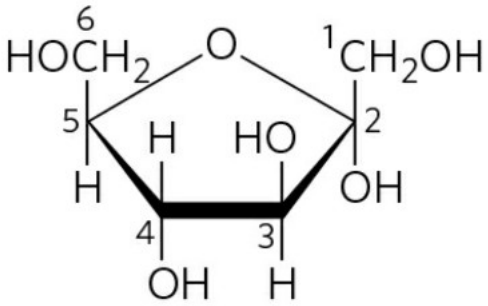
πψρανοσε

hemiacetal

吡喃（型）
葡萄糖

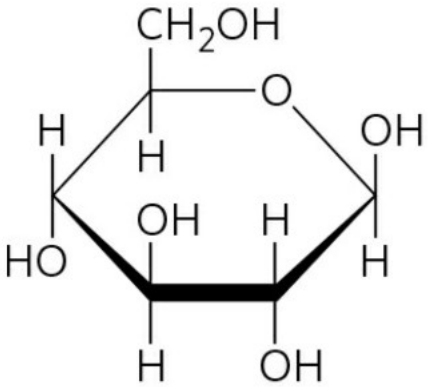


α-D-Glucopyranose

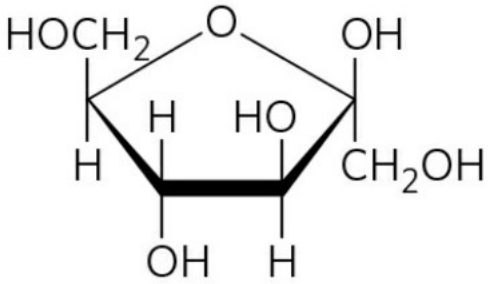


α-D-Fructofuranose

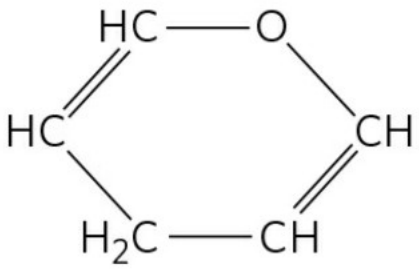
呋喃（型）
果糖



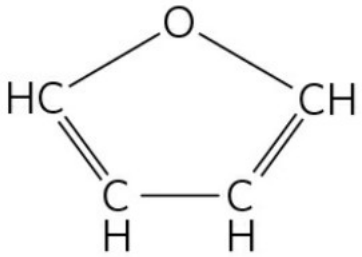
β-D-Glucopyranose



β-D-Fructofuranose



Pyran

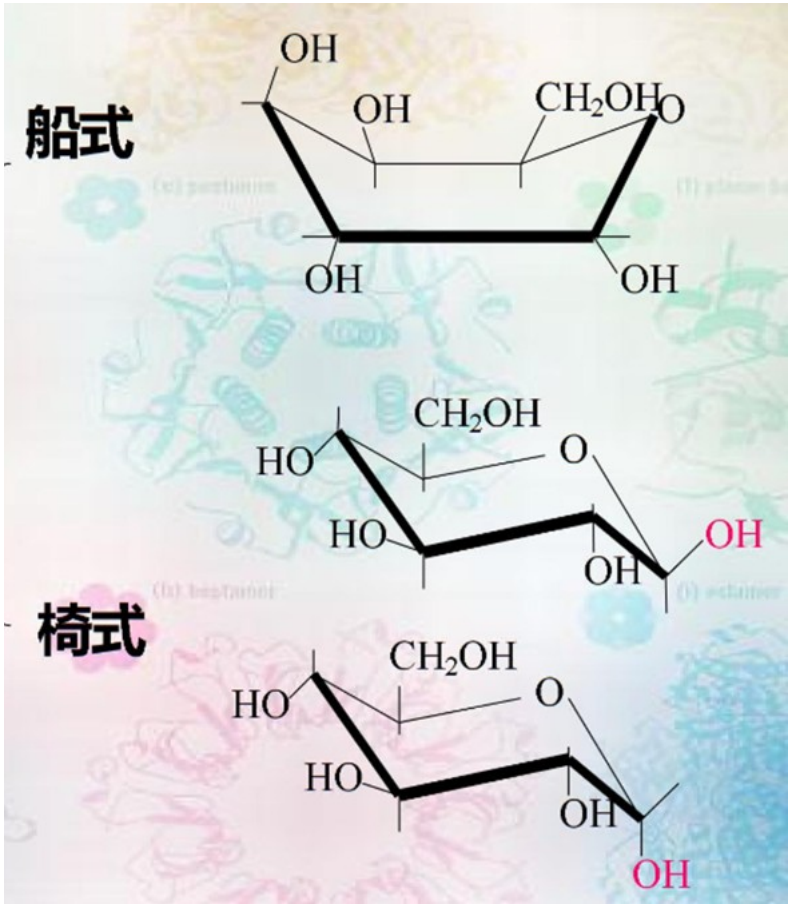


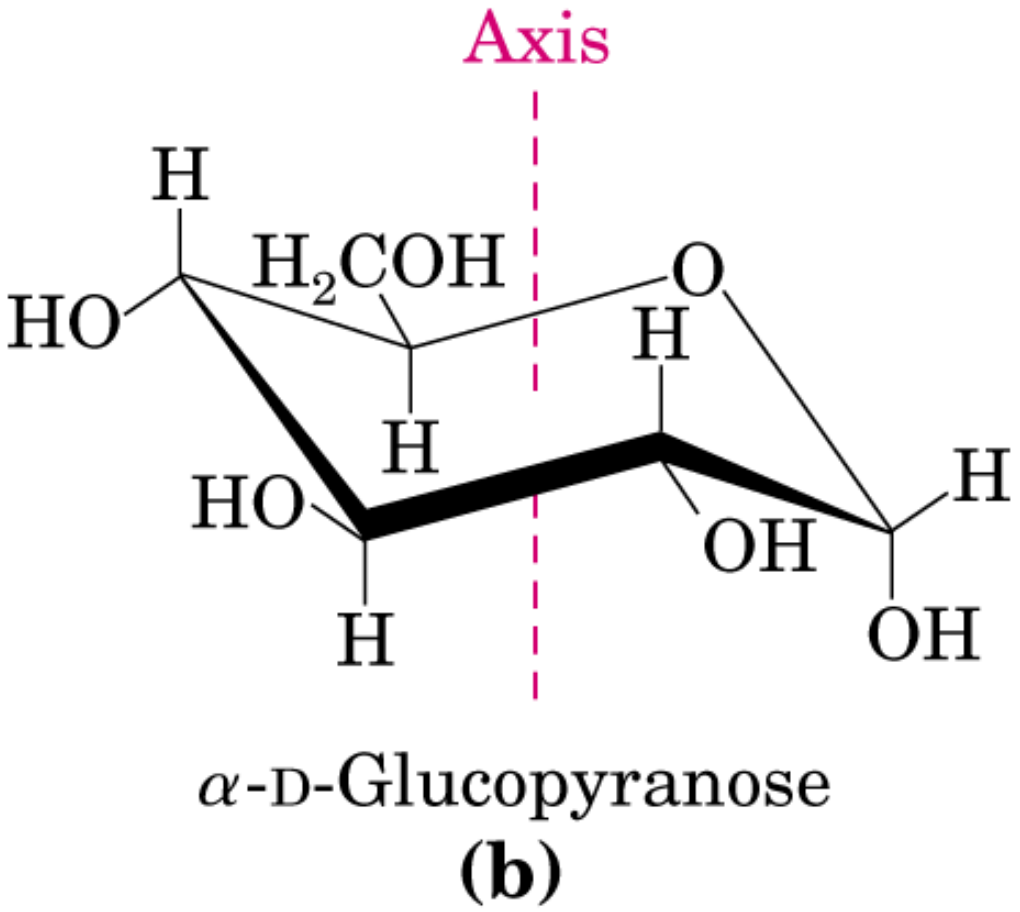
Furan

Biochem



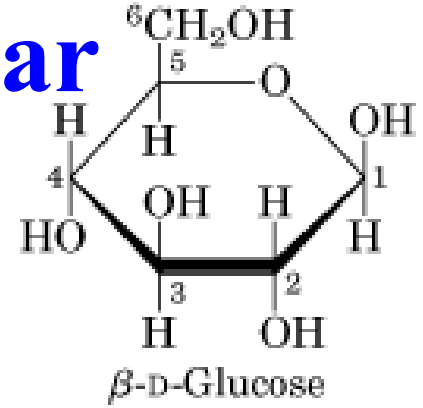
Chair form of the pyranose ring





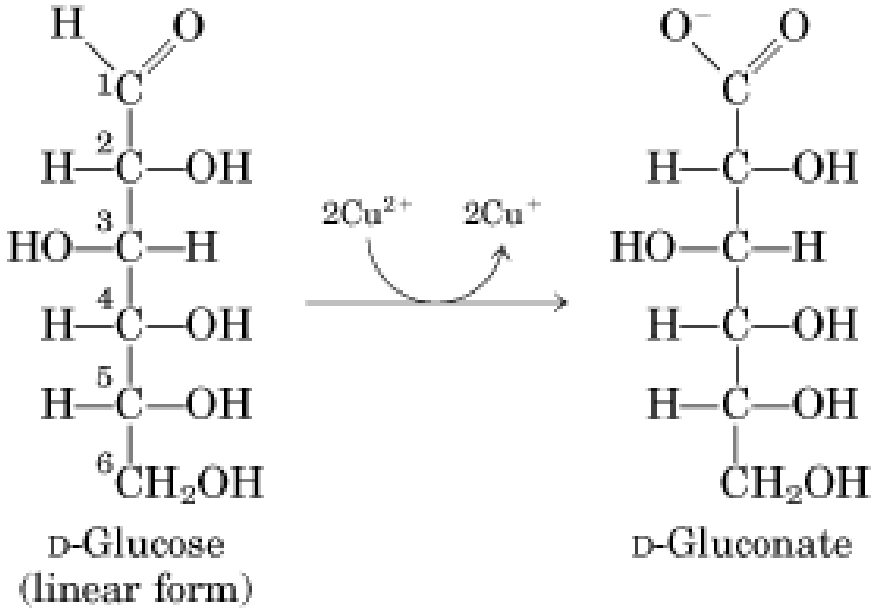
船式不稳定，椅式稳定

1.3 Reducing sugar



β-D-Glucose

A monosaccharide is a reducing sugar.



D-Glucose (linear form)

D-Gluconate

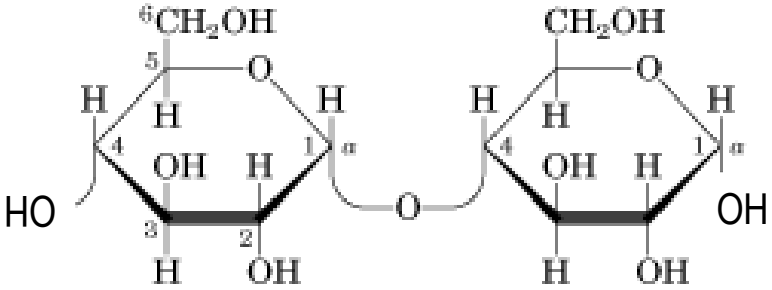
(a)

2. disaccharides

A disaccharide is formed when two monosaccharides become joined by a **glycosidic bond**.



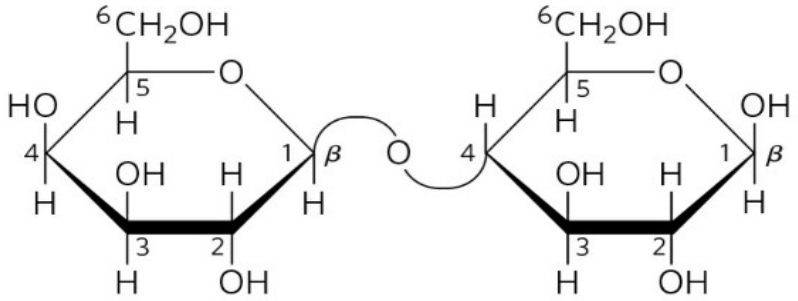
Nonreducing end



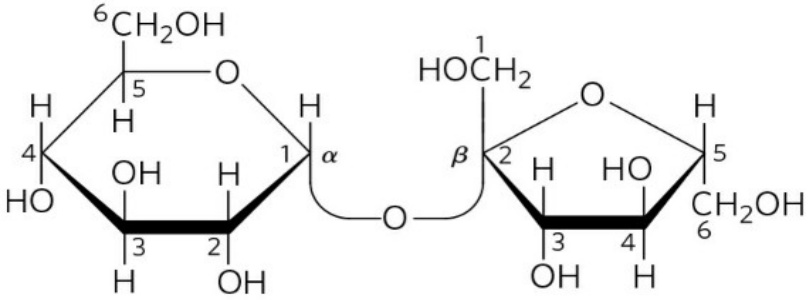
Reducing end

glycosidic bond

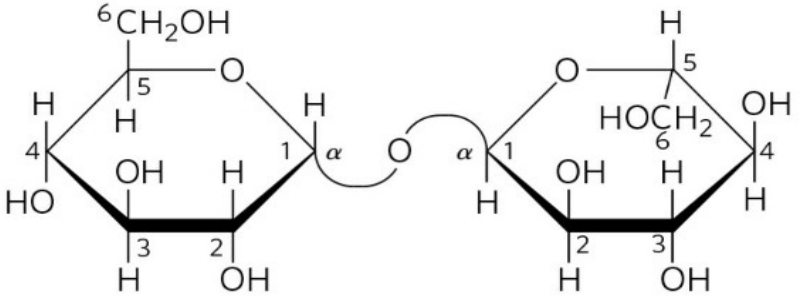
1↔1
1↔2
1→3
1→4
1→6
for glucose



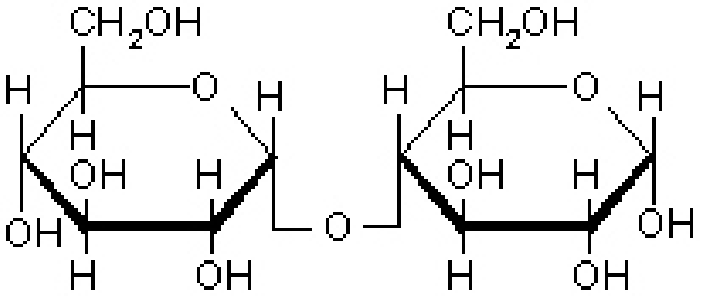
Lactose (β form)
β-D-galactopyranosyl-(1→4)-β-D-glucopyranose
Gal(β1→4)Glc



Sucrose
β-D-fructofuranosyl α-D-glucopyranoside
Fru(2β↔α1)Glc ≡ Glc(α1↔2β)Fru



Trehalose
α-D-glucopyranosyl α-D-glucopyranoside
Glc(α1↔1α)Glc



Maltose (α 1→4)

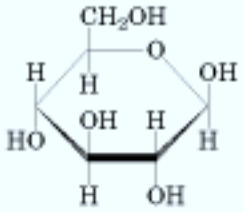


3. Sugar derivatives

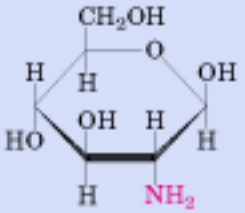
The hydroxyl groups of sugars can be replaced by other groups to form a wide range of biologically important molecules including phosphorylated sugars, amino sugars and nucleotides.



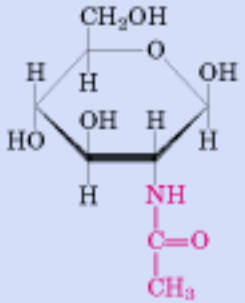
Glucose family



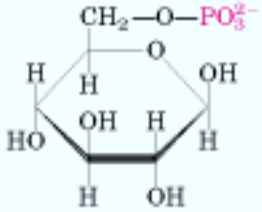
β-D-Glucose



β-D-Glucosamine



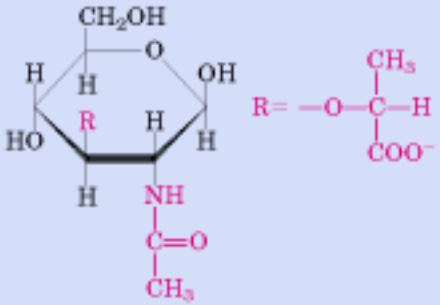
N-Acetyl-β-D-glucosamine



β-D-Glucose 6-phosphate

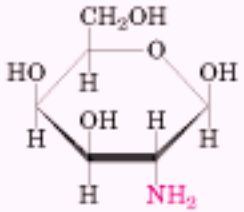


Muramic acid



N-Acetylmuramic acid

Amino sugars

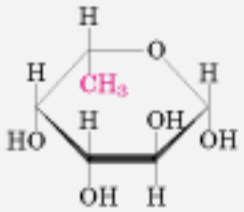


β-D-Galactosamine

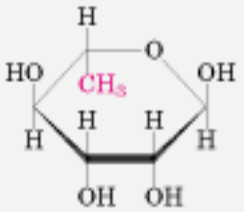


β-D-Mannosamine

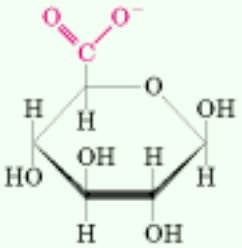
Deoxy sugars



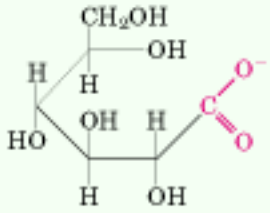
β-L-Fucose



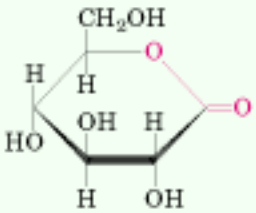
α-L-Rhamnose



β-D-Glucuronate

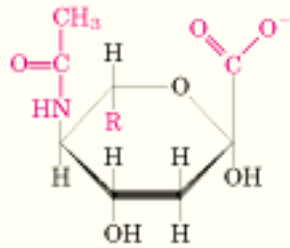


D-Gluconate

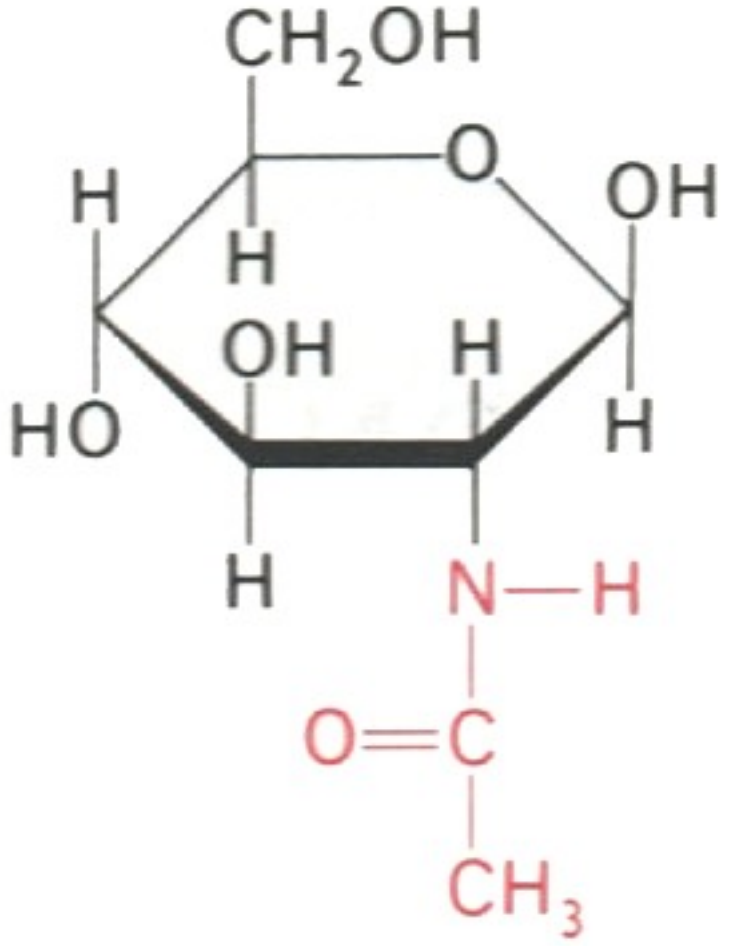


D-Glucono-δ-lactone

Acidic sugars



N-Acetylneuraminic acid
(sialic acid)

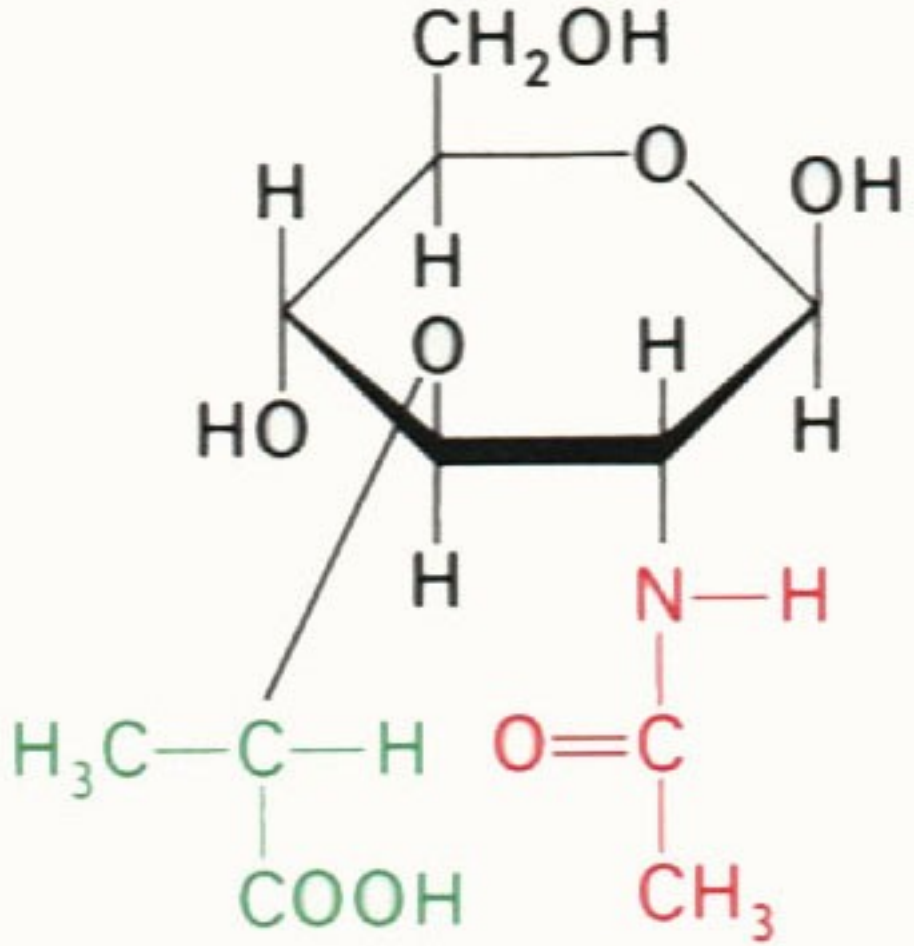


N-Acetylglucosamine
(NAG)

Biochemistry

N— 乙酰氨基葡萄糖

NAG



N-Acetylmuramic acid
(NAM)

N— 乙 酰 胞 壁 酸

Recall: Proteins are Glycosylated in ER and Golgi

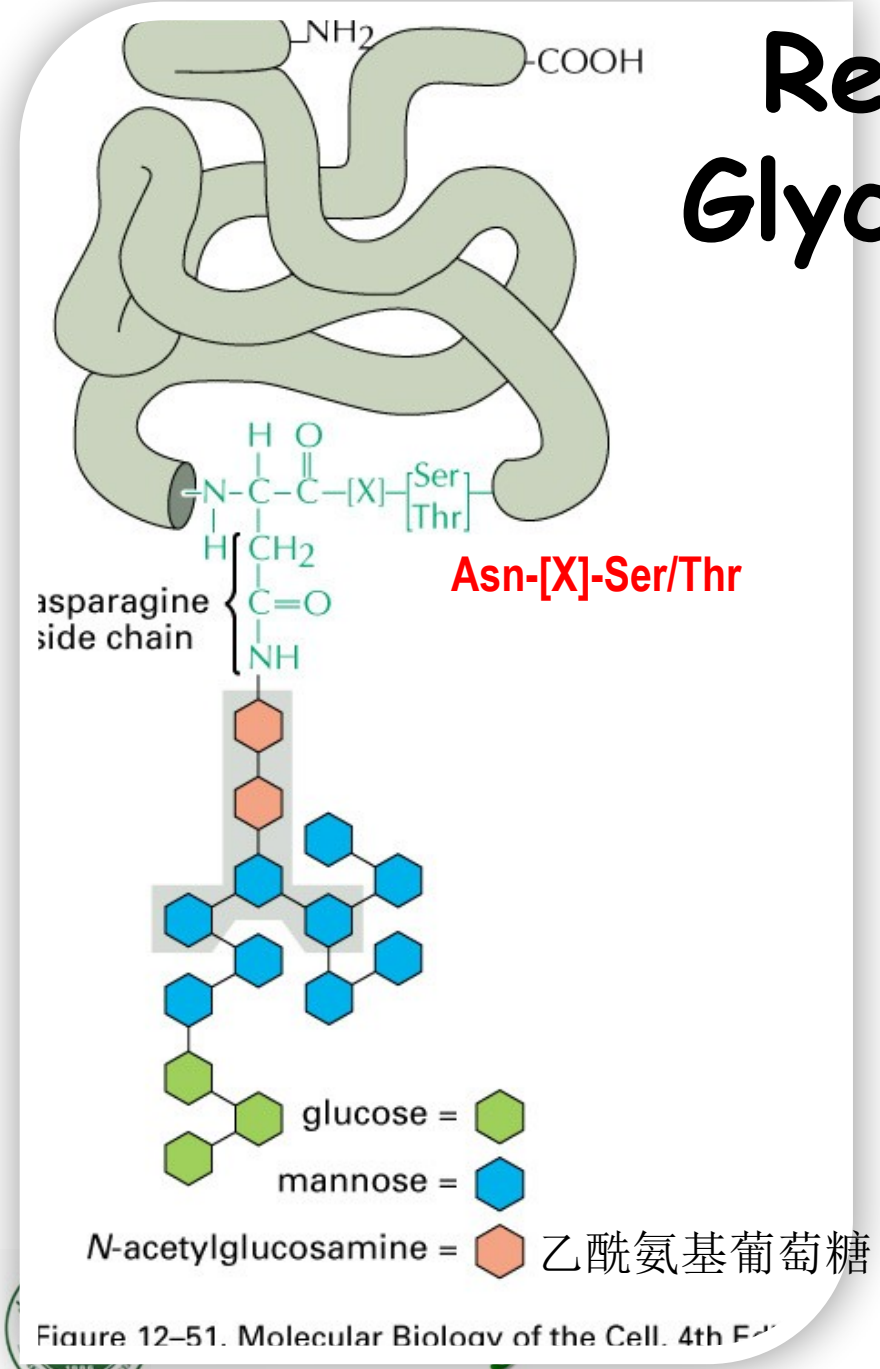


Figure 12–51. Molecular Biology of the Cell, 4th Ed.

-important for folding

-important for function

-important for targeting



4. nomenclature

The names of simple sugars and sugar derivatives can all be abbreviated.

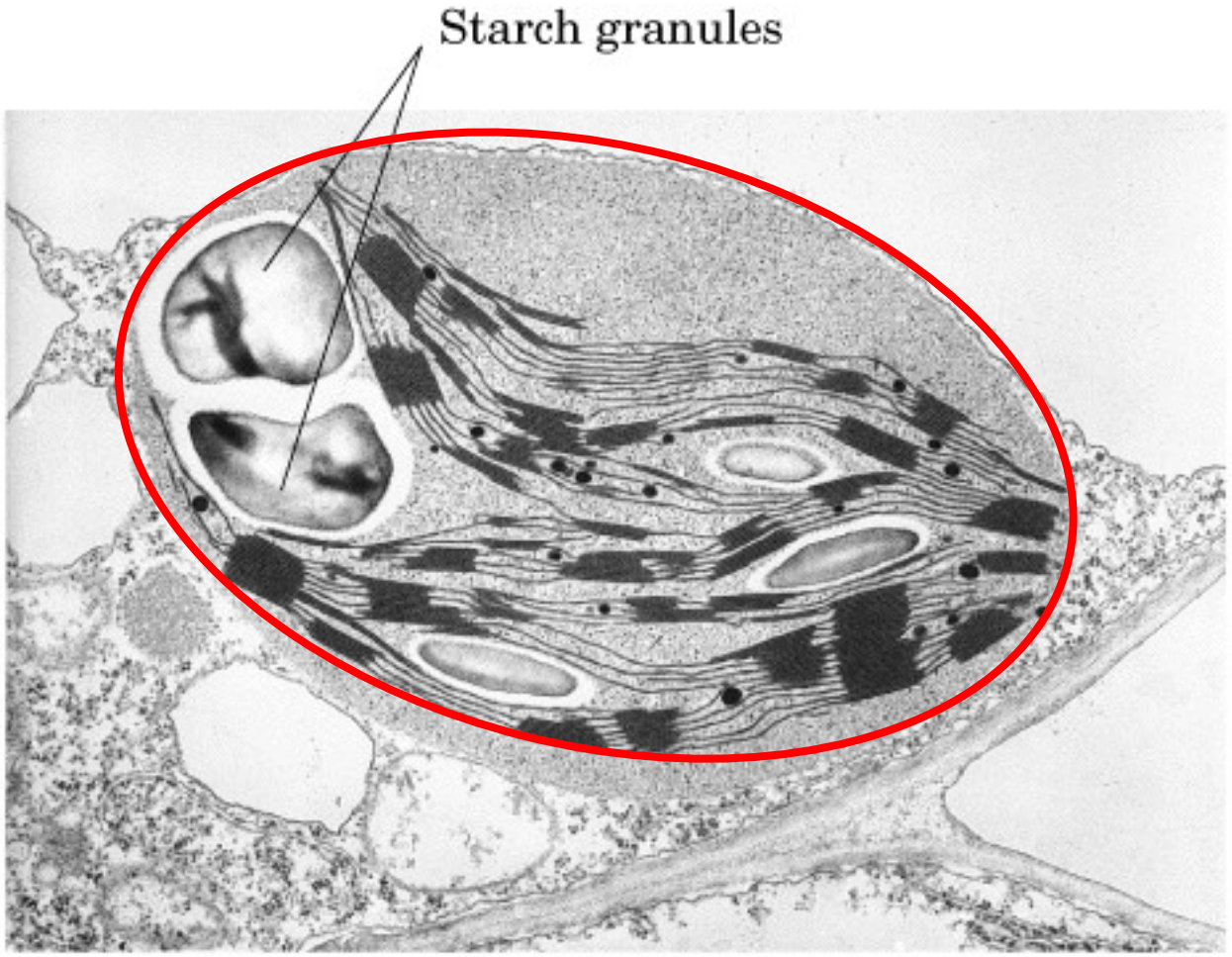
Glc	glucose
Fru	fructose



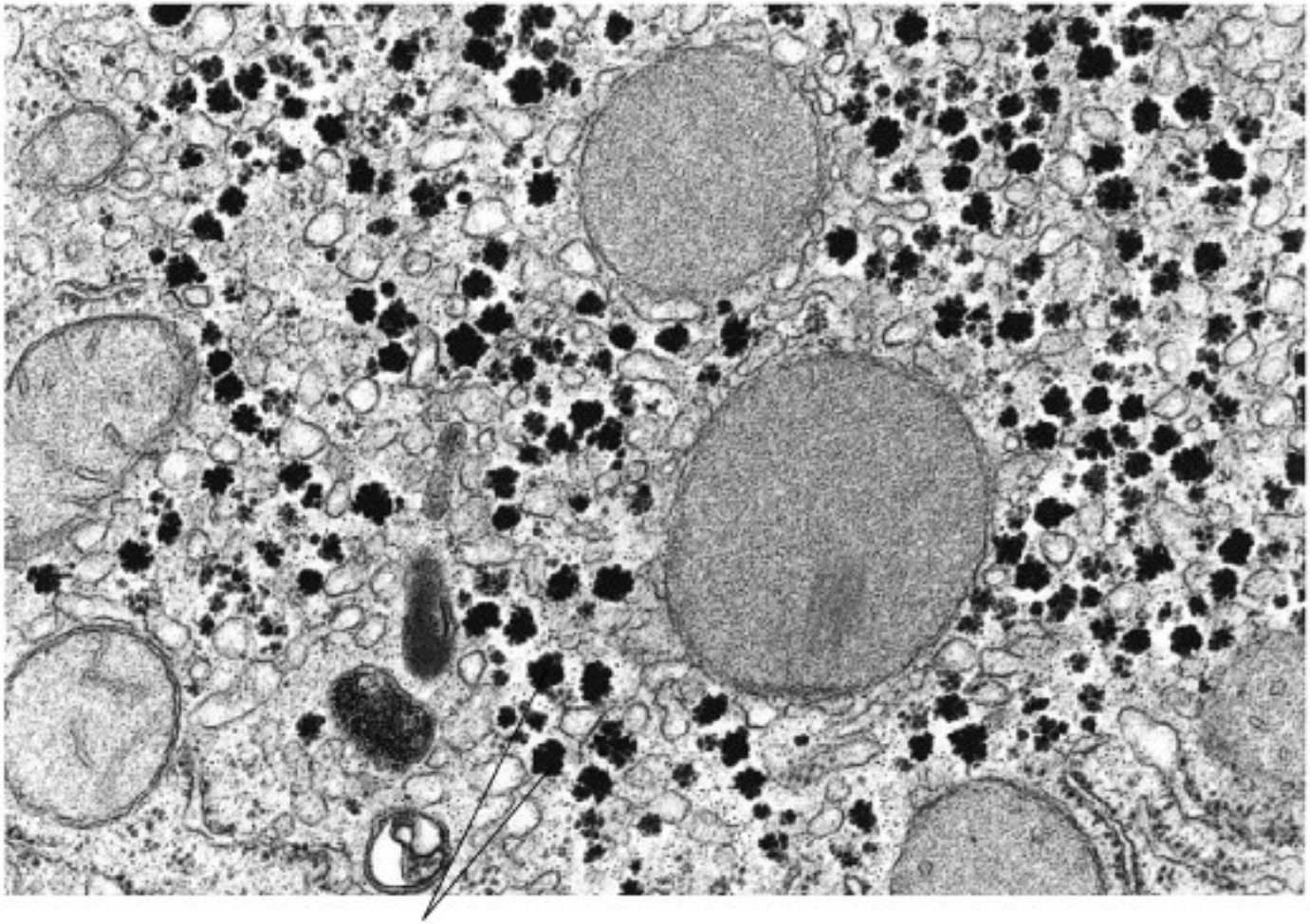
J2 Polysaccharides and oligosaccharides

Long chains of monosacharides joined together are collectively called polysaccharides. The major storage polysaccharides are **glycogen**, **starch** 。 and **dextran**. **Cellulose** is a structural polysaccharide found in plant cell wall.





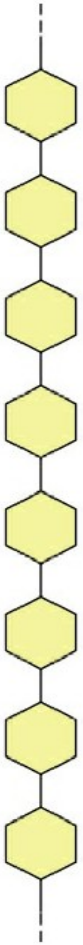
(a)



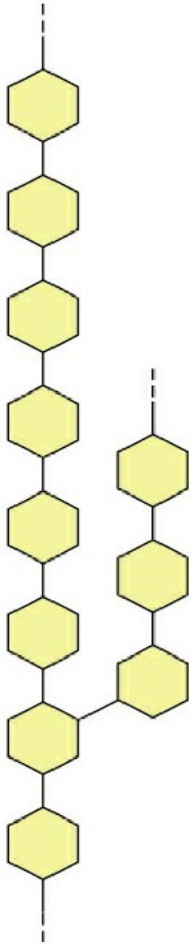
(b)

Homopolysaccharides

Unbranched

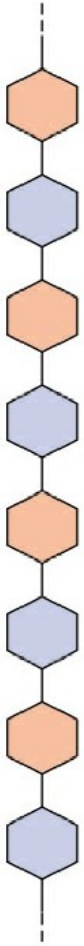


Branched

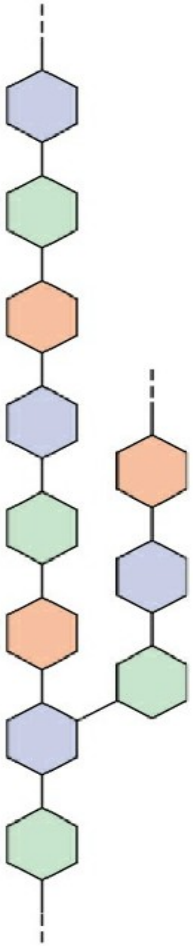


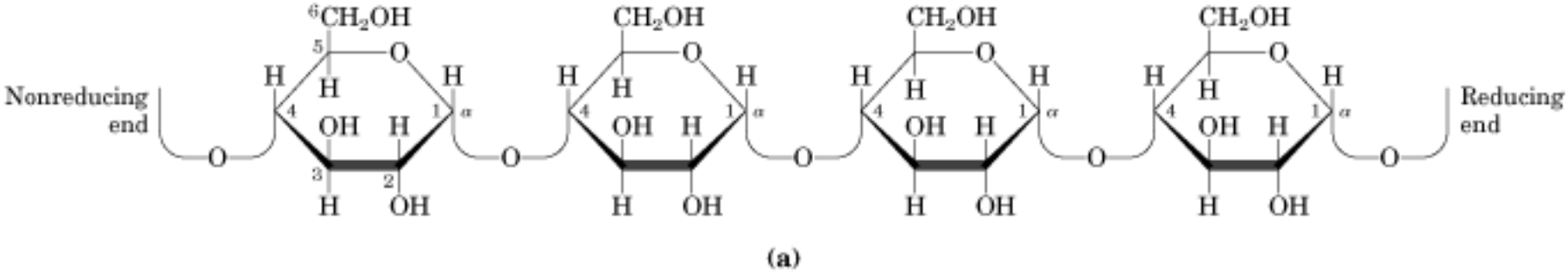
Heteropolysaccharides

Two monomer types, unbranched



Multiple monomer types, branched

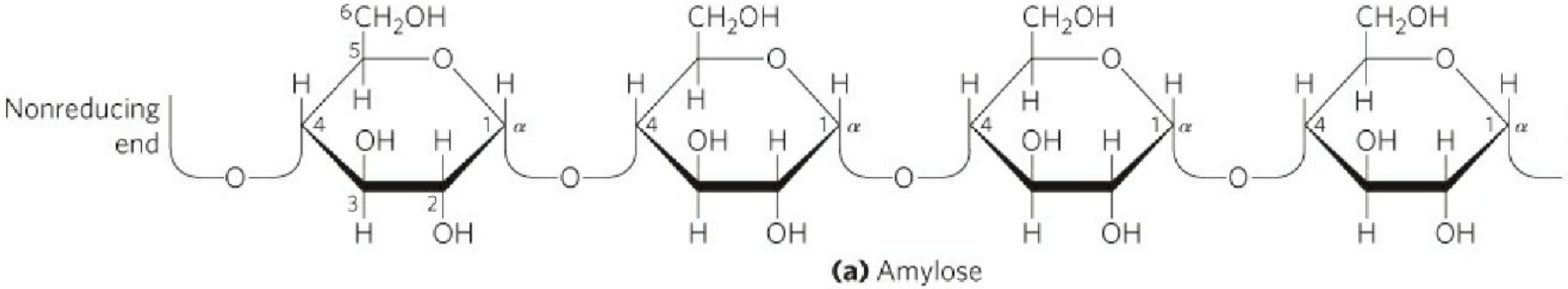


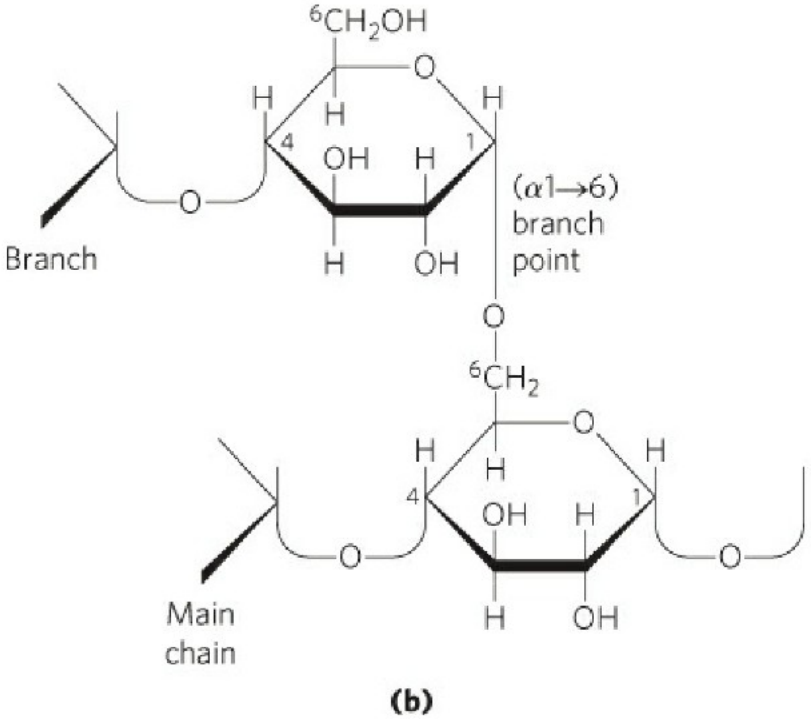


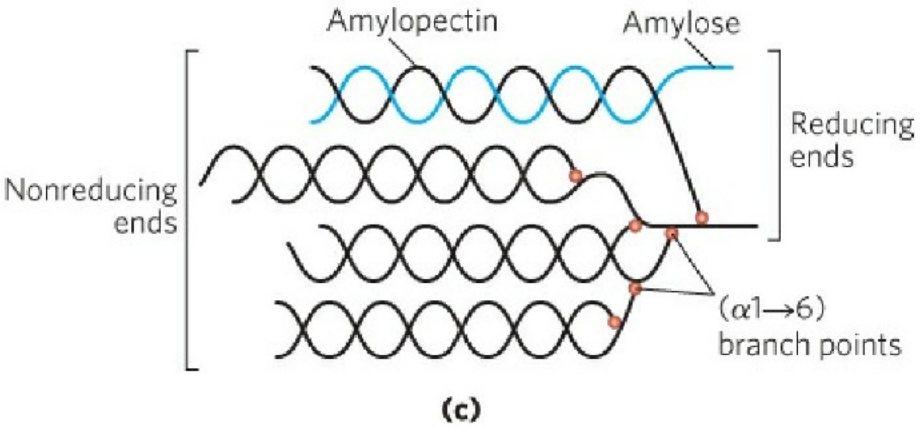
glycosidic bond

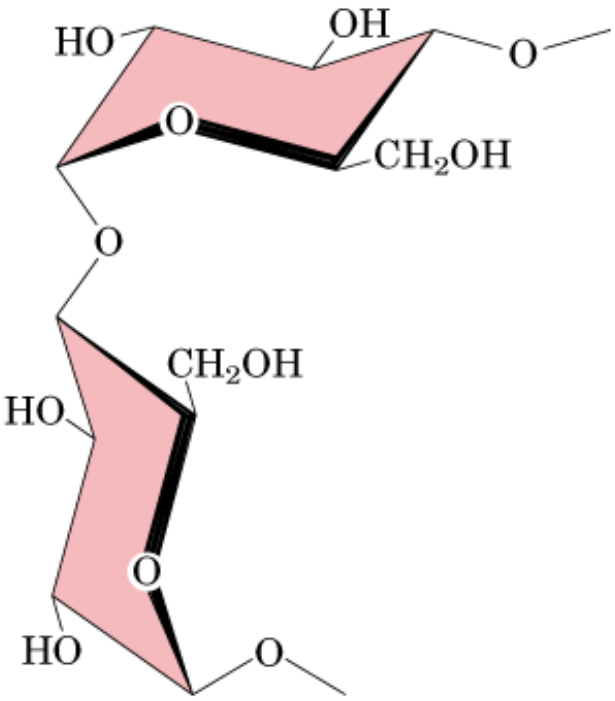
Starch: amylose, amylopectin in plants
glycogen in animals

α-D-glucose as monomer



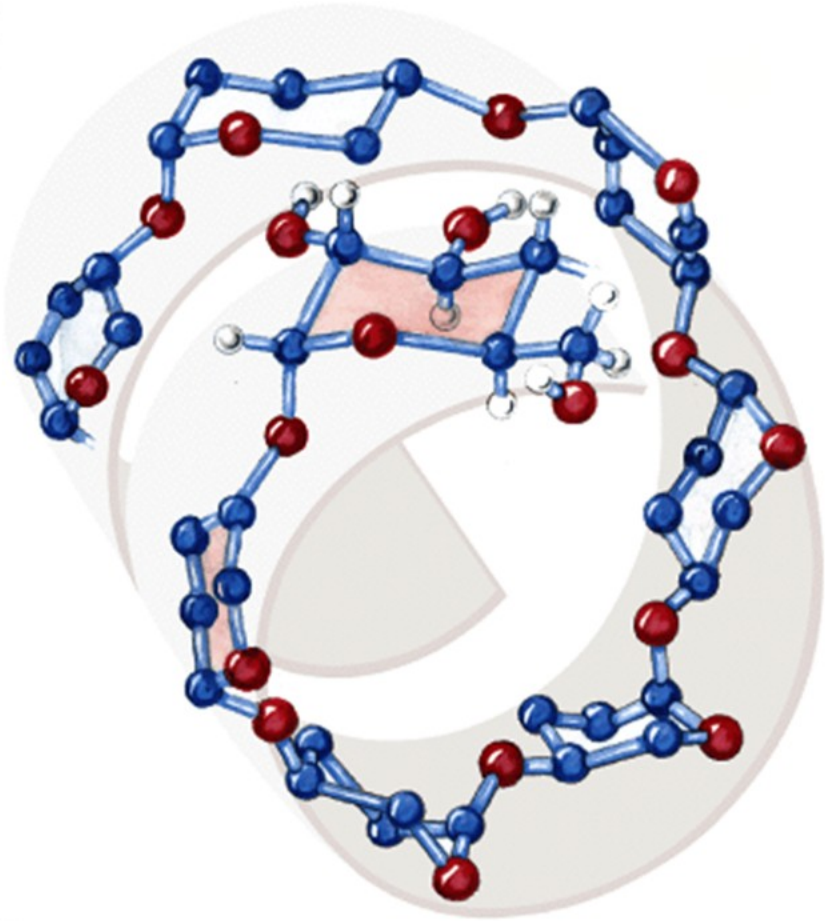




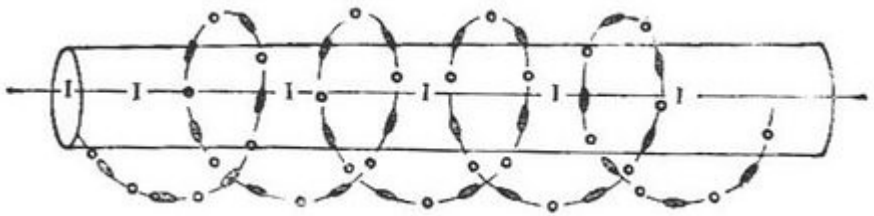


(α1→4)-linked D-glucose units

(a)



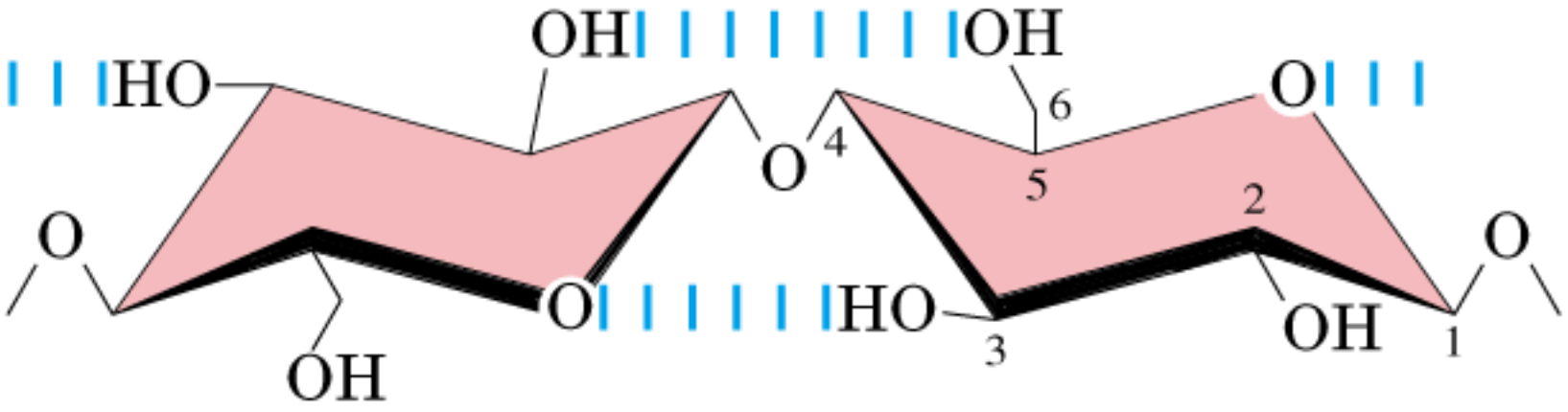
(b)



starch

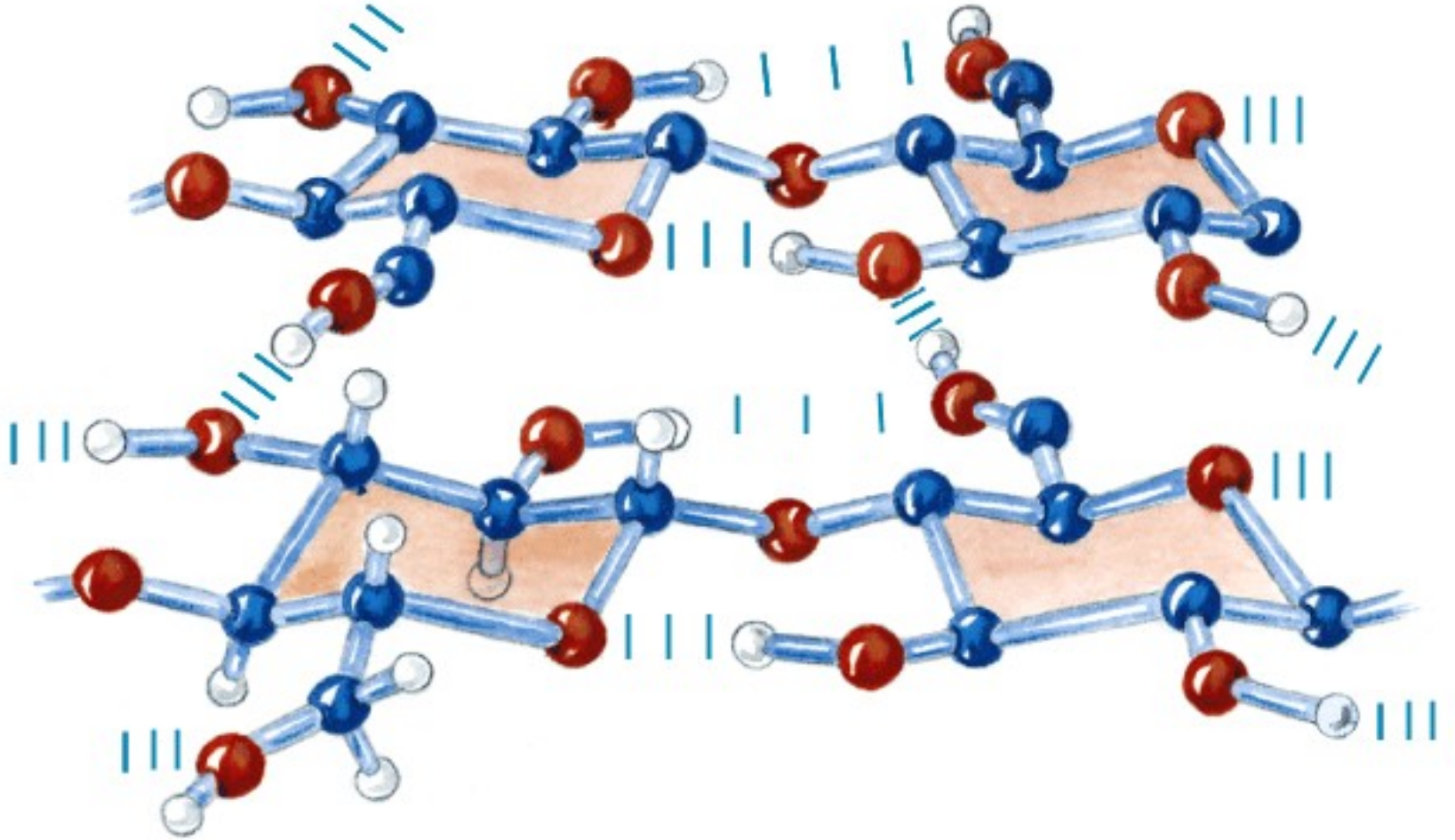
Biochemistry

β-D-glucose as monomer



(β1→4)-linked D-glucose units

(a)



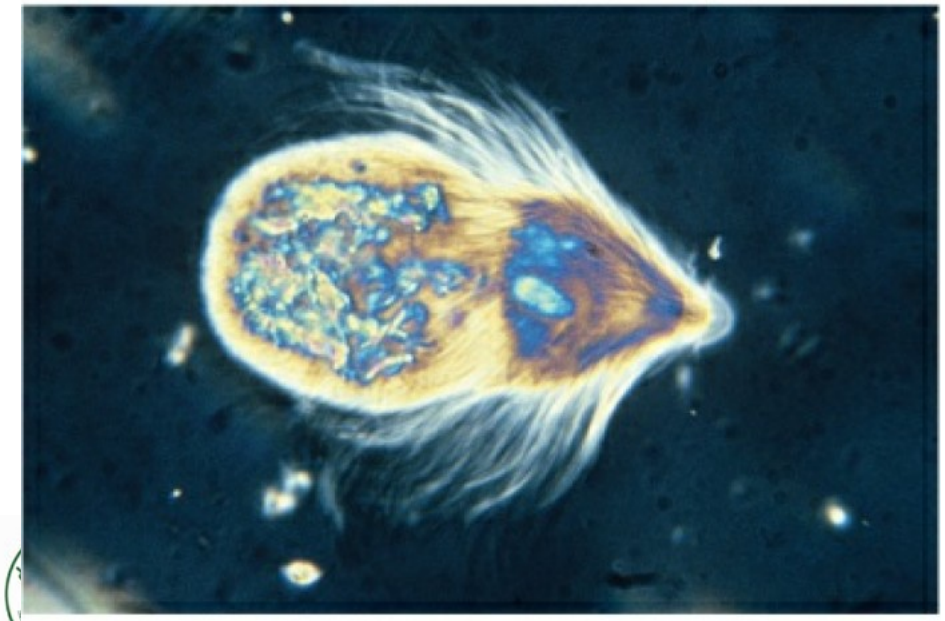
(b)

cellulose



(a)

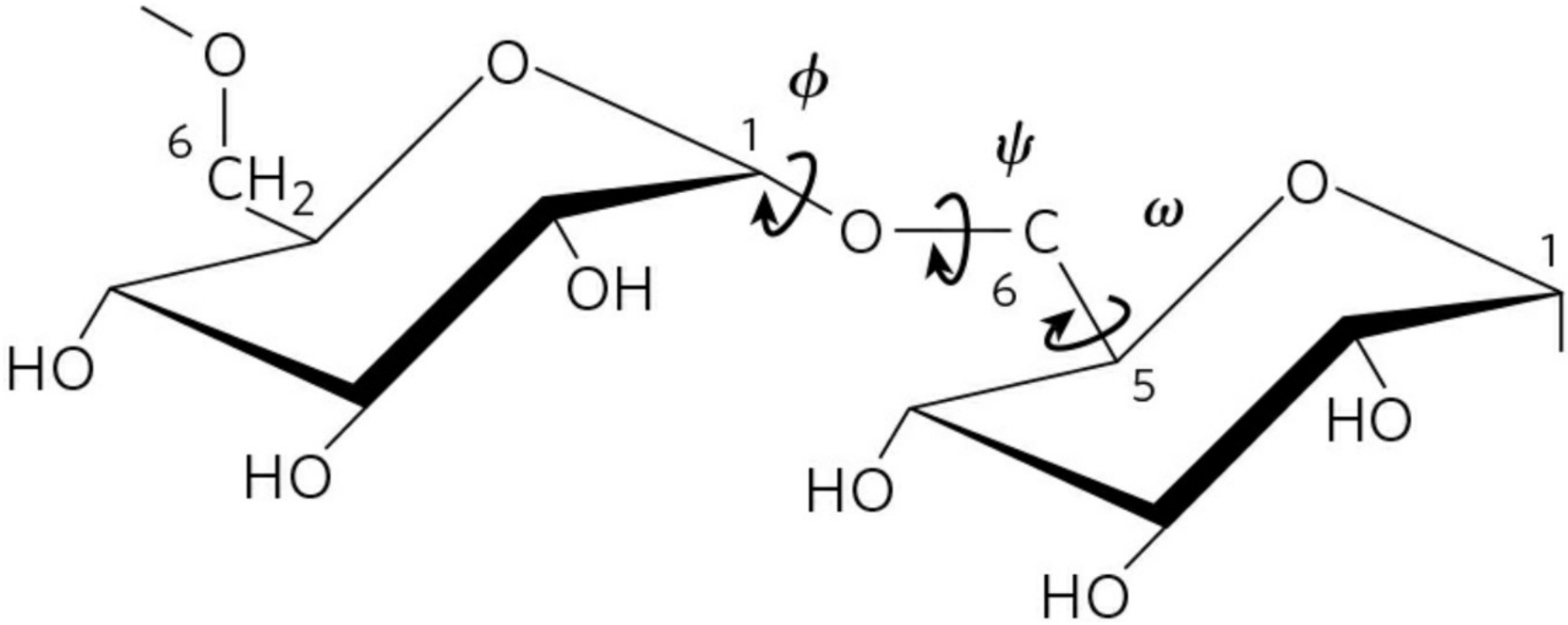
Termite



(b)

Trichonympha

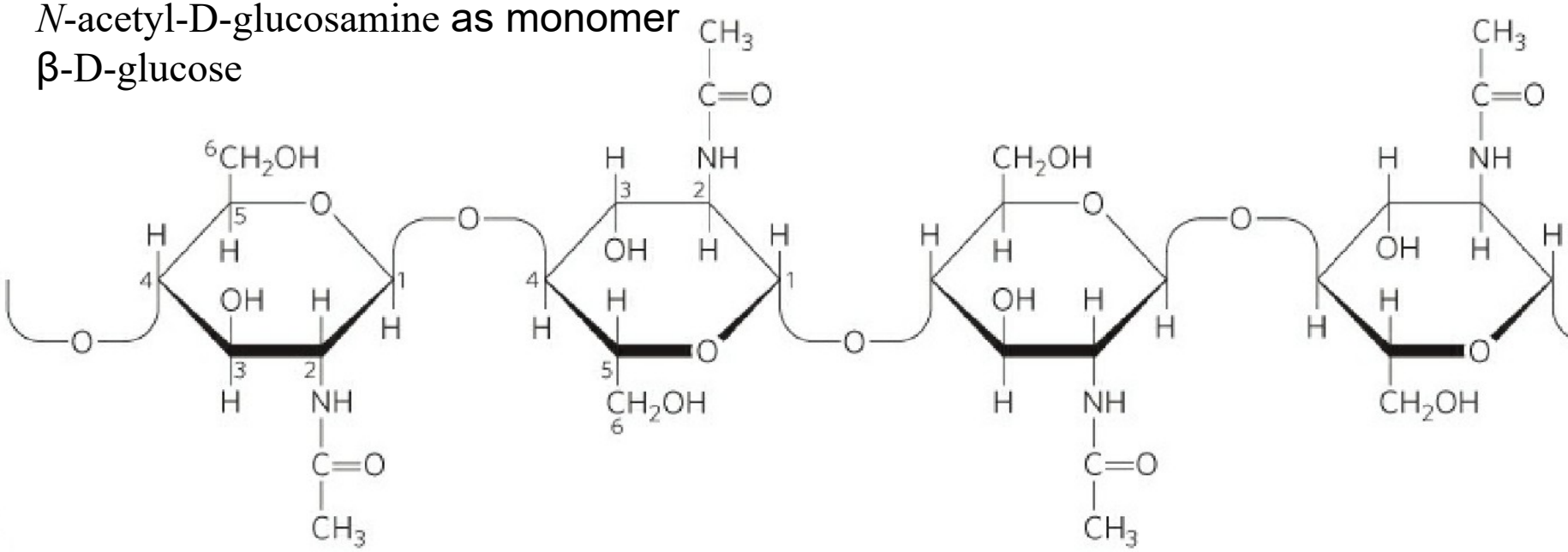
α-D-glucose as monomer



Dextran

(α1→6)Glc repeats (with (α1→3) branches, not shown)

N-acetyl-D-glucosamine as monomer
β-D-glucose



(a)

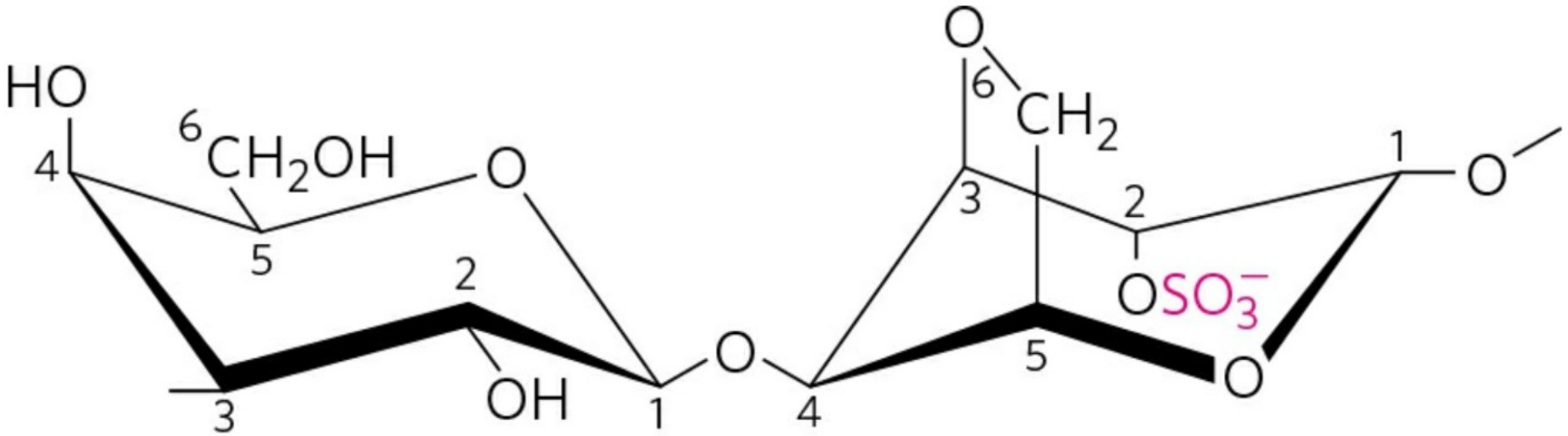


(b)

Chitin

a homopolymer of *N*-acetyl-D-glucosamine units in (β1→4) linkage.

β-D-galactose， *3,6-andydro α -L-galactose*， as monomer

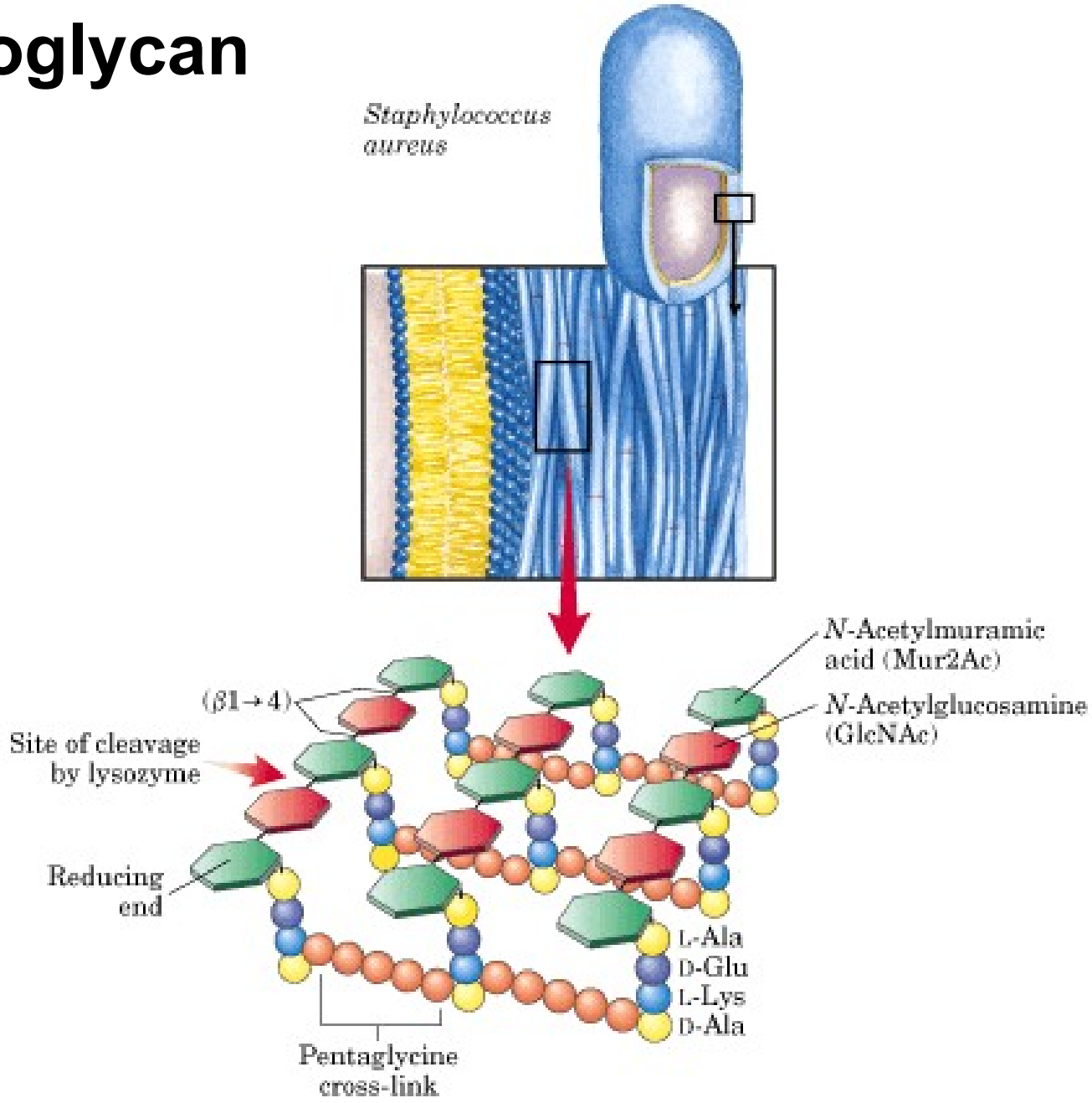


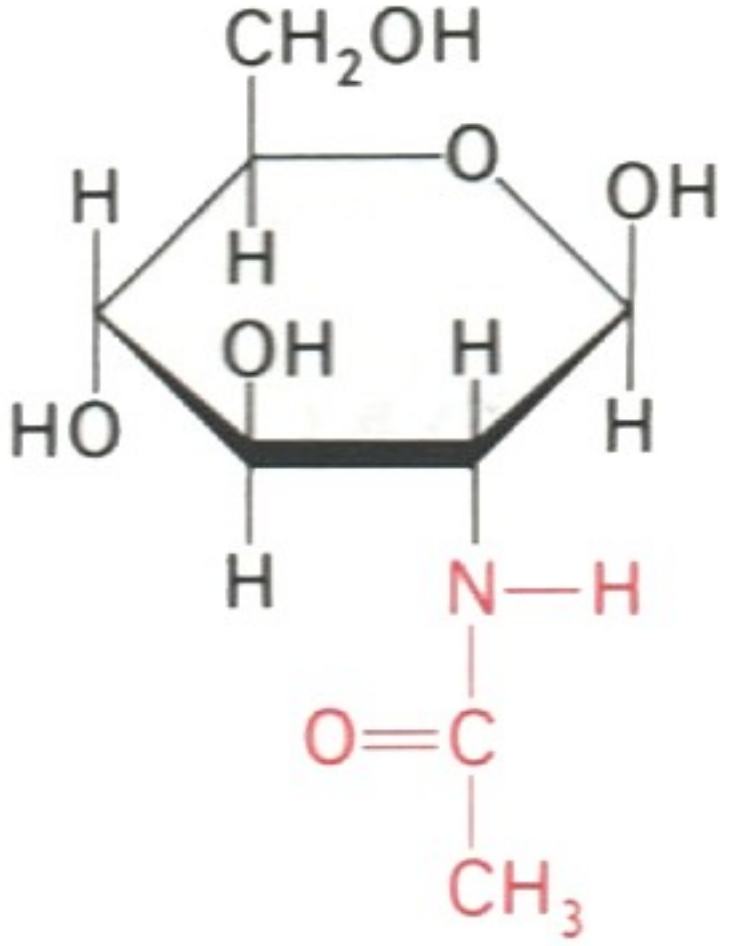
Agarose

3)D-Gal(β1→4)3,6-anhydro-L-Gal2S (α1

repeating units

peptidoglycan



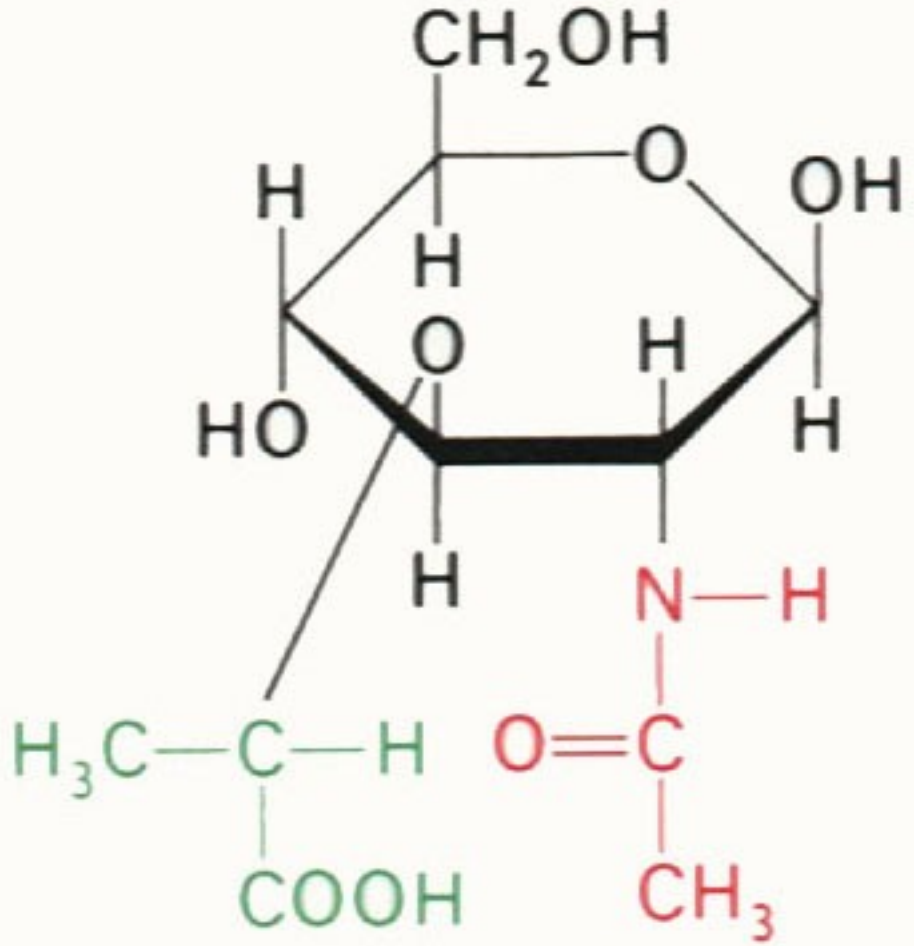


N-Acetylglucosamine
(NAG)

Biochemistry

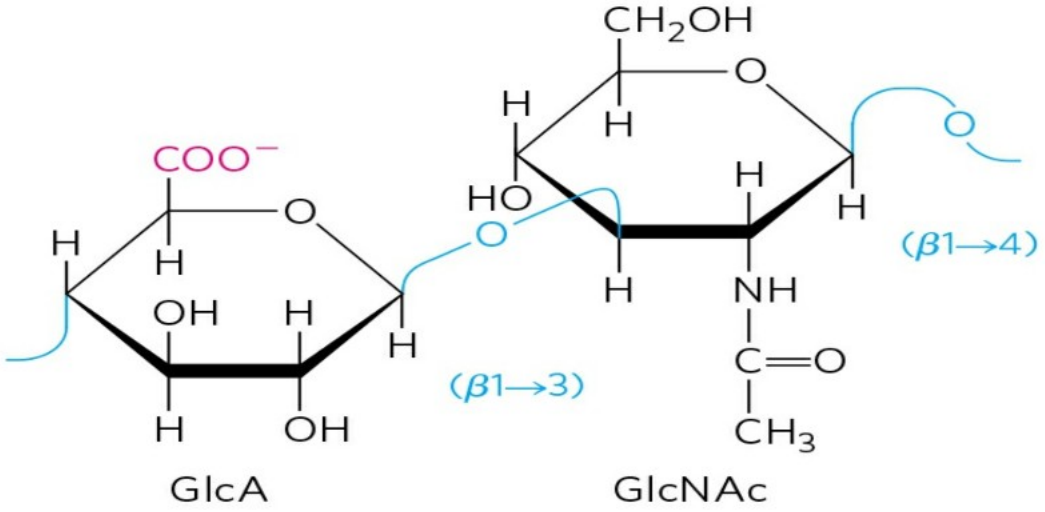
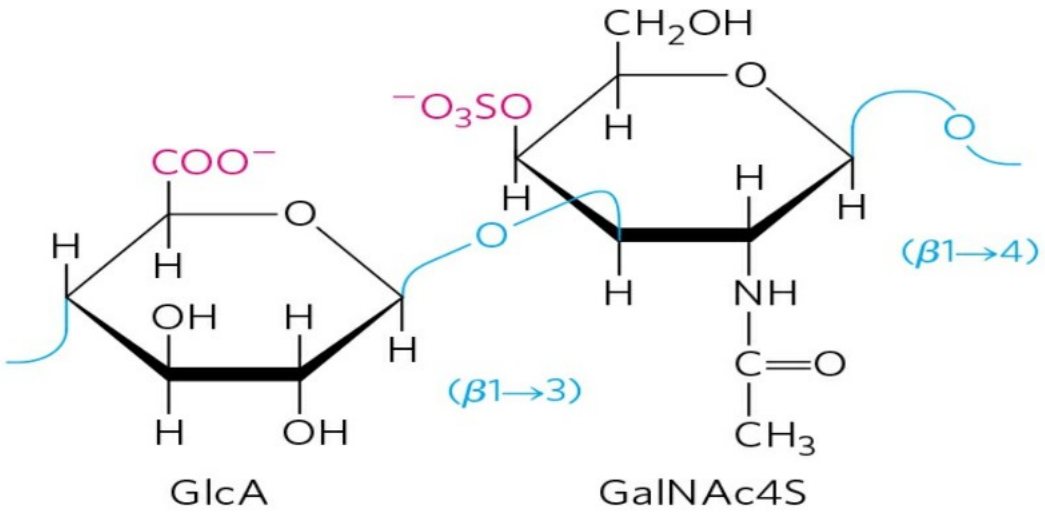
N— 乙酰氨基葡萄糖

NAG

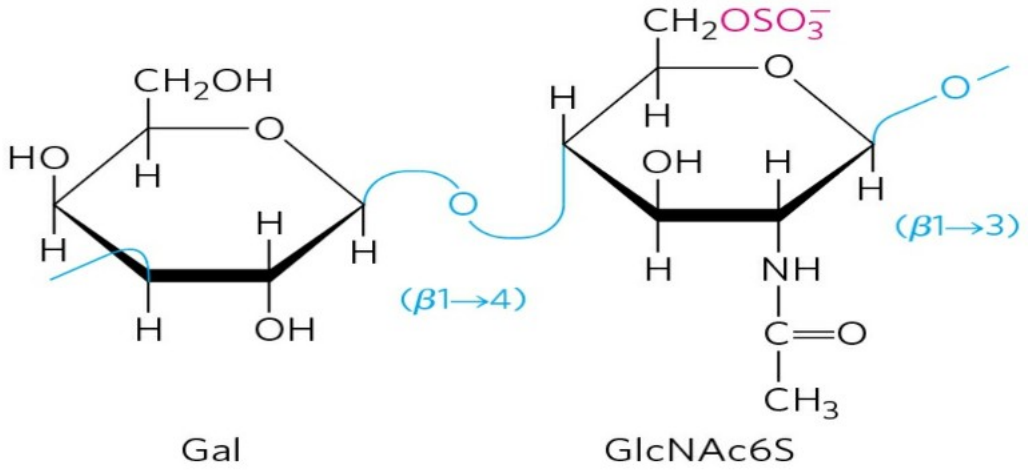


N-Acetylmuramic acid
(NAM)

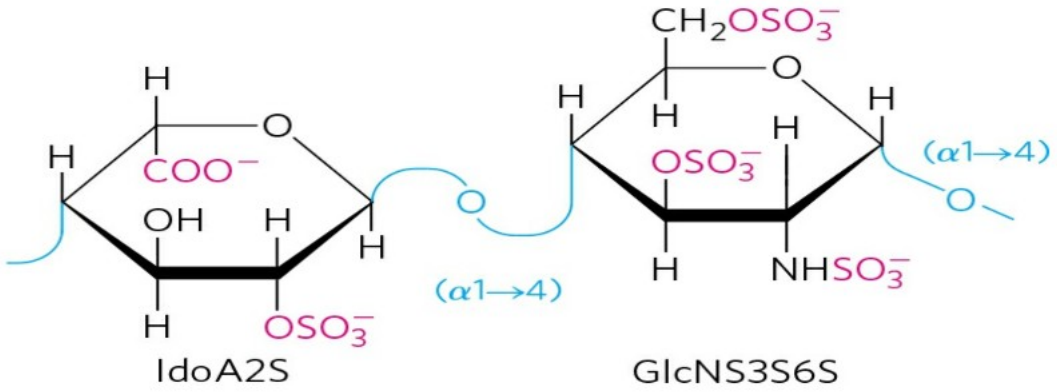
N— 乙 酰 胞 壁 酸

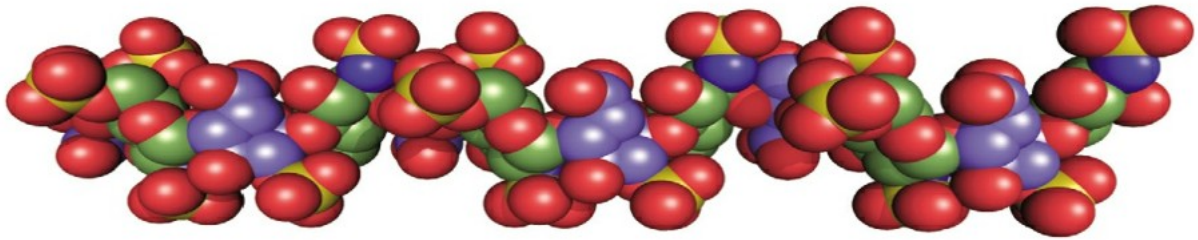
Glycosaminoglycan	Repeating disaccharide
Number of disaccharides per chain	
Hyaluronan ~50,000	
Chondroitin 4-sulfate 20–60	

Keratan sulfate
~25



Heparin
15–90





Heparin segment

TABLE 7-2 Structures and Roles of Some Polysaccharides

Polymer	Type[a]	Repeating unit[b]	Size (number of monosaccharide	Roles/significance
Starch				Energy storage: in plants
Amylose	Homo-	(α1→4) Glc, linear	50–5,000	
Amylopectin	Homo-	(α1→4) Glc, with (α1→6) Glc branches every 24–30 residues	Up to 10^6	
Glycogen	Homo-	(α1→4) Glc, with (α1→6) Glc branches every 8–12 residues	Up to 50,000	Energy storage: in bacteria and animal cells
Cellulose	Homo-	(β1→4) Glc	Up to 15,000	Structural: in plants, gives rigidity and strength to cell walls

Chitin	Homo-	(β1→4) GlcNAc	Very large	Structural: in insects, spiders, crustaceans, gives rigidity and strength to exoskeletons
Dextran	Homo-	(α1→6) Glc, with (α1→3) branches	Wide range	Structural: in bacteria, extracellular adhesive
Peptidoglycan	Hetero-; peptides attached	4)Mur2Ac(β1→4) GlcNAc(β1	Very large	Structural: in bacteria, gives rigidity and strength to cell envelope
Agarose	Hetero-	3)D-Gal (β1→4)3,6-anhydro-L-Gal(α1	1,000	Structural: in algae, cell wall material
Hyaluronan (a glycosamino-glycan)	Hetero-; acidic	4)GlcA (β1→3) GlcNAc(β1	Up to 100,000	Structural: in vertebrates, extracellular matrix of skin and connective tissue; viscosity and lubrication in joints

Carbohydrate metabolism
(Section J3---Glycolysis)

Glycolysis comes from the Greek *glykys*, meaning “sweet”, and *lysis*, meaning “splitting”.



1. The Development of Biochemistry and the Delineation of Glycolysis

1897, **Eduard Buchner** (Germany), accidental observation : sucrose was rapidly fermented into alcohol by cell-free yeast extract.

The accepted view that fermentation is tied to living cells was shaken and Biochemistry was born: *Metabolism became chemistry*!



1900s, Arthur Harden and William Young separated the yeast juice into two fractions: one heat-labile, nondialyzable enzymes and the other heat-stable, dialyzable *cozymase* (metal ions, ATP, ADP, NAD^+).

1910s-1930s, Gustav Embden and Otto Meyerhof (Germany), studied muscle and its extracts:

many reactions of lactic acid (muscle) and alcohol (yeast) fermentations were the same!

- Glycolysis was also known as ***Embden-Meyerhof pathway*** and was elucidated by the 1940s.





The Nobel Prize in Chemistry 1907

"for his biochemical researches and his discovery of cell-free fermentation"



Eduard Buchner

Germany

Landwirtschaftliche Hochschule (Agricultural College)
Berlin, Germany

1860 - 1917

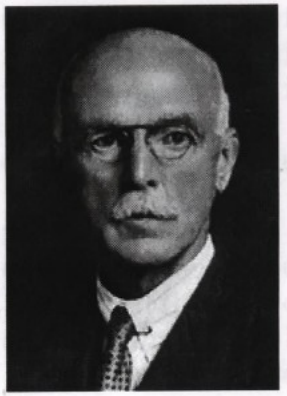




The Nobel Prize in Chemistry 1929

"for their investigations on the fermentation of sugar and fermentative enzymes"

Presentation Speech



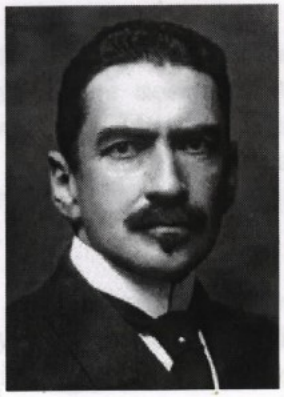
Sir Arthur Harden

Great Britain

London University
London, Great Britain

1865 - 1940

Biography



Hans von Euler-Chelpin

Sweden

Stockholm University
Stockholm, Sweden

1873 - 1964

Biography
Swedish Nobel Stamps



The Nobel Prize in Physiology or Medicine 1922

Presentation Speech

"for his discovery relating to the production of heat in the muscle"



Sir Archibald Vivian Hill

Great Britain

London University
London, Great Britain

1886 - 1977

Biography

"for his discovery of the fixed relationship between the consumption of oxygen and the metabolism of lactic acid in the muscle"



Otto Fritz Meyerhof

Germany

Kiel University
Kiel, Germany

1884 - 1951

Biography

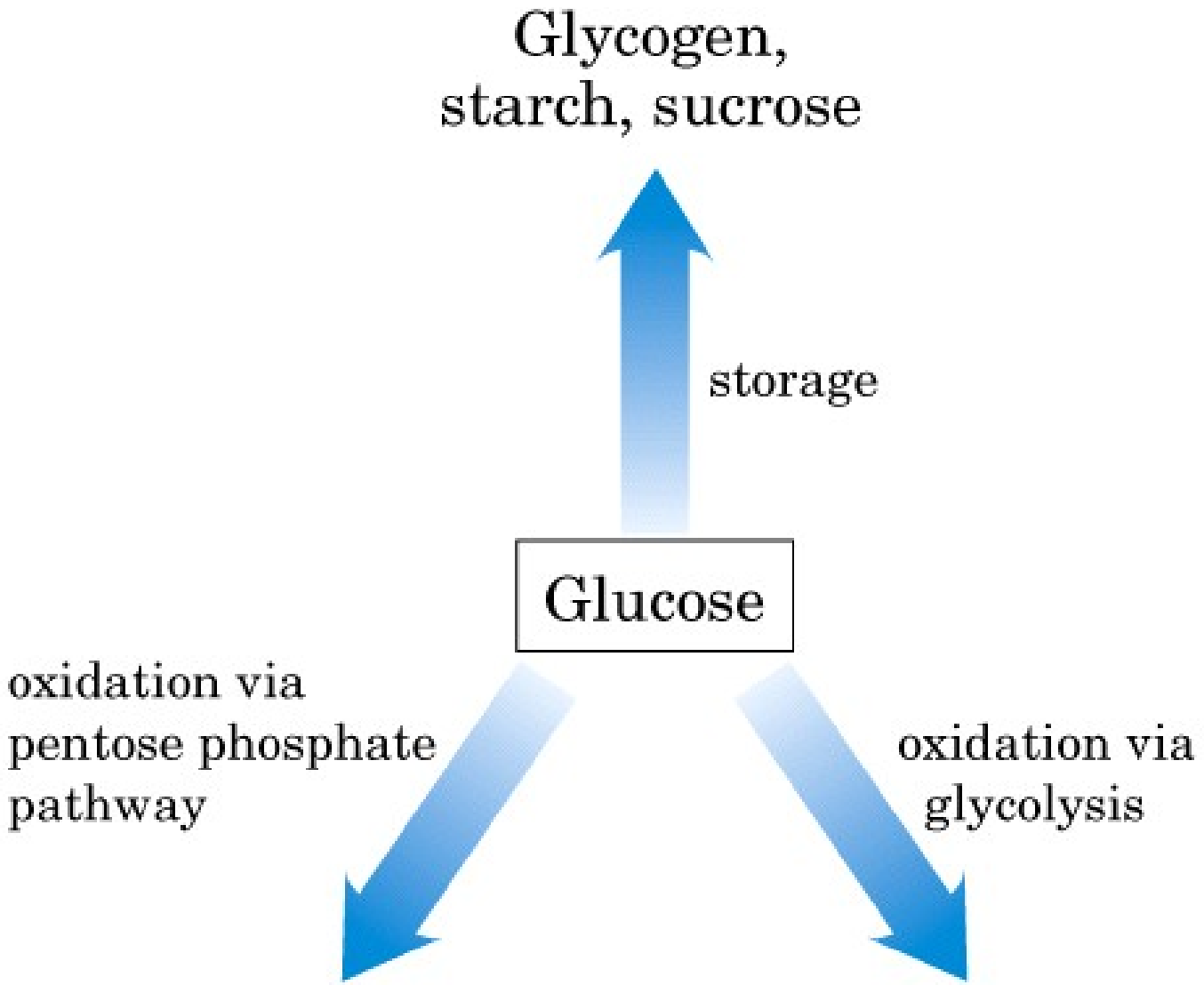
Gustav Embden

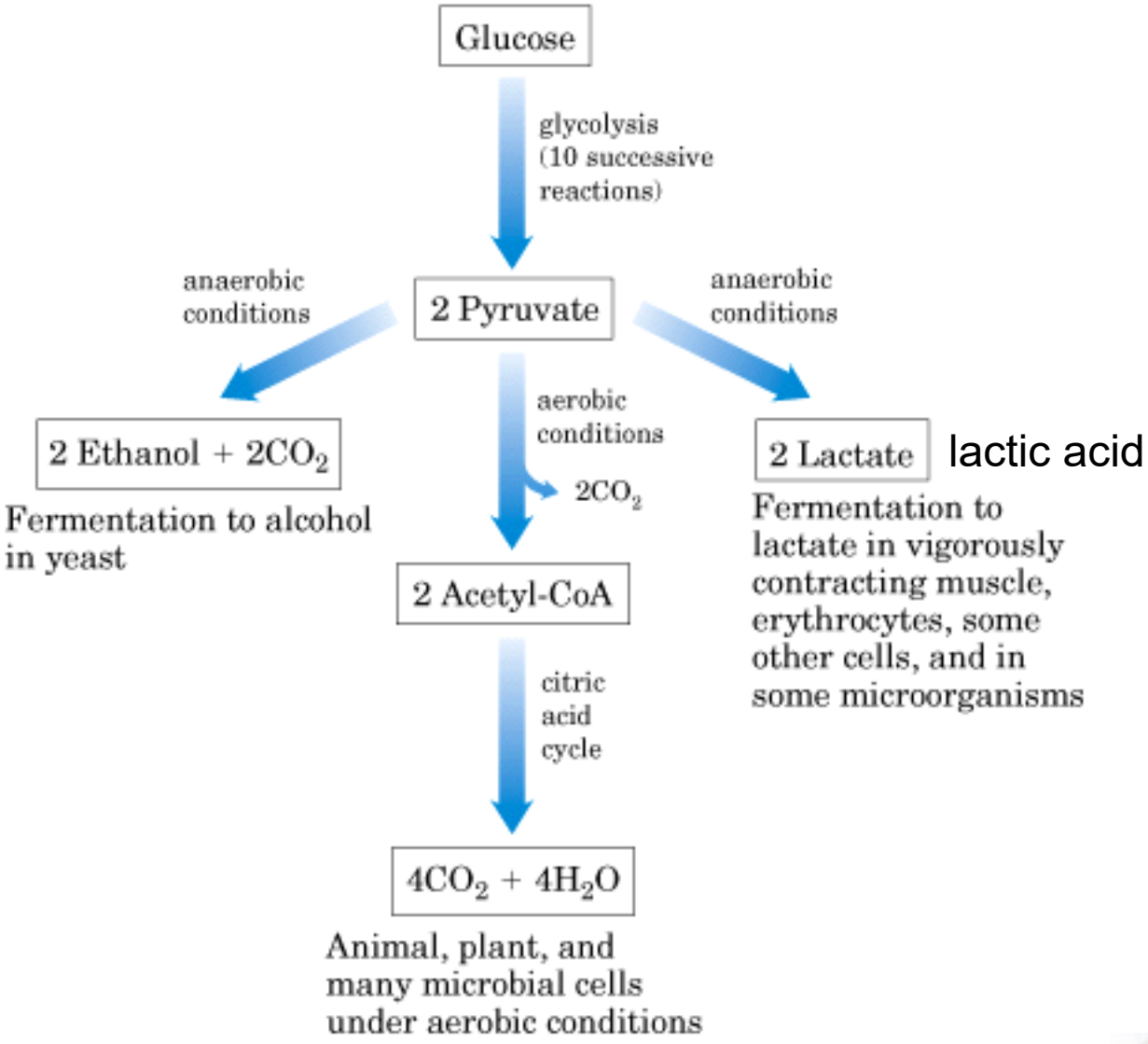
Awards [edit]

Although Embden was never awarded a Nobel prize, he was nominated 12 times over 8 years.[4]

Nobel Prize Academic Field	Year	Nominator(s)
Physiology or Medicine	1923	Karl Hürthle, Albrecht Julius Theodor Bethe, Ernst Schmitz
Physiology or Medicine	1924	Albrecht Kossel
Physiology or Medicine	1927	Albrecht Kossel
Physiology or Medicine	1929	L Seitz, Bernh. Fischer-Wasels, K Goldstein
Physiology or Medicine	1930	Ernst Schmitz
Physiology or Medicine	1932	A Dieudonné
Chemistry	1932	Hans von Euler-Chelpin
Physiology or Medicine	1933	E Cruickshank




Glycogen,
starch, sucrose
storage
Glucose
oxidation via
pentose phosphate
pathway
oxidation via
glycolysis
Ribose 5-phosphate
Pyruvate


Glucose
glycolysis
(10 successive
reactions)
anaerobic
conditions
2 Pyruvate
anaerobic
conditions
aerobic
conditions
$2CO_2$
2 Ethanol + $2CO_2$
Fermentation to alcohol
in yeast
2 Acetyl-CoA
2 Lactate
lactic acid
Fermentation to
lactate in vigorously
contracting muscle,
erythrocytes, some
other cells, and in
some microorganisms
citric
acid
cycle
$4CO_2 + 4H_2O$
Animal, plant, and
many microbial cells
under aerobic conditions

An overview on D-glucose

The major fuel of most organisms, $\Delta G'^{o}$ = –2840 kJ/mole if completely oxidized to CO_2 and H_2O via the glycolysis pathway, citric acid cycle and oxidative phosphorylation (generating ATP) .

Can also be oxidized to make NADPH and ribose-5-P via the pentose phosphate pathway （ PPP ） .

Can be stored in polymer form (glycogen or starch) or be converted to fat for long term storage.

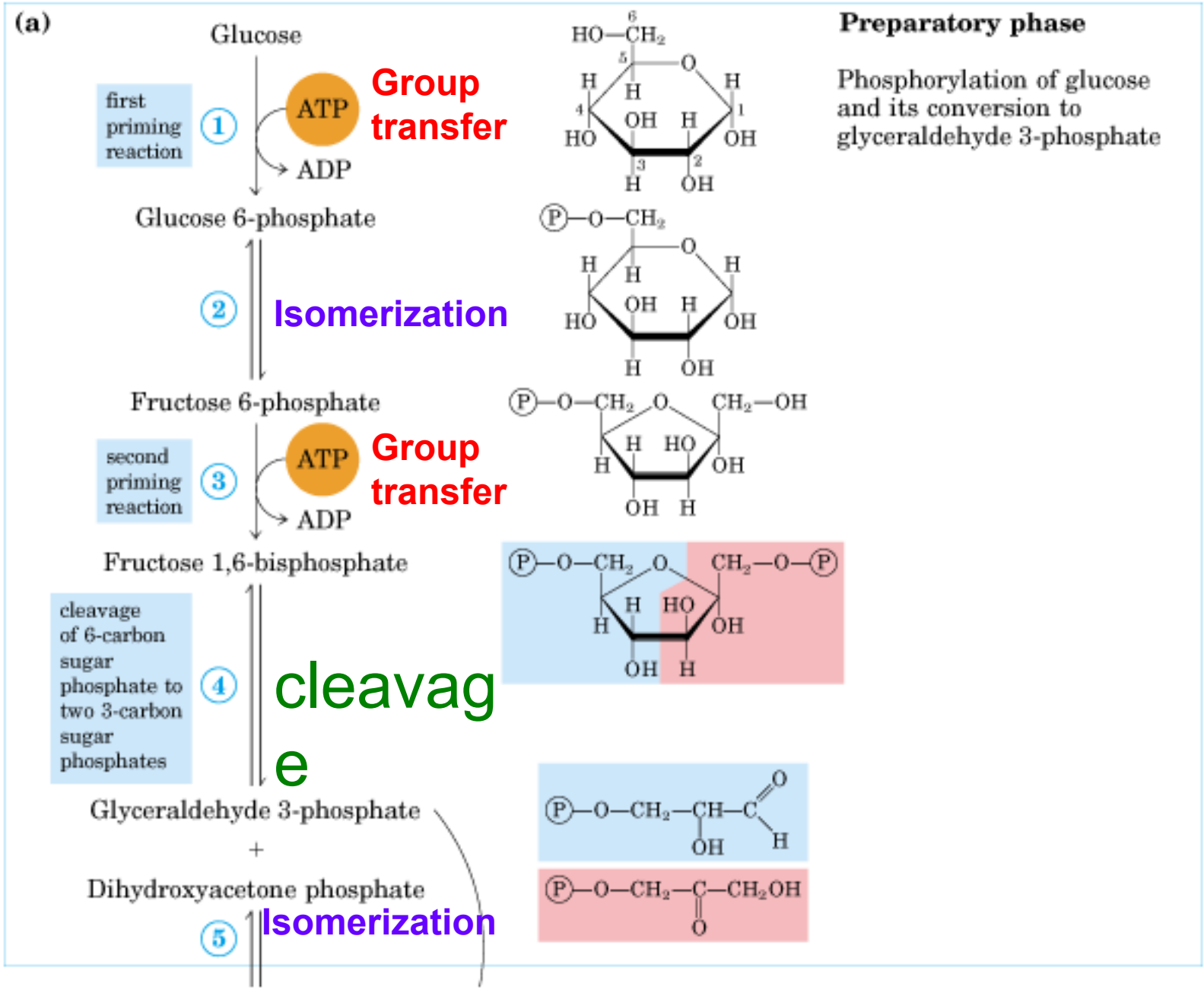
Is also a versatile precursor for carbon skeletons of almost all kinds of biomolecules, including amino acids, nucleotides, fatty acids, coenzymes and other metabolic intermediates.

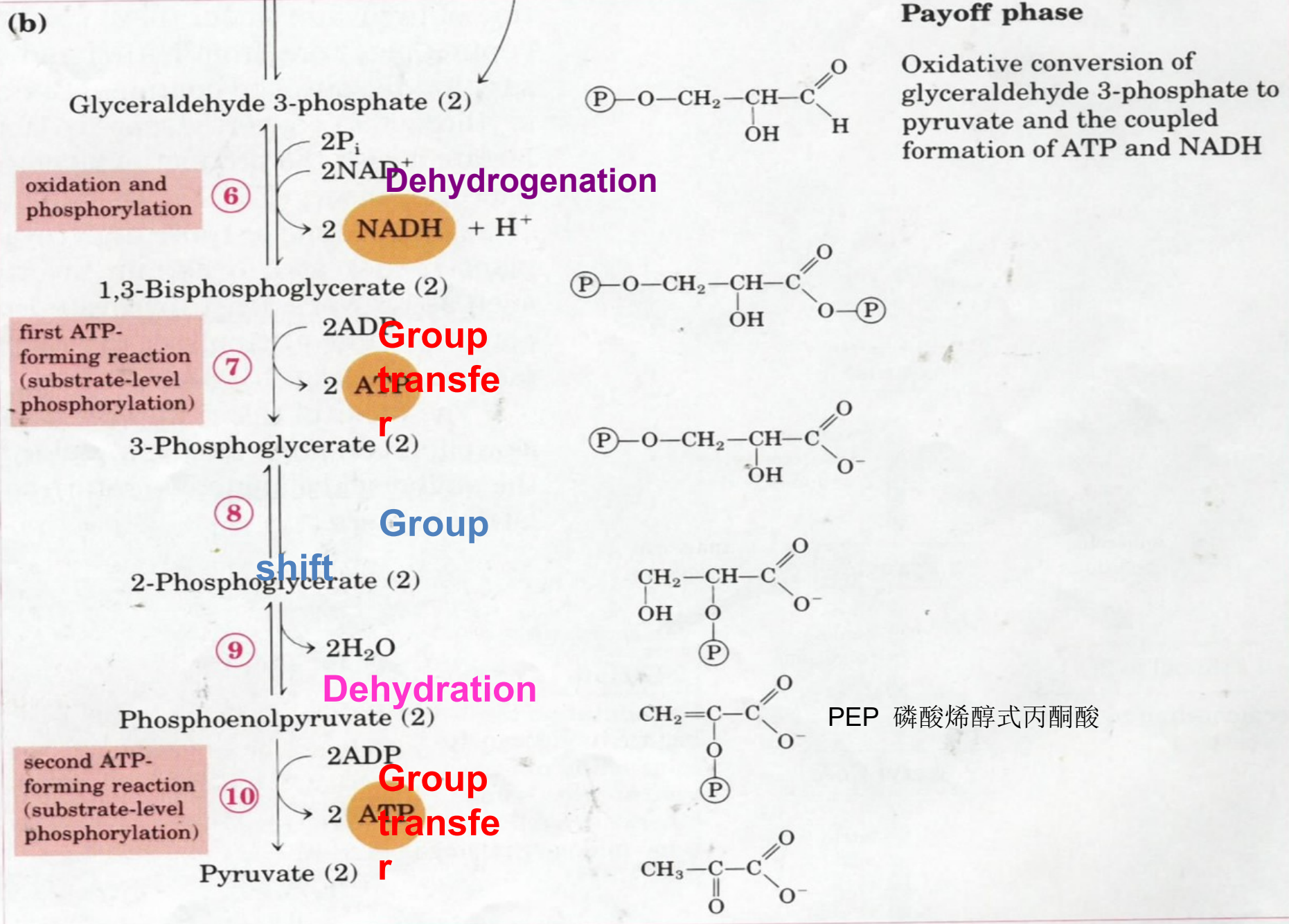


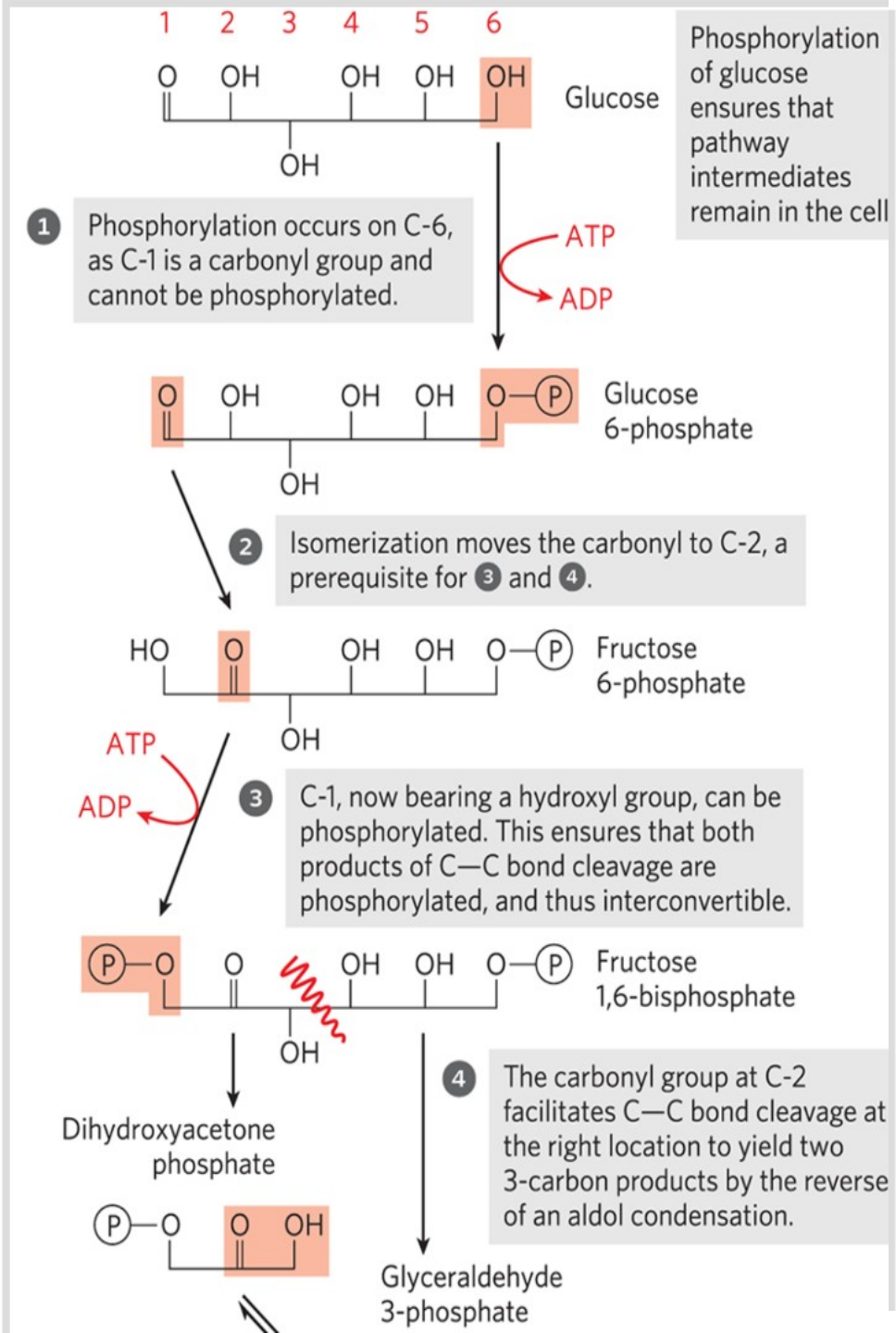
2. The overall glycolysis can be divided into two phases

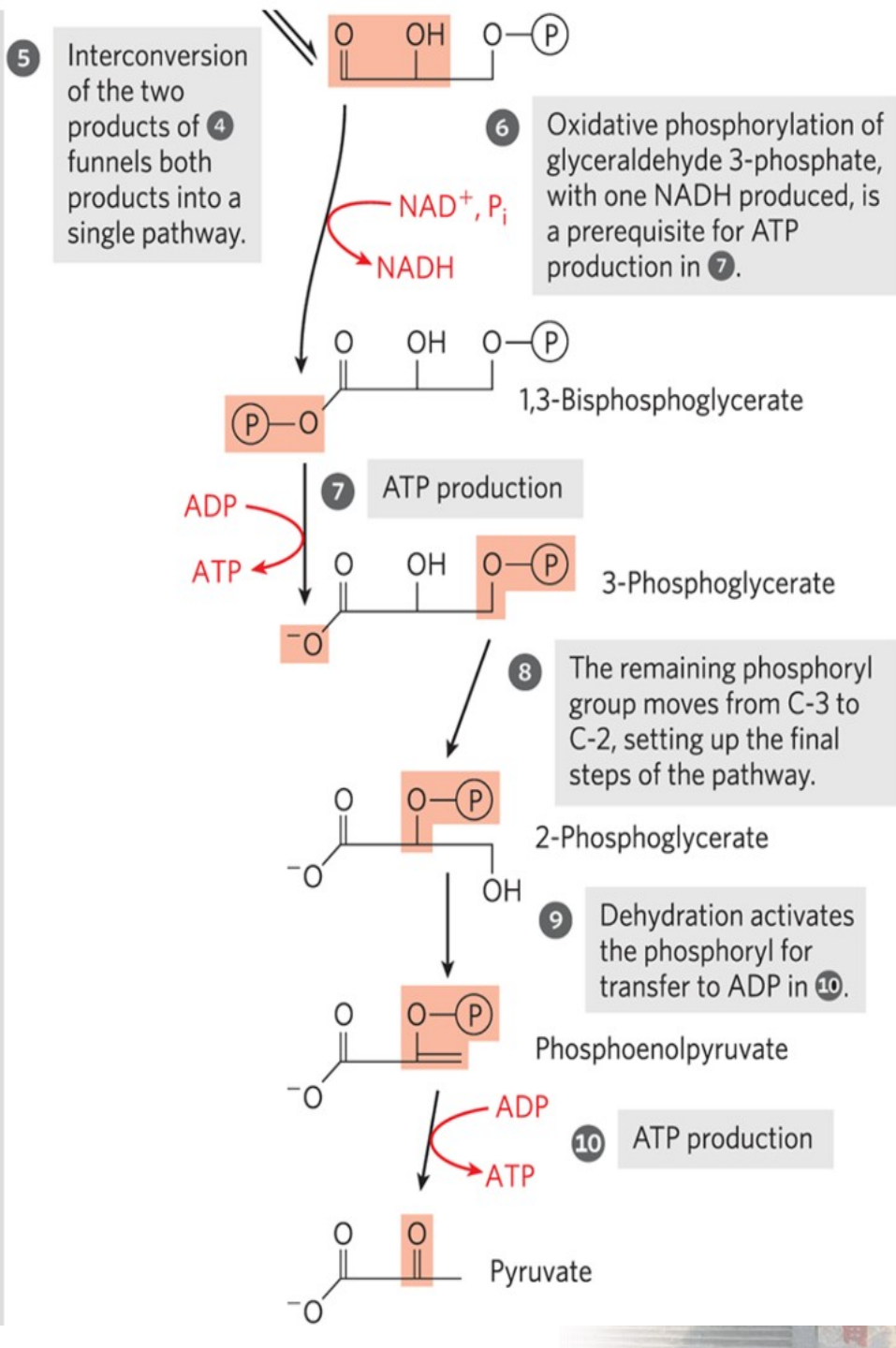
The hexose is first phossphorylated (thus activated) and then cleaved to produce two three-carbon intermediates at the preparatory phase, consuming ATP.
The three-carbon intermediates are then oxidized during the payoff phase, generating ATP and NADH.




(a)
Glucose
first priming reaction
1
ATP
ADP
Group transfer
Glucose 6-phosphate
2
Isomerization
Fructose 6-phosphate
second priming reaction
3
ATP
ADP
Group transfer
Fructose 1,6-bisphosphate
cleavage of 6-carbon sugar phosphate to two 3-carbon sugar phosphates
4
cleavage
Glyceraldehyde 3-phosphate
+
Dihydroxyacetone phosphate
5
Isomerization
Preparatory phase
Phosphorylation of glucose and its conversion to glyceraldehyde 3-phosphate


(b)
Payoff phase
Oxidative conversion of glyceraldehyde 3-phosphate to pyruvate and the coupled formation of ATP and NADH
Glyceraldehyde 3-phosphate (2)
2P$_i$
2NAD$^+$
oxidation and phosphorylation
6
Dehydrogenation
2 NADH + H$^+$
1,3-Bisphosphoglycerate (2)
2ADP
first ATP-forming reaction (substrate-level phosphorylation)
7
Group transfer
2 ATP
3-Phosphoglycerate (2)
8
Group shift
2-Phosphoglycerate (2)
9
2H$_2$O
Dehydration
Phosphoenolpyruvate (2)
PEP 磷酸烯醇式丙酮酸
2ADP
second ATP-forming reaction (substrate-level phosphorylation)
10
Group transfer
2 ATP
Pyruvate (2)

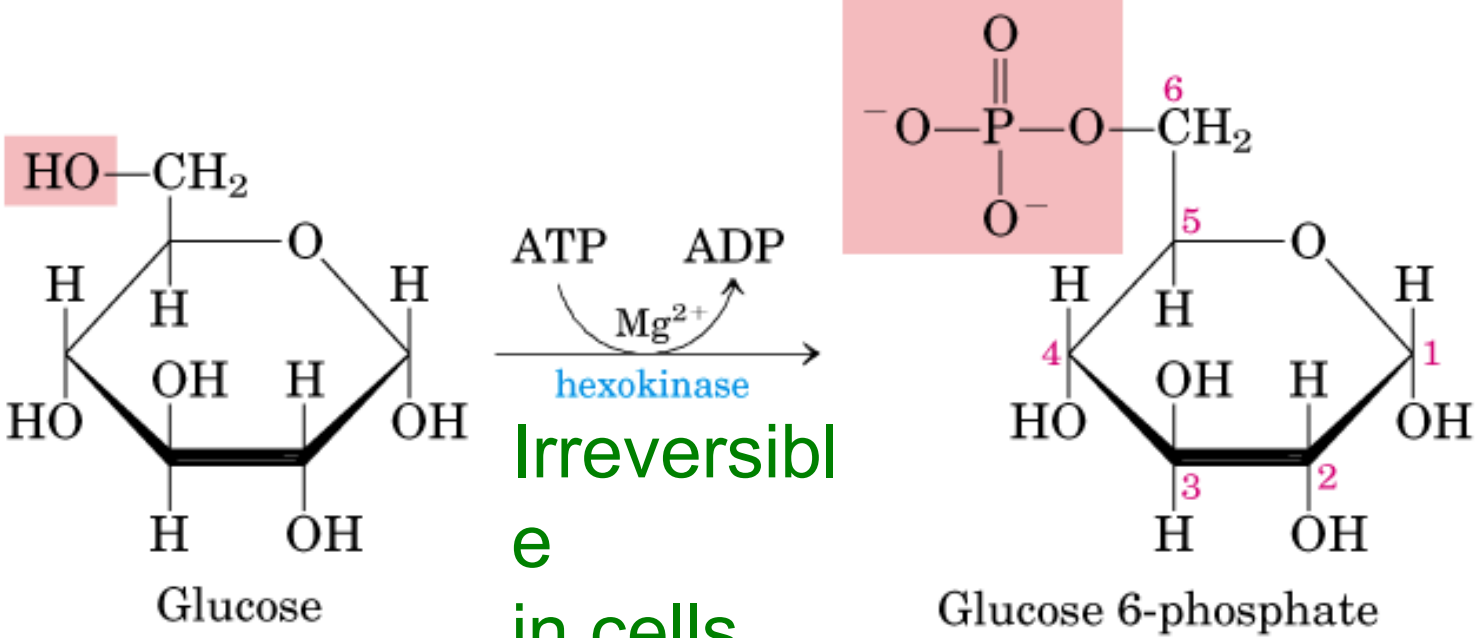

1 2 3 4 5 6
Glucose
Phosphorylation of glucose ensures that pathway intermediates remain in the cell
1 Phosphorylation occurs on C-6, as C-1 is a carbonyl group and cannot be phosphorylated.
ATP
ADP
Glucose 6-phosphate
2 Isomerization moves the carbonyl to C-2, a prerequisite for 3 and 4.
Fructose 6-phosphate
ATP
ADP
3 C-1, now bearing a hydroxyl group, can be phosphorylated. This ensures that both products of C—C bond cleavage are phosphorylated, and thus interconvertible.
Fructose 1,6-bisphosphate
Dihydroxyacetone phosphate
4 The carbonyl group at C-2 facilitates C—C bond cleavage at the right location to yield two 3-carbon products by the reverse of an aldol condensation.
Glyceraldehyde 3-phosphate


5 Interconversion of the two products of 4 funnels both products into a single pathway.
6 Oxidative phosphorylation of glyceraldehyde 3-phosphate, with one NADH produced, is a prerequisite for ATP production in 7.
NAD+, Pi
NADH
1,3-Bisphosphoglycerate
ADP
ATP
7 ATP production
3-Phosphoglycerate
8 The remaining phosphoryl group moves from C-3 to C-2, setting up the final steps of the pathway.
2-Phosphoglycerate
9 Dehydration activates the phosphoryl for transfer to ADP in 10.
Phosphoenolpyruvate
ADP
ATP
10 ATP production
Pyruvate

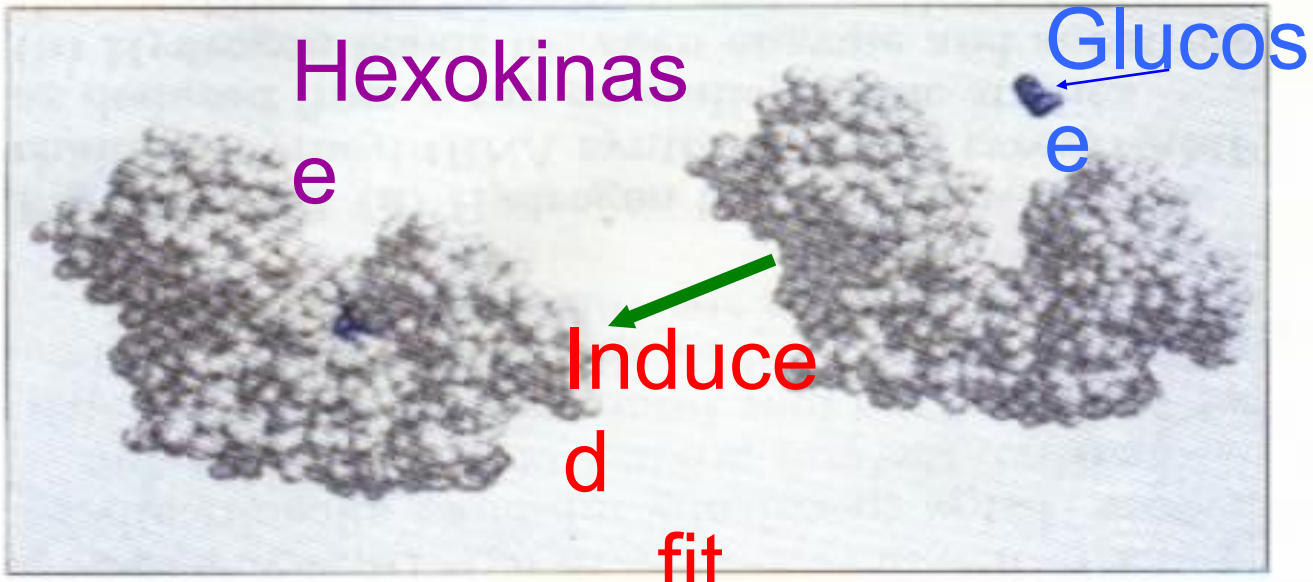
Hexokinase (also **glucokinase** in liver) catalyzes the first phosphorylation reaction on the pathway.

Phosphohexose isomerase (also called phosphoglucose isomerase) catalyzes the isomerization from glucose 6-P to fructose 6-P, converting an aldose to a ketose.





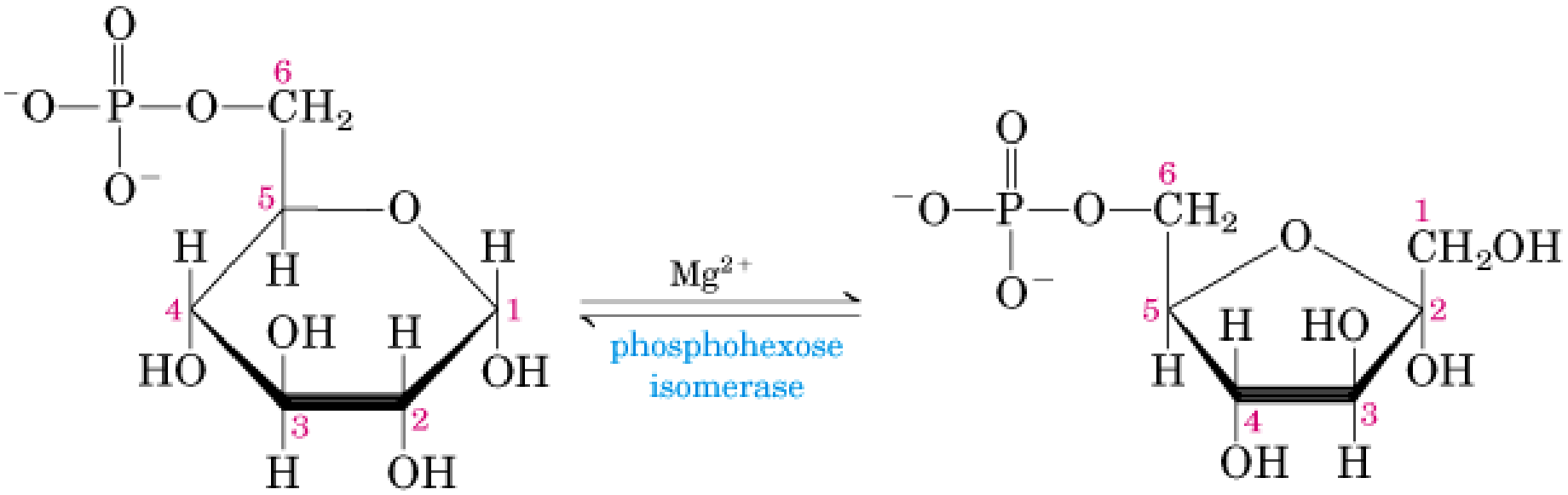
Irreversible in cells



Biochemis

An aldose

An ketose



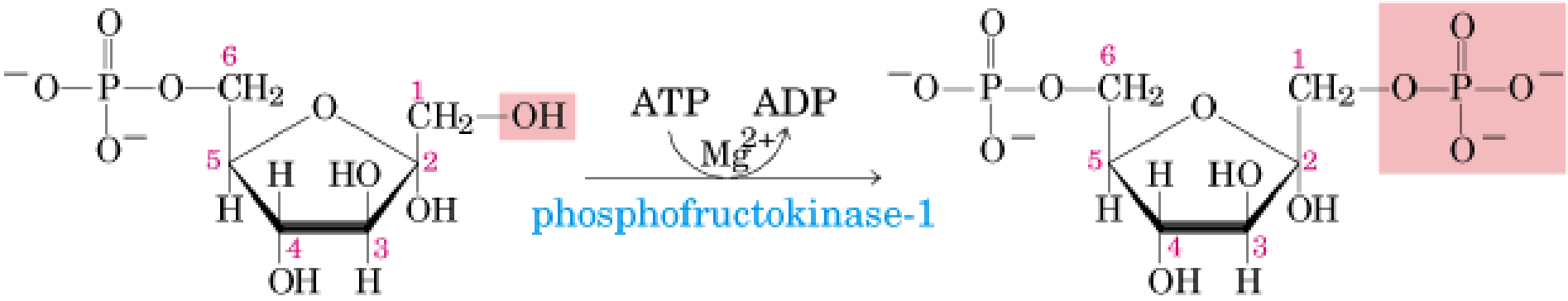
Glucose 6-phosphate

Fructose 6-phosphate

$\Delta G'^{\circ} = 1.7$ kJ/mol

Phosphofructokinase-1 (PFK-1, 磷酸果糖激酶 -1) then catalyzes the second phosphorylation step, converting fructose 6-P to fructose 1,6-bisphosphate; the overall rate of glycolysis is mainly controlled at this step; **PFK-1** is a highly regulatory enzyme; the plant PFK-1 makes use of PP_i, instead of ATP at this step.





Fructose 6-phosphate

Fructose 1,6-bisphosphate

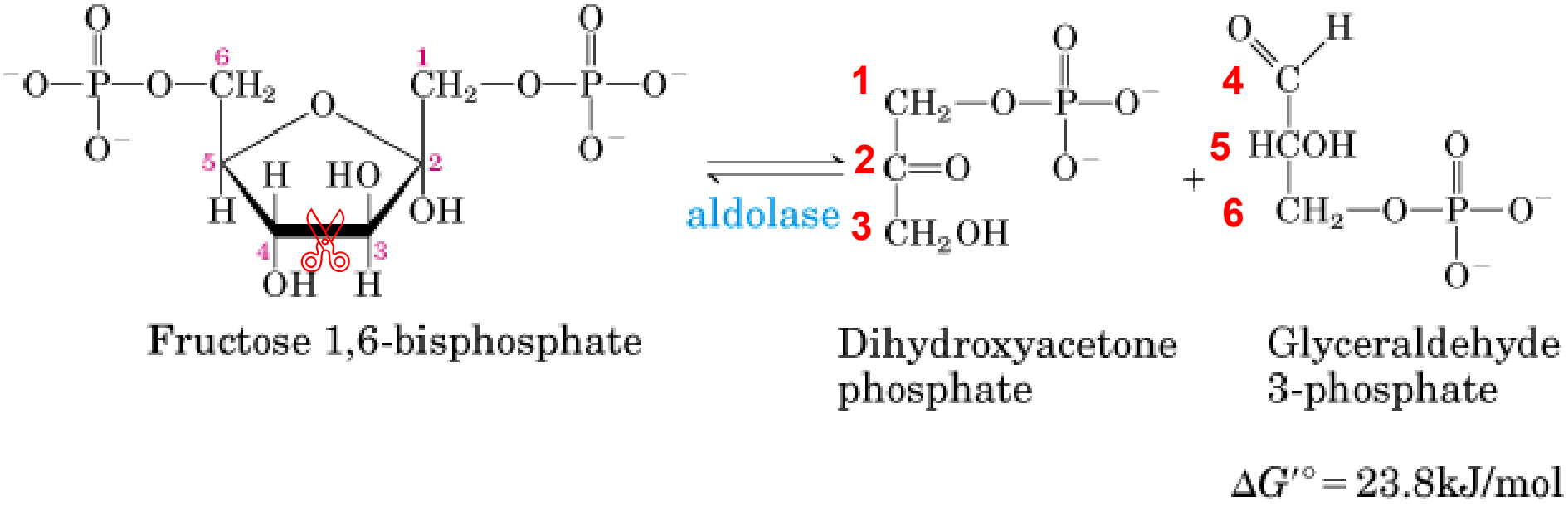
$\Delta G^{\circ} = -14.2$ kJ/mol

- **Aldolase**（醛缩酶）, named for the reverse reaction catalyzes the cleavage (“lysis”) of fructose 1,6-bisphosphate from the middle C-C bond to form two 3-carbon sugars, dihydroxyacetone phosphate and glyceraldehyde 3-phosphate; this is a reversal aldol condensation reaction.



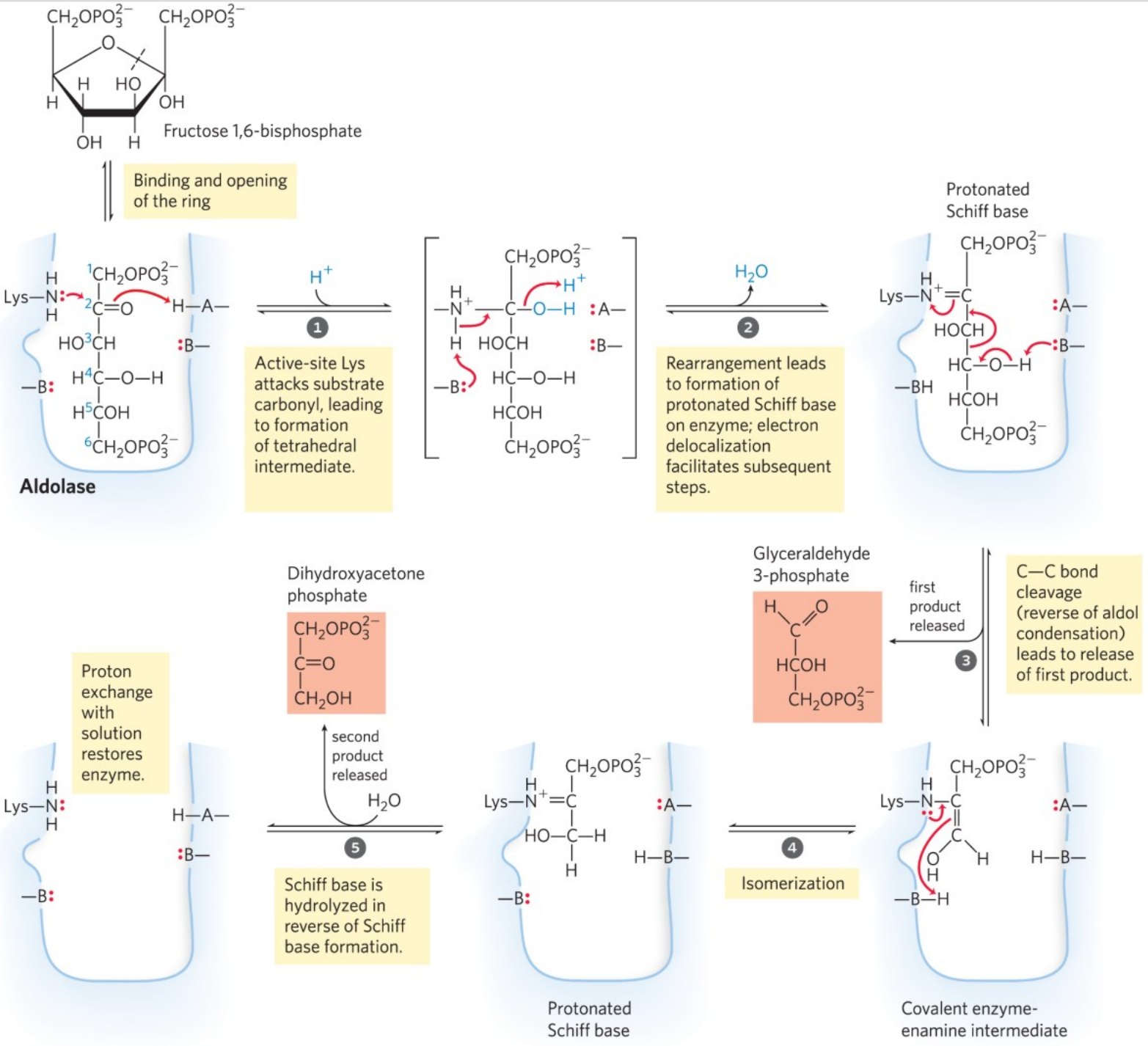
A ketone

An aldehyde



$\Delta G'^{\circ} = 23.8 \text{kJ/mol}$

The “lysis” step



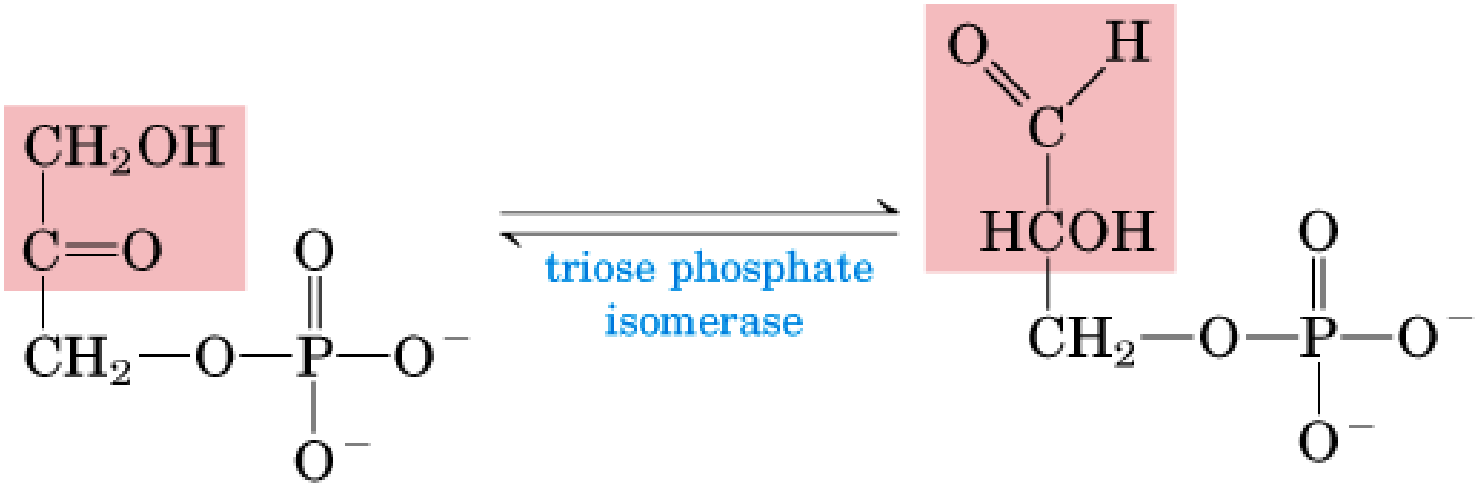
Nelson & Cox, *Lehninger Principles of Biochemistry*, 8e, © 2021 W. H. Freeman and Company

- **Triose phosphate isomerase** converts dihydroacetone phosphate to glyceraldehyde 3-phosphate; an intramolecular redox reaction (a hydrogen atom is transferred from C-1 to C-2).



A ketose

An aldose



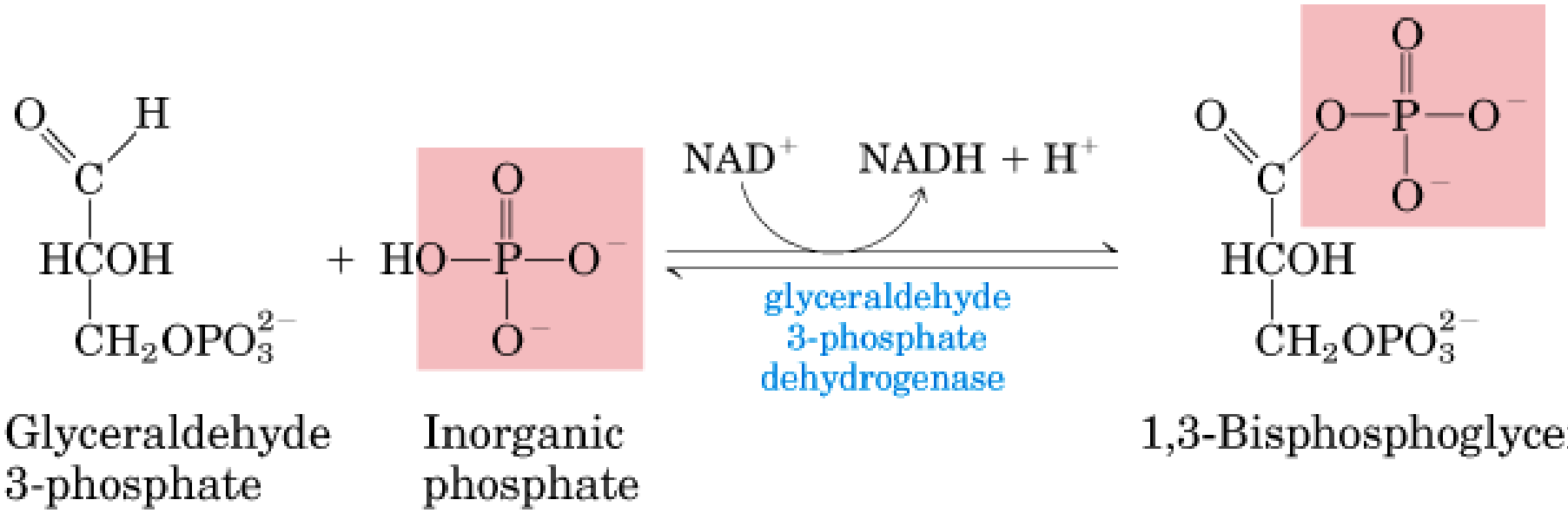
Dihydroxyacetone phosphate

Glyceraldehyde 3-phosphate

$\Delta G'^{\circ} = 7.5$ kJ/mol

Glyceraldehyde 3-phosphate dehydrogenase catalyzes first the oxidation and then the phosphorylation of glyceraldehyde 3-P to form glycerate 1,3-bisphosphate, an acyl phosphate（酰基磷酸）; the phosphate group linked to the carboxyl group via a anhydride bond has a high transfer potential.





Glyceraldehyde 3-phosphate

Inorganic phosphate

1,3-Bisphosphoglycerate

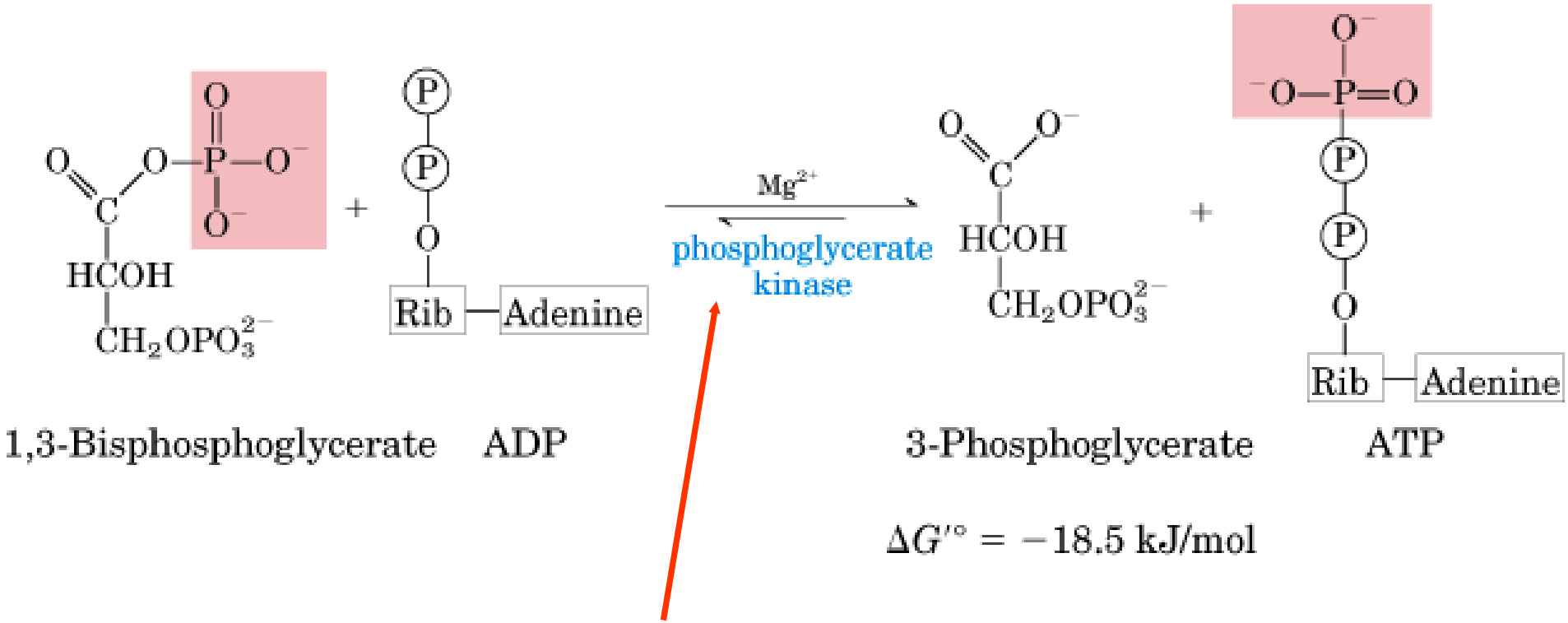
$\Delta G'^{\circ} = 6.3$ kJ/mol

acyl phosphate（酰基磷酸）

- **The phosphoglycerate kinase** catalyzes the direct transfer of the anhydride phosphate in 1,3-BPG （glycerate 1,3-bisphosphate） to an ADP to generate an ATP; this is called the **substrate-level phosphorylation**; 1,3-BPG is a high energy intermediate that leads to ATP formation.



Substrate-level phosphorylation For ATP generation

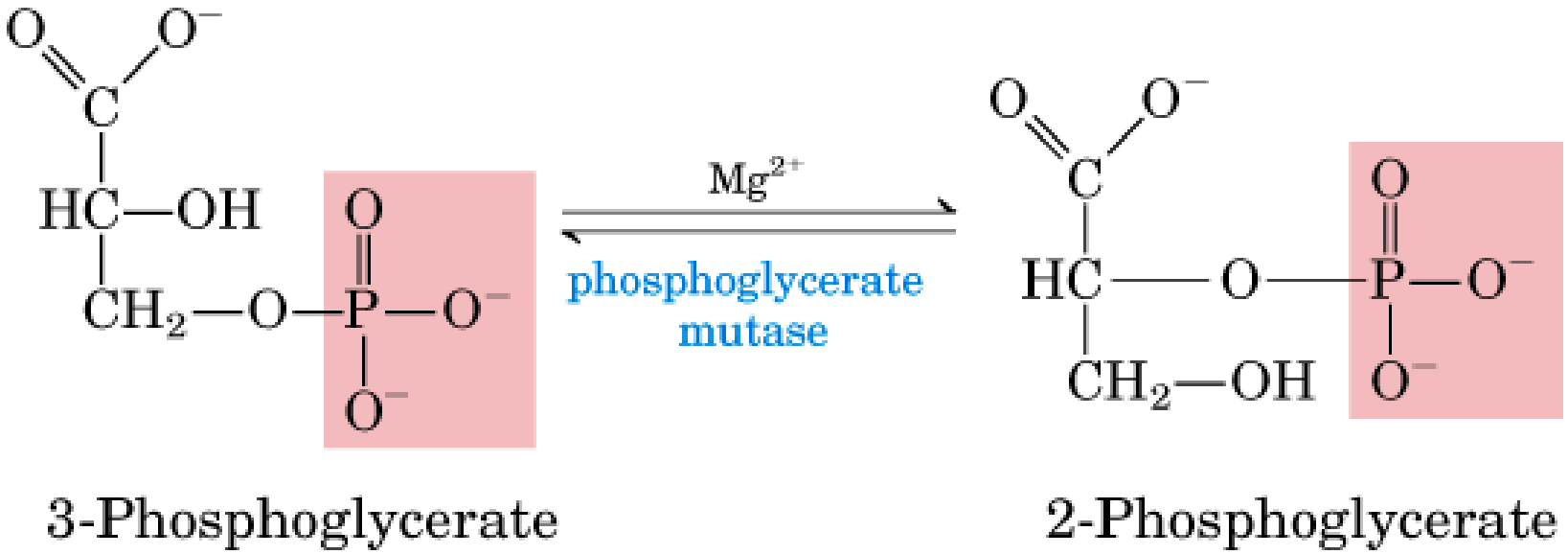


1,3-Bisphosphoglycerate ADP 3-Phosphoglycerate ATP

$\Delta G'^{\circ} = -18.5$ kJ/mol

Enzyme is named for the reverse reaction

The phosphoglycerate mutase catalyzes the shift of phosphoryl group on 3-phosphoglycerate from C-3 to C-2.

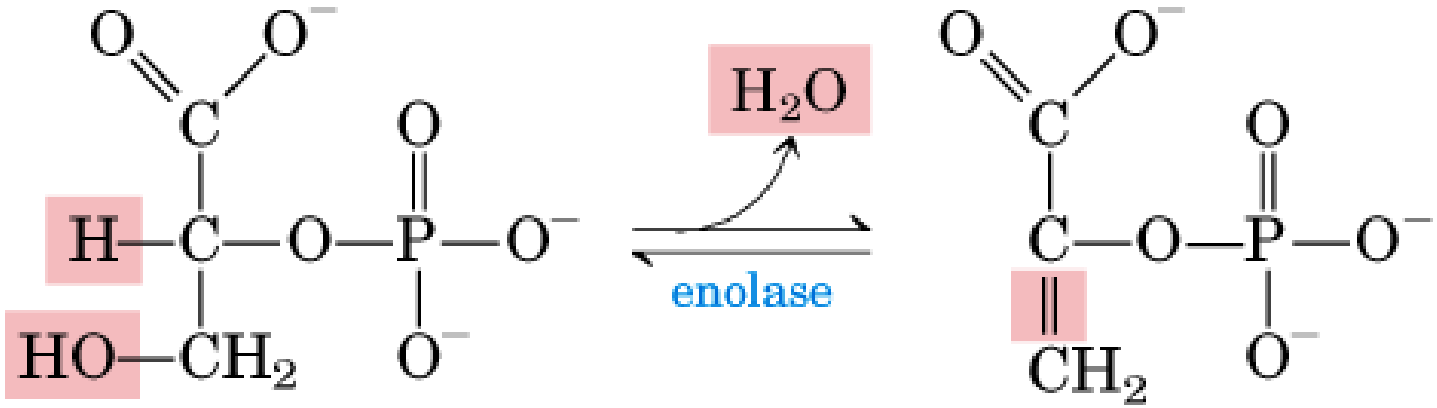


3-Phosphoglycerate

2-Phosphoglycerate

$\Delta G'^{\circ} = 4.4$ kJ/mol

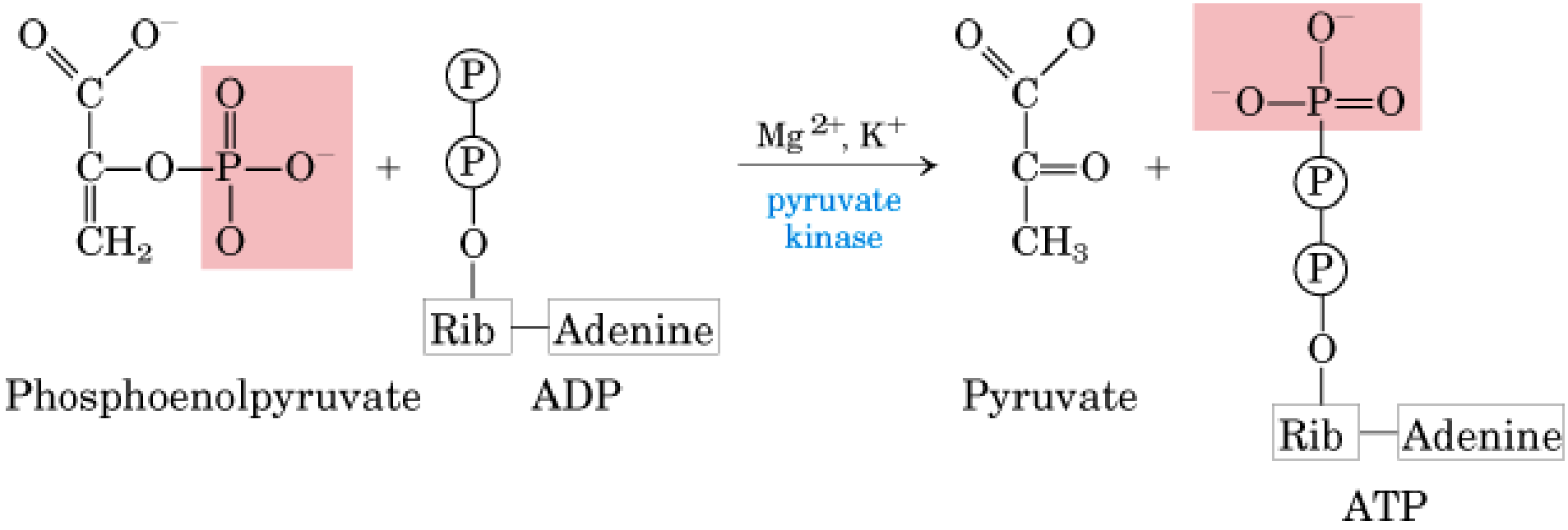
Enolase（烯醇酶）catalyzes the elimination of a H_2O from 2-phosphoglycerate to generate phosphoenolpyruvate (PEP).



$\Delta G'^{\circ} = 7.5$ kJ/mol

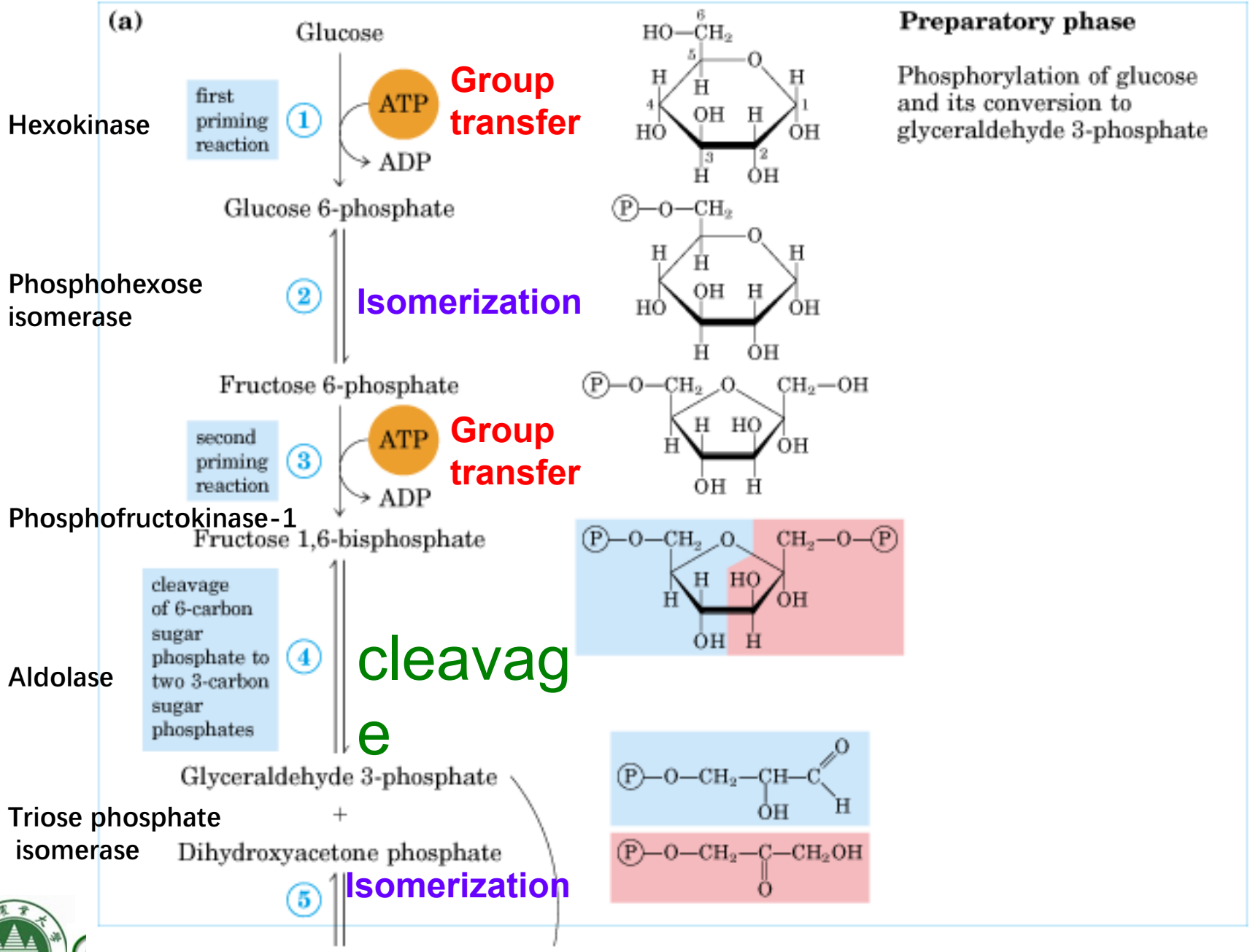
The pyruvate kinase (named for the reverse reaction) catalyzes the transfer of the phosphoryl group on PEP to ADP to form another molecule of ATP by “substrate-level phosphorylation”; enolpyruvate is formed and is quickly changed to pyruvate (丙酮酸).





Phosphoenolpyruvate ADP Pyruvate ATP

$\Delta G'^{\circ} = -31.4$ kJ/mol



Hexokinase

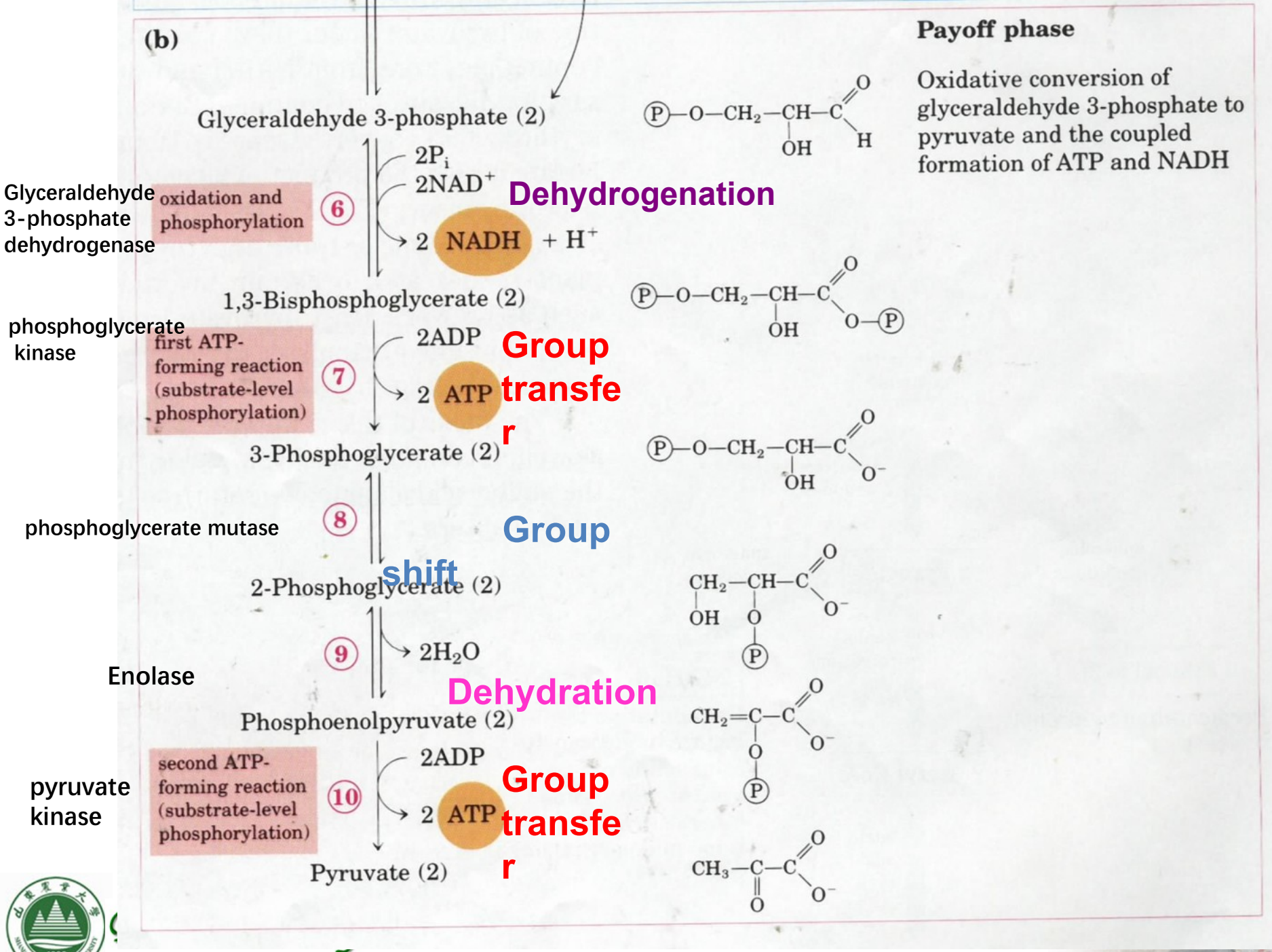
Phosphohexose isomerase

Phosphofructokinase-1

Aldolase

Triose phosphate isomerase

Biochemistry


(b)
Payoff phase
Oxidative conversion of glyceraldehyde 3-phosphate to pyruvate and the coupled formation of ATP and NADH
Glyceraldehyde 3-phosphate (2)
2P$_i$
2NAD$^+$
Dehydrogenation
Glyceraldehyde 3-phosphate dehydrogenase
oxidation and phosphorylation
6
2 NADH + H$^+$
1,3-Bisphosphoglycerate (2)
phosphoglycerate kinase
first ATP-forming reaction (substrate-level phosphorylation)
7
2ADP
2 ATP
Group transfer
3-Phosphoglycerate (2)
phosphoglycerate mutase
8
Group shift
2-Phosphoglycerate (2)
9
2H$_2$O
Enolase
Dehydration
Phosphoenolpyruvate (2)
pyruvate kinase
second ATP-forming reaction (substrate-level phosphorylation)
10
2ADP
2 ATP
Group transfer
Pyruvate (2)

A net gain of two ATP, two NADH, two pyruvates are resulted when a glucose molecule is oxidized via the glycolysis pathway:

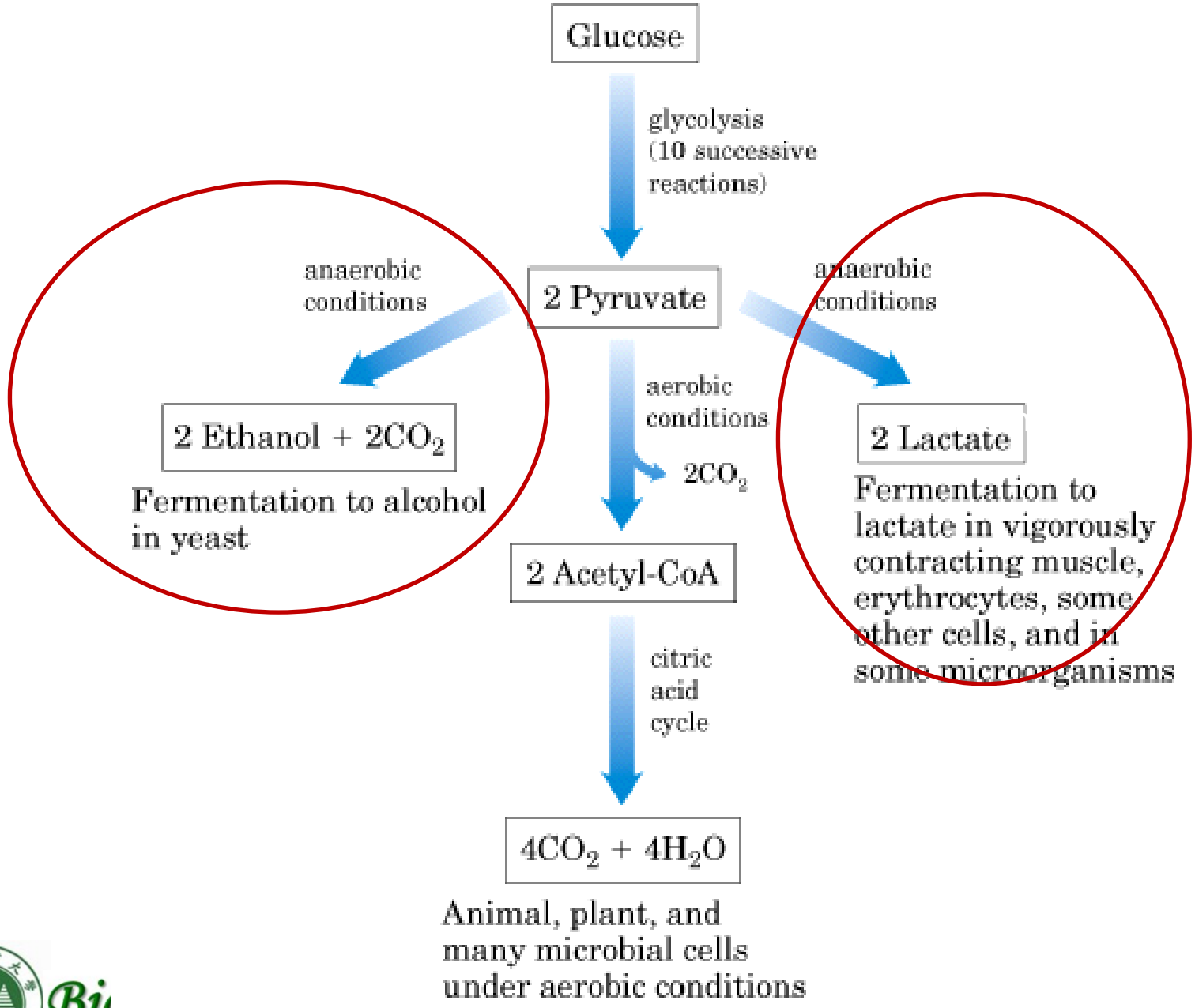
$$\text{Glucose} + 2\,\text{ADP} + 2P_i + 2NAD^+ \rightarrow 2\,\text{pyruvate} + 2ATP + 2H_2O + 2NADH + 2H^+$$



4. Fermentation: pyruvate is converted to lactic acid or ethanol under anaerobic conditions

This occurs to regenerate NAD^+ for the glycolysis pathway to continue when O_2 lacks.

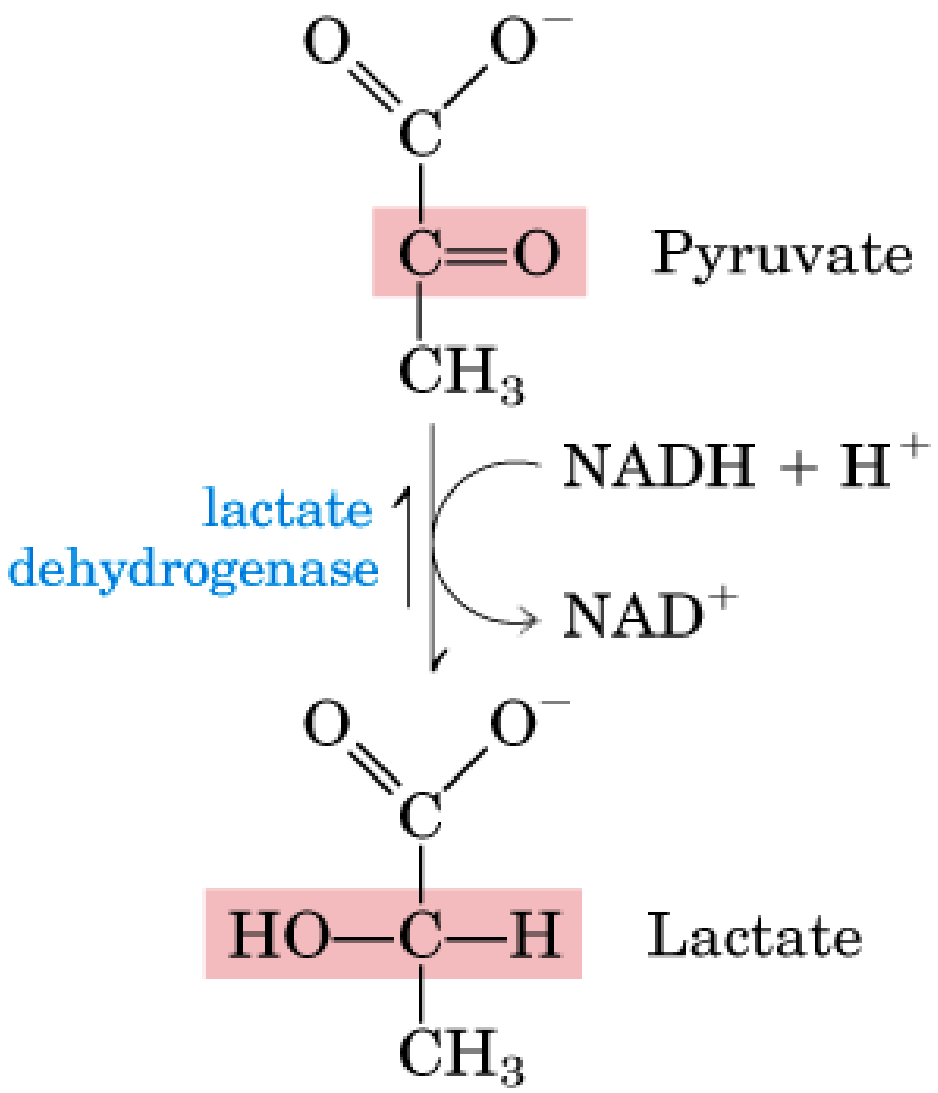



Glucose
glycolysis (10 successive reactions)
2 Pyruvate
anaerobic conditions
2 Ethanol + 2CO2
Fermentation to alcohol in yeast
aerobic conditions
2CO2
2 Acetyl-CoA
citric acid cycle
4CO2 + 4H2O
Animal, plant, and many microbial cells under aerobic conditions
anaerobic conditions
2 Lactate
Fermentation to lactate in vigorously contracting muscle, erythrocytes, some other cells, and in some microorganisms

- Lactic acid fermentation (occurring in very active muscle, some bacteria): pyruvate is reduced by NADH, catalyzed by lactate dehydrogenase.
- The lactate produced in muscle can be converted back to glucose by gluconeogenesis in the liver of vertebrates.



Pyruvate is reduced to lactate when O_2 lacks in a reaction catalyzed by **lactate Dehydrogenase.**

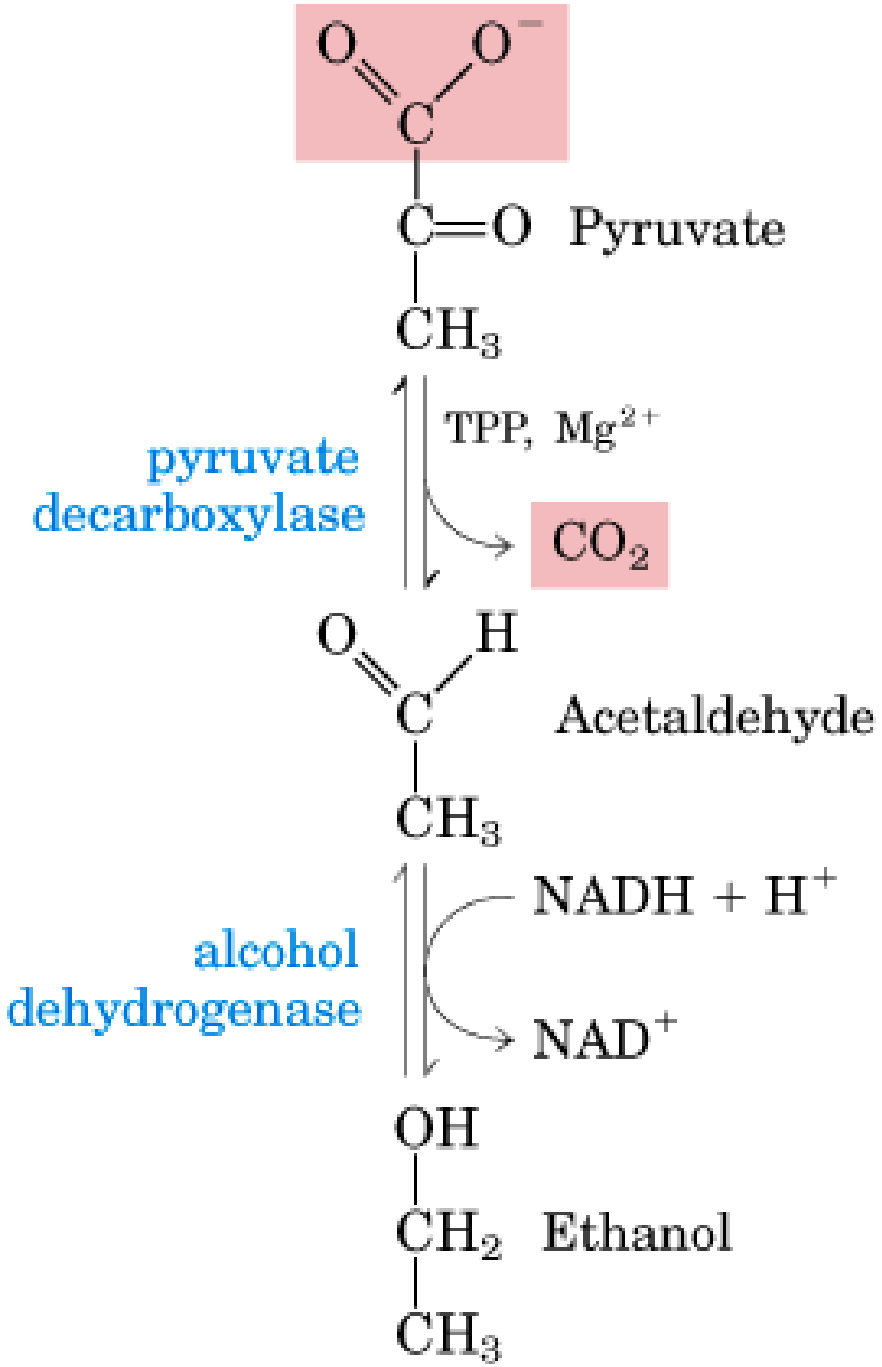


$$\Delta G'^{\circ} = -25.1 \text{ kJ/mol}$$

- Ethanol fermentation (occurring in yeast and other microorganisms): pyruvate is first decarboxylated and then reduced by NADH, catalyzed by pyruvate decarboxylase and alcohol dehydrogenase respectively.



Pyruvate can be decarboxylated and reduced to form ethanol in some micro-organisms



5. Energy yield

- Two ATP are used in glycolysis and four ATPs are synthesized for each molecule of glucose so that the net yield is two ATPs per glucose.
- Under aerobic conditions, the two NADH molecules arising from glycolysis also yield energy via oxidative phosphorylation.

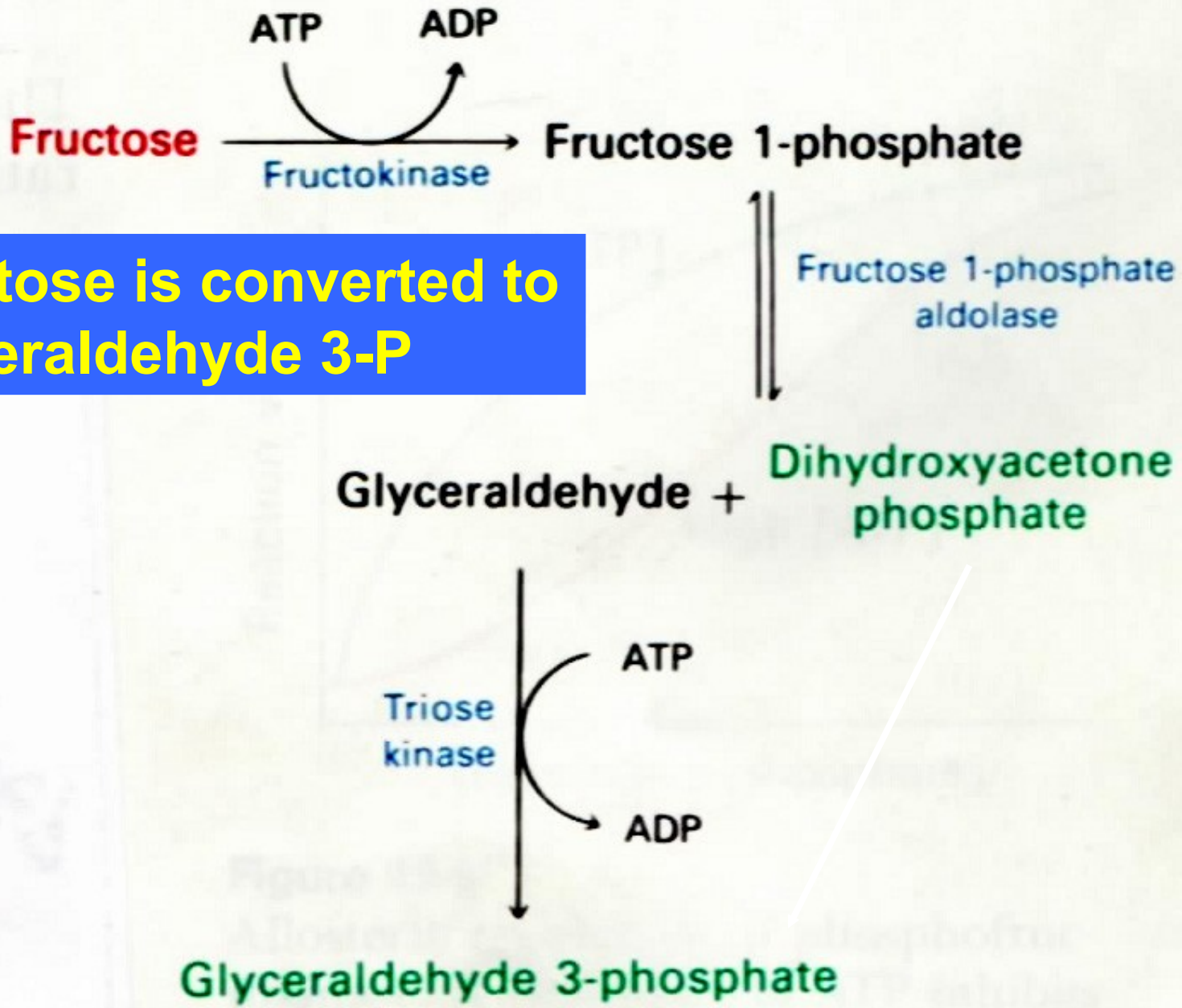


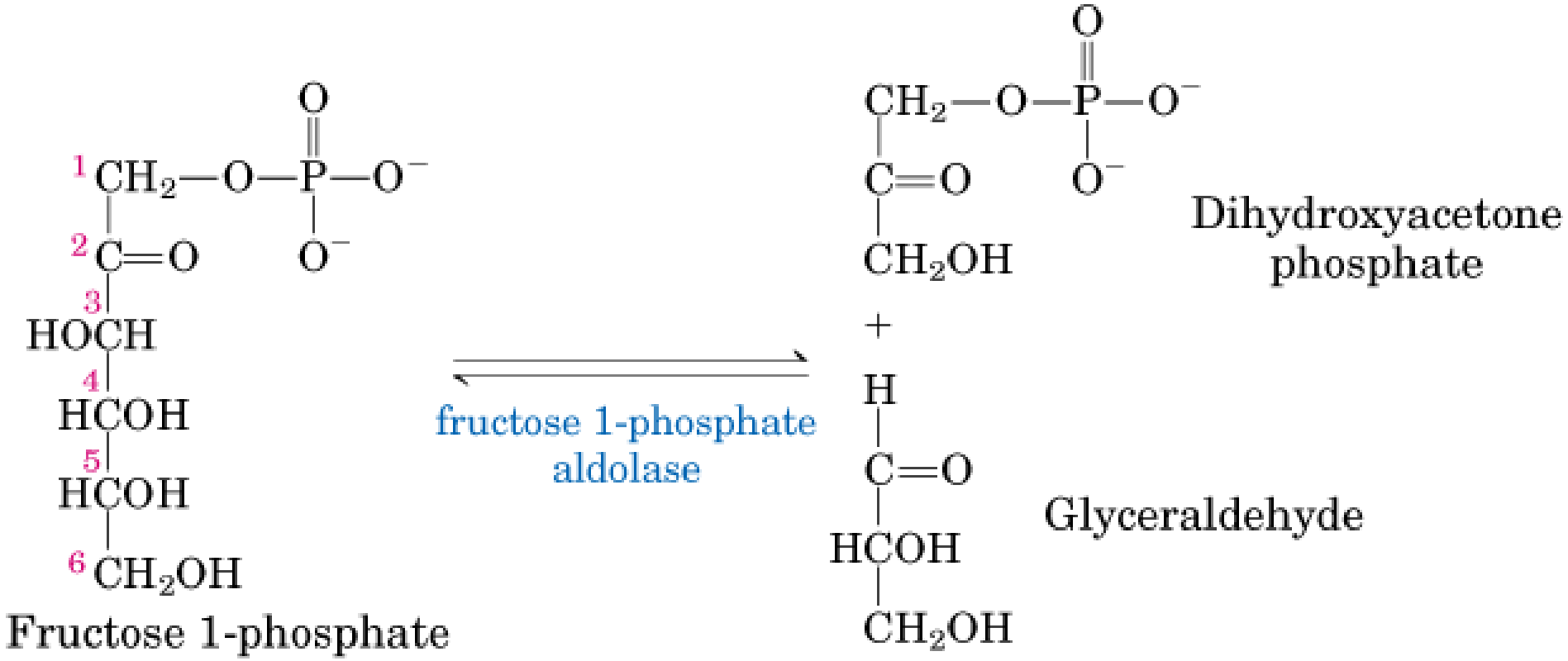
6. Other hexoses are also oxidized via the glycolysis pathway

They are also first primed by phosphorylation (at C-1 or C-6).
Fructose is primed and cleaved to form dihydroxyacetone phosphate and glyceraldehyde, which are further converted to glyceraldehyde 3-P.
Galactose is first converted to Glc-1-P via a UDP-galactose intermediate and UDP-glucose intermediate, then to Glc-6-P.



One fructose is converted to two glyceraldehyde 3-P




fructose 1-phosphate aldolase
Fructose 1-phosphate
Dihydroxyacetone phosphate
Glyceraldehyde

6. Regulation of glycolysis

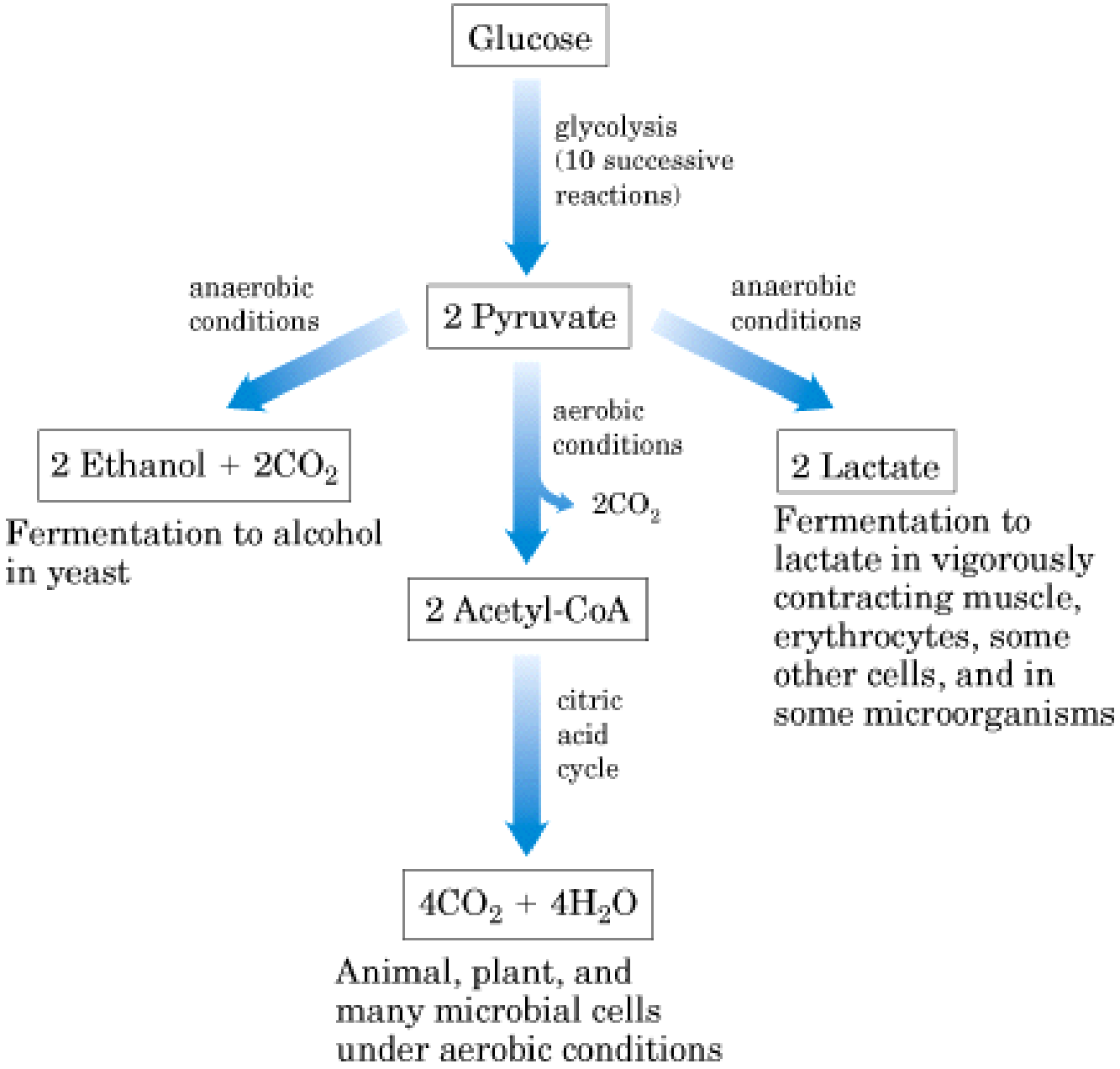
Phosphofructokinase-1 (PFK-1) is the main point of regulation for controlling the rate of glycolysis. PFK is allosterically inhibited by ATP, but this inhibition is relieved by AMP. Citrate also inhibits PFK. The build up of F-6-P stimulates the formation of 2,6-FBP that in turn stimulates PFK. Hexokinase and pyruvate kinase are additional control sites.



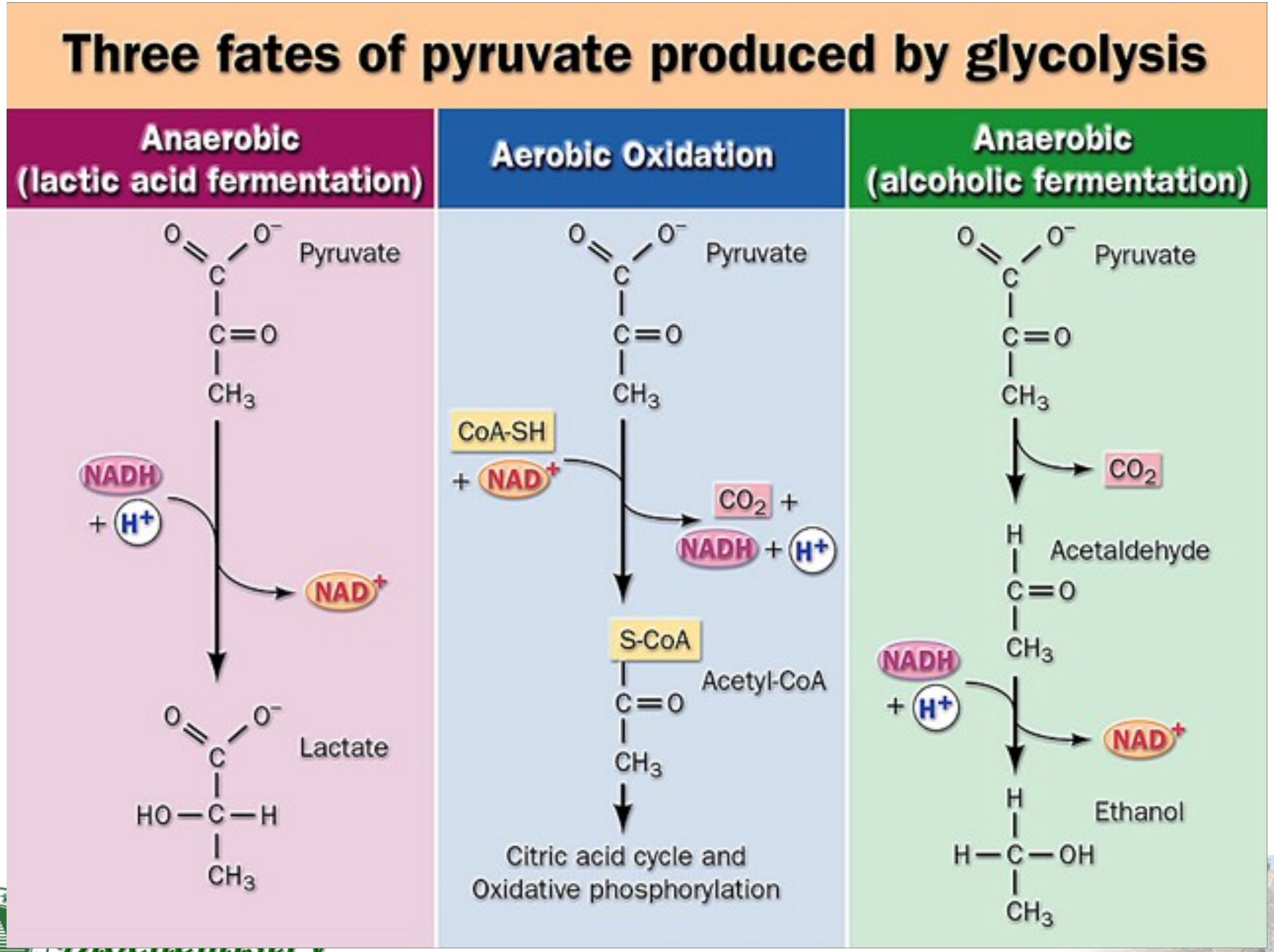
Summary

D-glucose is a commonly used fuel and versatile precursor in almost all organisms. Glucose is first converted into two three-carbon pyruvates via the ten-step glycolysis pathway without directly consuming O_2 and with a net production of two ATP molecules by substrate-level phosphorylation. Limited amount of energy can be released by oxidizing glucose under anaerobic conditions by fermentation.




Glucose
glycolysis (10 successive reactions)
2 Pyruvate
anaerobic conditions
anaerobic conditions
aerobic conditions
$2CO_2$
2 Ethanol + $2CO_2$
Fermentation to alcohol in yeast
2 Lactate
Fermentation to lactate in vigorously contracting muscle, erythrocytes, some other cells, and in some microorganisms
2 Acetyl-CoA
citric acid cycle
$4CO_2 + 4H_2O$
Animal, plant, and many microbial cells under aerobic conditions




Three fates of pyruvate produced by glycolysis
Anaerobic (lactic acid fermentation)
Aerobic Oxidation
Anaerobic (alcoholic fermentation)
Pyruvate
NADH
+ H+
NAD+
Lactate
HO—C—H
CH3
CoA-SH
+ NAD+
CO2 +
NADH + H+
S-CoA
Acetyl-CoA
Citric acid cycle and
Oxidative phosphorylation
CO2
Acetaldehyde
Ethanol
H—C—OH

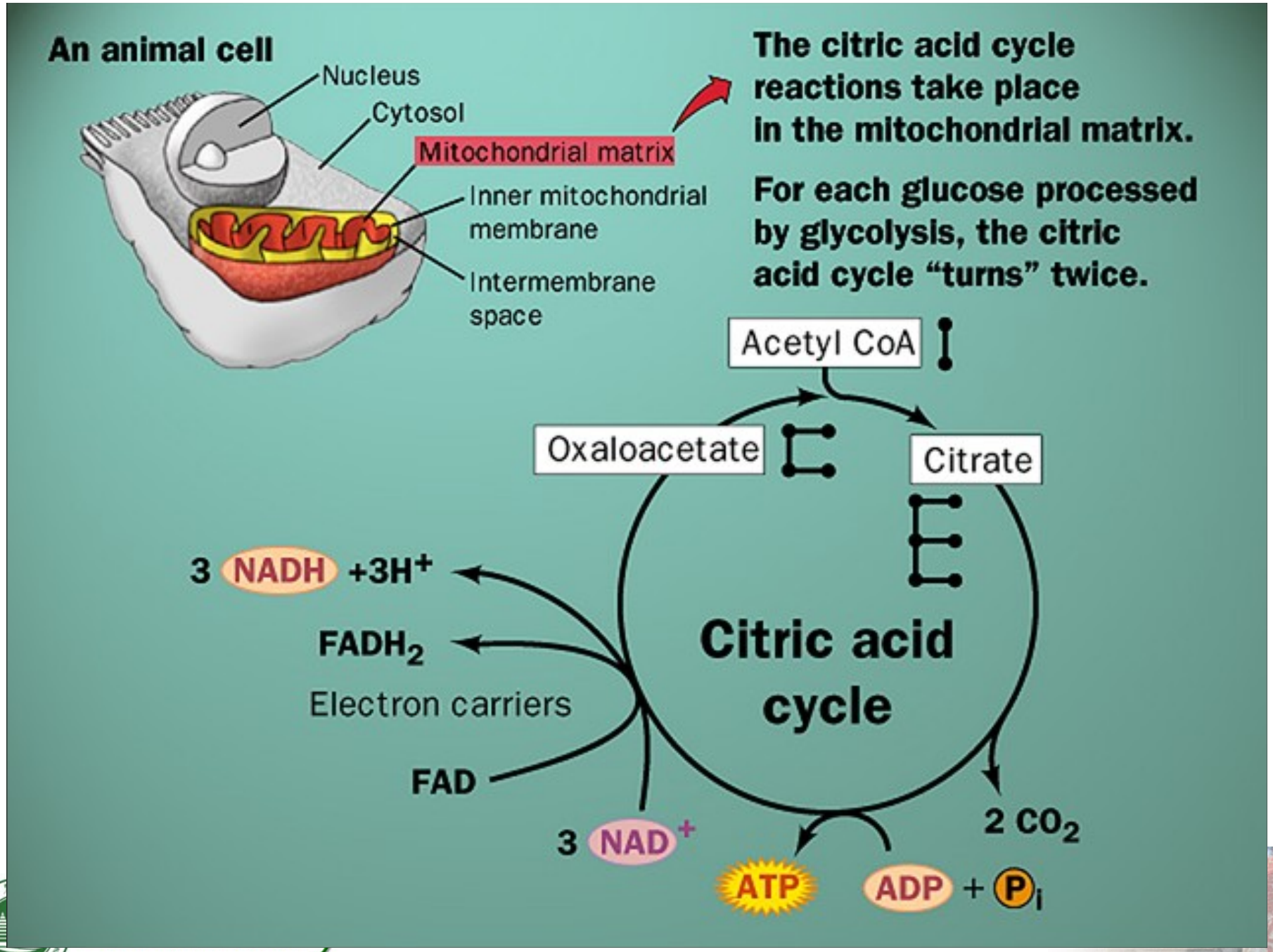


The Citric Acid Cycle (L1)

The common pathway leading to complete oxidation of carbohydrates, fatty acids, and amino acids to CO_2.

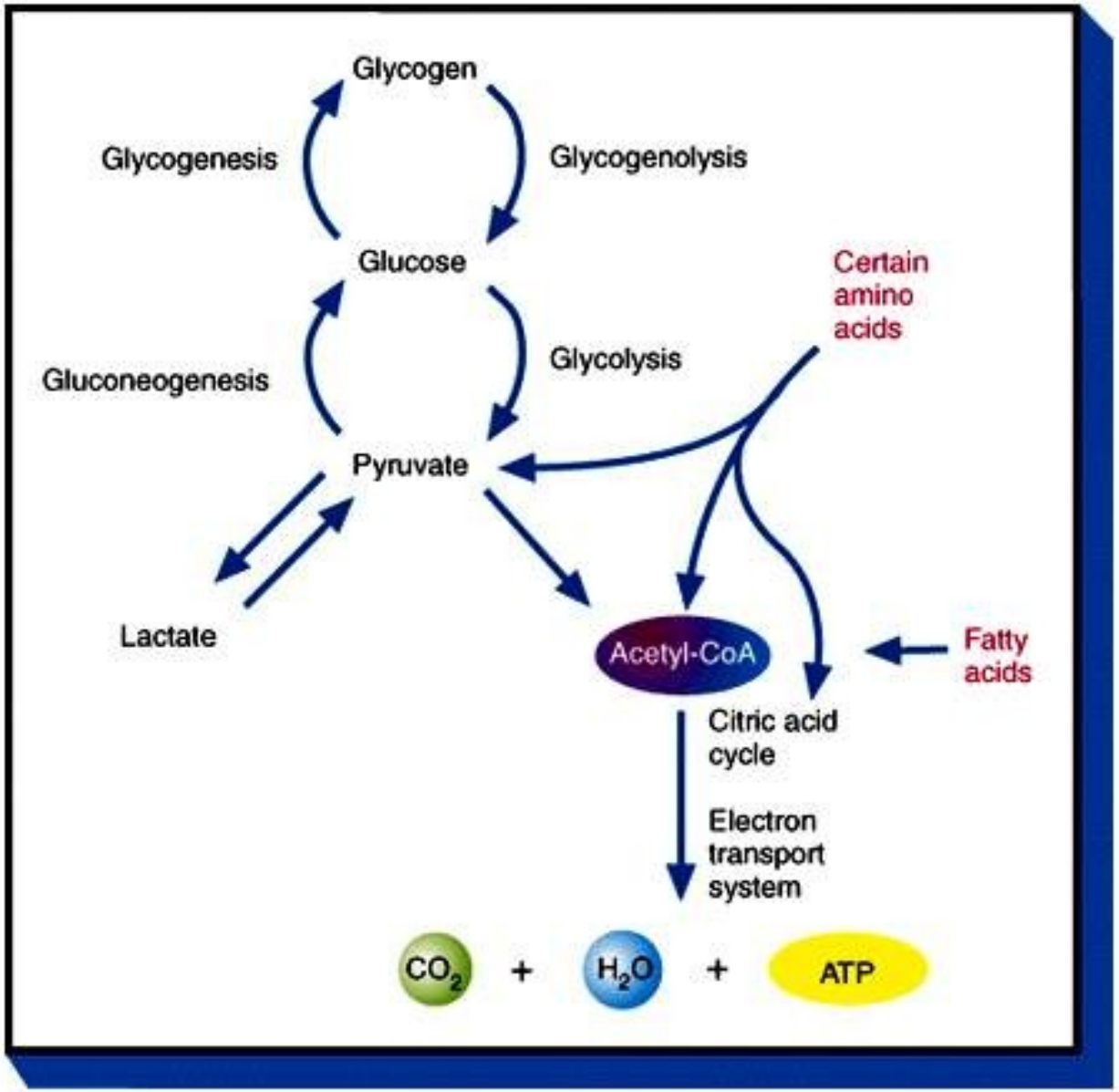
A pathway providing many precursors for biosynthesis

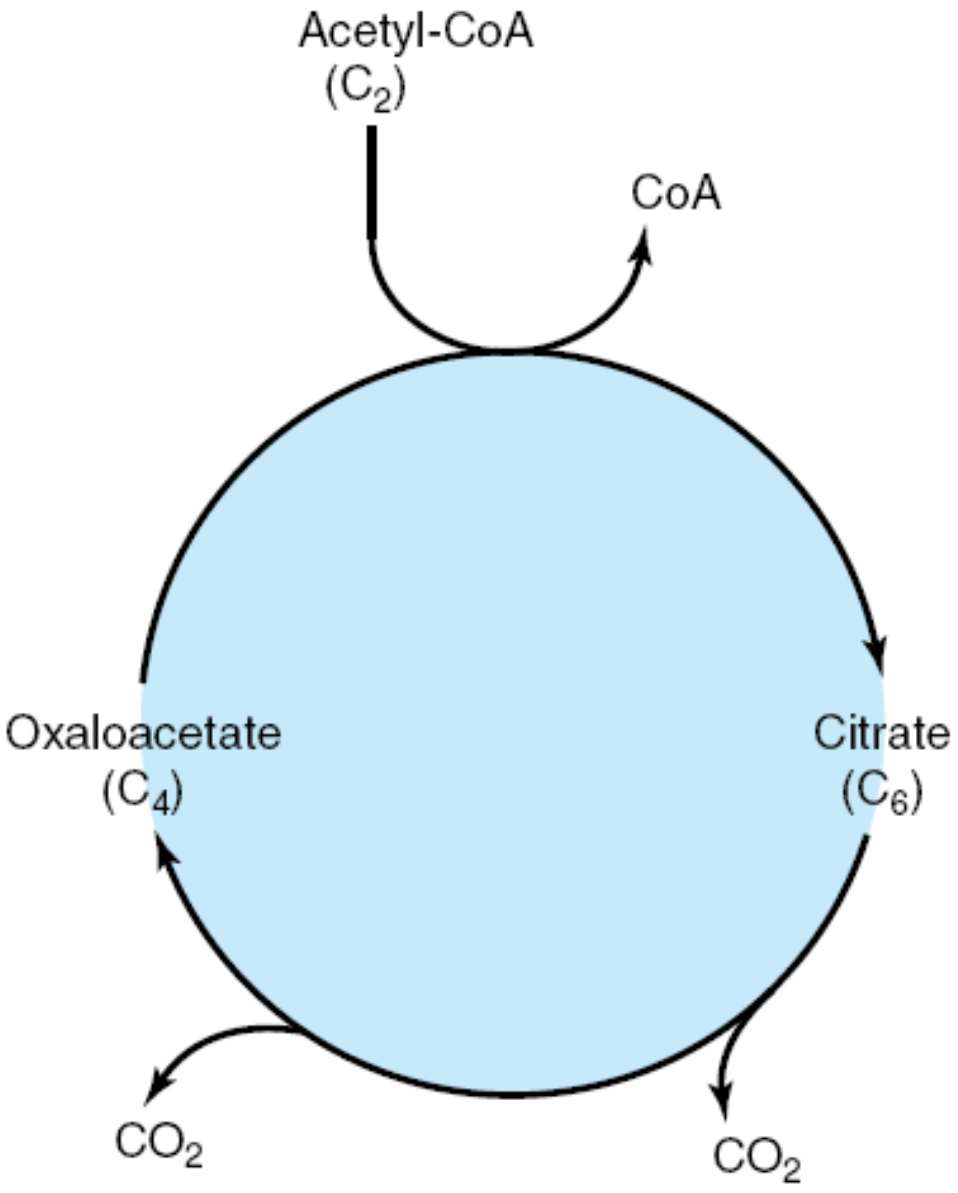



An animal cell
Nucleus
Cytosol
Mitochondrial matrix
Inner mitochondrial membrane
Intermembrane space
The citric acid cycle reactions take place in the mitochondrial matrix.
For each glucose processed by glycolysis, the citric acid cycle "turns" twice.
Acetyl CoA
Oxaloacetate
Citrate
3 NADH +3H+
FADH2
Electron carriers
FAD
Citric acid cycle
3 NAD+
2 CO2
ATP
ADP + Pi

Biochemistry by McKee/McKee, ©1996 McGraw-Hill Higher Education

Overview of major pathways in carbohydrate metabolism.
M Figure 8.1




Acetyl-CoA
(C_2)
CoA
Oxaloacetate
(C_4)
Citrate
(C_6)
CO_2
CO_2

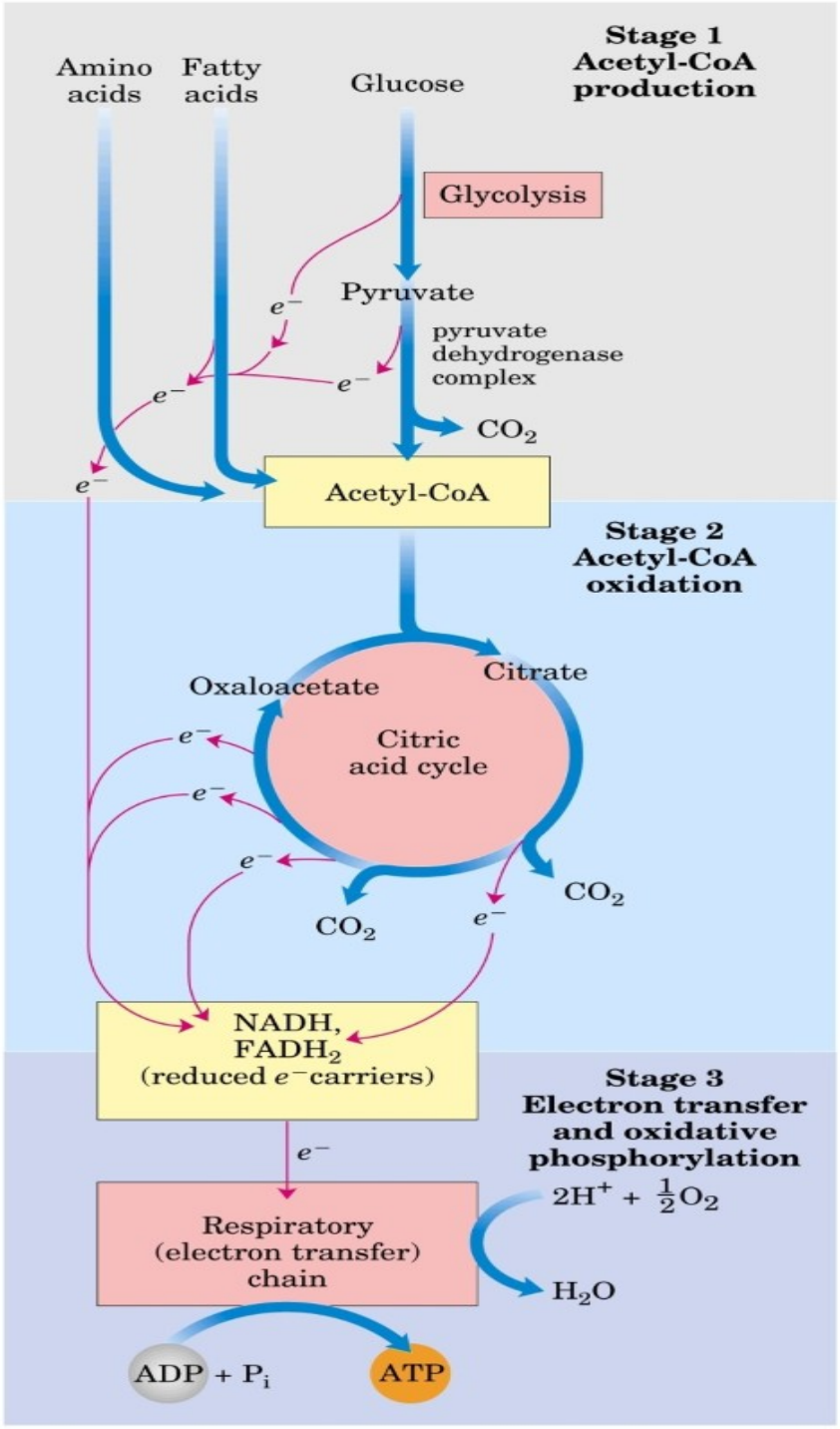
1. The cellular respiration can be divided into three stages

Stage I **All the fuel molecules** are oxidized to generate a common two-carbon unit, acetyl-CoA.

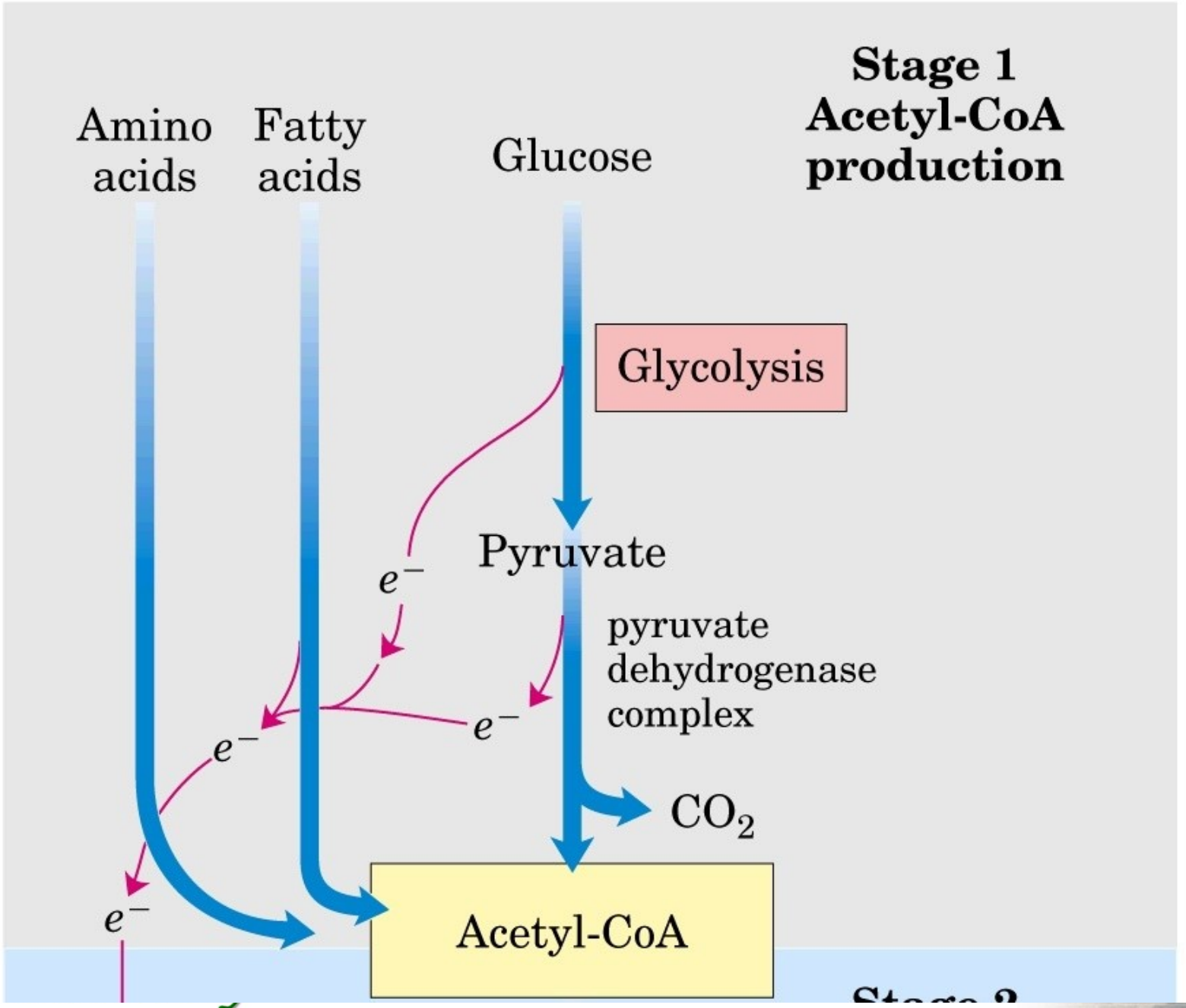
Stage II The acetyl-CoA is completely **oxidized into CO_2,** with electrons collected **by NAD and FAD** via a cyclic pathway (named as the **citric acid cycle, Krebs cycle, or tricarboxylic acid cycle** （**TCA**）).

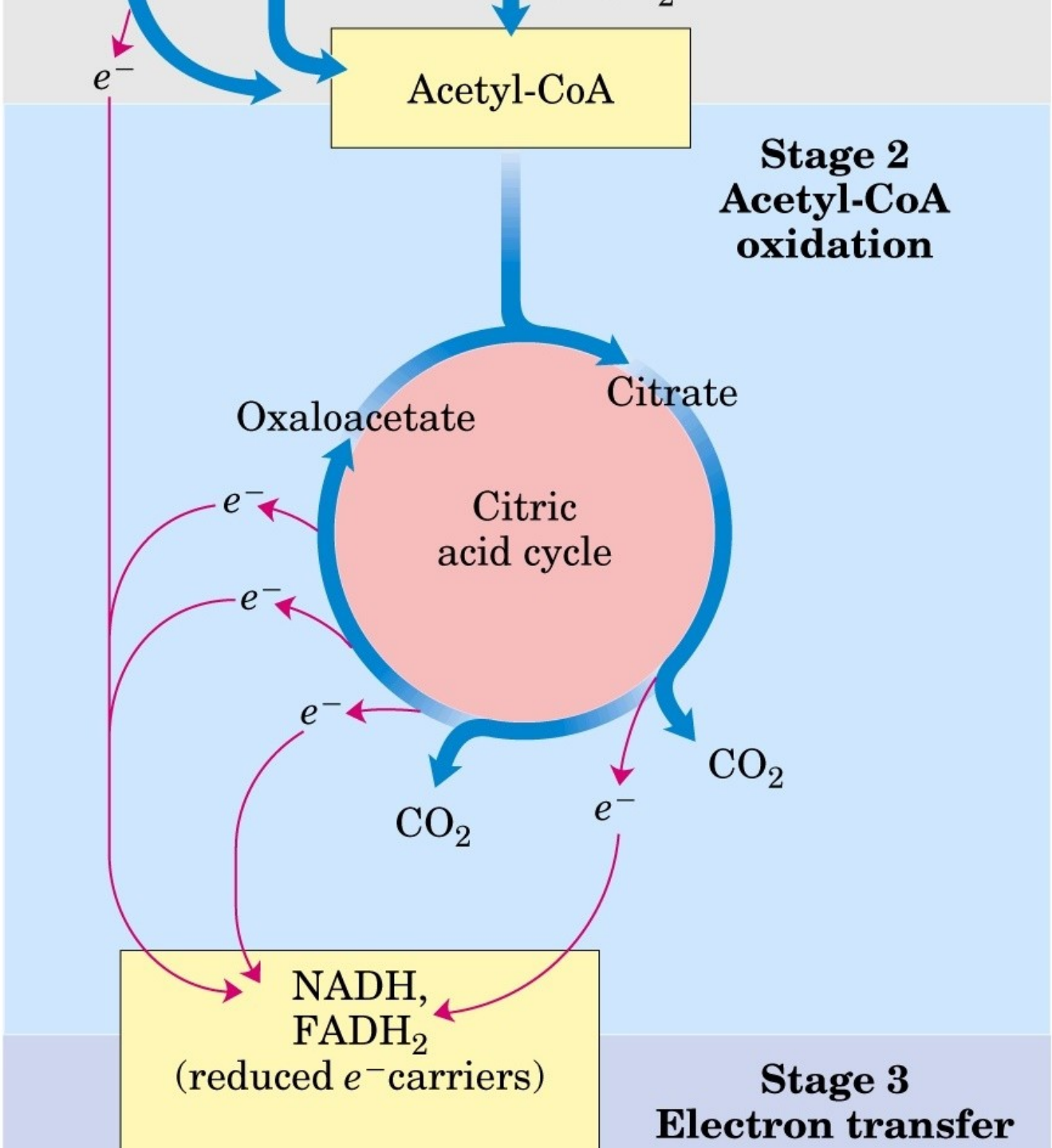
Stage III Electrons of NADH and $FADH_2$ are **transferred** to O_2 via a series carriers, producing H_2O and a H^+ gradient, which will promote ATP formation.

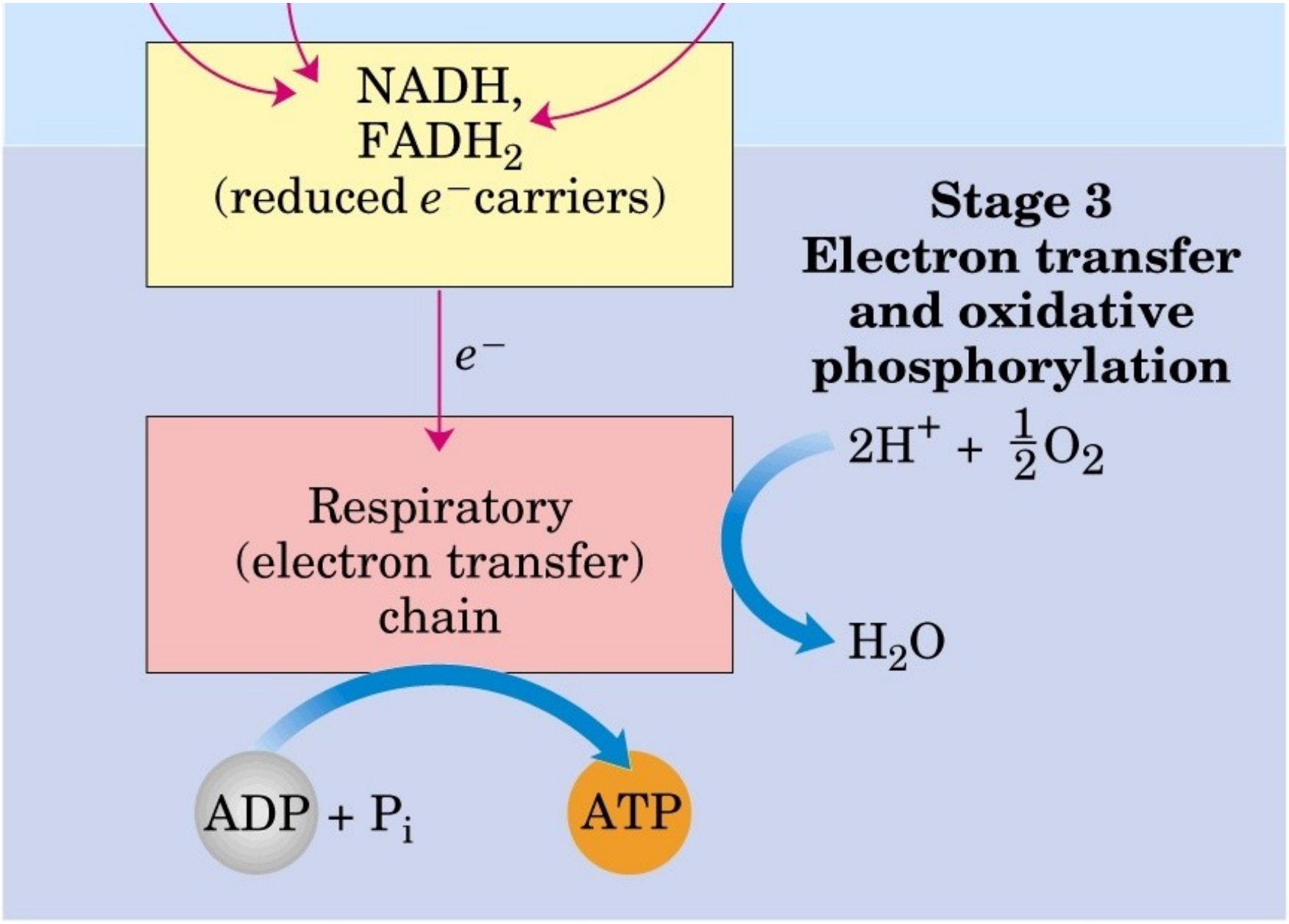



Stage 1
Acetyl-CoA
production
Amino
acids
Fatty
acids
Glucose
Glycolysis
Pyruvate
pyruvate
dehydrogenase
complex
e^-
e^-
e^-
e^-
CO_2
Acetyl-CoA
Stage 2
Acetyl-CoA
oxidation
Citrate
Oxaloacetate
Citric
acid cycle
e^-
e^-
e^-
e^-
CO_2
CO_2
NADH,
$FADH_2$
(reduced e^- carriers)
Stage 3
Electron transfer
and oxidative
phosphorylation
e^-
$2H^+ + \frac{1}{2}O_2$
Respiratory
(electron transfer)
chain
H_2O
ADP + P_i
ATP

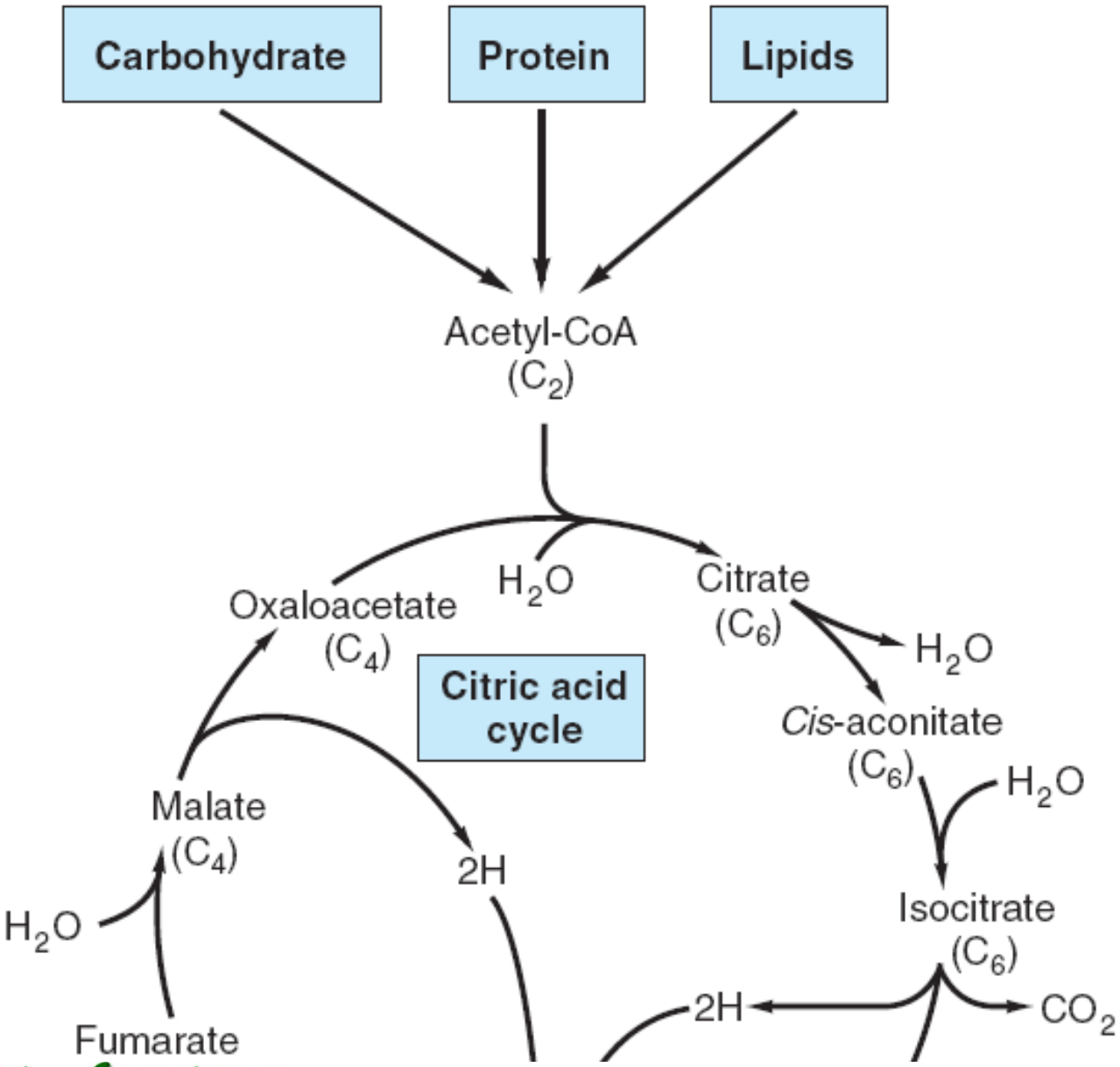
Biochemist


Stage 1
Acetyl-CoA
production
Amino
acids
Fatty
acids
Glucose
Glycolysis
Pyruvate
pyruvate
dehydrogenase
complex
e^-
e^-
e^-
e^-
CO_2
Acetyl-CoA

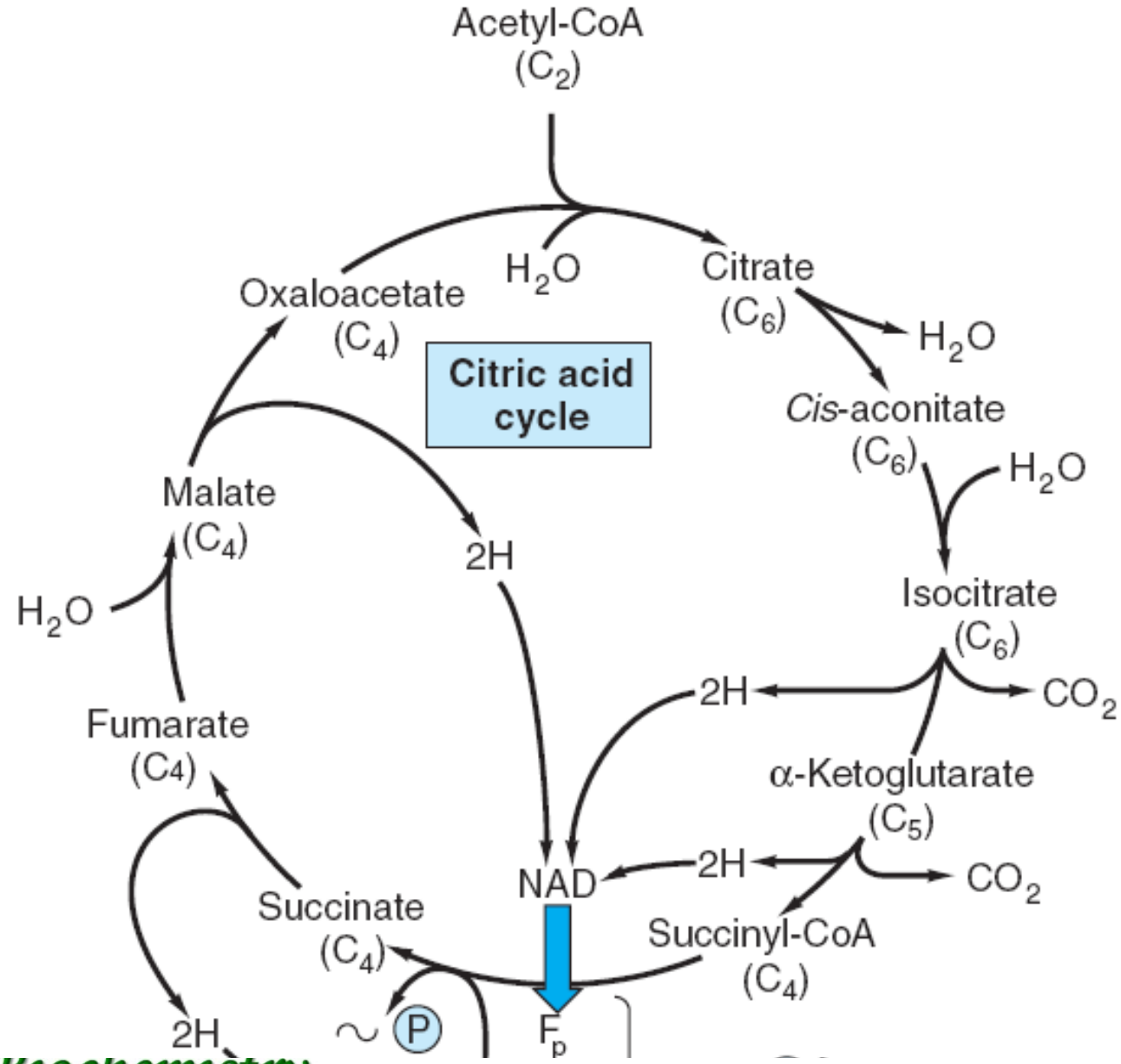

Acetyl-CoA
Stage 2
Acetyl-CoA
oxidation
Citrate
Oxaloacetate
Citric
acid cycle
e^-
e^-
e^-
e^-
CO_2
CO_2
NADH,
$FADH_2$
(reduced e^- carriers)
Stage 3
Electron transfer


NADH,
$FADH_2$
(reduced e^- carriers)
e^-
Respiratory
(electron transfer)
chain
ADP + P_i
ATP
Stage 3
Electron transfer
and oxidative
phosphorylation
$2H^+ + \frac{1}{2}O_2$
H_2O

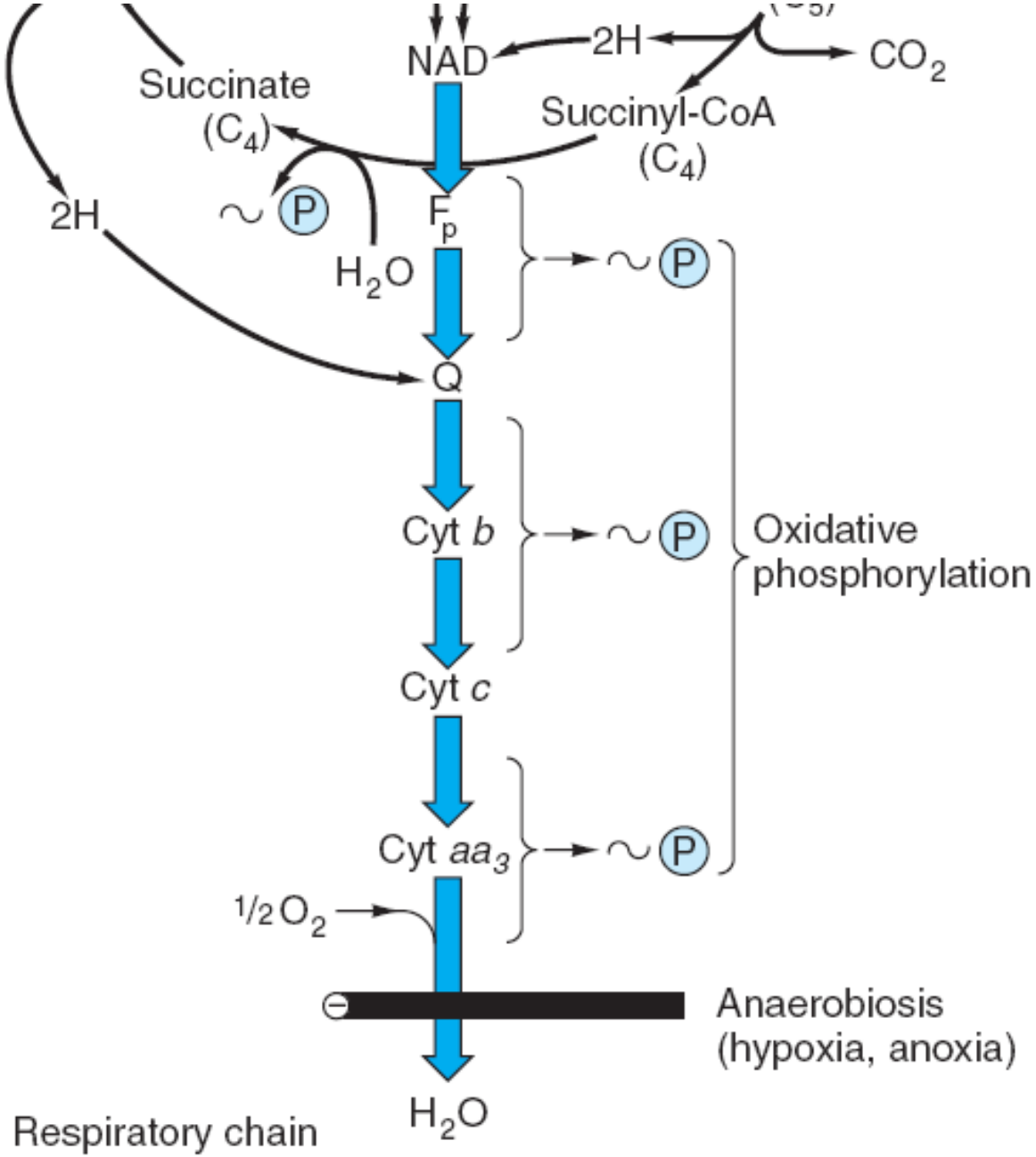
Stage I


Carbohydrate
Protein
Lipids
Acetyl-CoA
(C_2)
H_2O
Oxaloacetate
(C_4)
Citrate
(C_6)
H_2O
Citric acid cycle
Cis-aconitate
(C_6)
H_2O
Malate
(C_4)
2H
H_2O
Isocitrate
(C_6)
2H
CO_2
Fumarate

Stage II


Acetyl-CoA
(C2)
H2O
Citrate
(C6)
H2O
Cis-aconitate
(C6)
H2O
Isocitrate
(C6)
2H
CO2
α-Ketoglutarate
(C5)
2H
CO2
Succinyl-CoA
(C4)
NAD
2H
Succinate
(C4)
2H
Fumarate
(C4)
H2O
Malate
(C4)
2H
Oxaloacetate
(C4)
Citric acid cycle
P
Fp

Stage III



Respiratory chain

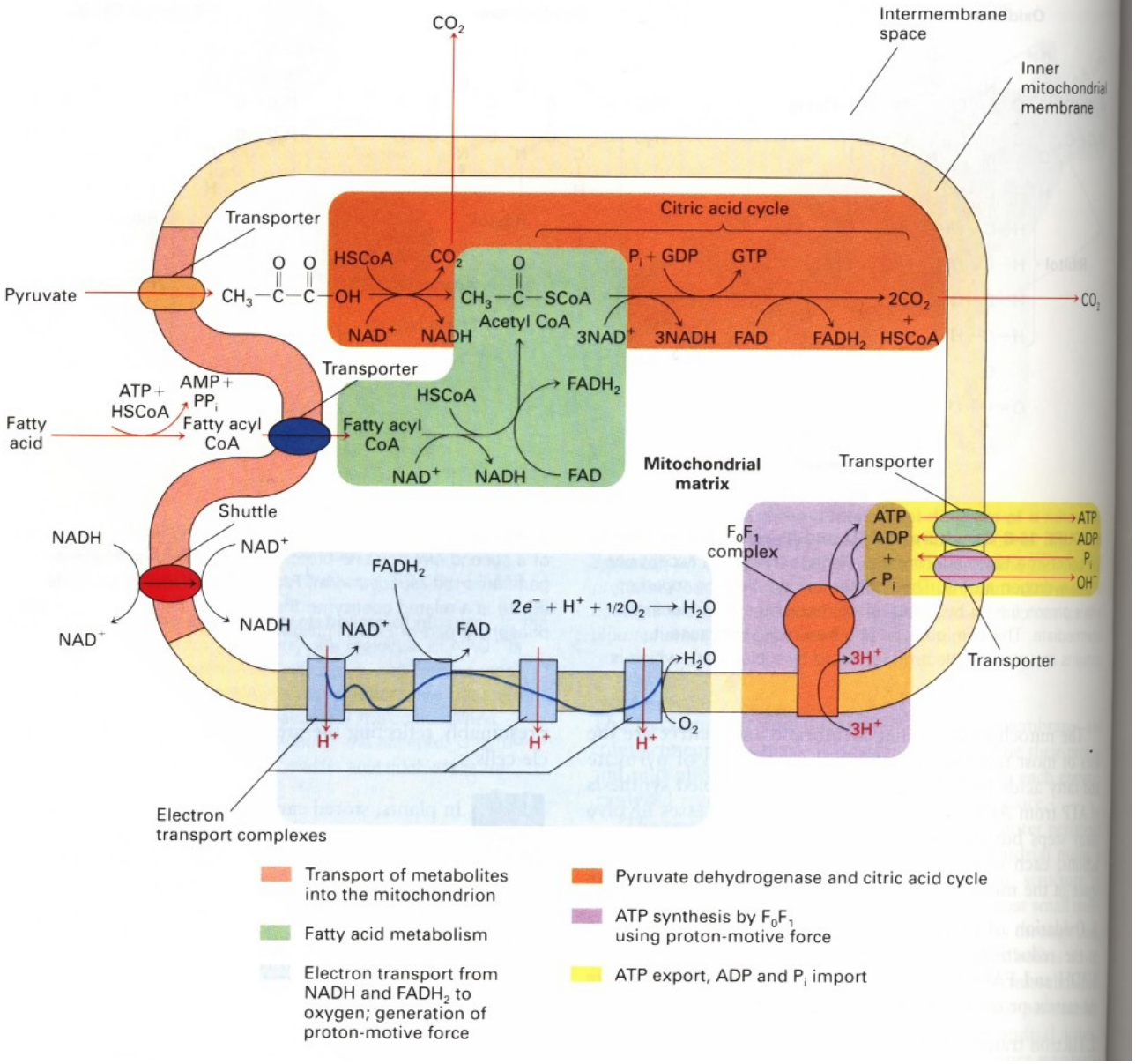


Hans Krebs
1900 – 1981

Biochemist



Mitochondria is the major site for fuel oxidation to generate ATP.



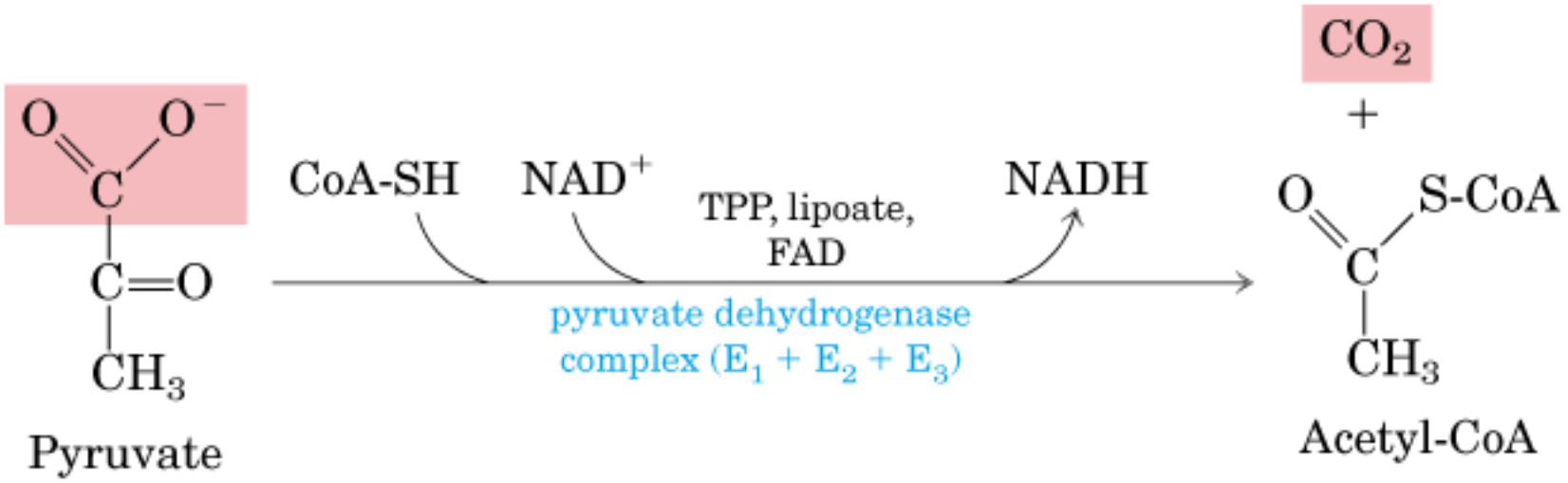
Biochemistry



2. Pyruvate is oxidized to acetyl-CoA by the catalysis of pyruvate dehydrogenase complex

Pyruvate is first transported into mitochondria via a specific transporter on the inner membrane. Pyruvate is converted to acetyl-CoA and CO_2 by oxidative decarboxylation.





$\Delta G'^{\circ} = -33.4$ kJ/mol

The oxidative decarboxylation of pyruvate in mitochondria: producing acetyl-CoA and CO_2.



- The **pyruvate dehydrogenase complex** is a huge multimeric assembly of three kinds of enzymes, having 60 subunits in bacteria and more in mammals.
- **pyruvate dehydrogenase (E_1)**
- **dihydrolipoyl transacetylase (E_2)**
- **dihydrolipoyl dehydrogenase (E_3)**



丙酮酸脱氢酶复合体

E1： 丙酮酸脱氢酶组分（ pyruvate dehydrogenase ）

E2 ： 二氢硫辛酸转乙酰基酶

（ dihydrolipoyl transacetylase ）

E3 ： 二氢硫辛酸脱氢酶

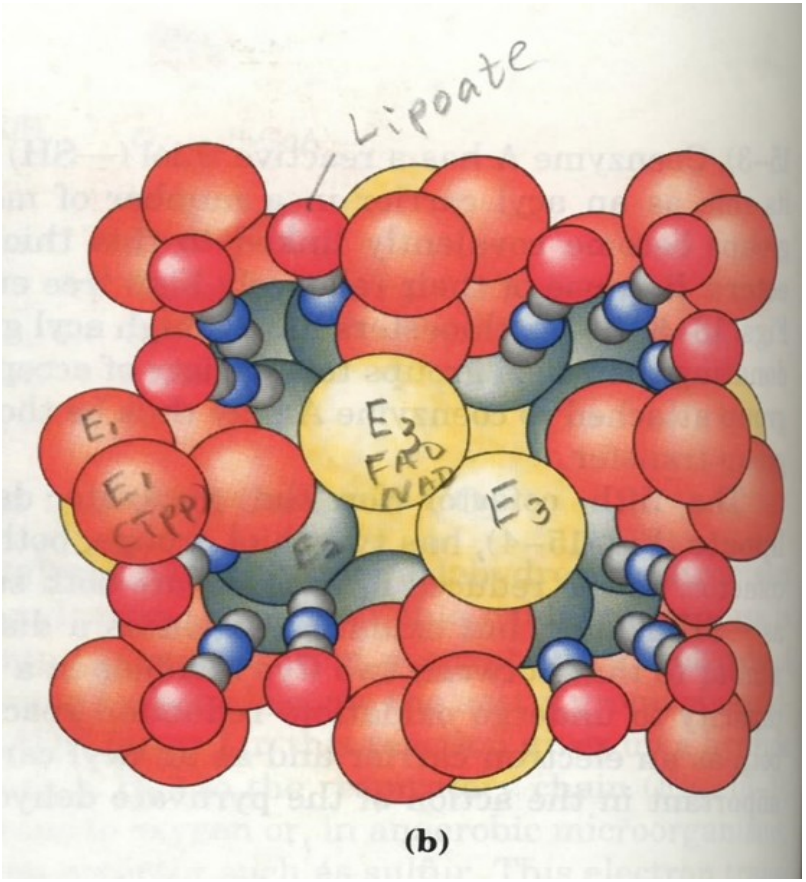
（ dihydrolipoyl dehydrogenase ）

辅因子： CoA-SH, NADH, 硫辛酸

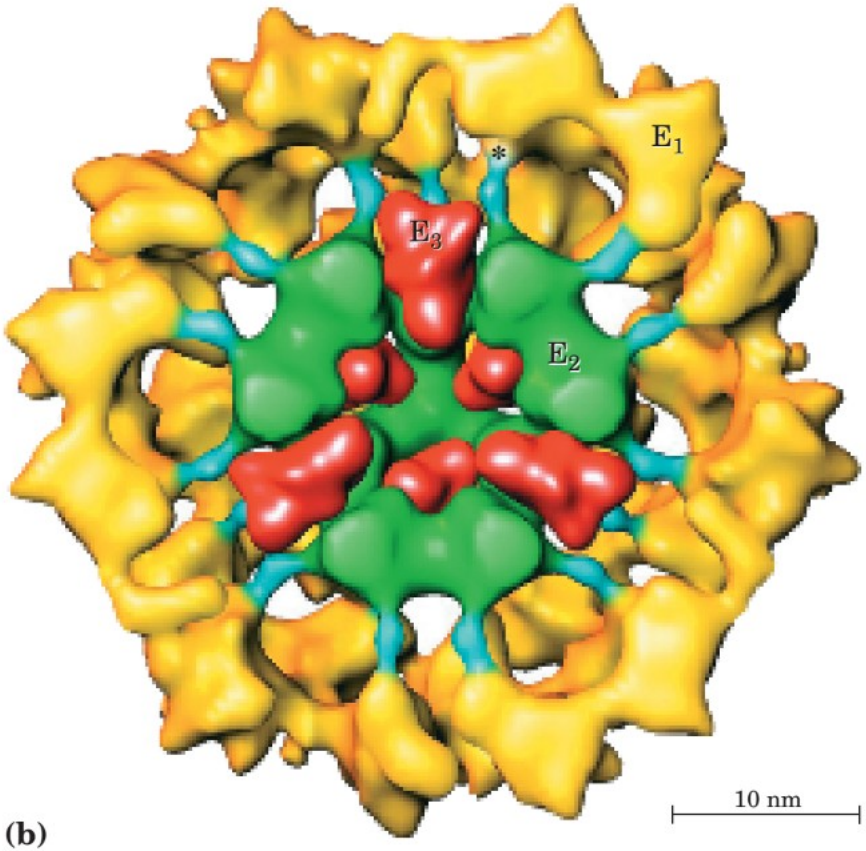
FAD， TPP，

Mg^{2+}

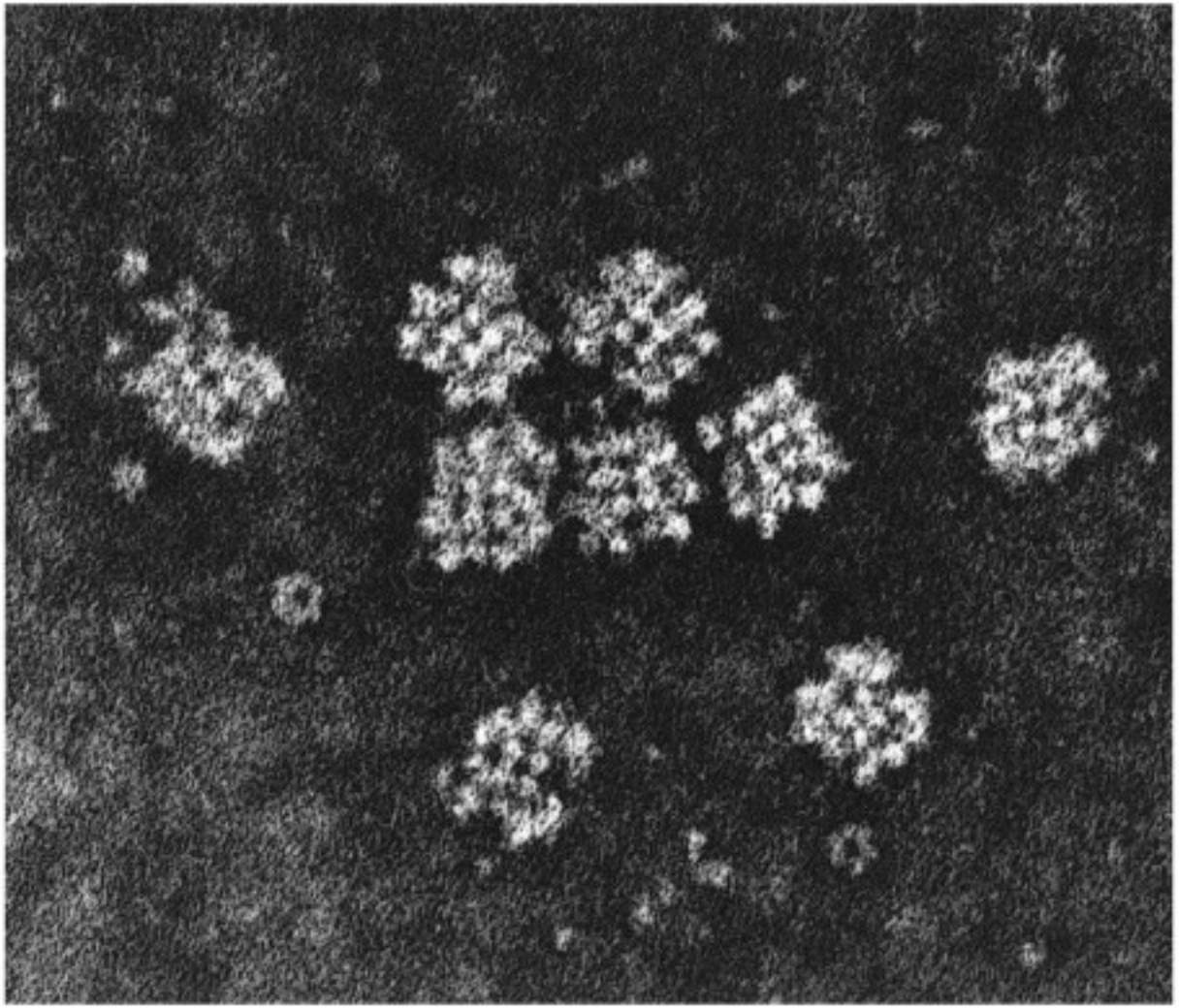


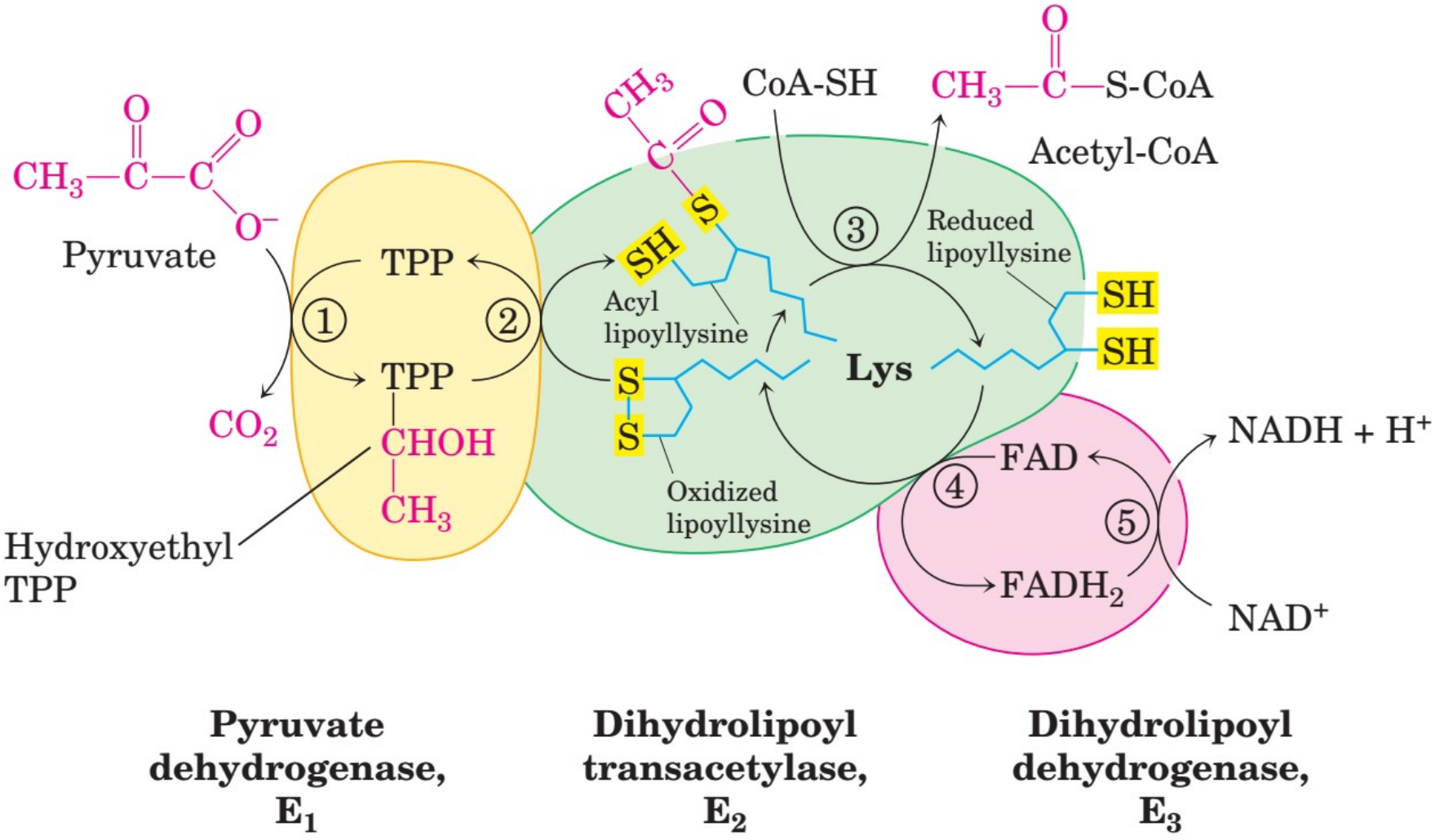


(b)



(b)

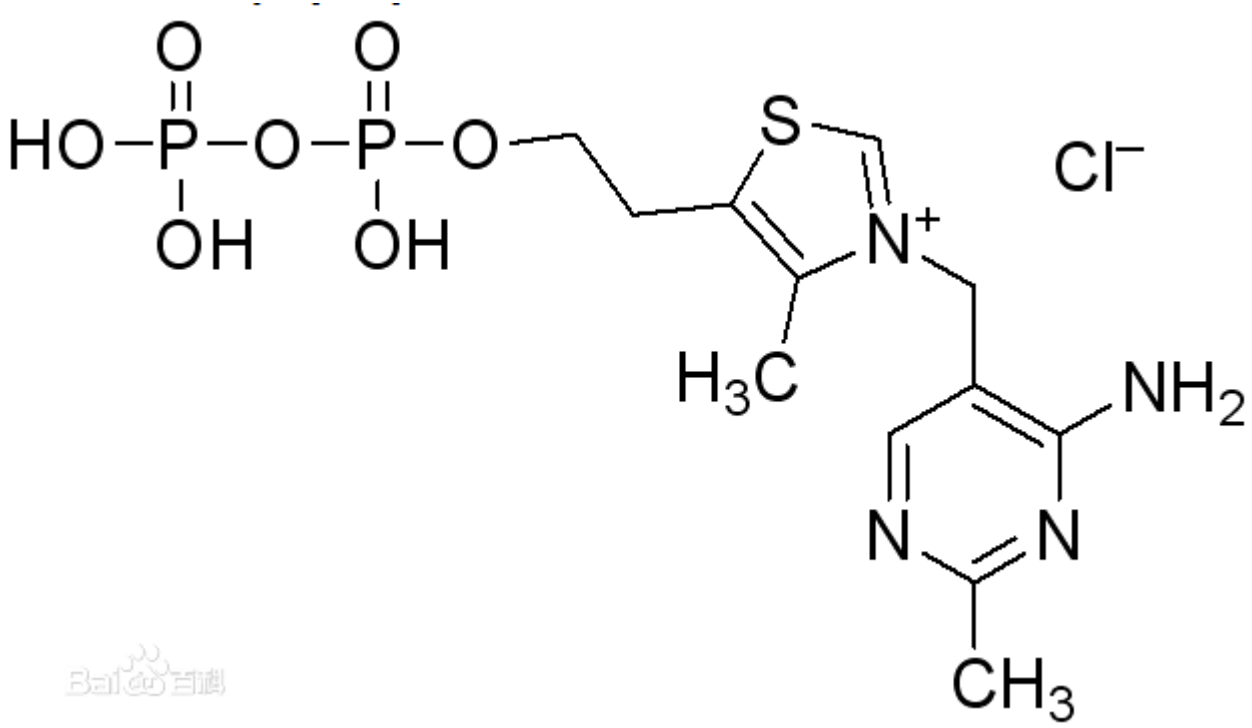

0.05 μm

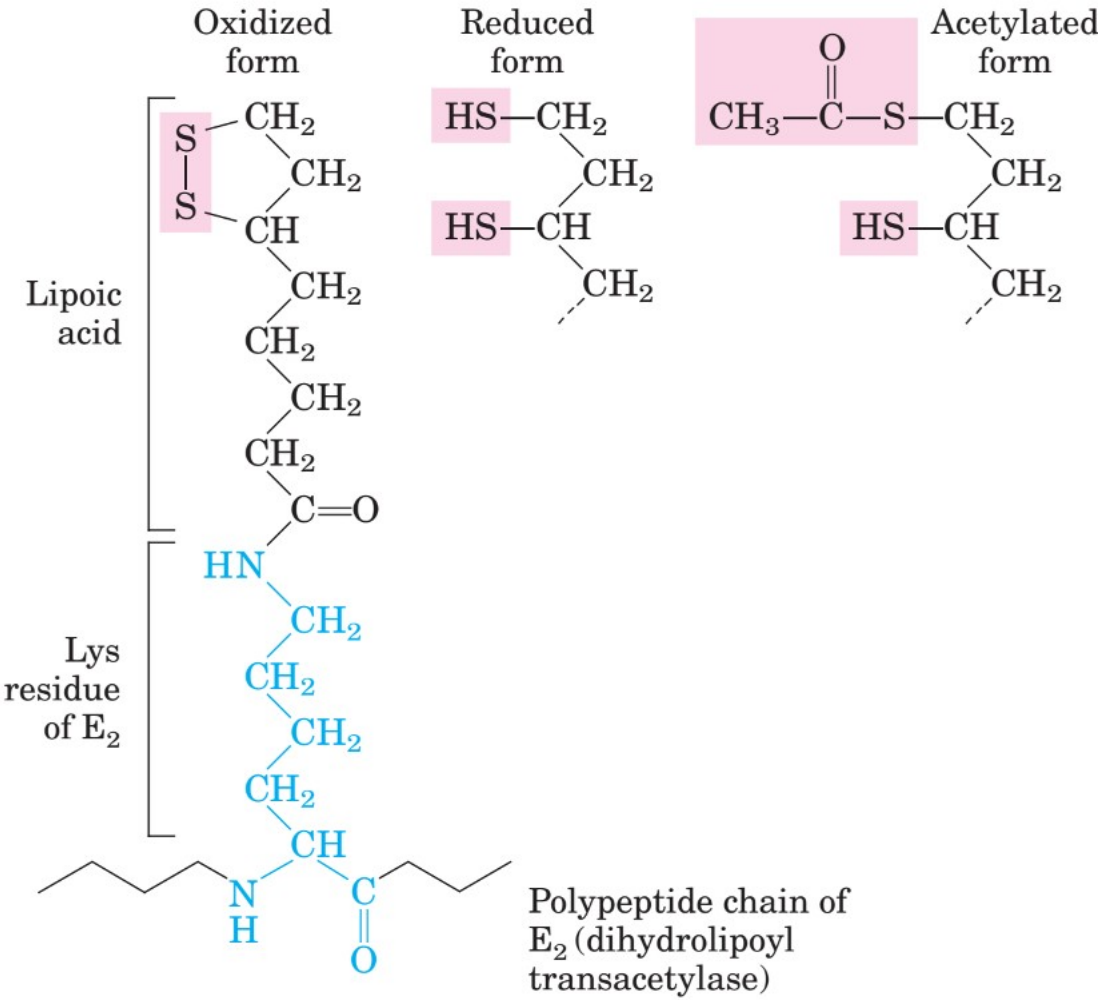


The oxidative decarboxylation of pyruvate is catalyzed by a multienzyme complex: pyruvate dehydrogenase complex.

τηιαμινε πψροπηοσπηατε (ΤΠΠ)
焦磷酸硫胺素

Confusing with
base T Thymine

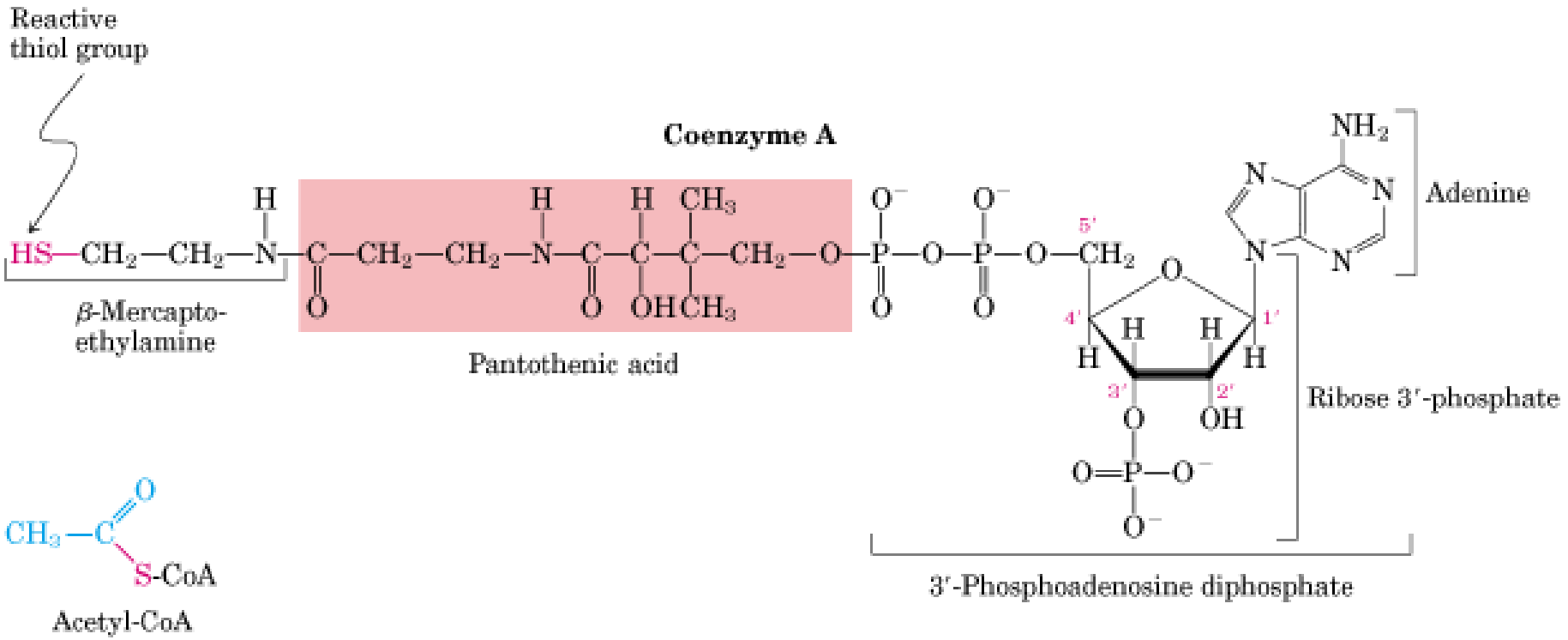




Λιποιχ αχιδ (λιποατε) ιν αμιδε λινκαγε ωιτη α Λψσ ρεσιδυε.

硫辛酸

Biochemistry



Coenzyme A (CoA-SH): discovered in 1945 by Lipmann, one of the "carrier molecules", deliver activated acyl groups (with 2-24 Carbons) for degradation or biosynthesis.

3. The complete oxidation of pyruvate in animal tissues was proposed to undergo via a cyclic pathway

Hans Krebs proposed the “citric acid cycle” for the complete oxidation of pyruvate in animal tissues in 1937 (he wrongly hypothesized that pyruvate condenses with oxaloacetate in his original proposal).

The citric acid cycle was confirmed to be universal in cells by *in vitro* studies with purified enzymes and *in vivo* studies with radio isotopes (“radio isotope tracer experiments”).

Krebs was awarded the Nobel prize in medicine in 1953 for revealing the citric acid cycle (thus also called the Krebs cycle).



The Nobel Prize in Physiology or Medicine 1953

Presentation Speech

"for his discovery of the citric acid cycle"



Sir Hans Adolf Krebs

Great Britain

Sheffield University
Sheffield, Great Britain

1900 - 1981

Biography

4. The acetyl group (carried by CoA) is completely oxidized to CO_2 via the citric acid cycle

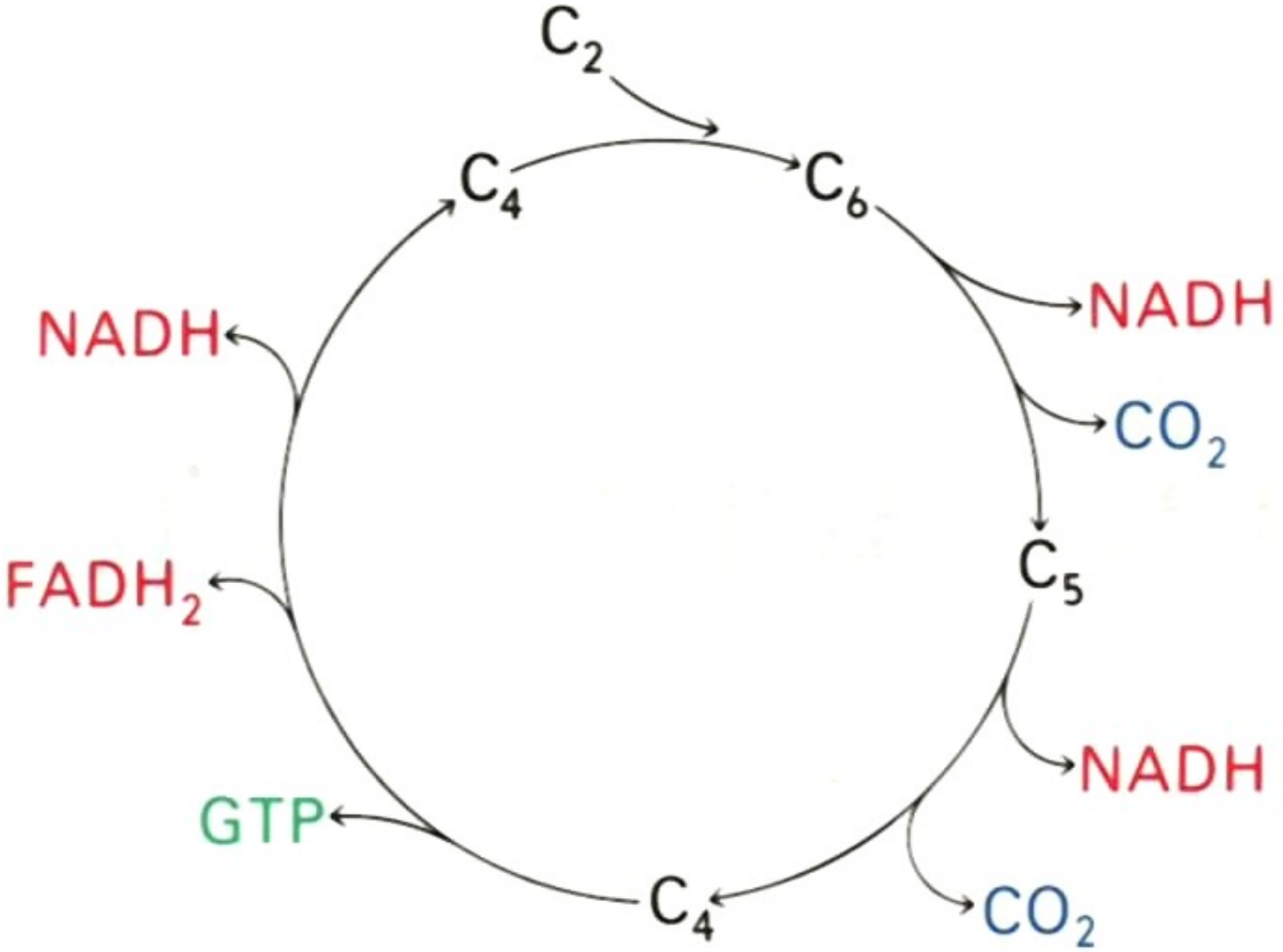
The 4-carbon oxaloacetate (草酰乙酸) acts as the "carrier" for the oxidation.

The two carbons released as 2 CO_2 in the first cycle of oxidation are not from the acetyl-CoA just joined.

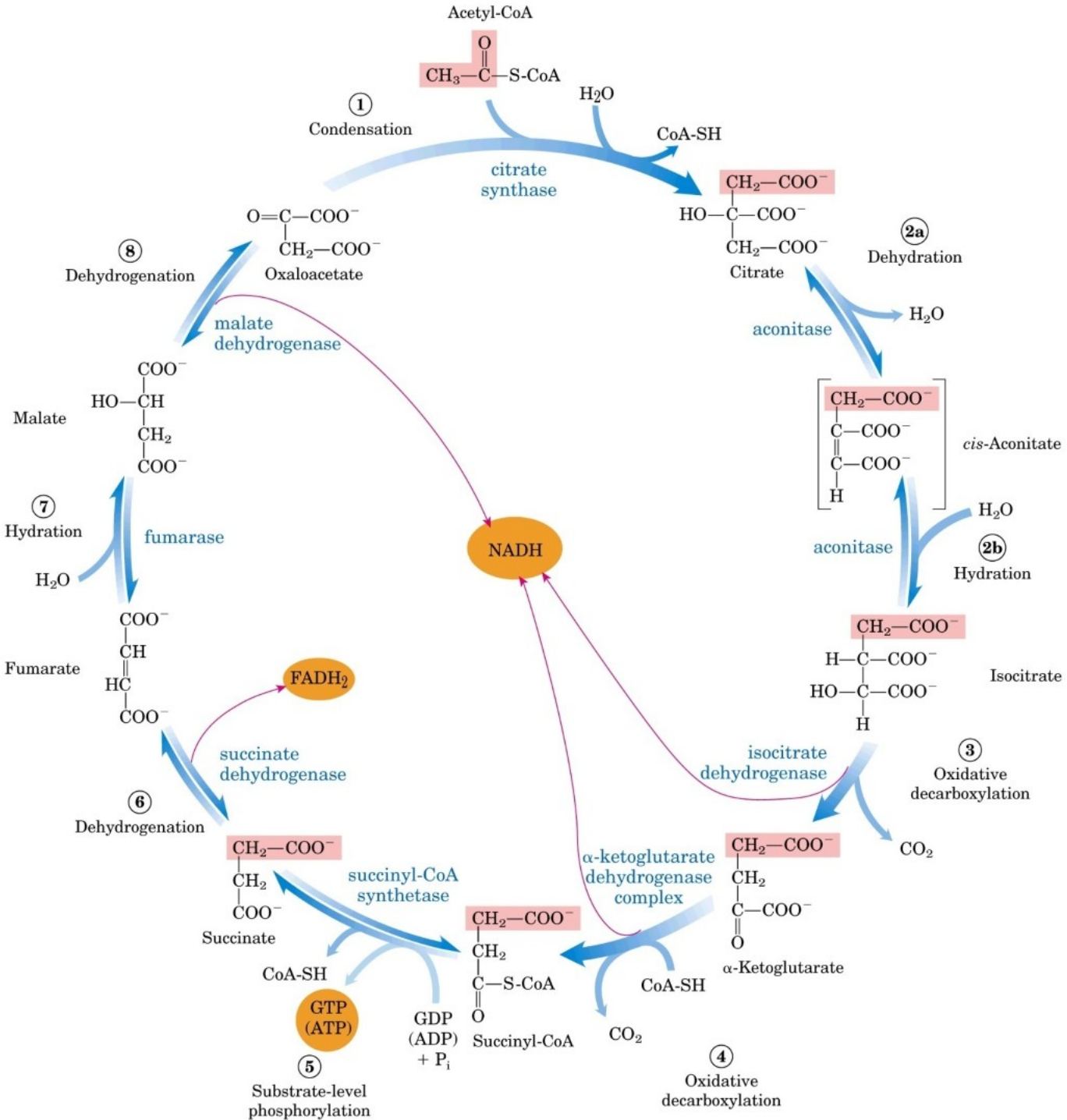
The 8 electrons released are collected by three NAD^+ and one FAD.

One molecule of ATP (or GTP) is produced per cycle by substrate-level phosphorylation.




C_2
C_4
C_6
NADH
CO_2
C_5
NADH
CO_2
C_4
GTP
$FADH_2$
NADH

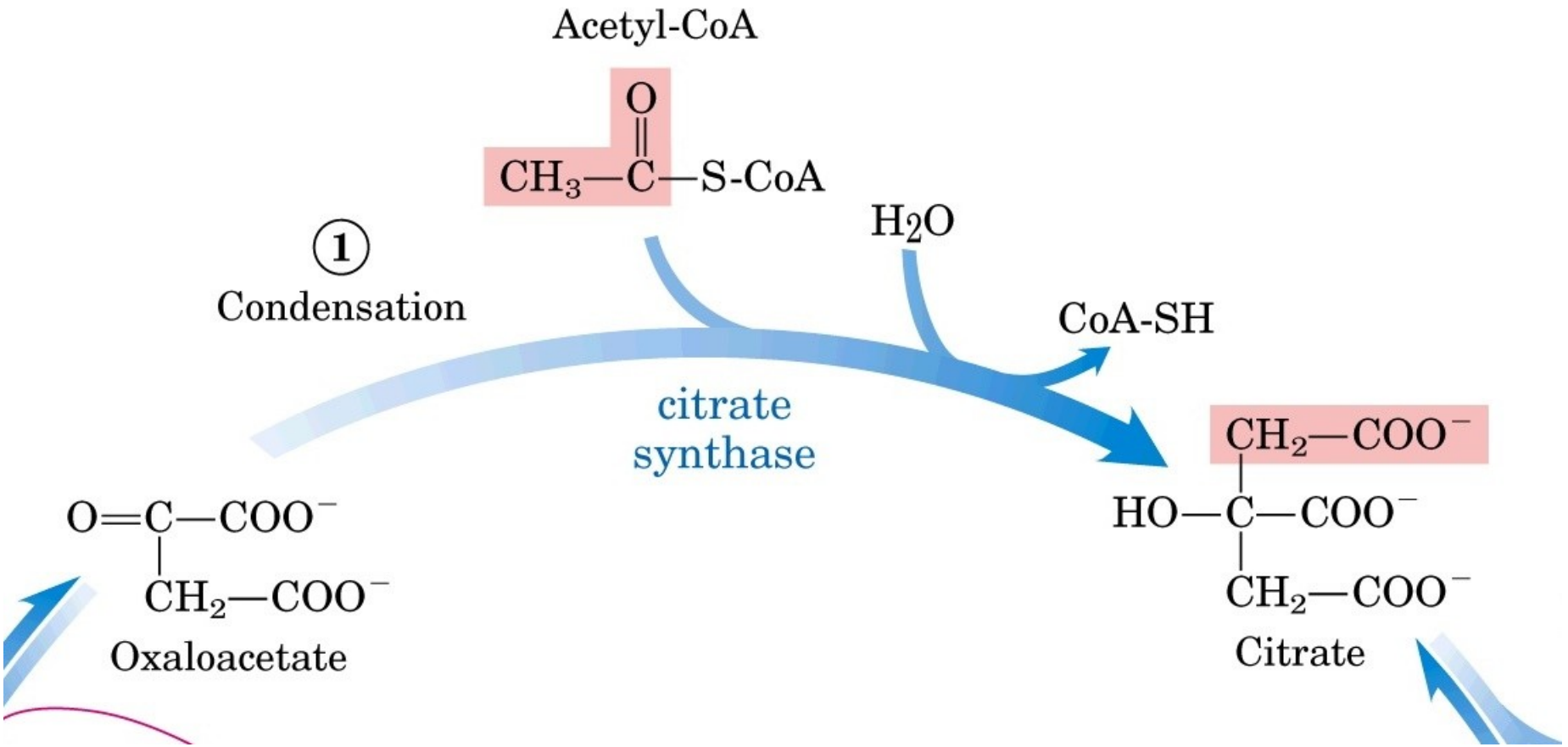
The citric acid cycle

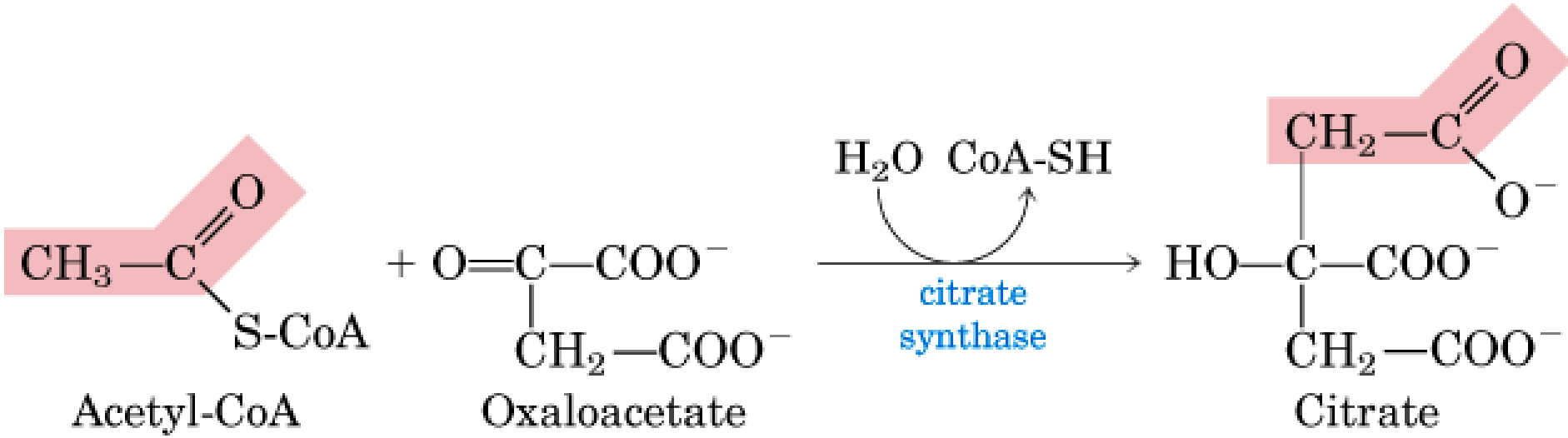


5. The citric acid cycle consists of eight successive reactions

Step 1
Citrate (6C) is formed from the irreversible condensation of acetyl CoA (2C) and oxaloacetate (4C) – catalyzed by citrate synthase.




Acetyl-CoA
O
CH3—C—S-CoA
H2O
1
Condensation
CoA-SH
citrate
synthase
O=C—COO⁻
CH2—COO⁻
Oxaloacetate
CH2—COO⁻
HO—C—COO⁻
CH2—COO⁻
Citrate

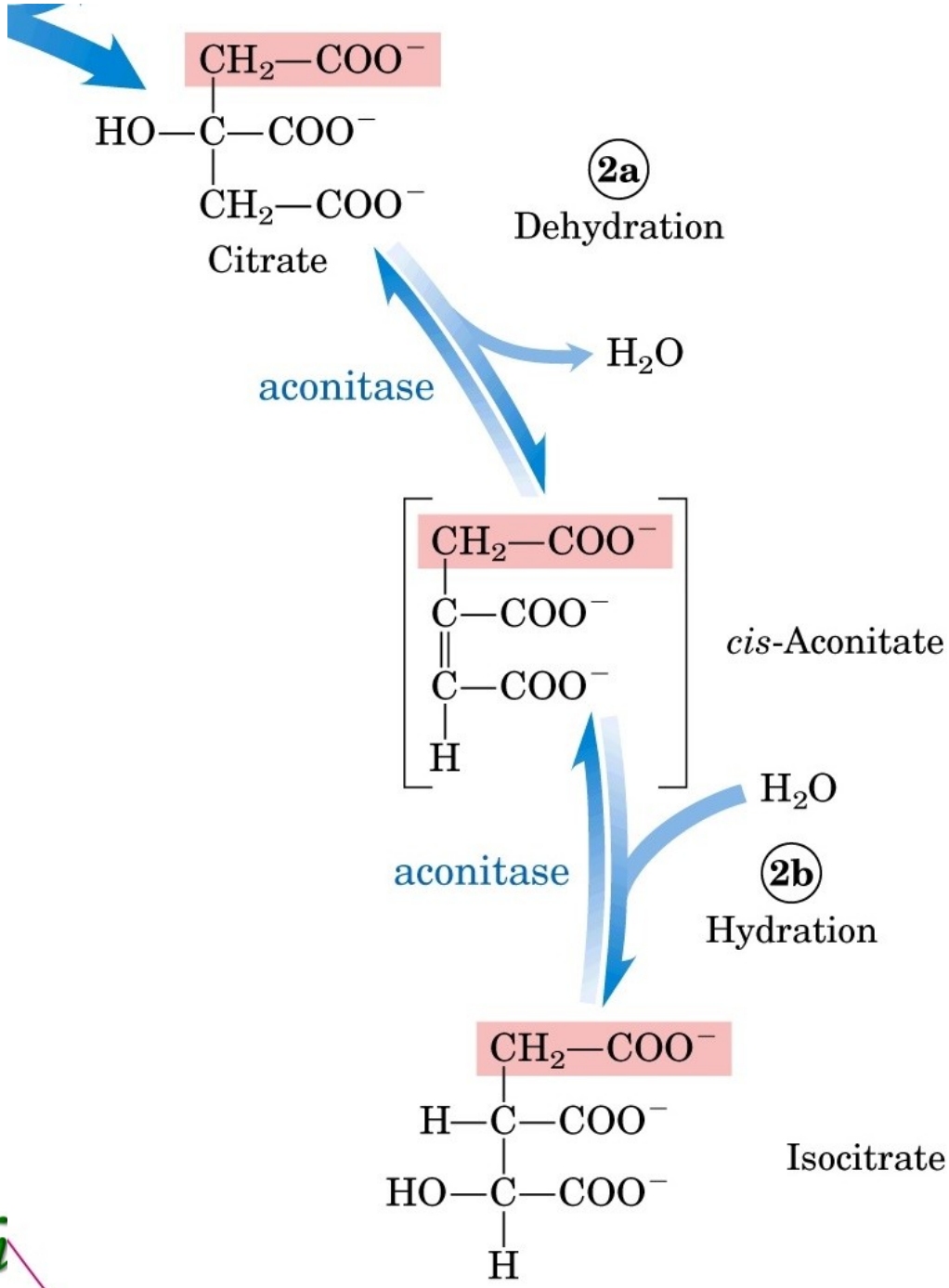


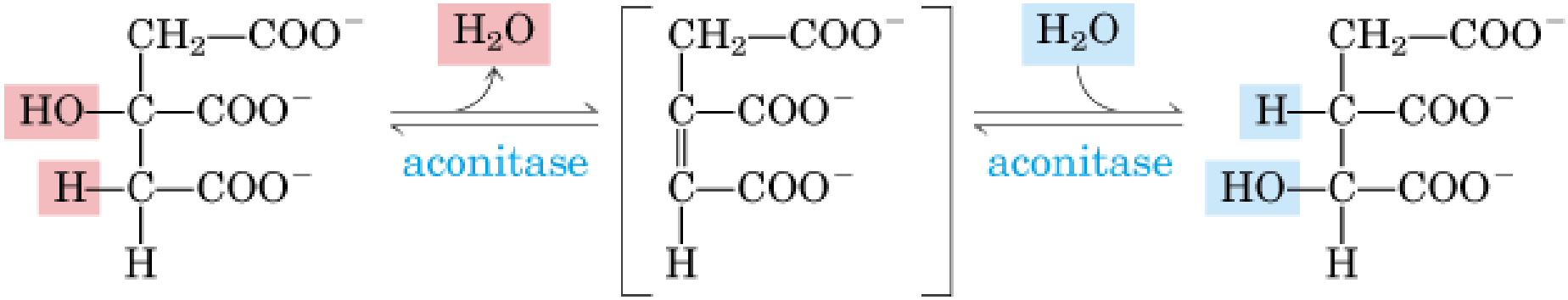
Acetyl-CoA Oxaloacetate Citrate

$\Delta G'^{\circ} = -32.2$ kJ/mol

Step 2 Citrate is isomerized into isocitrate via a dehydration step followed by a hydration step; *cis*-aconitate (顺乌头酸) is an intermediate during this transformation, thus the catalytic enzyme is named as aconitase.




CH2—COO⁻
HO—C—COO⁻
CH2—COO⁻
Citrate
2a
Dehydration
H2O
aconitase
CH2—COO⁻
C—COO⁻
C—COO⁻
H
cis-Aconitate
H2O
aconitase
2b
Hydration
CH2—COO⁻
H—C—COO⁻
HO—C—COO⁻
H
Isocitrate

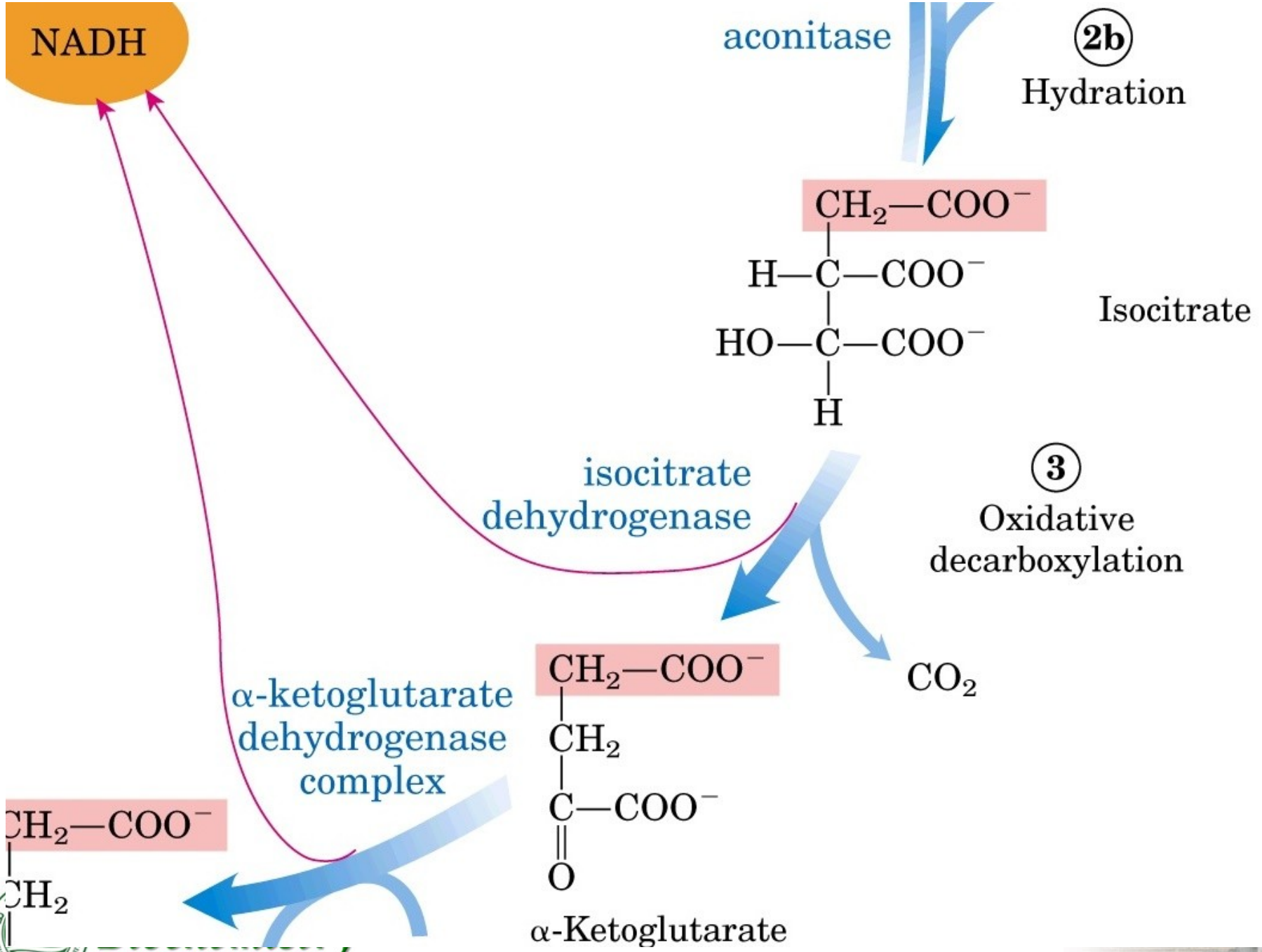


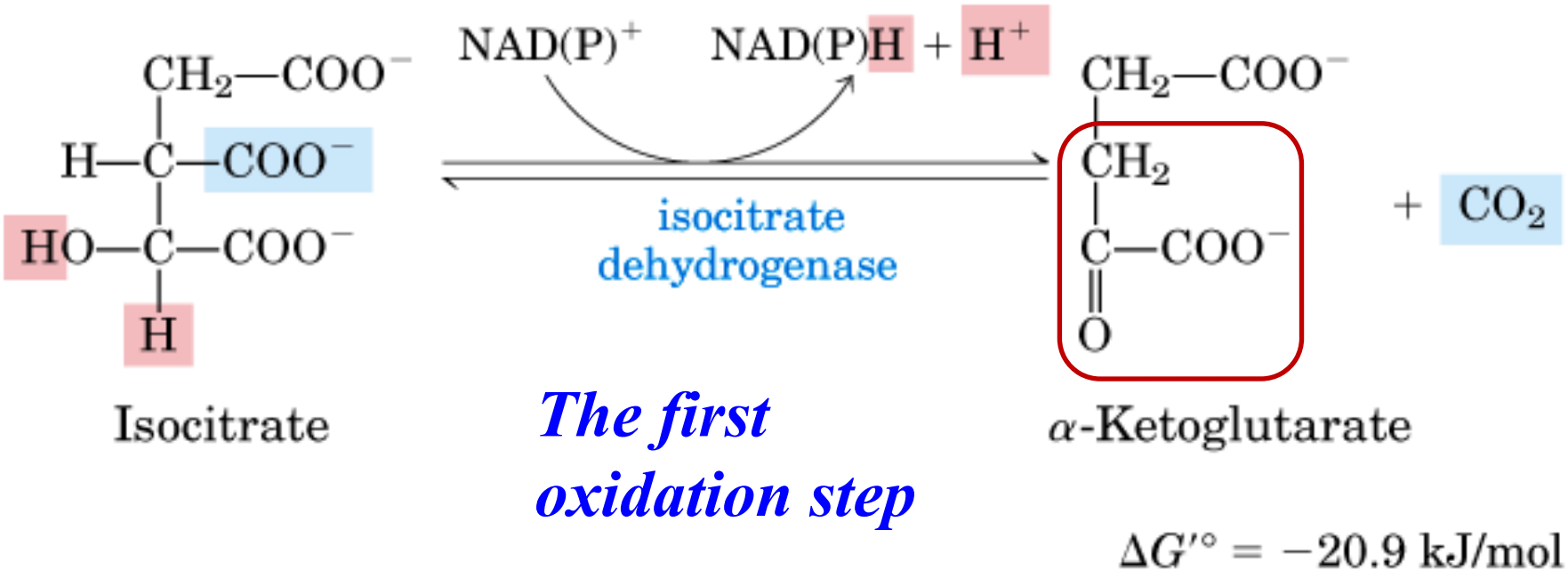
Citrate *cis*-Aconitate Isocitrate

$\Delta G'^{\circ} = 13.3$ kJ/mol

- **Step 3** Isocitrate is **first oxidized** and **then decarboxylated** to form **a-ketoglutarate** (α－酮戊二酸); two electrons are collected by NAD^+; The reaction is catalyzed by isocitrate dehydrogenase.




NADH
aconitase
2b
Hydration
CH2—COO−
H—C—COO−
HO—C—COO−
H
Isocitrate
isocitrate
dehydrogenase
3
Oxidative
decarboxylation
CO2
α-ketoglutarate
dehydrogenase
complex
CH2—COO−
CH2
C—COO−
O
α-Ketoglutarate
CH2—COO−
CH2



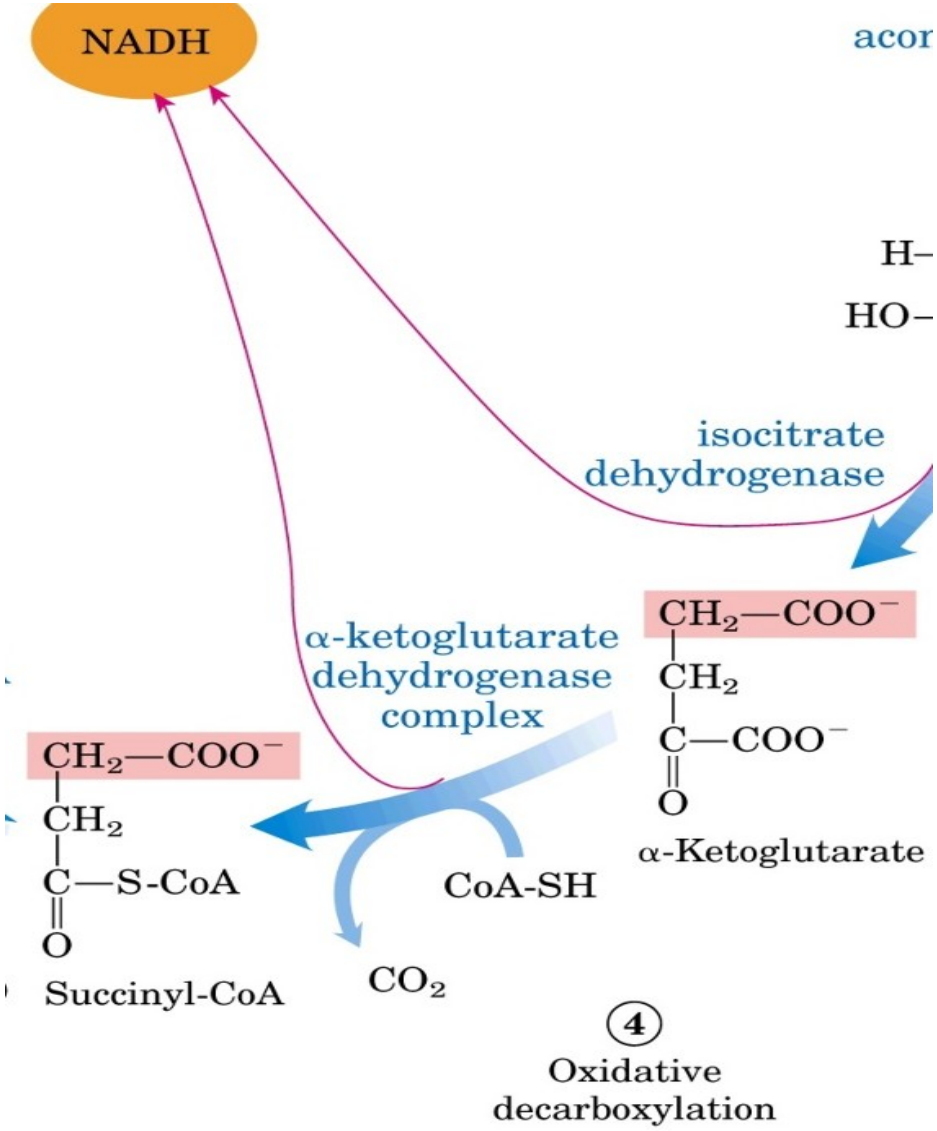
The first oxidation step

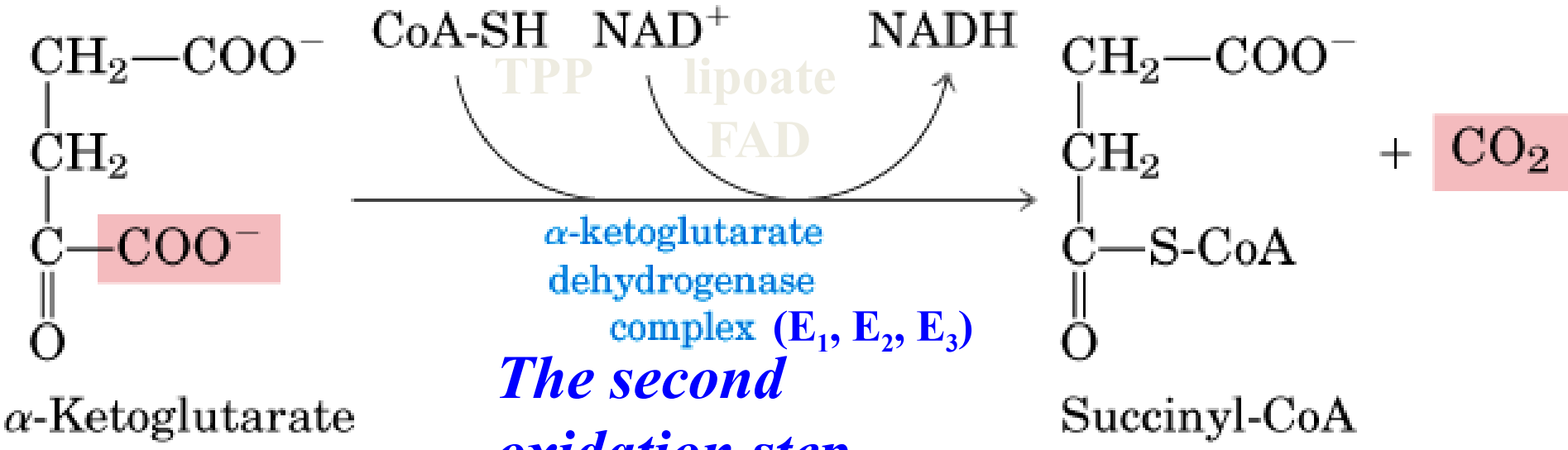
Isocitrate is converted to α-ketoglutarate via an oxidative decarboxylation step, generating NADH CO_2.

Step 4 α-ketoglutarate undergoes another round of oxidative decarboxylation; **decarboxylated first**, **then oxidized** to form **succinyl-CoA** (琥珀酰辅酶 A); The reaction is catalyzed by **α-ketoglutarate dehydrogenase complex**; reactions and enzymes closely **resemble pyruvate dehydrogenase** complex (with similar E_1 and E_2, identical E_3).



Step 4


NADH
acor
H–
HO–
isocitrate
dehydrogenase
α-ketoglutarate
dehydrogenase
complex
CH2—COO⁻
CH2
C—COO⁻
O
α-Ketoglutarate
CH2—COO⁻
CH2
C—S-CoA
O
Succinyl-CoA
CoA-SH
CO2
4
Oxidative
decarboxylation



The second oxidation step

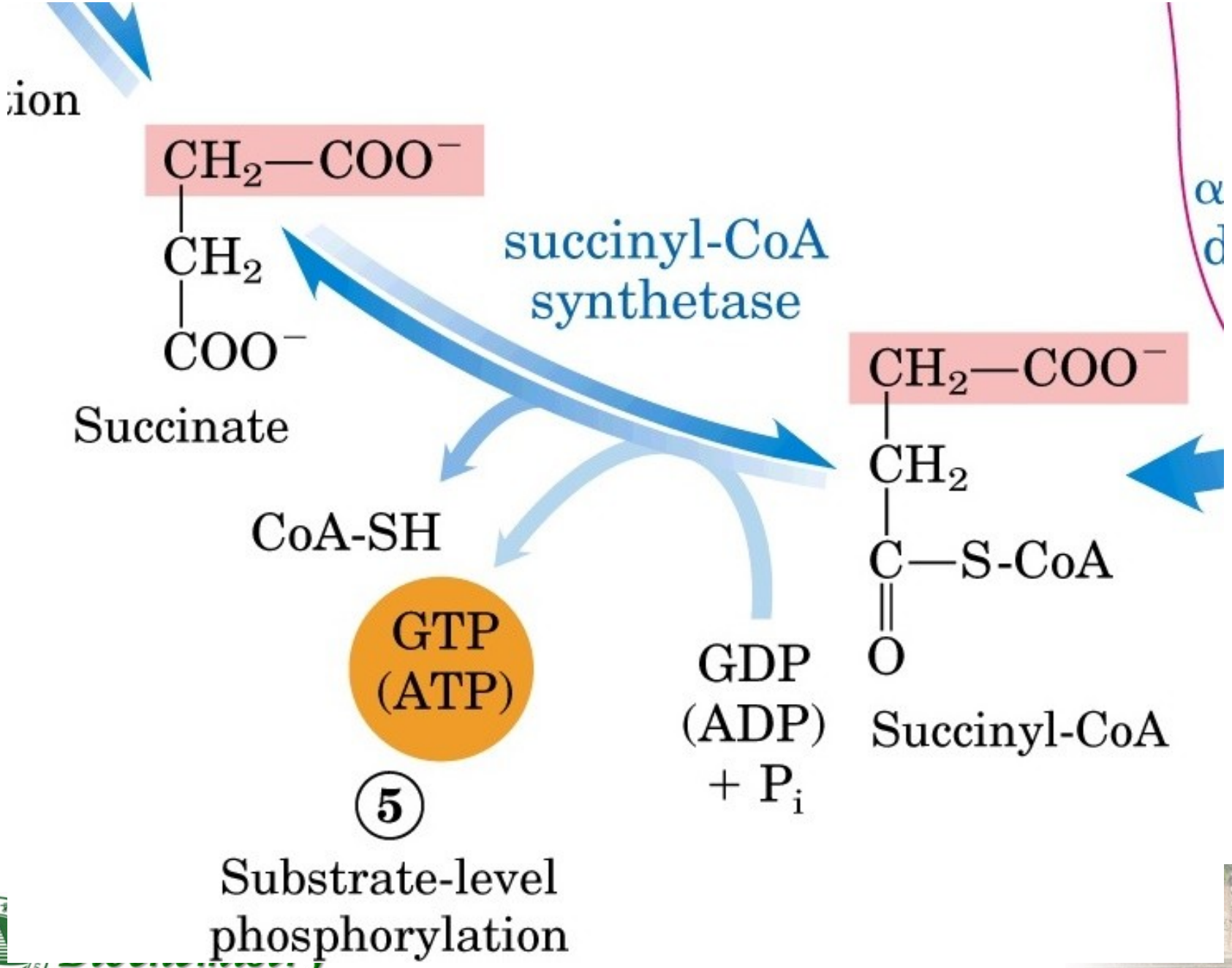
$\Delta G'^{\circ} = -33.5$ kJ/mol

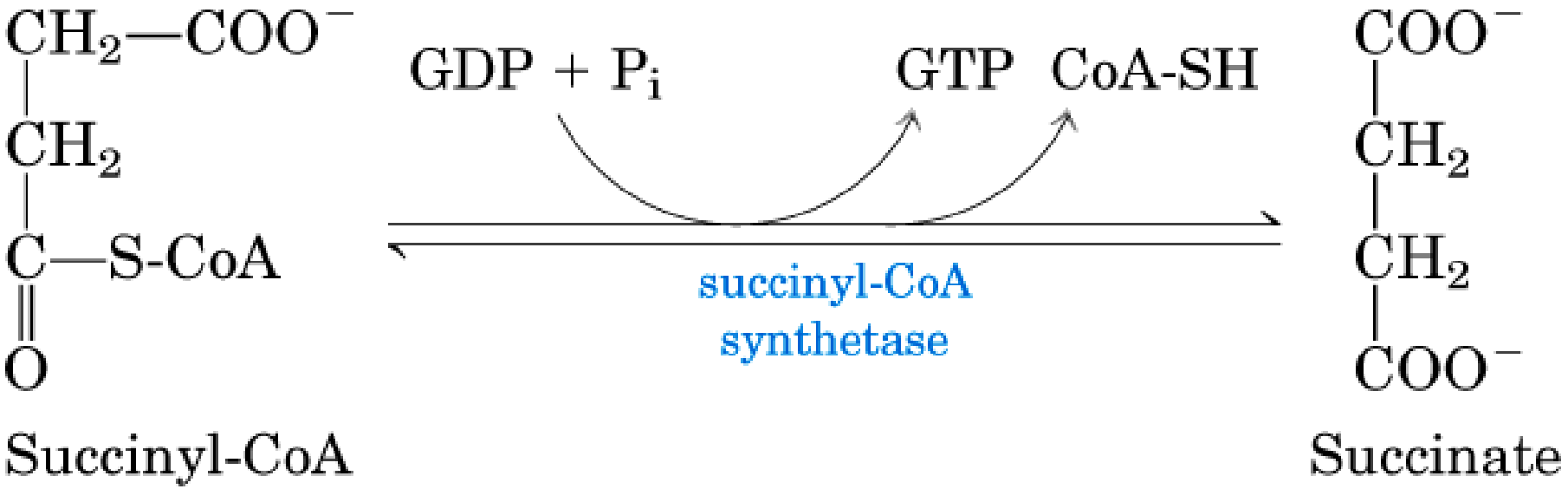
The α-ketoglutarate dehydrogenase complex closely resembles the pyruvate dehyrogenase complex in structure and function.



- **Step 5** Succinyl-CoA is hydrolyzed to succinate (琥珀酸); the free energy is harvested by a GDP or an ADP to form a GTP or an ATP by substrate-level phosphorylation.




ion
CH2—COO−
CH2
COO−
Succinate
succinyl-CoA
synthetase
α
d
CH2—COO−
CH2
C—S-CoA
O
Succinyl-CoA
CoA-SH
GTP
(ATP)
GDP
(ADP)
+ Pi
5
Substrate-level
phosphorylation



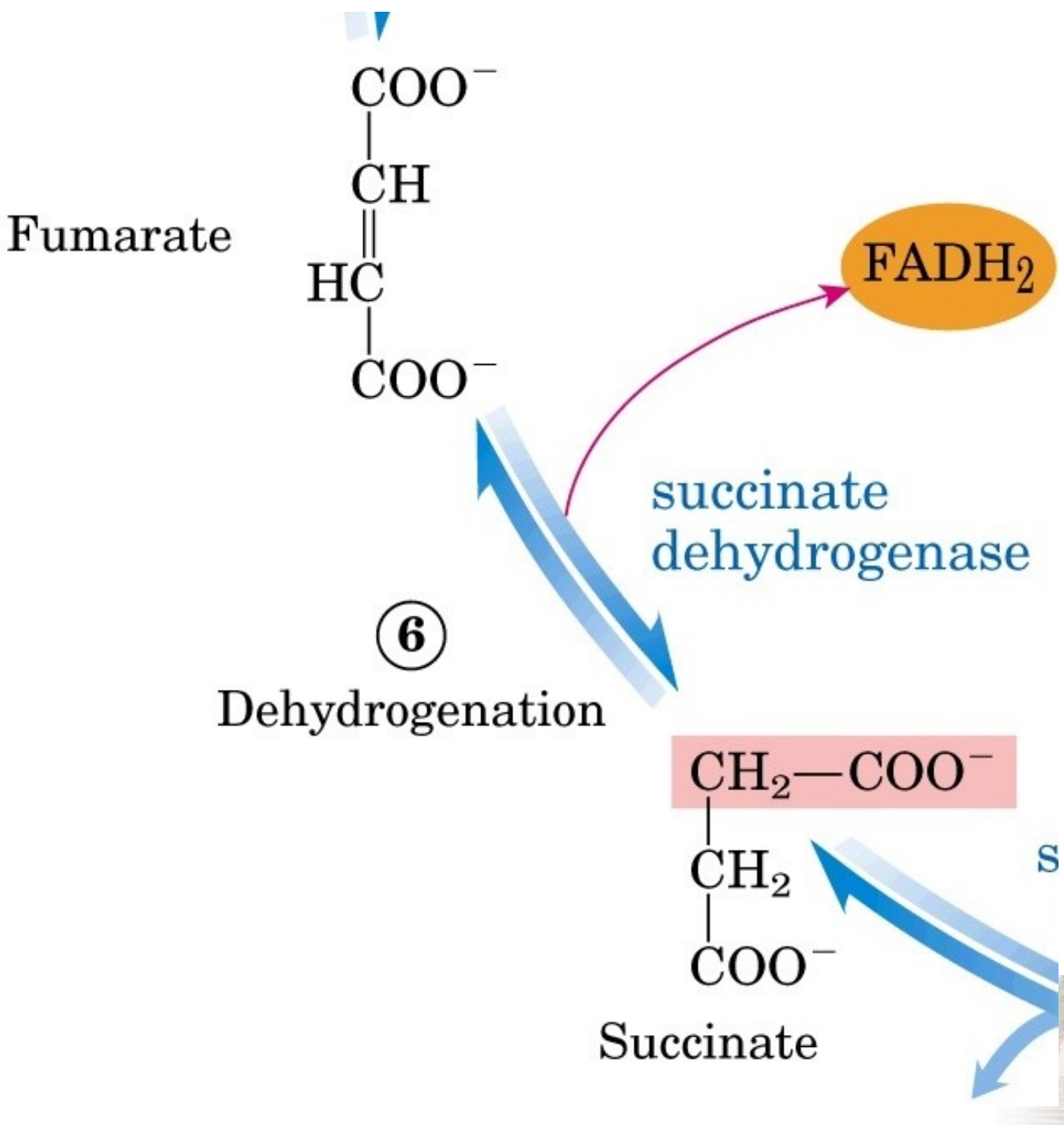
Succinyl-CoA

Succinate

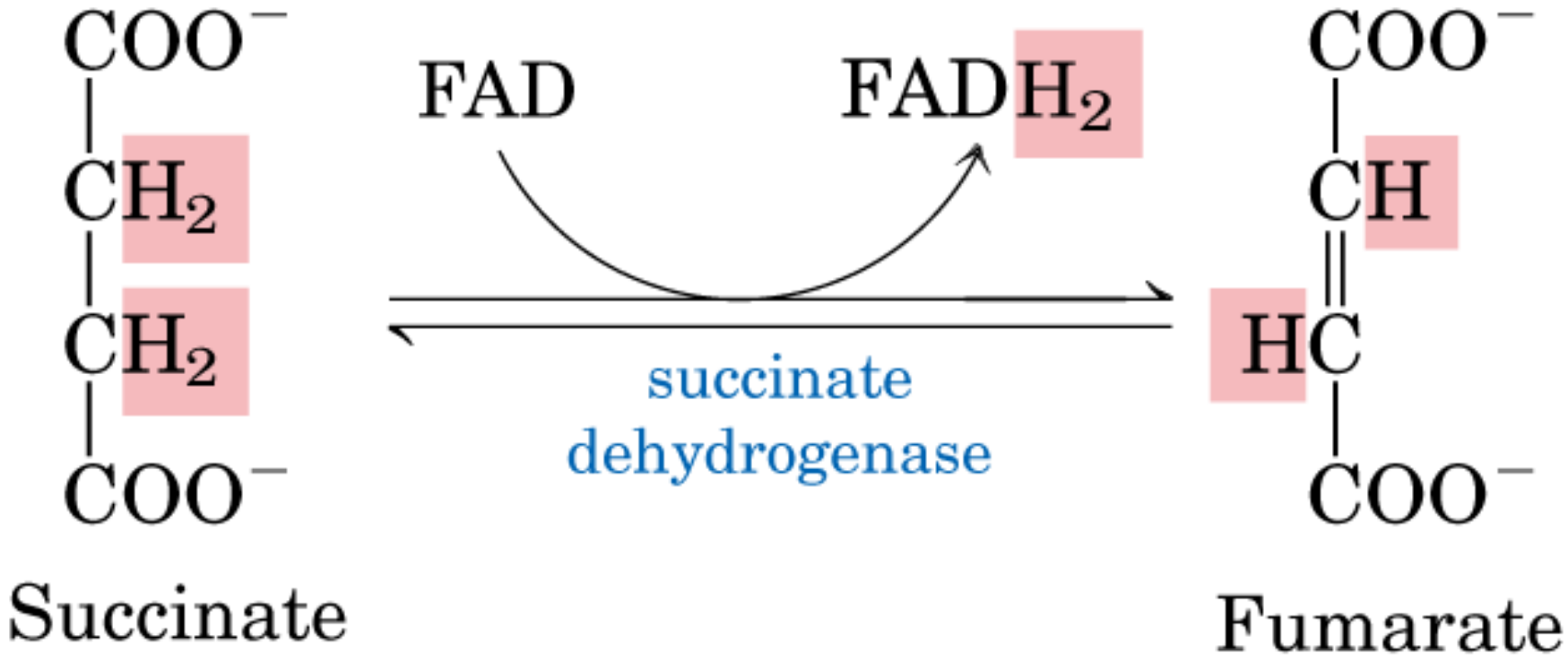
$$\Delta G'^{\circ} = -2.9 \text{ kJ/mol}$$

Step 6 Succinate is oxidized to **fumarate** (延胡索酸或反丁烯二酸); catalyzed by a **flavoprotein succinate dehydrogenase** (with a covalently bound FAD); malonate (丙二酸) is a strong competitive inhibitor of the enzyme, that will block the whole cycle.



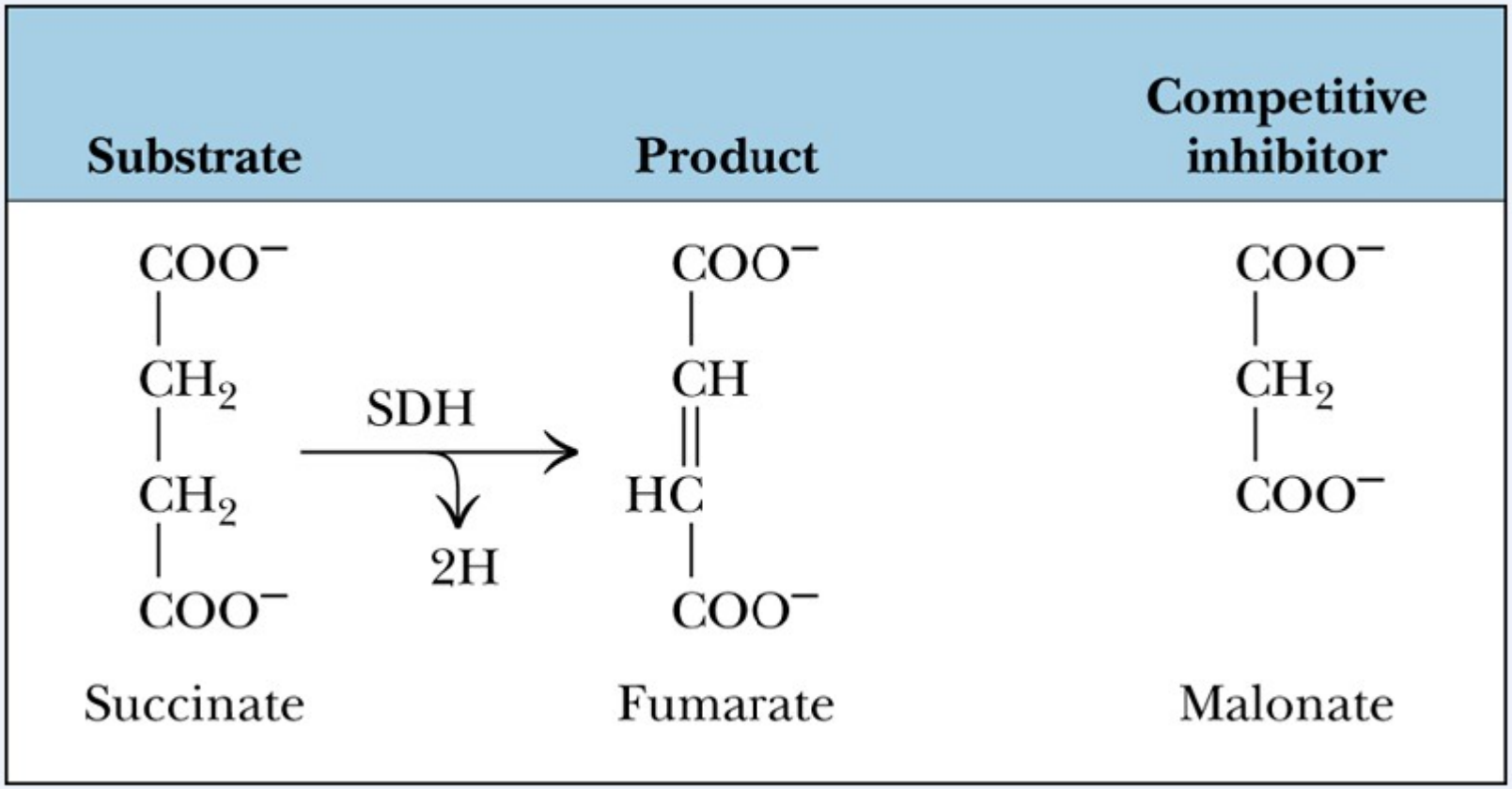

Fumarate
COO⁻
CH
HC
COO⁻
FADH2
succinate
dehydrogenase
6
Dehydrogenation
CH2—COO⁻
CH2
COO⁻
Succinate

The third oxidation step



$$\Delta G'^{\circ} = 0 \text{ kJ/mol}$$

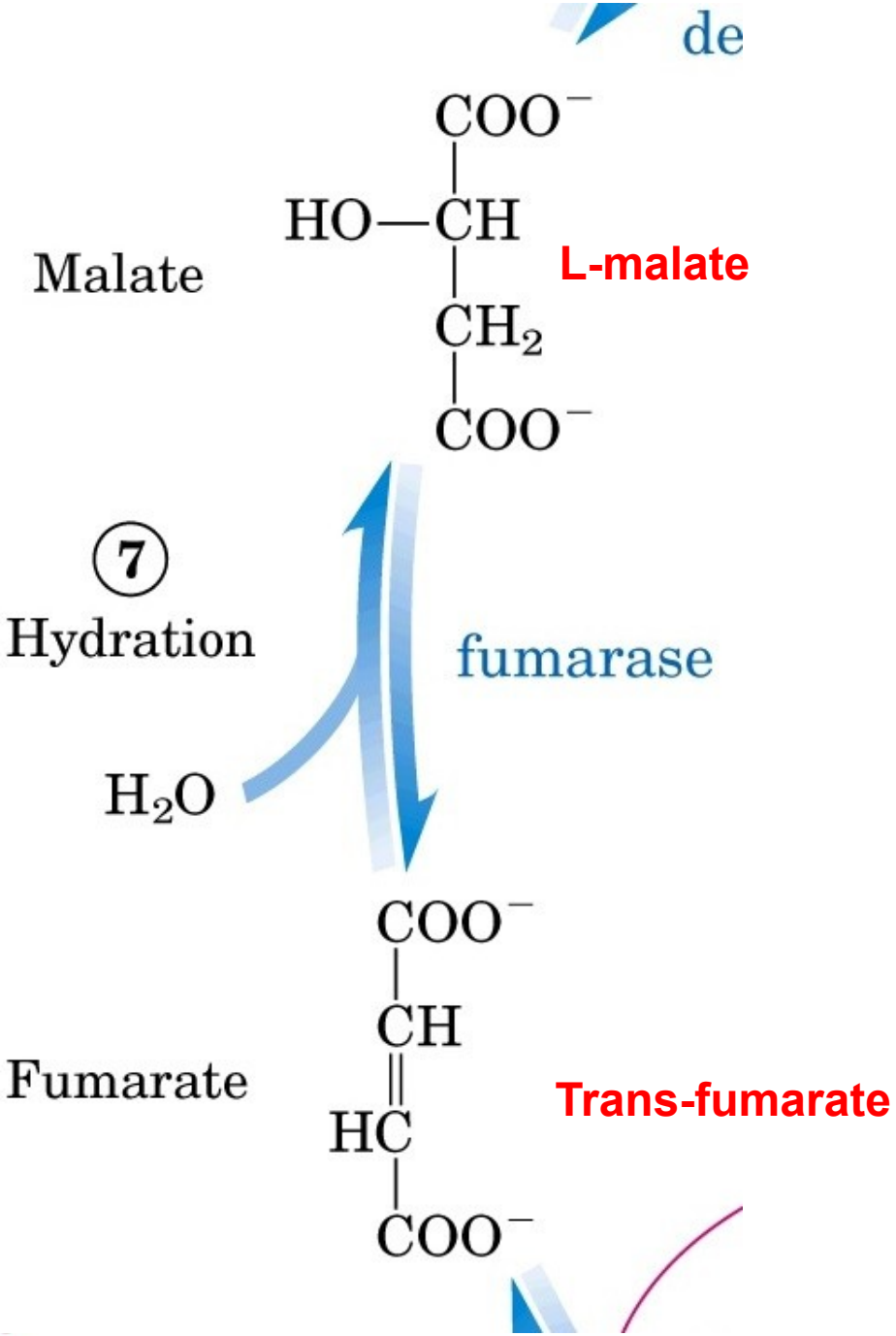
From Section C Enzyme

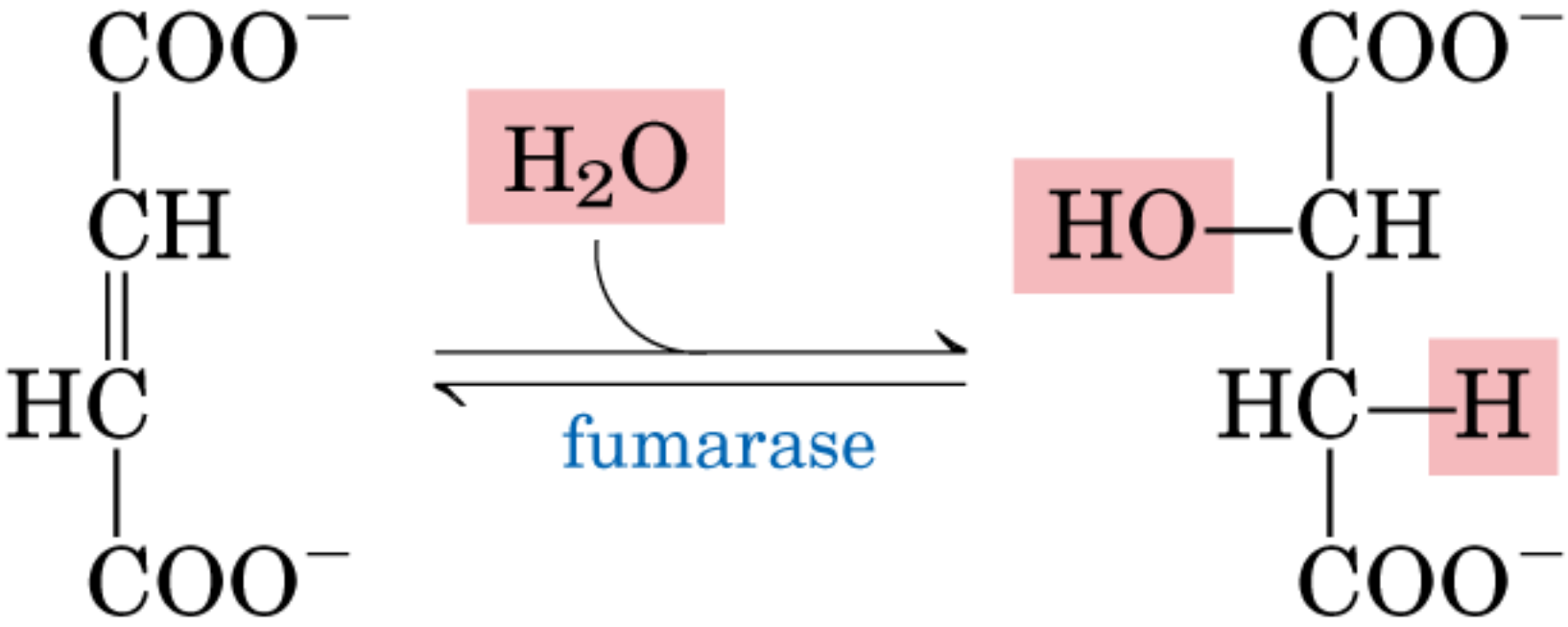


Inhibition of succinate dehydrogenase by malonate

Step 7 Fumarate is hydrated to **L-malate** by the action of **fumarase**; the enzyme is highly stereospecific, only **act on the *trans* and L isomers**, not on the *cis* and D isomers .




de
COO⁻
HO—CH
CH₂
COO⁻
Malate
L-malate
7
Hydration
fumarase
H₂O
COO⁻
CH
HC
COO⁻
Fumarate
Trans-fumarate



Fumarate

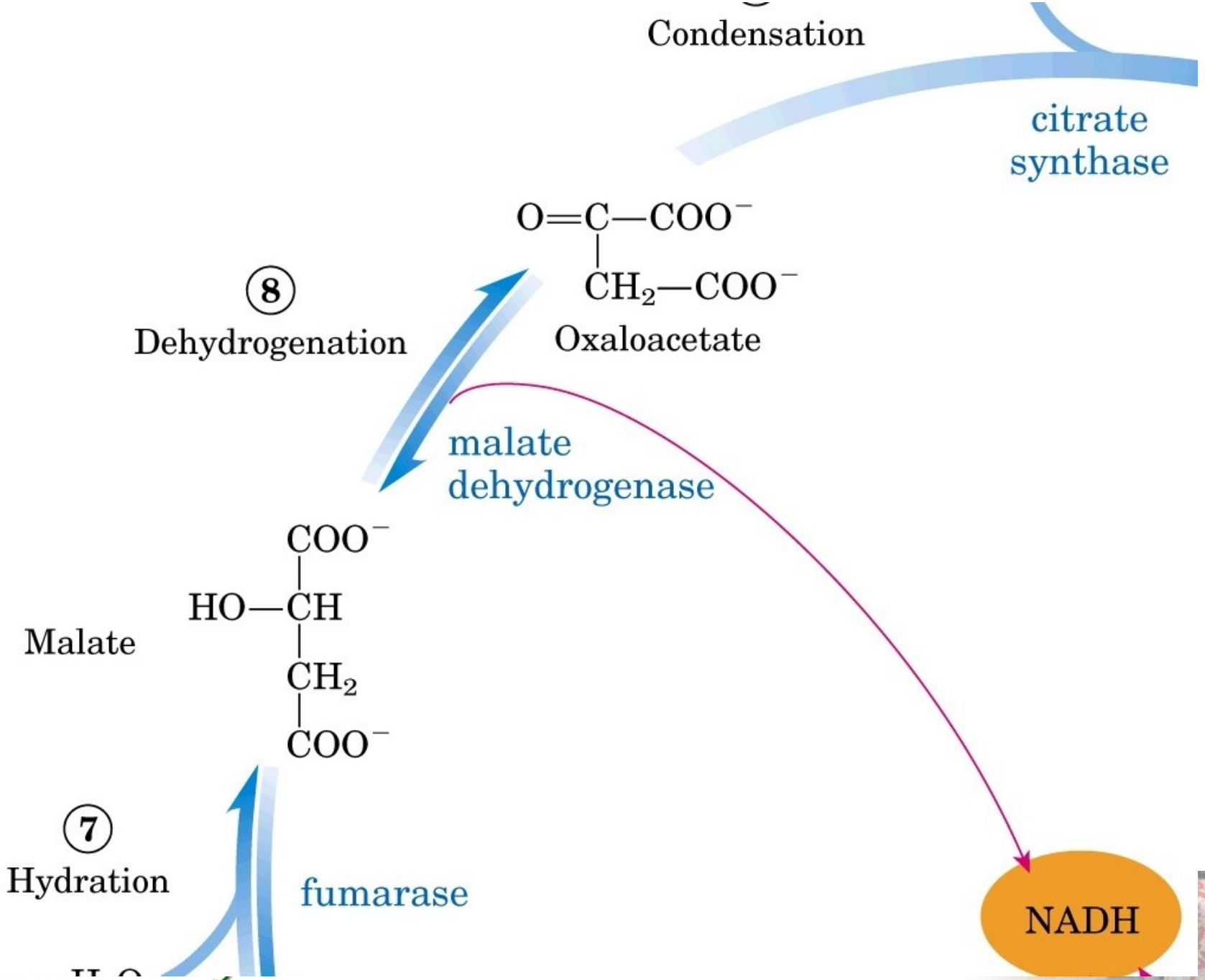
L-Malate

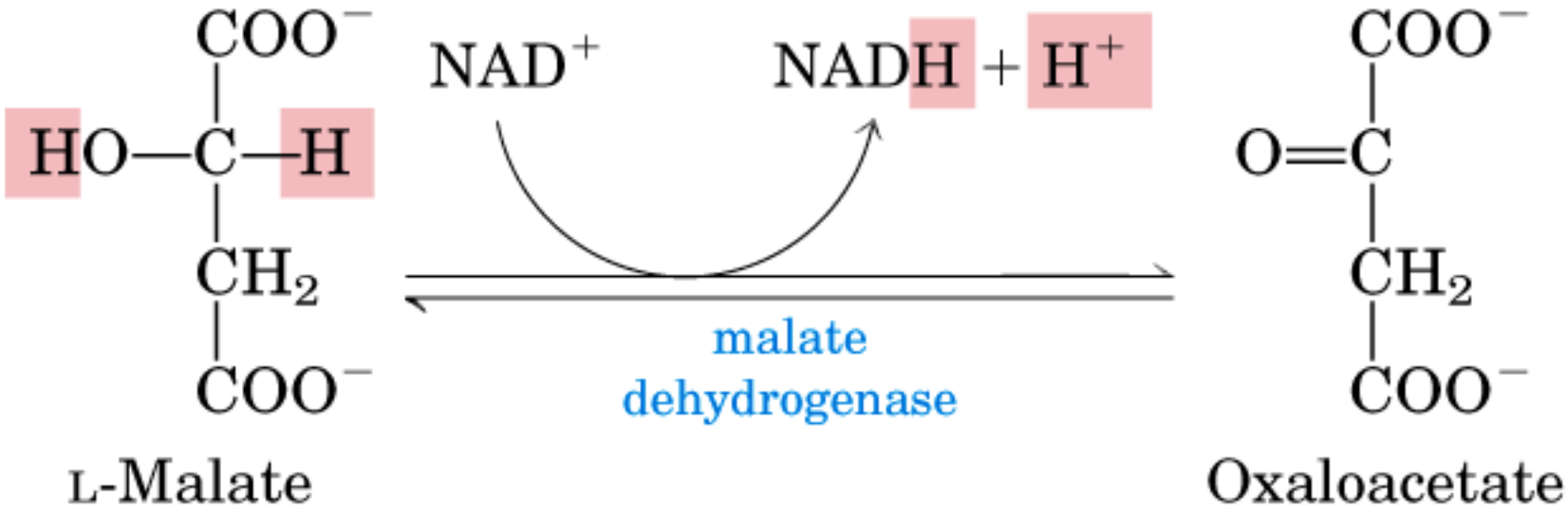
Trans-fumarate

$$\Delta G'^{\circ} = -3.8 \text{ kJ/mol}$$

Step 8 **Oxaloacetate** is regenerated by the oxidation of L-malate; this reaction is catalyzed by malate dehydrogenase with two electrons collected by NAD^+.

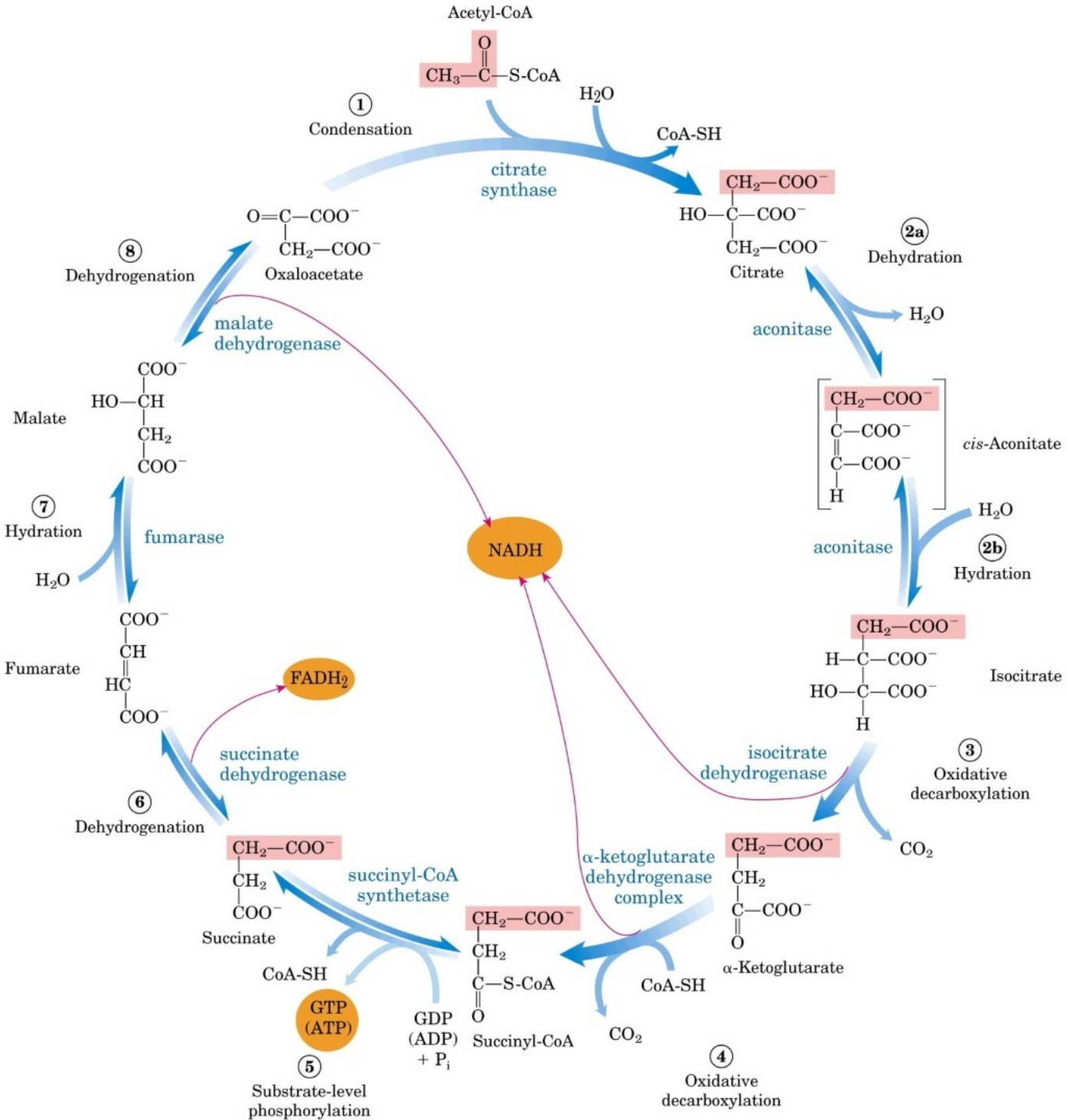



Condensation
citrate synthase
O=C—COO⁻
CH₂—COO⁻
Oxaloacetate
8
Dehydrogenation
malate dehydrogenase
COO⁻
HO—CH
CH₂
COO⁻
Malate
7
Hydration
fumarase
NADH



$\Delta G'^{\circ} = 29.7$ kJ/mol

Oxaloacetate is regenerated at the end


Acetyl-CoA
CH3—C—S-CoA
O
1
Condensation
H2O
CoA-SH
citrate synthase
CH2—COO⁻
HO—C—COO⁻
CH2—COO⁻
Citrate
2a
Dehydration
H2O
aconitase
CH2—COO⁻
C—COO⁻
C—COO⁻
H
cis-Aconitate
H2O
aconitase
2b
Hydration
CH2—COO⁻
H—C—COO⁻
HO—C—COO⁻
H
Isocitrate
3
Oxidative decarboxylation
CO2
isocitrate dehydrogenase
NADH
CH2—COO⁻
CH2
C—COO⁻
O
α-Ketoglutarate
α-ketoglutarate dehydrogenase complex
CH2—COO⁻
CH2
C—S-CoA
O
Succinyl-CoA
CO2
CoA-SH
4
Oxidative decarboxylation
succinyl-CoA synthetase
CoA-SH
GTP (ATP)
GDP (ADP) + Pi
5
Substrate-level phosphorylation
CH2—COO⁻
CH2
COO⁻
Succinate
6
Dehydrogenation
succinate dehydrogenase
FADH2
COO⁻
CH
HC
COO⁻
Fumarate
7
Hydration
H2O
fumarase
COO⁻
HO—CH
CH2
COO⁻
Malate
8
Dehydrogenation
malate dehydrogenase
O=C—COO⁻
CH2—COO⁻
Oxaloacetate

6. The complete oxidation of one glucose may yield as many as 32 ATP

All the NADH and $FADH_2$ will eventually pass their electrons to O_2 after being transferred through a series of electron carriers.

The complete oxidation of each NADH molecule leads to the generation of 2.5 ATP, and $FADH_2$ of about 1.5 ATP.



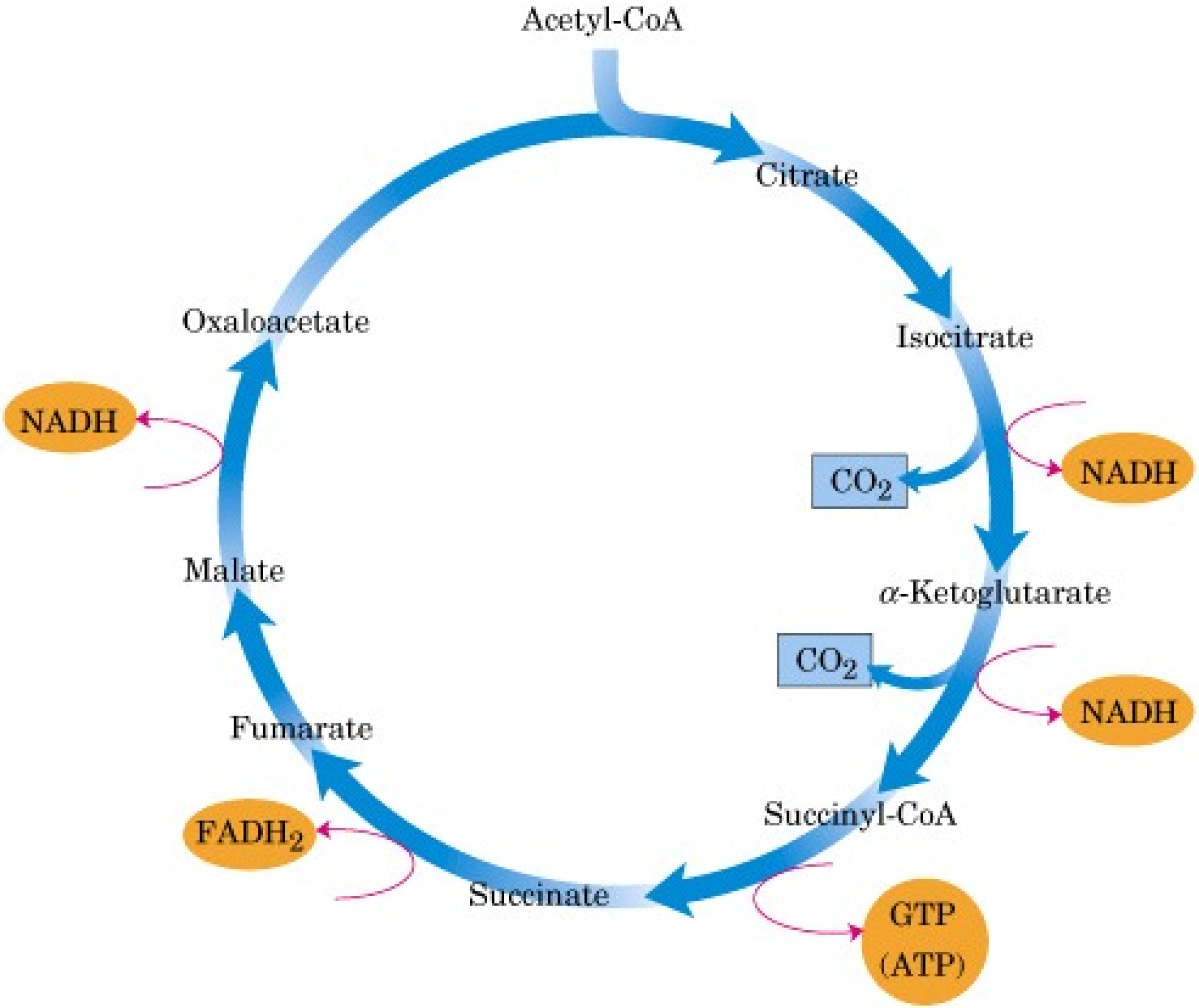

Acetyl-CoA
Citrate
Isocitrate
NADH
CO_2
α-Ketoglutarate
CO_2
NADH
Succinyl-CoA
GTP
(ATP)
Succinate
$FADH_2$
Fumarate
Malate
NADH
Oxaloacetate

table 16–1

Stoichiometry of Coenzyme Reduction and ATP Formation in the Aerobic Oxidation of Glucose via Glycolysis, the Pyruvate Dehydrogenase Reaction, the Citric Acid Cycle, and Oxidative Phosphorylation

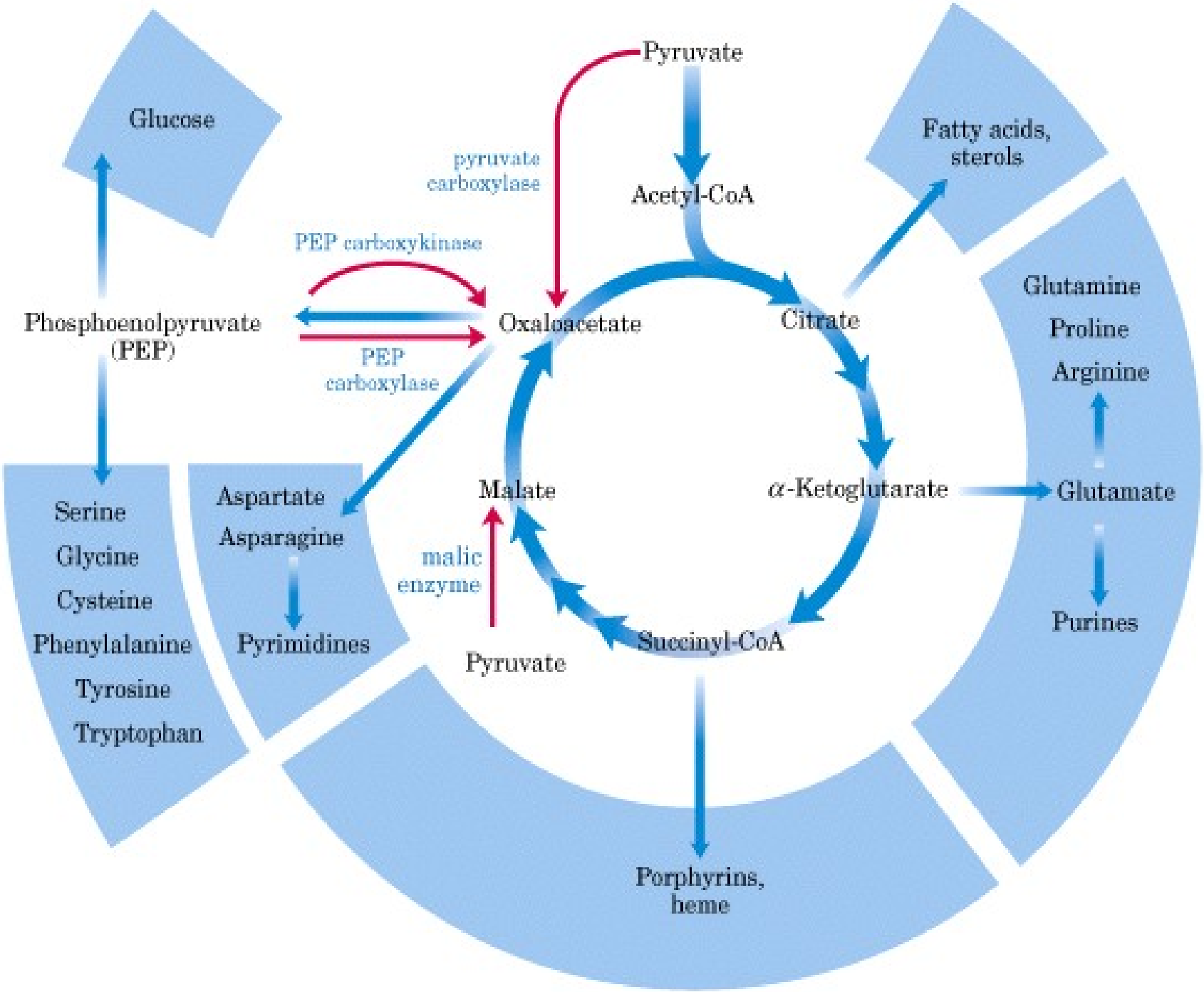
Reaction	Number of ATP or reduced coenzymes directly formed	Number of ATP ultimately formed*
Glucose ⟶ glucose 6-phosphate	−1 ATP	−1
Fructose 6-phosphate ⟶ fructose 1,6-bisphosphate	−1 ATP	−1
2 Glyceraldehyde 3-phosphate ⟶ 2 1,3-bisphosphoglycerate	2 NADH	3–5
2 1,3-Bisphosphoglycerate ⟶ 2 3-phosphoglycerate	2 ATP	2
2 Phosphoenolpyruvate ⟶ 2 pyruvate	2 ATP	2
2 Pyruvate ⟶ 2 acetyl-CoA	2 NADH	5
2 Isocitrate ⟶ 2 α-ketoglutarate	2 NADH	5
2 α-Ketoglutarate ⟶ 2 succinyl-CoA	2 NADH	5
2 Succinyl-CoA ⟶ 2 succinate	2 ATP (or 2 GTP)	2
2 Succinate ⟶ 2 fumarate	2 $FADH_2$	3
2 Malate ⟶ 2 oxaloacetate	2 NADH	5
Total		30–32

*This is calculated as 2.5 ATP per NADH and 1.5 ATP per $FADH_2$. A negative value indicates consumption.

7. The citric acid intermediates are important sources for biosynthetic precursors

The citric acid cycle is the hub of intermediary metabolism serving both the catabolic and anabolic processes.
It provides precursors for the biosynthesis of glucose, amino acids, nucleotides, fatty acids, sterols, heme groups, etc.




Glucose
Pyruvate
pyruvate carboxylase
Acetyl-CoA
Fatty acids, sterols
PEP carboxykinase
Phosphoenolpyruvate (PEP)
Oxaloacetate
Citrate
PEP carboxylase
Glutamine
Proline
Arginine
Serine
Glycine
Cysteine
Phenylalanine
Tyrosine
Tryptophan
Aspartate
Asparagine
Malate
α-Ketoglutarate
Glutamate
malic enzyme
Pyrimidines
Pyruvate
Succinyl-CoA
Purines
Porphyrins, heme

回补反应： 当中间产物用于其它物质的合成，须补充 OAA，以使循环继续

table 16–2

Anaplerotic Reactions

Reaction	Tissue(s)/organism(s)
Pyruvate + HCO_3^- + ATP $\xrightleftharpoons{\text{pyruvate carboxylase}}$ oxaloacetate + ADP + P_i	Liver, kidney
Phosphoenolpyruvate + CO_2 + GDP $\xrightleftharpoons{\text{PEP carboxykinase}}$ oxaloacetate + GTP	Heart, skeletal muscle
Phosphoenolpyruvate + HCO_3^- $\xrightleftharpoons{\text{PEP carboxylase}}$ oxaloacetate + P_i	Higher plants, yeast, bacteria
Pyruvate + HCO_3^- + NAD(P)H $\xrightleftharpoons{\text{malic enzyme}}$ malate + $NAD(P)^+$	Widely distributed in eukaryotes and prokaryotes



8. Net conversion of acetate to carbohydrates is allowed via the glyoxylate cycle（乙醛酸循环）

Net conversion of acetate to four-carbon citric acid cycle intermediates occurs via the glyoxylate cycle, found **in plants**, certain invertebrates, and some microorganisms (including *E. coli* and yeast).

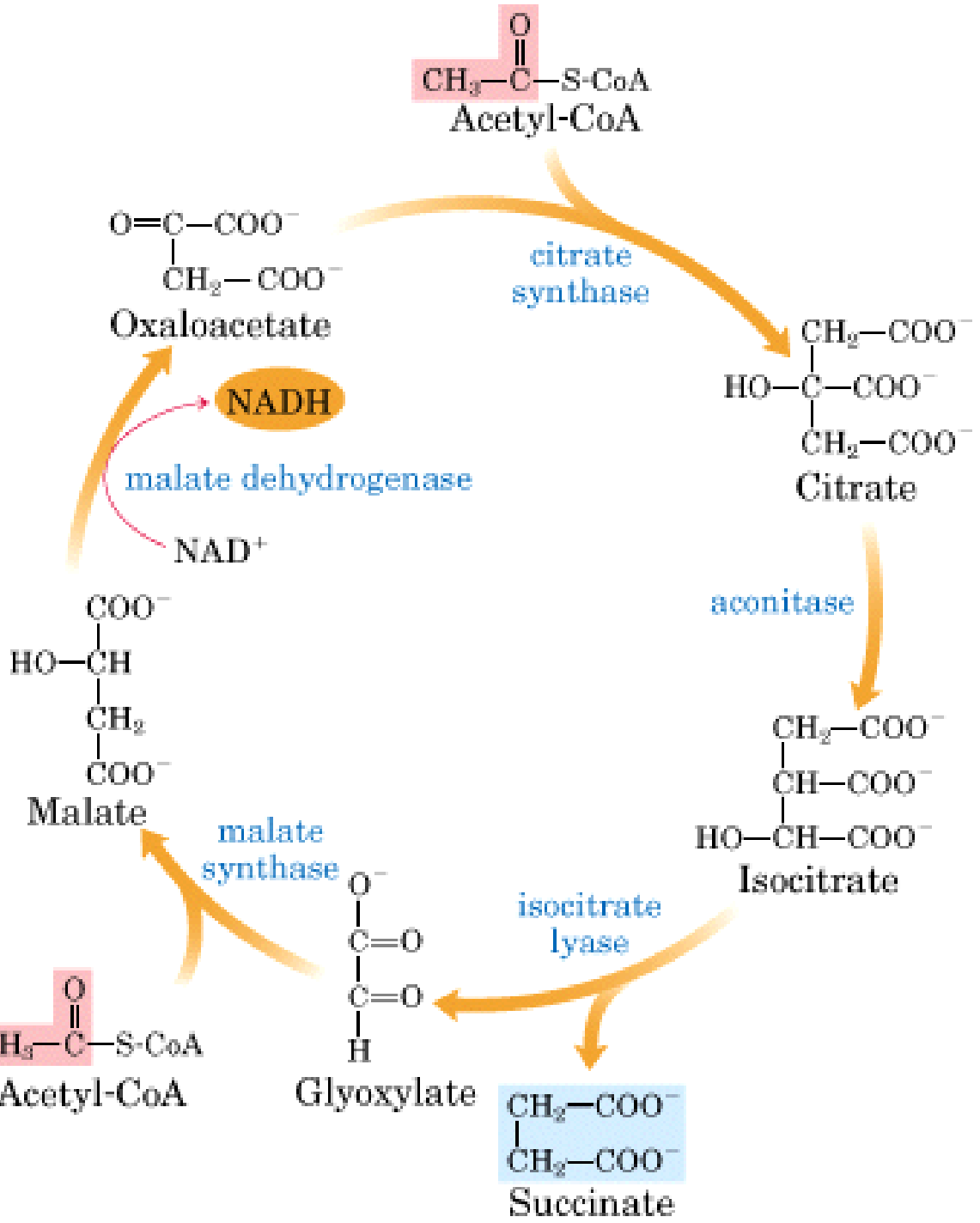
The glyoxylate cycle shares three steps and bypasses the rest, including the two decarboxylation steps, of the citric acid cycle.



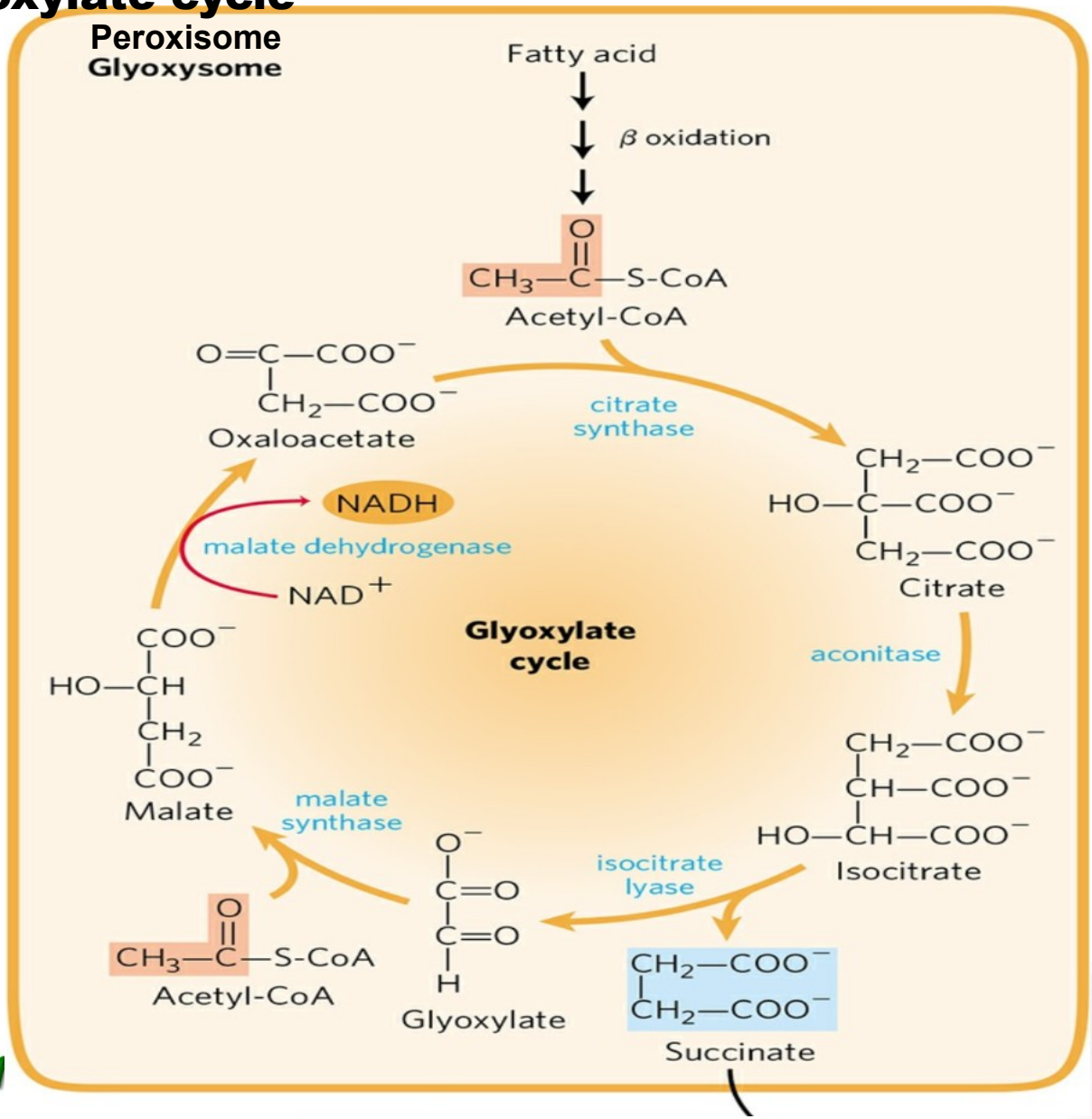
Two acetyl-CoA molecules enter each glyoxylate cycle with a net production of **one succinate**. **Isocitrate lyase and malate synthase** are the two bypassing enzymes, converting isocitrate to malate via the glyoxylate intermediate, releasing a succinate on the way.

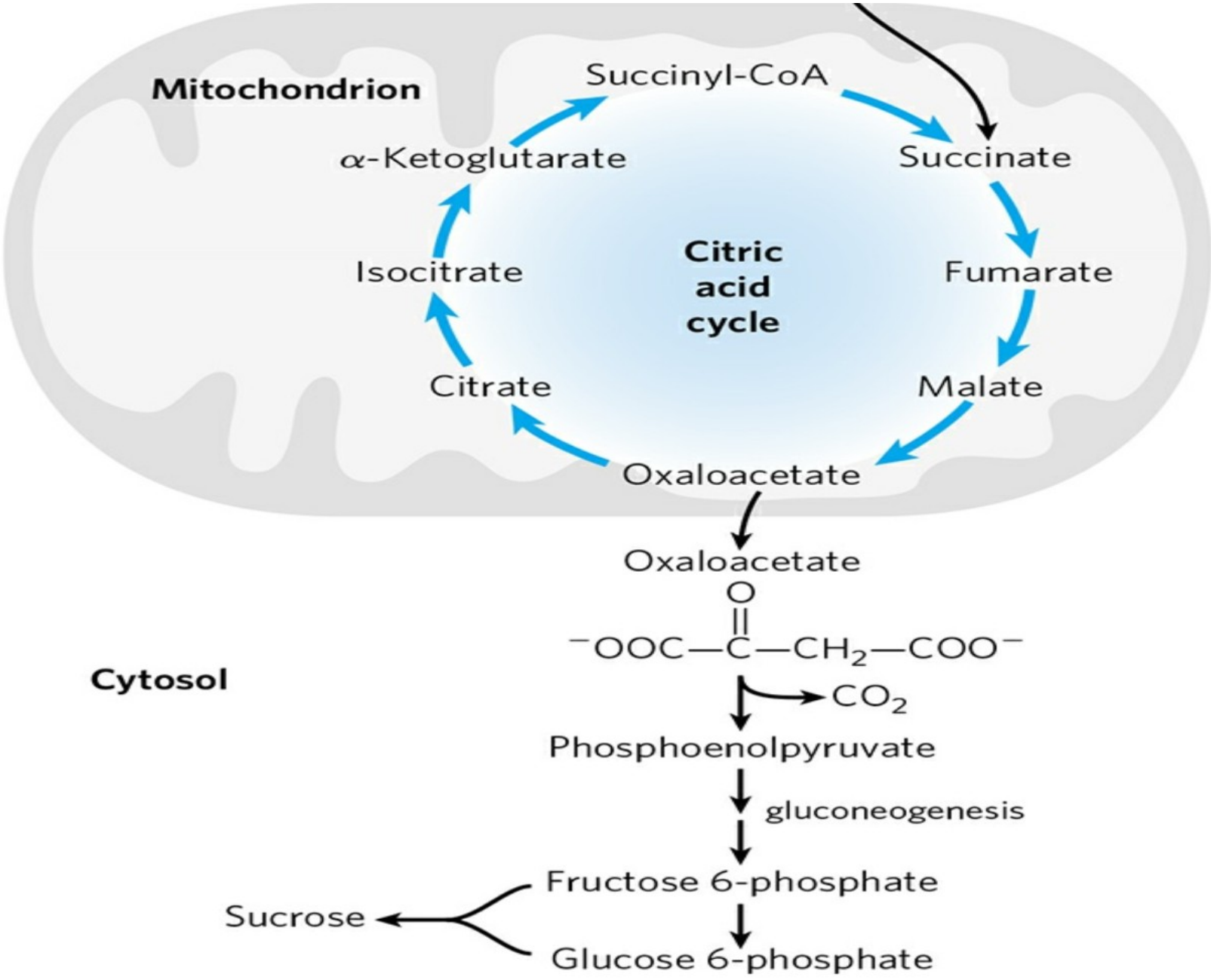


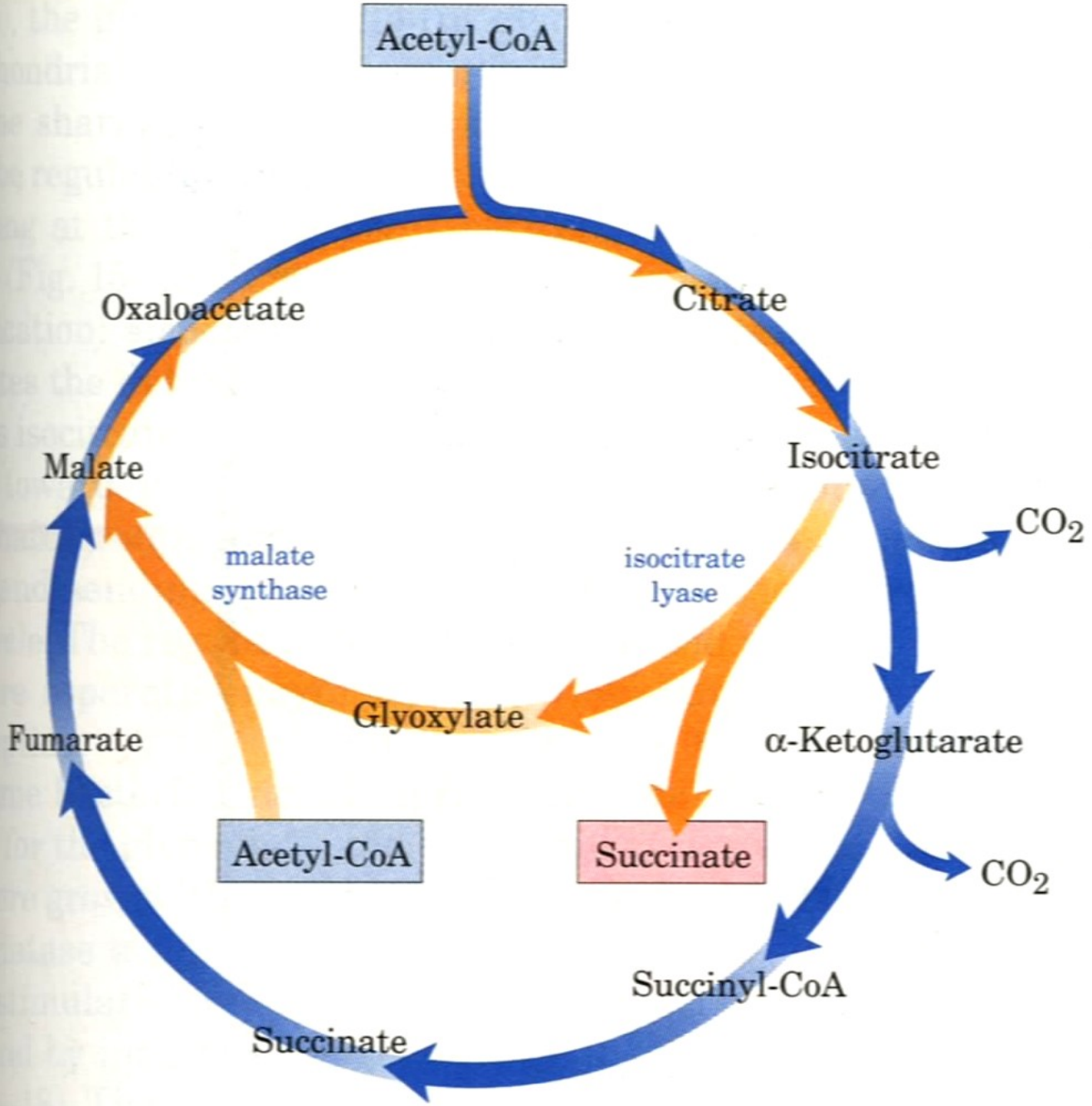
The glyoxylate cycle



Conversion of stored fatty acids to sucrose in germinating seeds through the glyoxylate cycle



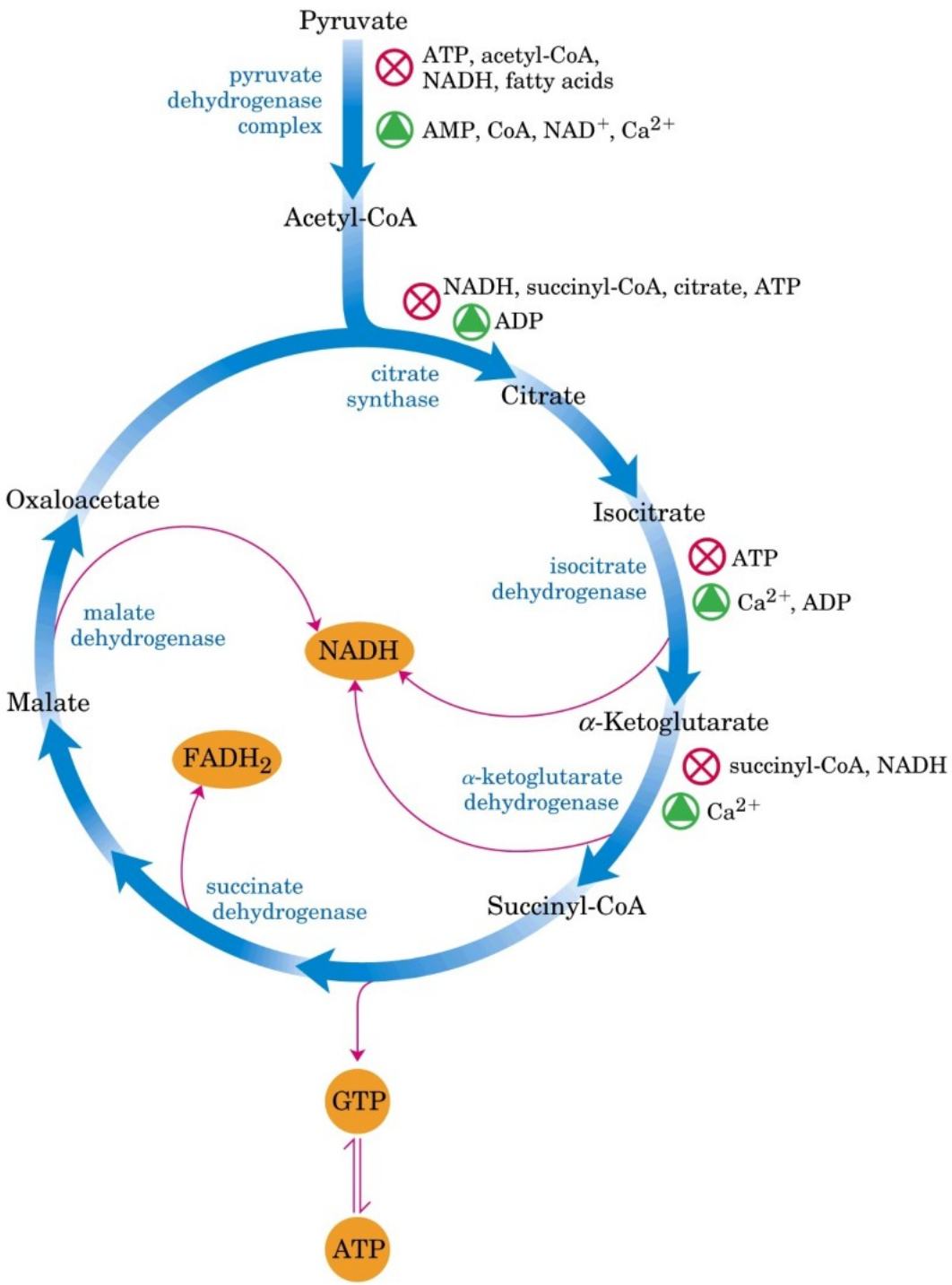

Mitochondrion
Succinyl-CoA
α-Ketoglutarate
Succinate
Isocitrate
Citric acid cycle
Fumarate
Citrate
Malate
Oxaloacetate
Oxaloacetate
O
⁻OOC—C—CH₂—COO⁻
CO₂
Cytosol
Phosphoenolpyruvate
gluconeogenesis
Fructose 6-phosphate
Sucrose
Glucose 6-phosphate


Acetyl-CoA
Citrate
Isocitrate
CO_2
α-Ketoglutarate
CO_2
Succinyl-CoA
Succinate
Fumarate
Malate
Oxaloacetate
malate synthase
isocitrate lyase
Glyoxylate
Acetyl-CoA
Succinate

9. The pyruvate dehydrogenase complex in vertebrates is regulated alloseterically and covalently

10. The rate of the citric acid cycle is controlled at three exergonic irreversible steps




Pyruvate
pyruvate dehydrogenase complex
ATP, acetyl-CoA, NADH, fatty acids
AMP, CoA, NAD+, Ca2+
Acetyl-CoA
NADH, succinyl-CoA, citrate, ATP
ADP
citrate synthase
Citrate
Isocitrate
isocitrate dehydrogenase
ATP
Ca2+, ADP
α-Ketoglutarate
α-ketoglutarate dehydrogenase
succinyl-CoA, NADH
Ca2+
Succinyl-CoA
succinate dehydrogenase
FADH2
Malate
malate dehydrogenase
NADH
Oxaloacetate
GTP
ATP

Biochem

Citrate synthase, isocitrate dehydrogenase and α-ketoglutarate dehydrogenase;

Inhibited by **product** feedback (citrate, succinyl-CoA) and **high energy charge** (ATP, NADH);

Activated by a **low energy charge** (ADP) or a signal for energy requirement (**Ca^{2+}**).



Summary

Pyruvate is converted to acetyl-CoA by the action of pyruvate dehydrogenase complex, a huge enzyme complex.

Acetyl-CoA is converted to 2 CO_2 via the eight-step citric acid cycle, generating three NADH, one $FADH_2$, and one ATP (by substrate-level phophorylation).

Intermediates of citric acid cycle are drawn off to synthesize many other biomolecules, including fatty acids, steroids, amino acids, heme, pyrimidines, and glucose.



The activity of pyruvate dehydrogenase complex is regulated by allosteric effectors and reversible phosphorylations.

Net conversion of fatty acids to glucose can occur in germinating seeds, some invertebrates and some bacteria via the glycoxylate cycle, which shares three steps with the citric acid cycle but bypasses the two decarboxylation steps, converting two molecules of acetyl-CoA to one succinate.

Acetyl-CoA is partitioned into the glyoxylate cycle and citric acid cycle via a coordinately regulation of the isocitrate dehydrogenase and isocitrate lyase.

J5 Pentose phosphate pathway
(戊糖磷酸途径)

Overview
Main reactions of the pathway
Control of the pathway



Pentose phosphate pathway converts glucose to specialized products needed by the cells

Glc-6-P is first dehydrogenated by a $NADP^+$-containing dehydrogenase to form 6-phosphogluconate (6-磷酸葡萄糖酸).

6-phosphogluconate then undergoes a oxidative decarboxylation to form D-ribulose 5-P, generating another molecule of NADPH.

D-ribulose 5-P is then converted to ribose 5-P.



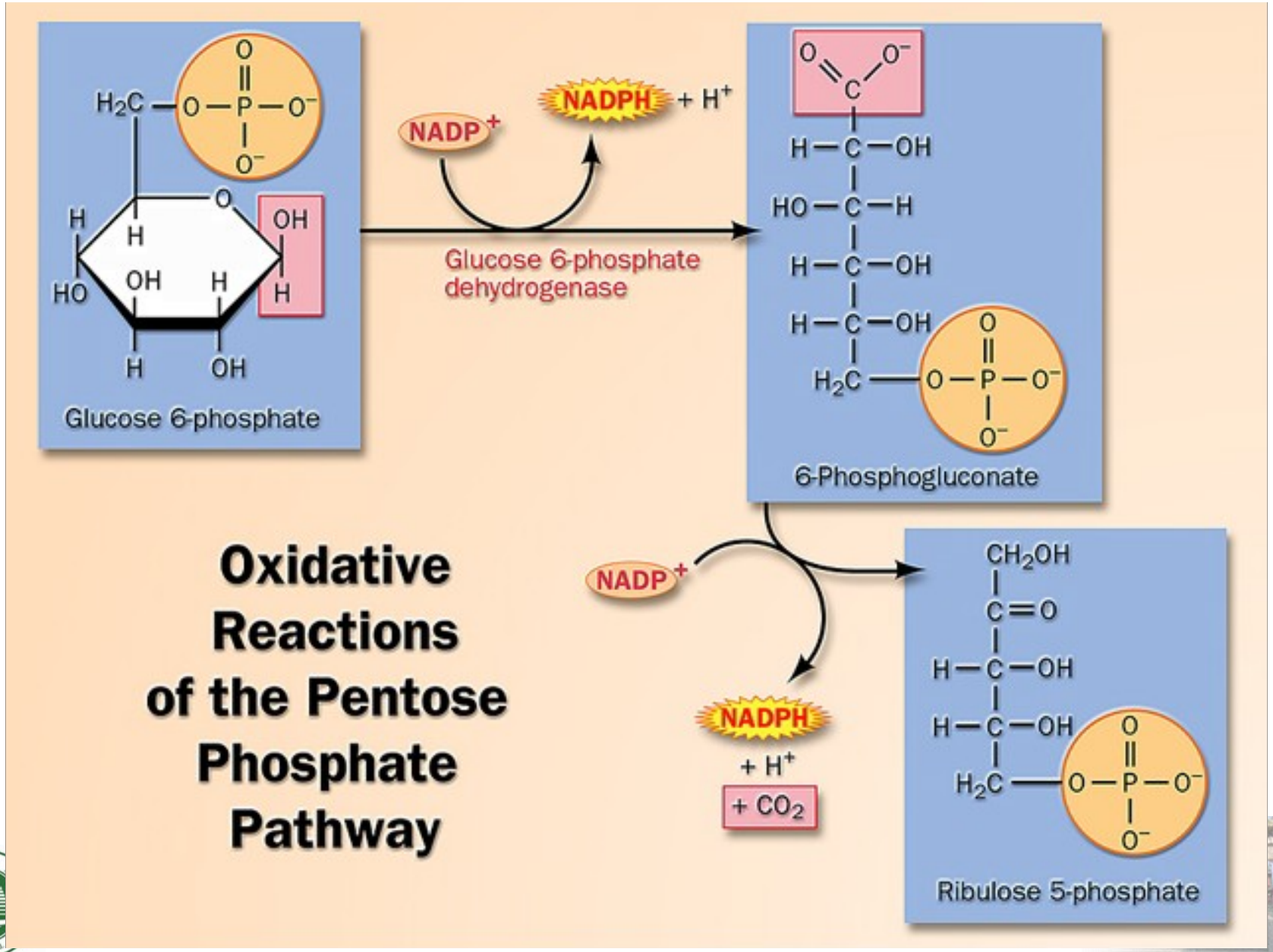
When NADPH is the primary requirement in the cell, the pentose phosphates are recycled into Glc-6-P via a series of rearrangements of the carbon skeleton, catalyzed by transketolase and transaldolase (no cofactor involved).

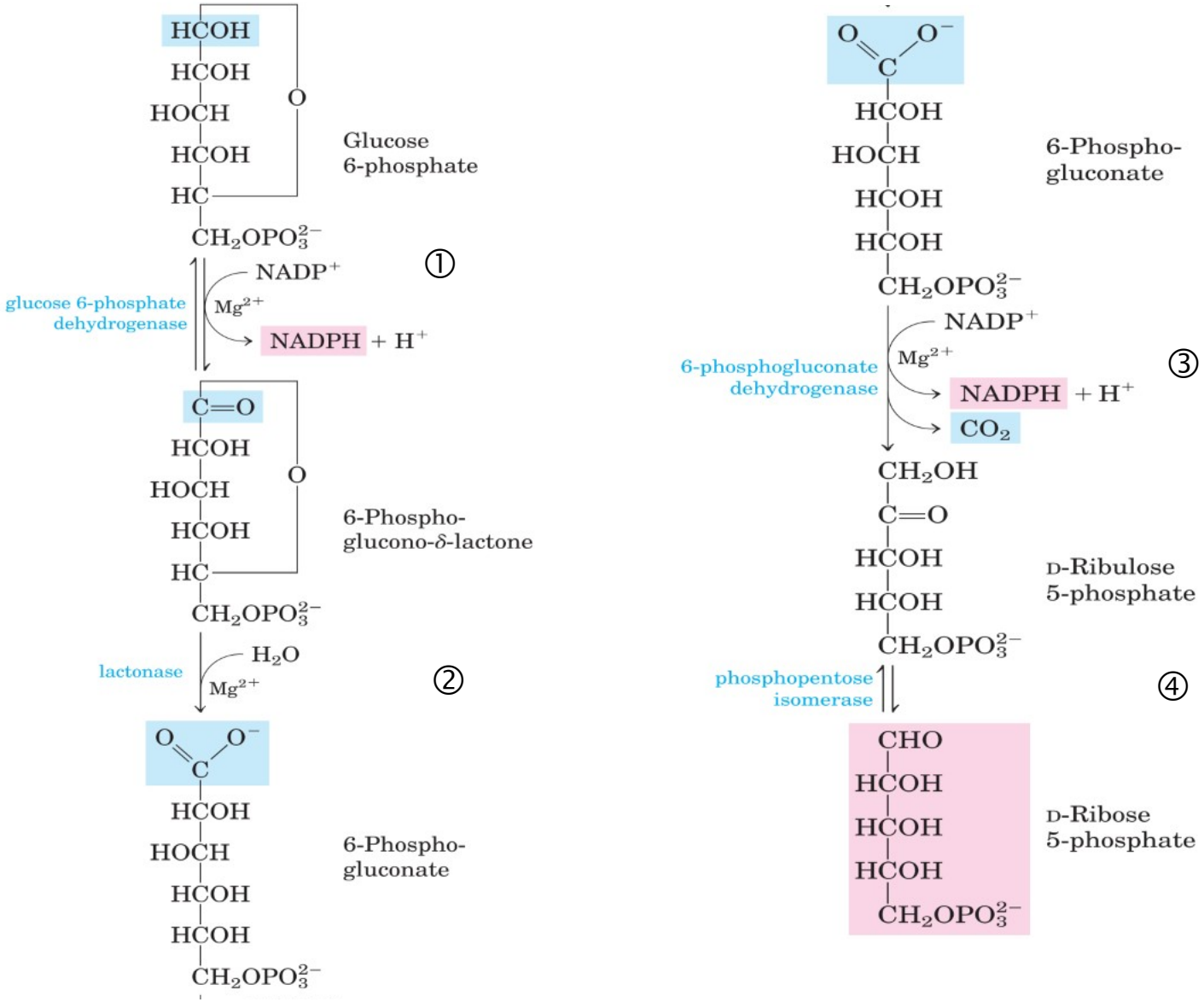
Six five-carbon sugar phosphates are converted to five six-carbon sugar phosphates.

The reverse of this rearrangement, regeneration of six five-carbon sugar phosphate from five six-carbon sugar phosphate occurs in the *Calvin cycle* (for photosynthetic fixation of CO_2 in plants).



Stage 1 Oxidative reactions

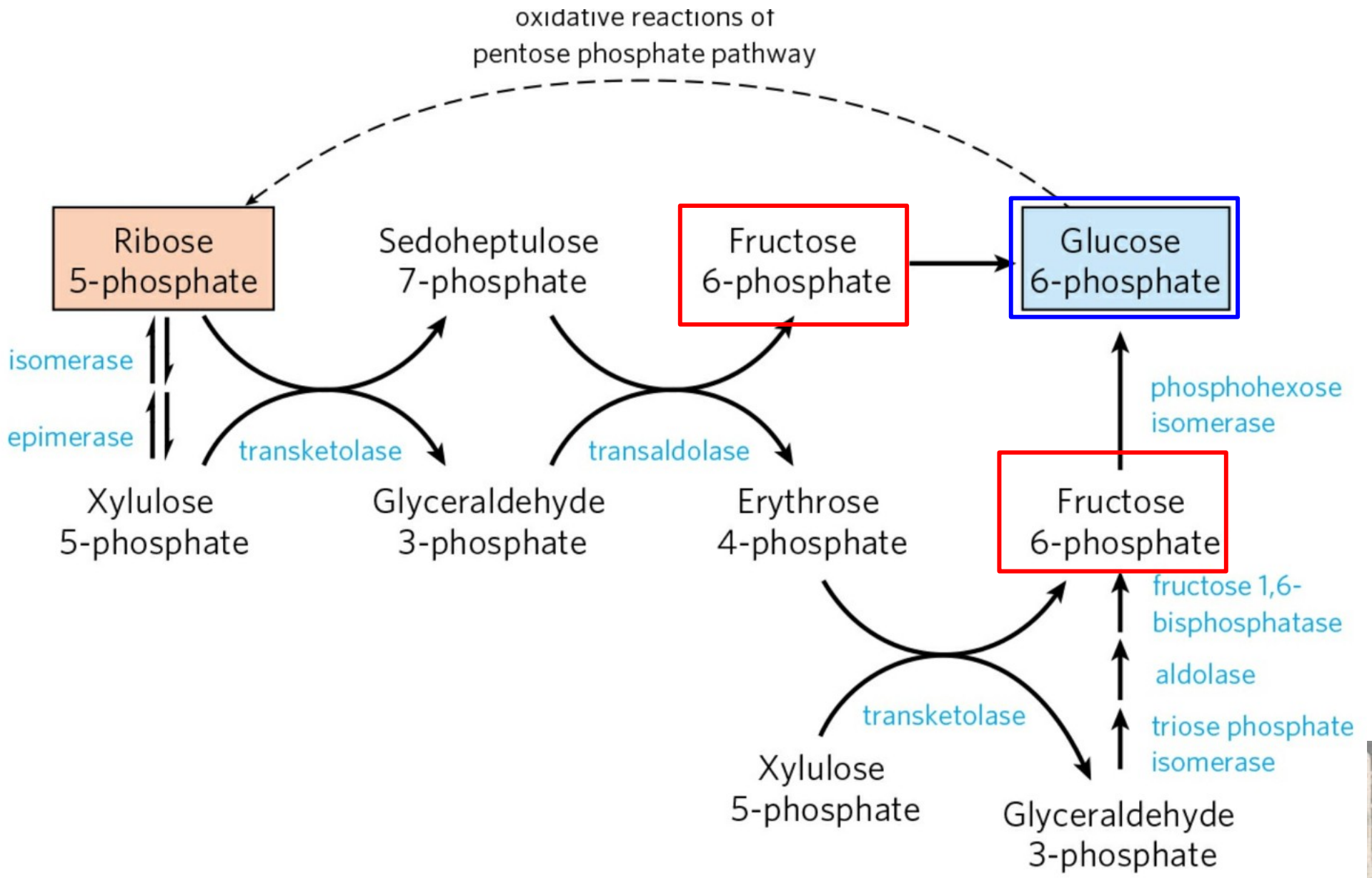


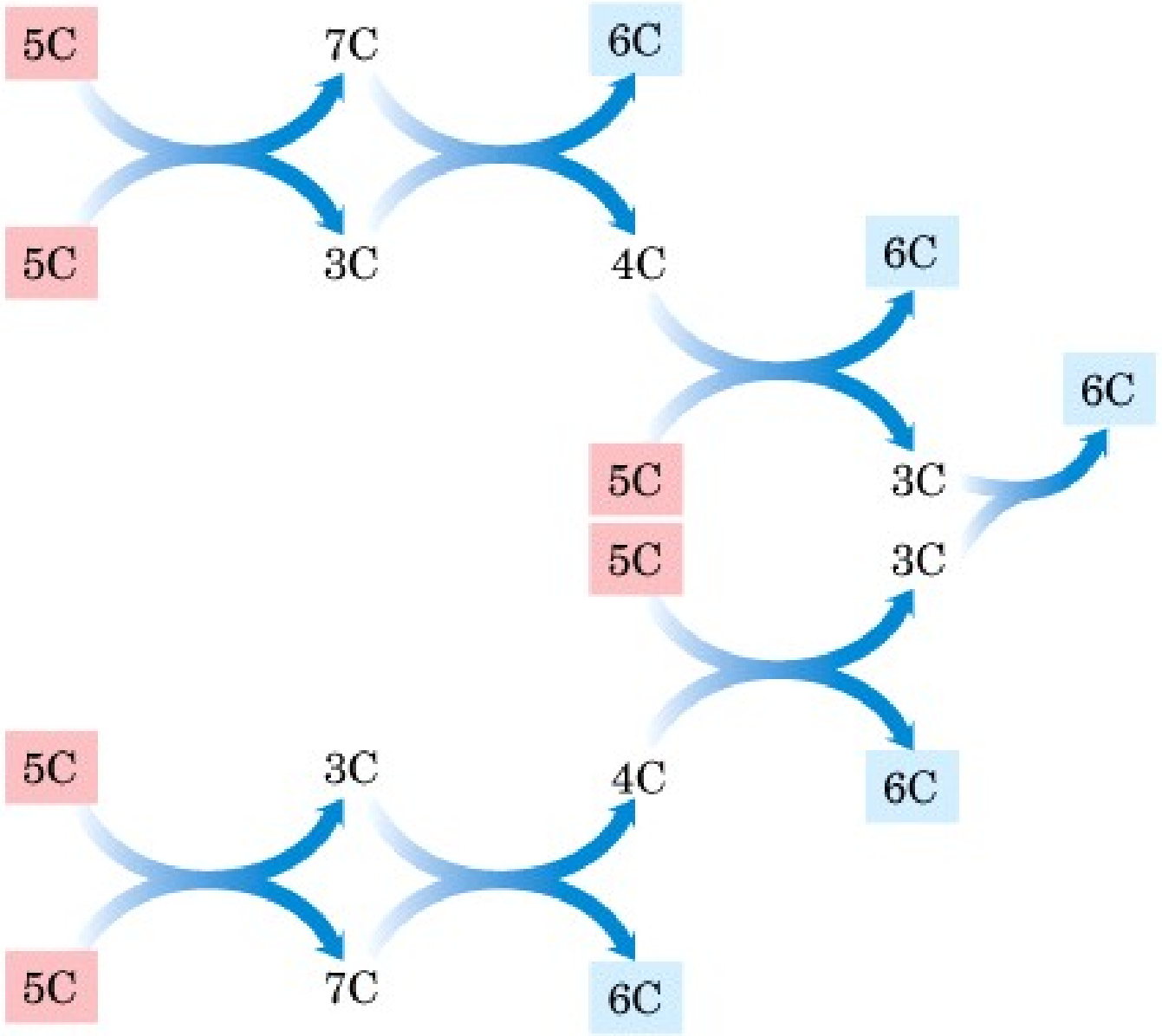


ΦΙΓΥΡΕ Οξιδατιωε ρεαχτιονσ οφ τηε πεντοσε πηοσπηατε πατηωαψ.

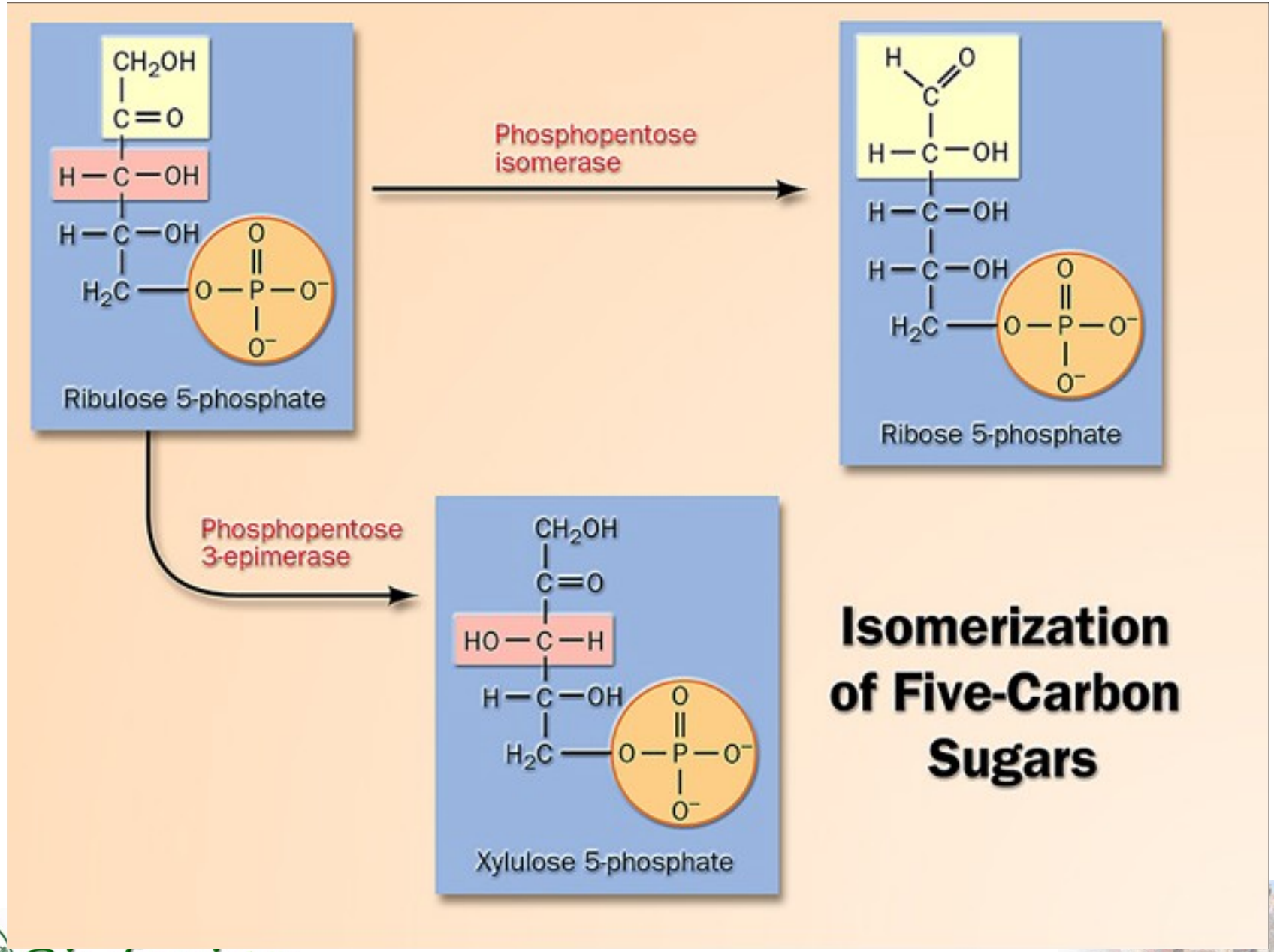
Τηε ενδ προδυχτσ αρε ριβοσε 5–πηοσπηατε, XO_2, ανδ ΝΑΔΠΗ.

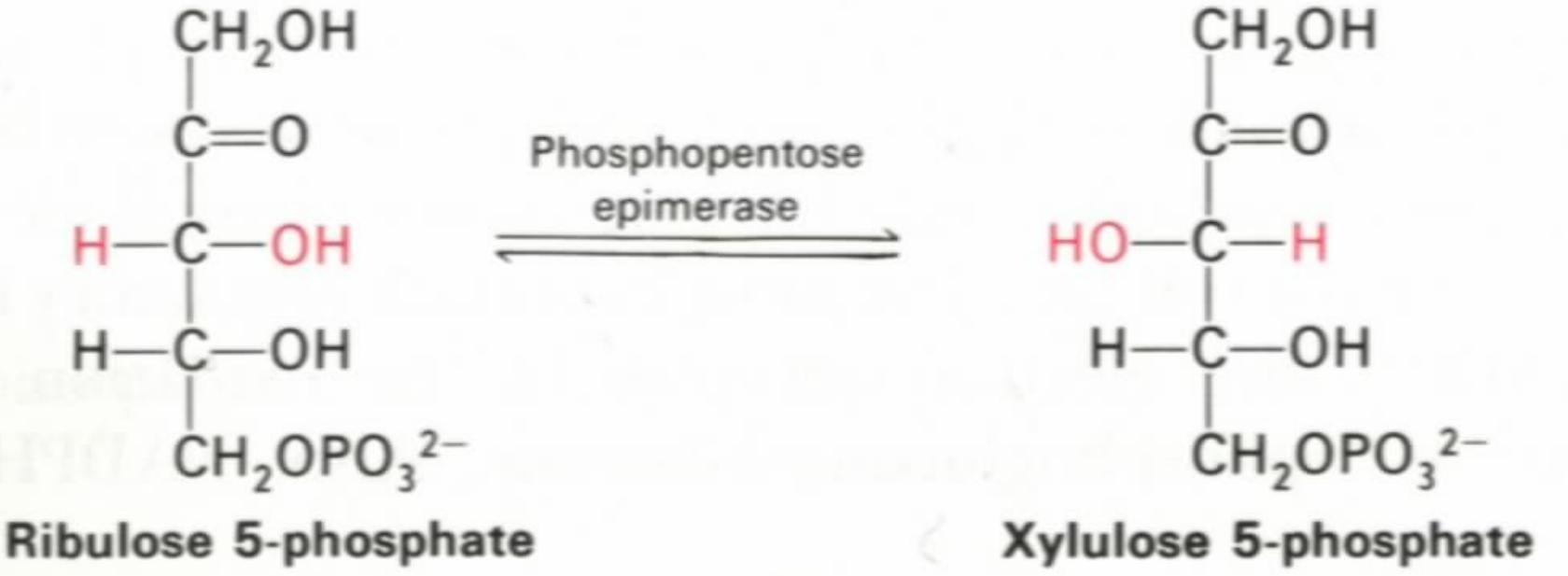
Stage 2: The regeneration of six-carbon Glucose 6-P from five-carbon Ribose 5-P





(b)


Ribulose 5-phosphate
Phosphopentose isomerase
Ribose 5-phosphate
Phosphopentose 3-epimerase
Xylulose 5-phosphate
Isomerization of Five-Carbon Sugars



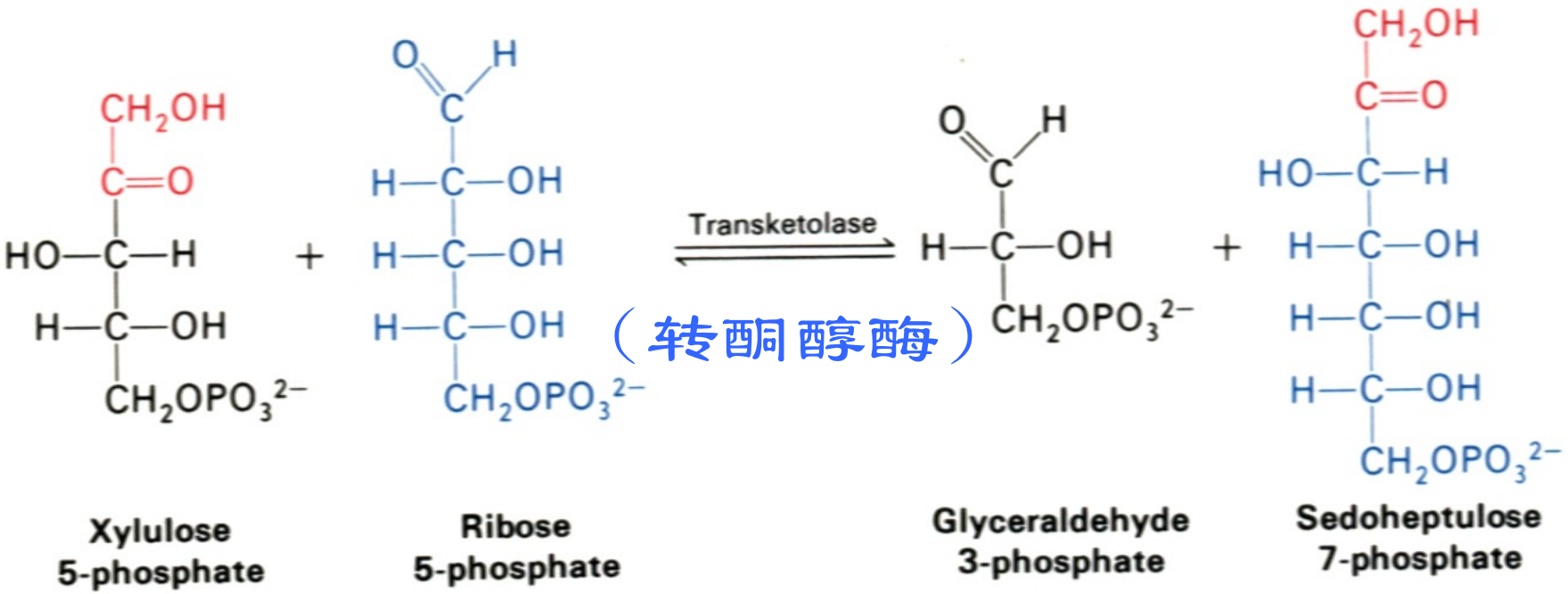
Ribulose 5-phosphate

核酮糖 5- 磷酸

Xylulose 5-phosphate

木酮糖 5- 磷酸

Ribulose 5-P is first isomerized to form xylulose 5-P to initiate the regeneration of glucose 6-P.

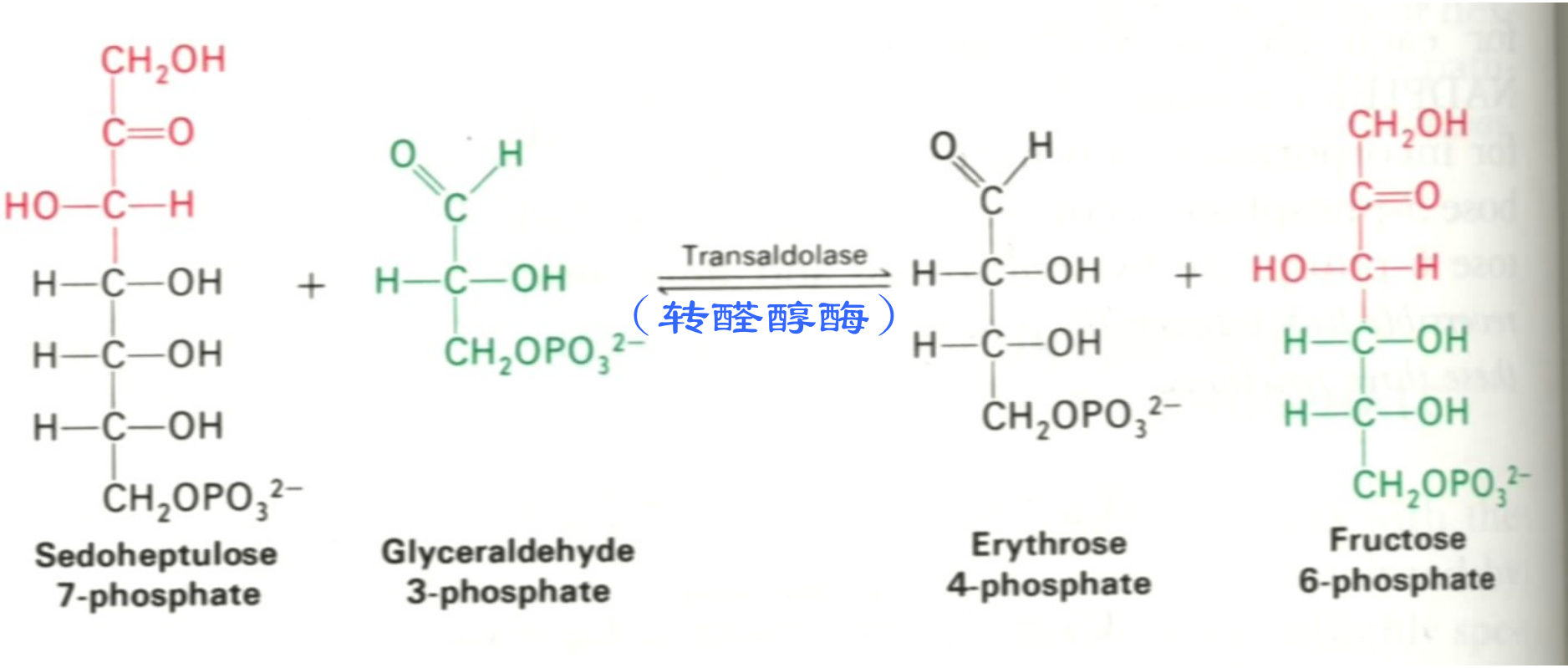


（转酮醇酶）

7- 磷酸景天庚酮糖

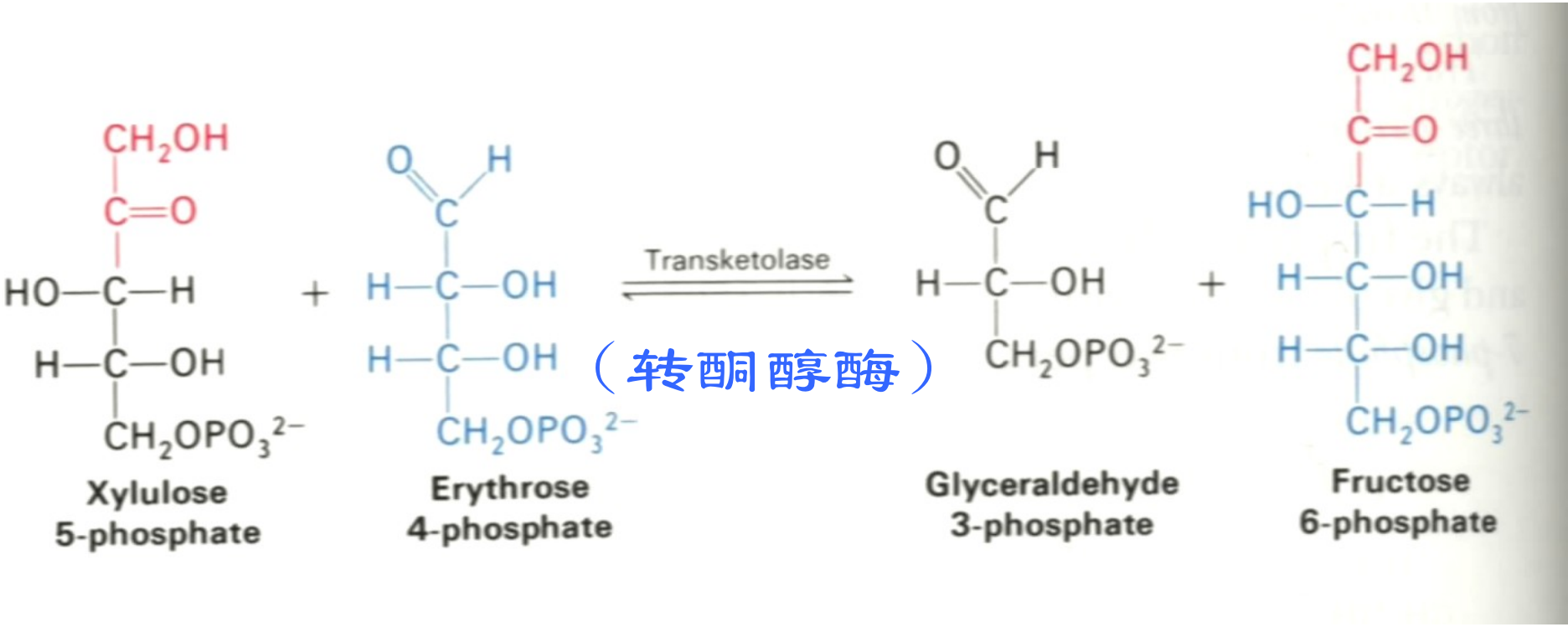
Donor (ketose)

Acceptor (aldose)

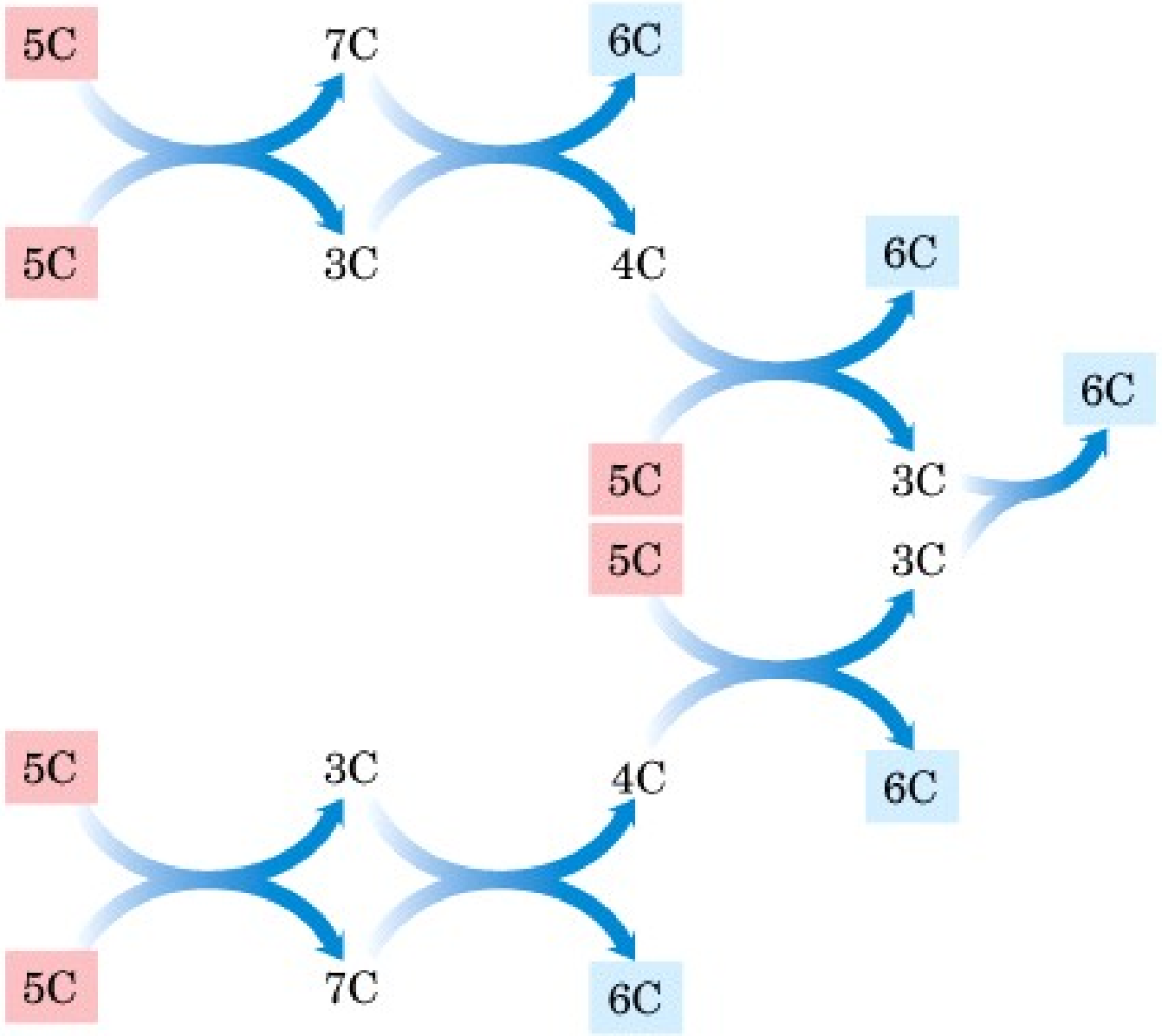

Transaldolase
（转醛醇酶）
Sedoheptulose 7-phosphate
Glyceraldehyde 3-phosphate
Erythrose 4-phosphate
Fructose 6-phosphate

景天庚酮糖

赤藓糖

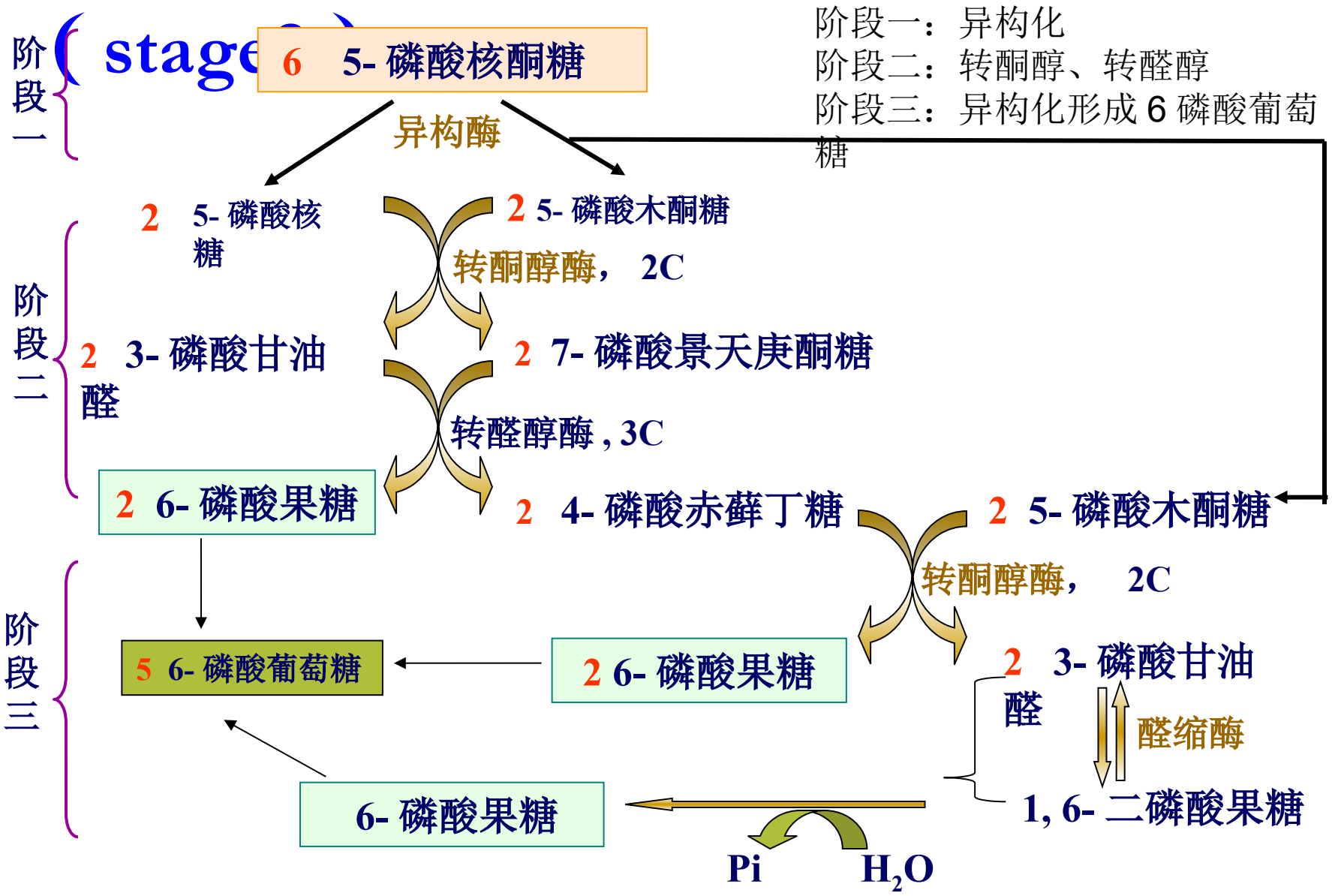

Transketolase
Xylulose 5-phosphate
Erythrose 4-phosphate
Glyceraldehyde 3-phosphate
Fructose 6-phosphate

（转酮醇酶）



(b)

PPP 途径的非氧化分子重排阶段



The Pentose Phosphate Pathway

- Generates reducing power as NADPH
- Adjusts to meet the cell's varying needs for R5P, NADPH, and ATP

- Counter the damaging effects of oxygen radicals

Overall:

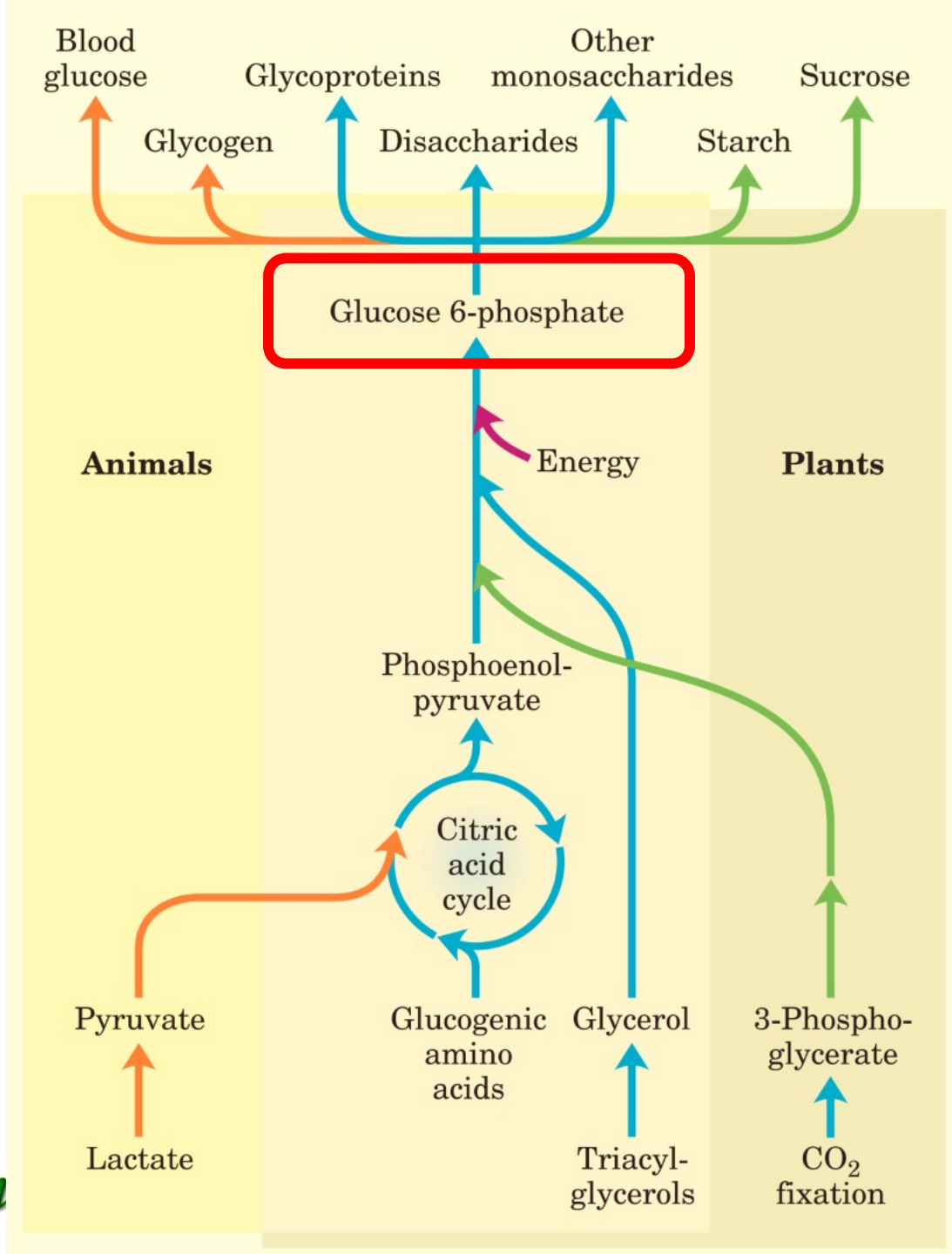
$$6\ \text{G-6-P} + 12\text{NADP}^+ + 7\ \text{H}_2\text{O} \longrightarrow 5\ \text{G-6-P} + 6\text{CO}_2 + 12\text{NADPH} + 12\text{H}^+$$



Gluconeogenesis
(J4)

Gluconeogenesis: The universal pathway for synthesis of glucose.




Blood glucose
Glycoproteins
Other monosaccharides
Sucrose
Glycogen
Disaccharides
Starch
Glucose 6-phosphate
Animals
Energy
Plants
Phosphoenol-pyruvate
Citric acid cycle
Pyruvate
Glucogenic amino acids
Glycerol
3-Phospho-glycerate
Lactate
Triacyl-glycerols
CO_2 fixation

Biochem

1. Carbohydrates are synthesized from simple precursors via gluconeogenesis

A few three-carbon compounds (including lactate, pyruvate, glycerol, and 3-phosphoglycerate) serve as the major precursors for carbohydrate (glucose) biosynthesis, or gluconeogenesis.

The reactions of gluconeogenesis are essentially the same in different organisms.

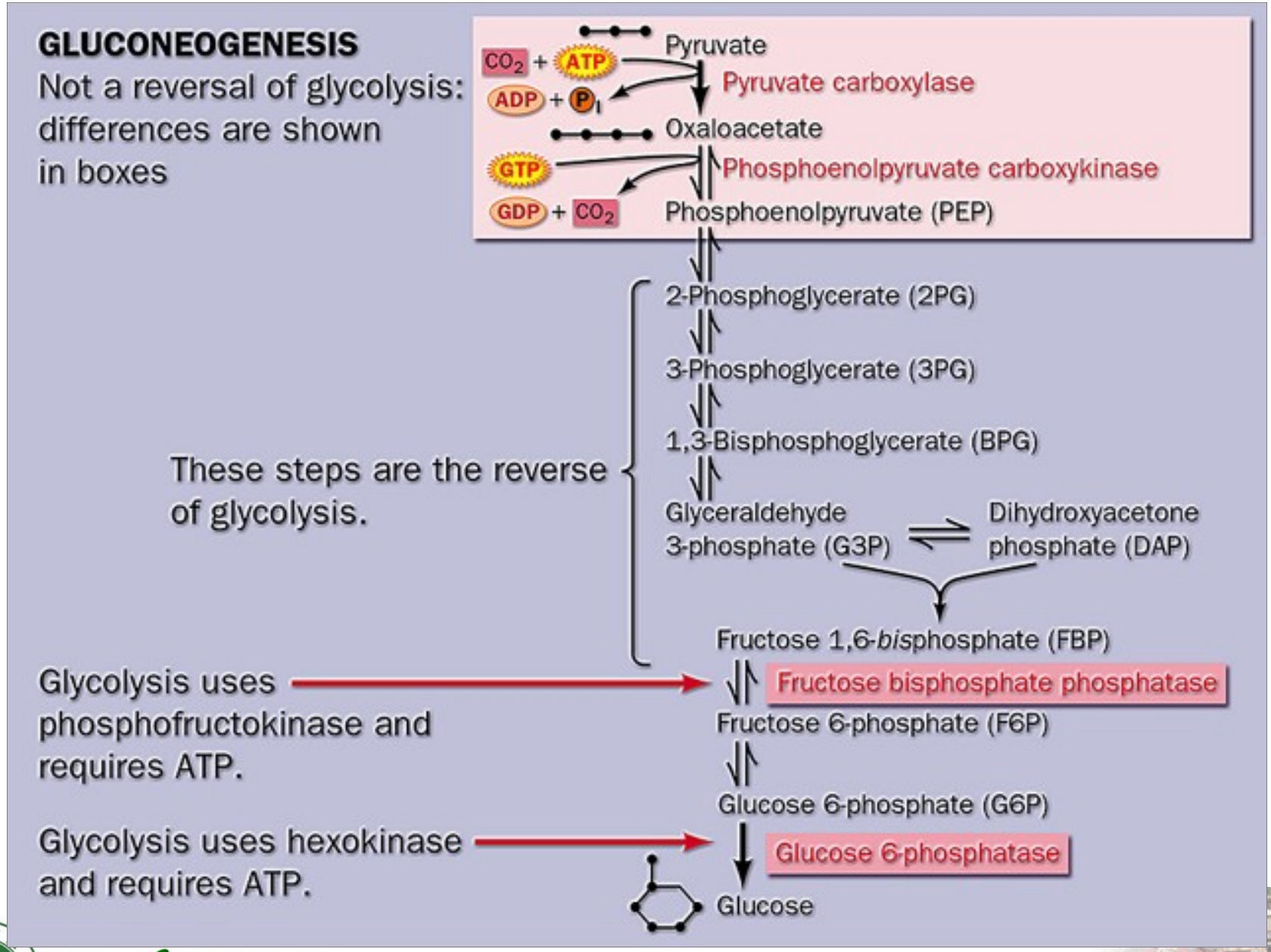
The conversion of pyruvate to glucose is the central pathway in gluconeogenesis.



2. The opposing pathways of glycolysis and gluconeogenesis have 3 reactions different and 7 reactions in common

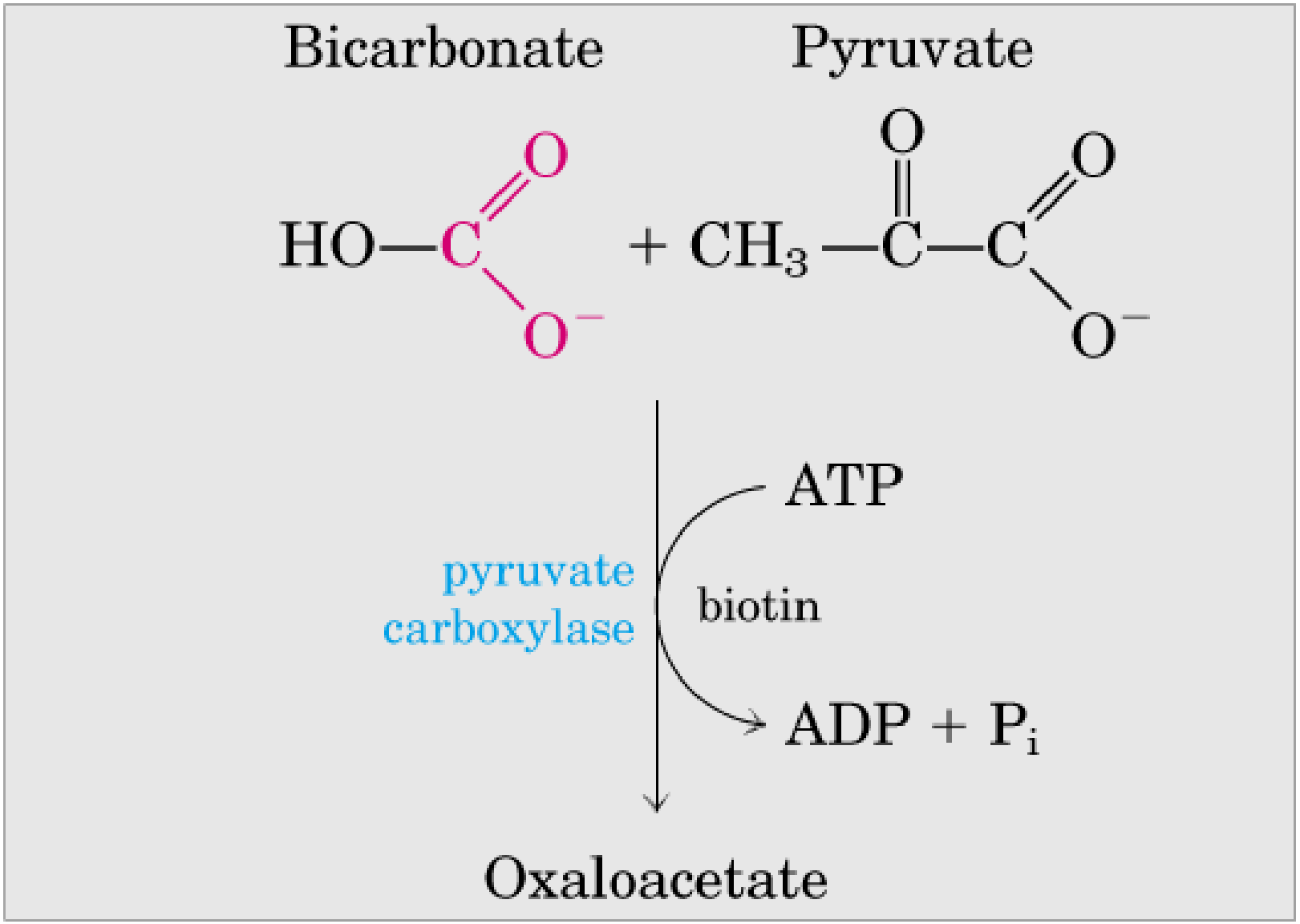
The reversible reactions between pyruvate and glucose are shared by gluconeogenesis and glycolysis, but the irreversible reactions are different ("bypassed" in gluconeogenesis).



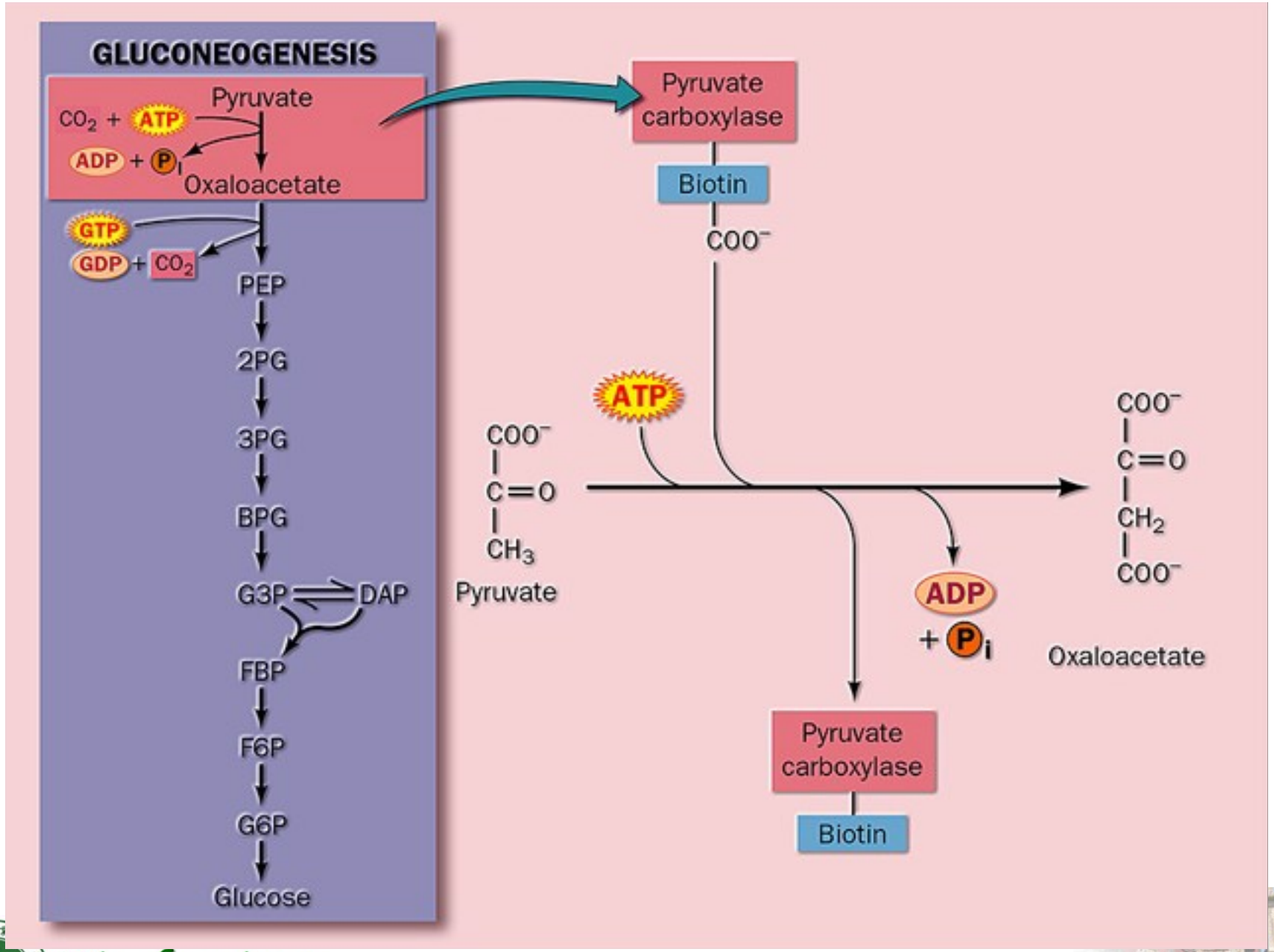

GLUCONEOGENESIS
Not a reversal of glycolysis: differences are shown in boxes
Pyruvate
CO2 + ATP
Pyruvate carboxylase
ADP + Pi
Oxaloacetate
GTP
Phosphoenolpyruvate carboxykinase
GDP + CO2
Phosphoenolpyruvate (PEP)
2-Phosphoglycerate (2PG)
3-Phosphoglycerate (3PG)
1,3-Bisphosphoglycerate (BPG)
These steps are the reverse of glycolysis.
Glyceraldehyde 3-phosphate (G3P)
Dihydroxyacetone phosphate (DAP)
Fructose 1,6-bisphosphate (FBP)
Glycolysis uses phosphofructokinase and requires ATP.
Fructose bisphosphate phosphatase
Fructose 6-phosphate (F6P)
Glucose 6-phosphate (G6P)
Glycolysis uses hexokinase and requires ATP.
Glucose 6-phosphatase
Glucose

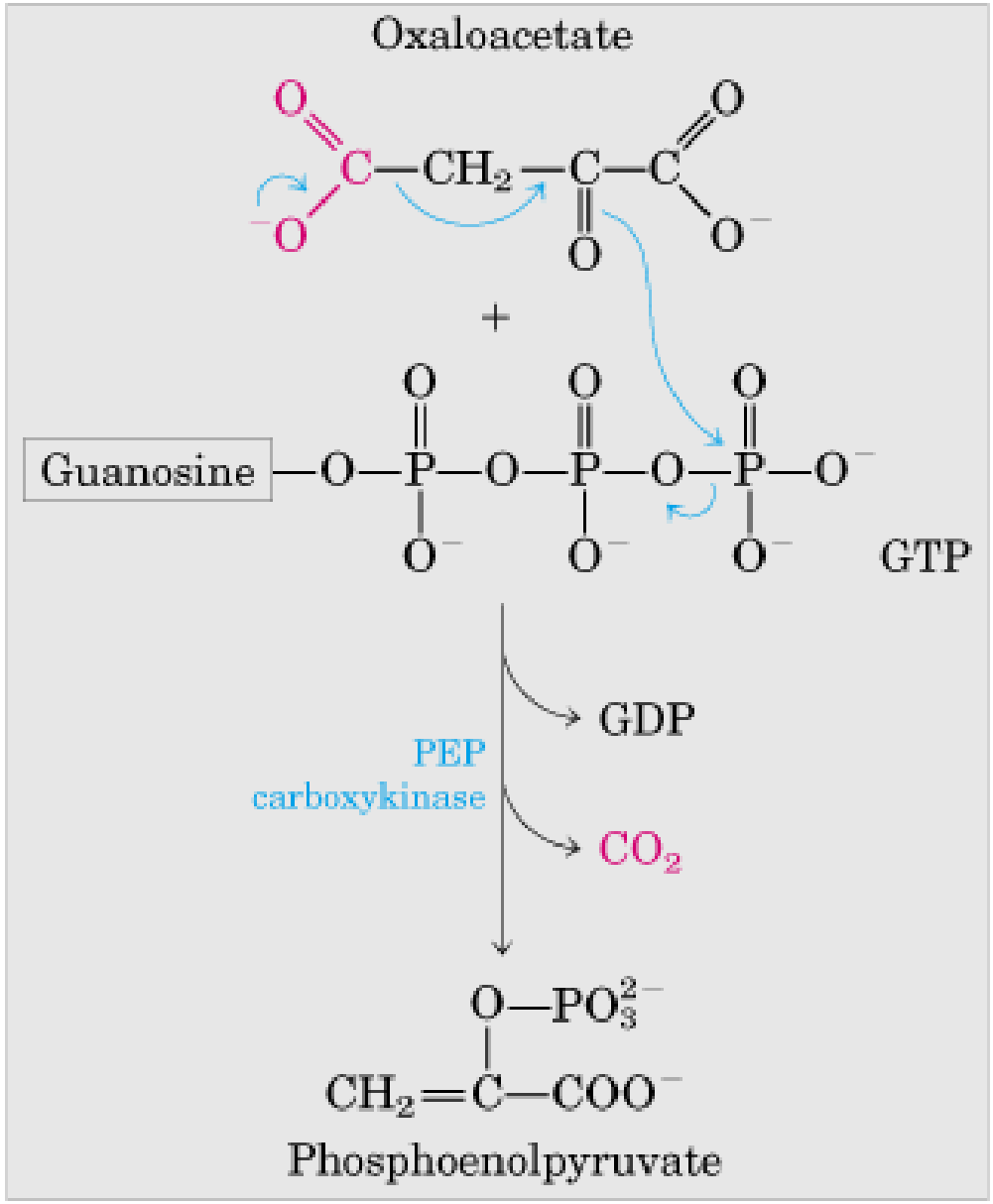
3. Pyruvate is converted to phosphoenoylpyruvate (PEP) via two Exergonic Reactions




Bicarbonate
Pyruvate
HO—C
O
O⁻
+ CH3—C—C
O
O
O⁻
ATP
pyruvate carboxylase
biotin
ADP + Pi
Oxaloacetate

(a)

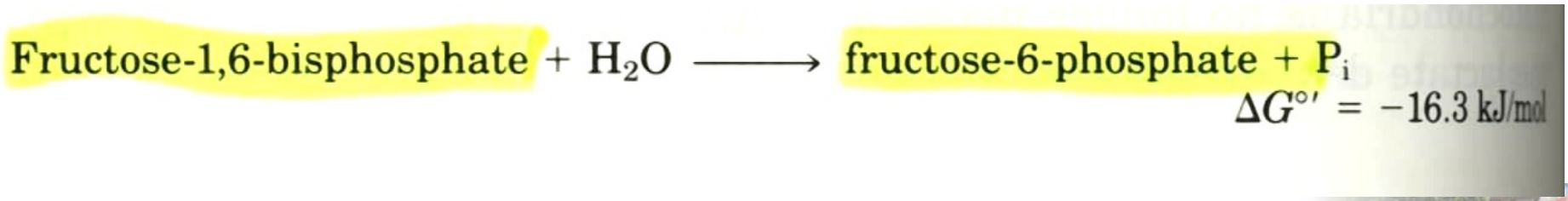

GLUCONEOGENESIS
Pyruvate
CO_2 + ATP
ADP + P_i
Oxaloacetate
GTP
GDP + CO_2
PEP
2PG
3PG
BPG
G3P
DAP
FBP
F6P
G6P
Glucose
Pyruvate carboxylase
Biotin
COO^-
ATP
COO^-
$C=O$
CH_3
Pyruvate
ADP
+ P_i
COO^-
$C=O$
CH_2
COO^-
Oxaloacetate
Pyruvate carboxylase
Biotin


Oxaloacetate
Guanosine
GTP
GDP
PEP
carboxykinase
CO_2
Phosphoenolpyruvate

(b)

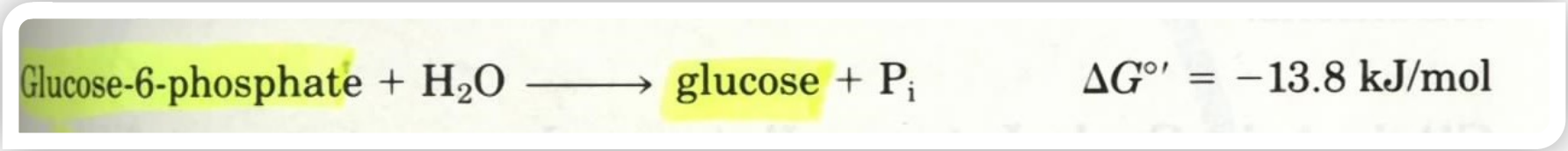
4. Conversion of fructose 1,6-bisphosphate to fructose 6-phosphate is the second bypassing step

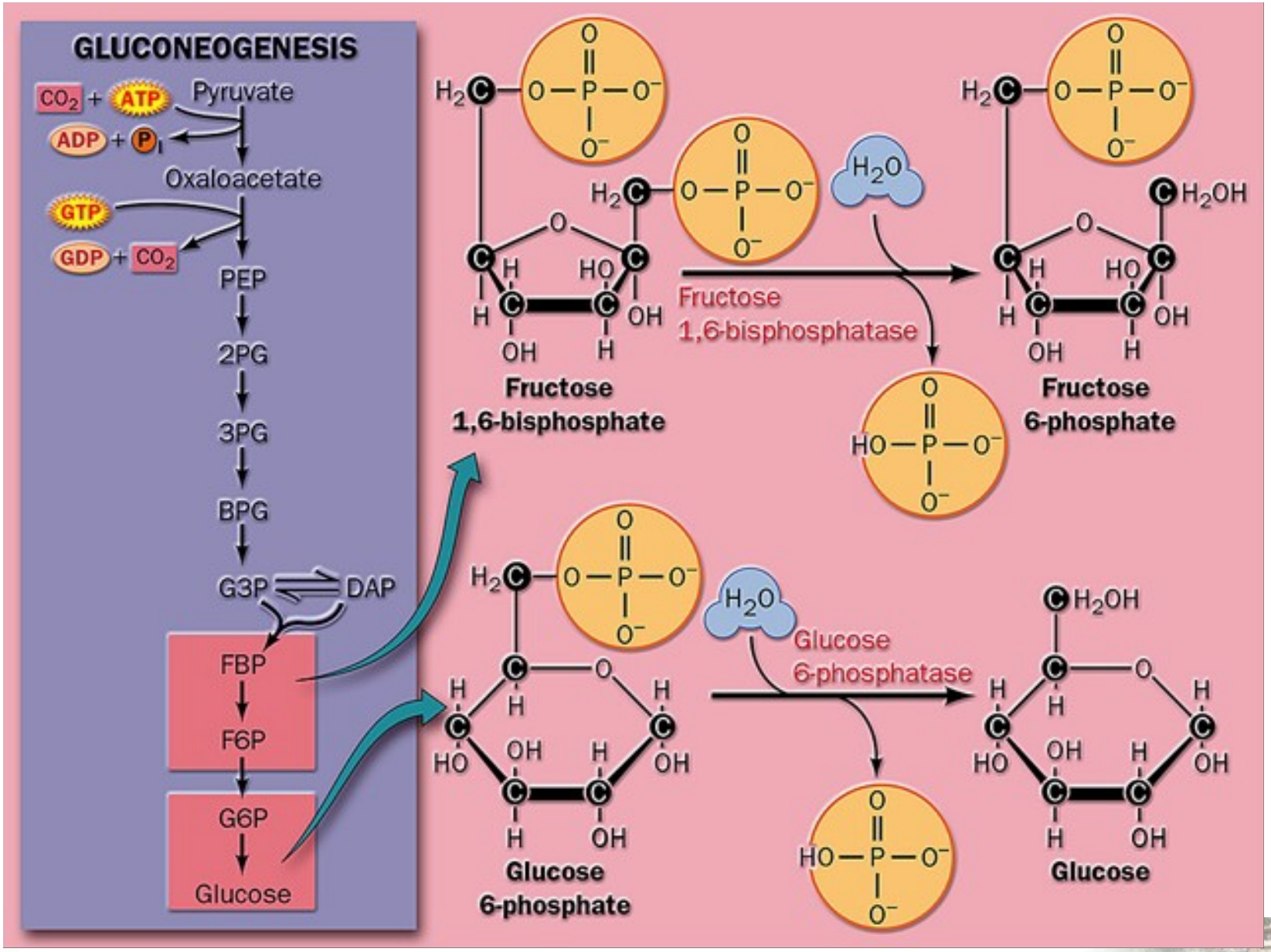
The reaction is catalyzed by Mg^{2+} -dependent fructose 1,6-bisphosphatase (instead of phosphofructokiase-1).

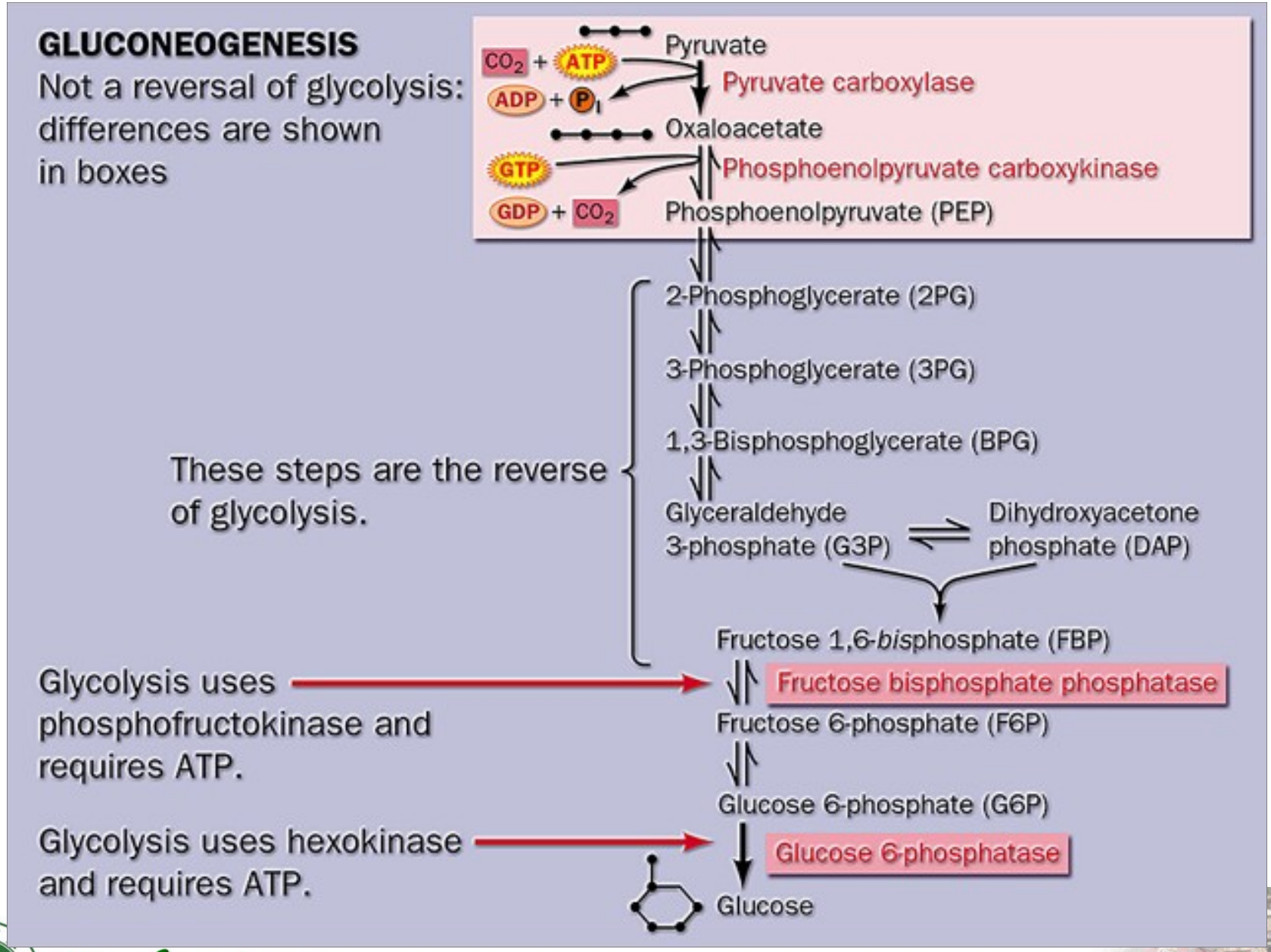


5. The conversion of glucose 6-phosphate to glucose is the last bypassing step

The reaction is catalyzed by glucose 6-phosphatase (instead of hexokiase).




GLUCONEOGENESIS
CO2 + ATP
Pyruvate
ADP + Pi
Oxaloacetate
GTP
GDP + CO2
PEP
2PG
3PG
BPG
G3P
DAP
FBP
F6P
G6P
Glucose
Fructose 1,6-bisphosphate
Fructose 1,6-bisphosphatase
H2O
Fructose 6-phosphate
Glucose 6-phosphate
Glucose 6-phosphatase
H2O
Glucose


GLUCONEOGENESIS
Not a reversal of glycolysis: differences are shown in boxes
Pyruvate
CO2 + ATP
Pyruvate carboxylase
ADP + Pi
Oxaloacetate
GTP
Phosphoenolpyruvate carboxykinase
GDP + CO2
Phosphoenolpyruvate (PEP)
2-Phosphoglycerate (2PG)
3-Phosphoglycerate (3PG)
1,3-Bisphosphoglycerate (BPG)
These steps are the reverse of glycolysis.
Glyceraldehyde 3-phosphate (G3P)
Dihydroxyacetone phosphate (DAP)
Fructose 1,6-bisphosphate (FBP)
Glycolysis uses phosphofructokinase and requires ATP.
Fructose bisphosphate phosphatase
Fructose 6-phosphate (F6P)
Glucose 6-phosphate (G6P)
Glycolysis uses hexokinase and requires ATP.
Glucose 6-phosphatase
Glucose

$$2\ \text{Pyruvate} + 4\text{ATP} + 2\text{GTP} + 2\text{NADH} + 2\text{H}^+ + 4\text{H}_2\text{O} \longrightarrow$$
$$\text{glucose} + 4\text{ADP} + 2\text{GDP} + 6\text{P}_\text{i} + 2\text{NAD}^+$$



6. More energy is consumed in gluconeogenesis than produced in glycolysis

Six high-energy phosphate groups are required when two molecules of pyruvates are converted to one glucose via gluconeogenesis pathway. Two molecules of ATP are produced when one glucose molecule is converted to two pyruvate molecules via glycolysis pathway.



Summary

Gluconeogenesis, the synthesis of glucose from 3-carbon compounds (mainly pyruvate) is highly conserved in all organisms.

Gluconeogenesis shares most of the reactions occurring in glycolysis, but bypassing the three irreversible reactions (using different enzymes).

Gluconeogenesis consumes more energy than glycolysis releases.

J 6 Polysaccharide Metabolism

（ catabolism, anabolism ）

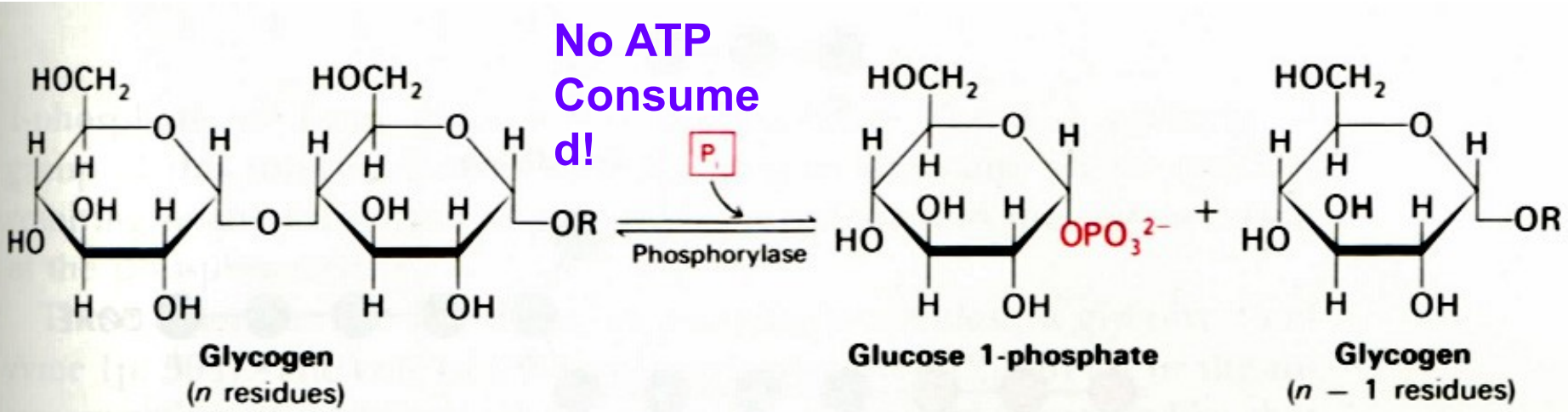
Glycogen Starch Sucrose

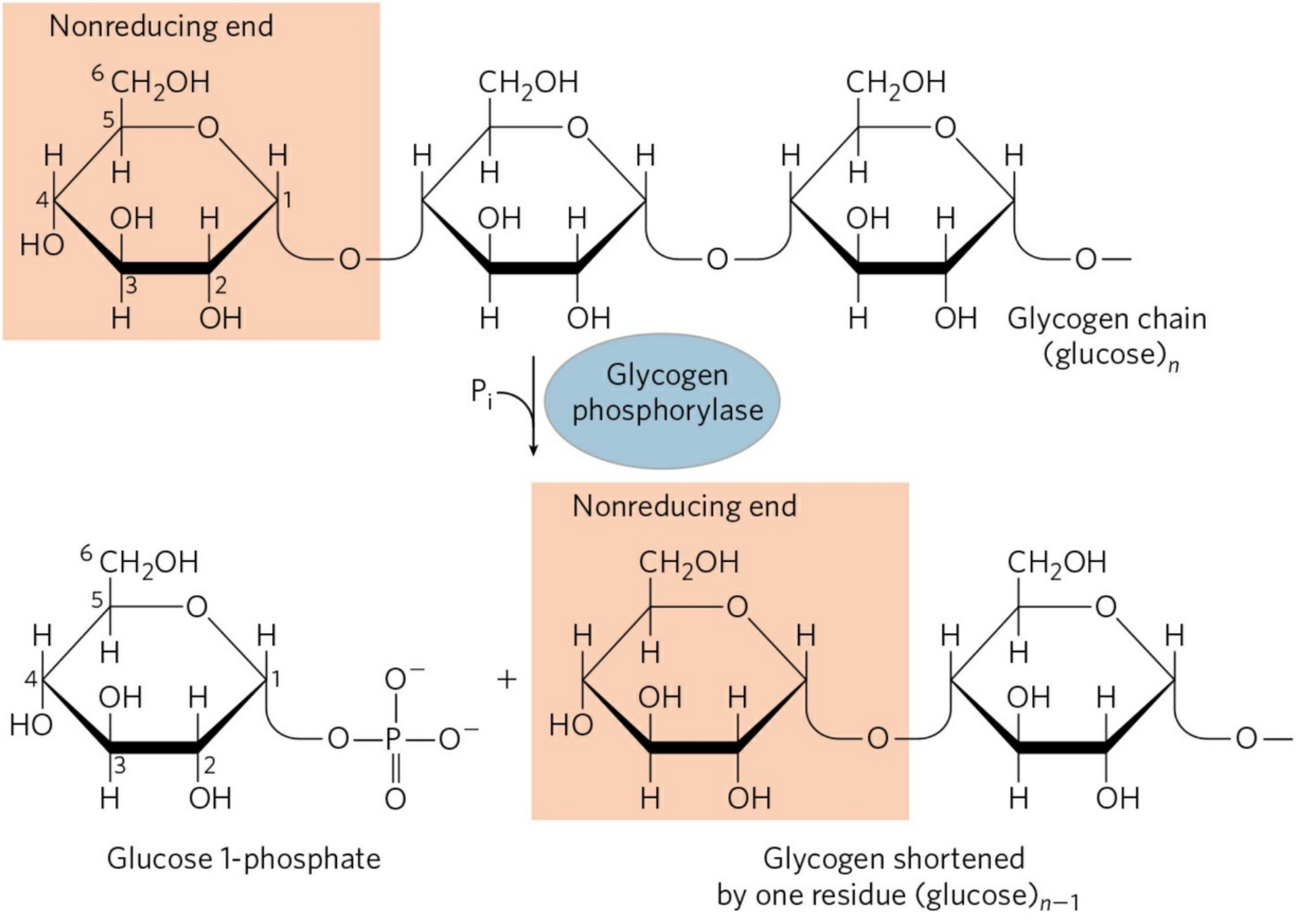


1. Glycogen in cells is first converted to Glc-1-P for oxidative degradation

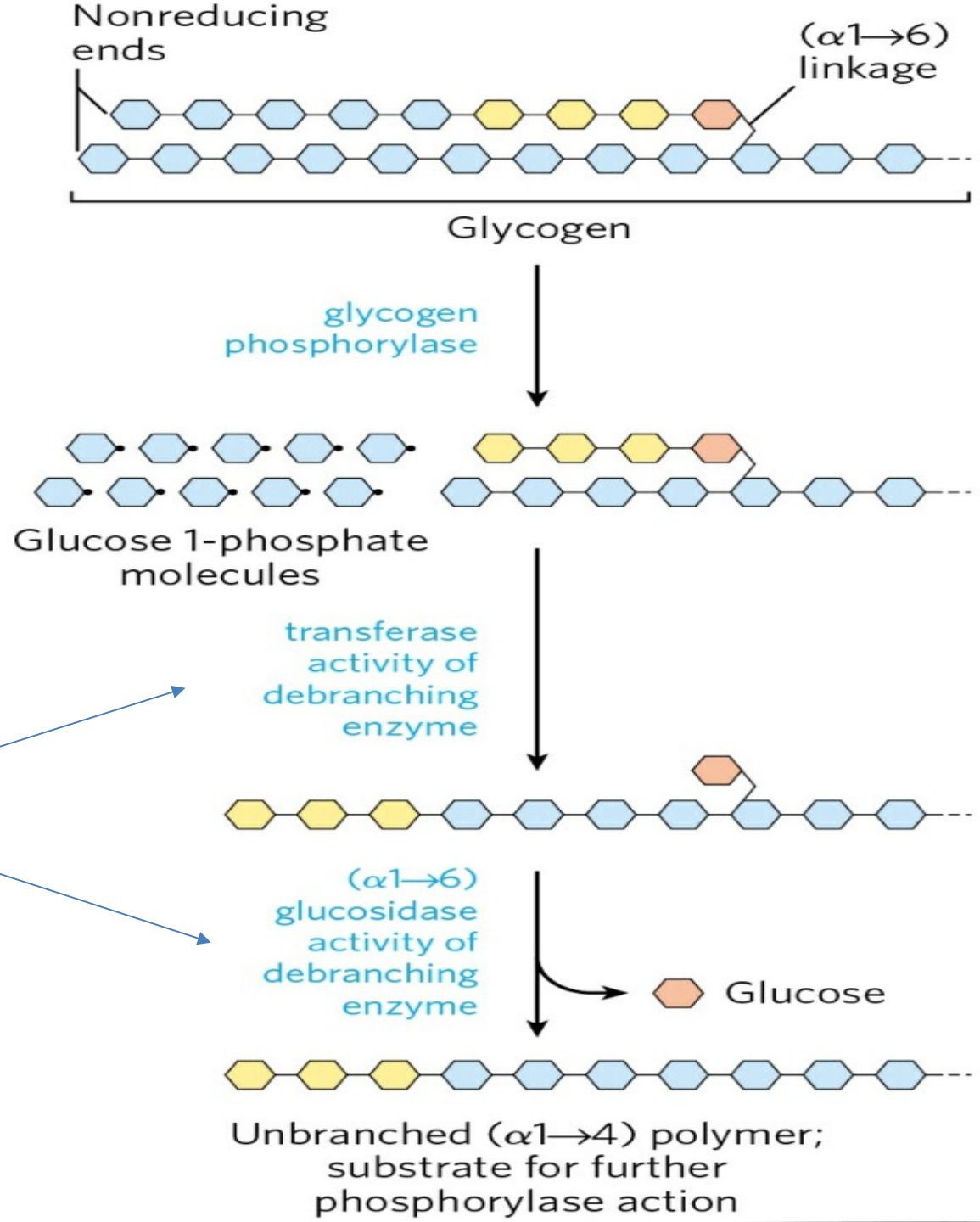
The glucose unit at the nonreducing terminal of glycogen is removed as **Glc-1-P** via phosphorolysis: The (α1 4) glycosidic bond is attacked by an inorganic phosphate).




No ATP Consumed!
P_i
Phosphorylase
OPO_3^{2-}
Glycogen
(*n* residues)
Glucose 1-phosphate
Glycogen
(*n* – 1 residues)


Nonreducing end
Glycogen chain
(glucose)$_n$
P_i
Glycogen
phosphorylase
Nonreducing end
Glucose 1-phosphate
Glycogen shortened
by one residue (glucose)$_{n-1}$

Glycogen breakdown near an (α1→6) branch point.



a bifunctional debranching enzyme

2. Dietary poly- and disaccharides are hydrolyzed to monosaccharides in the digestive system

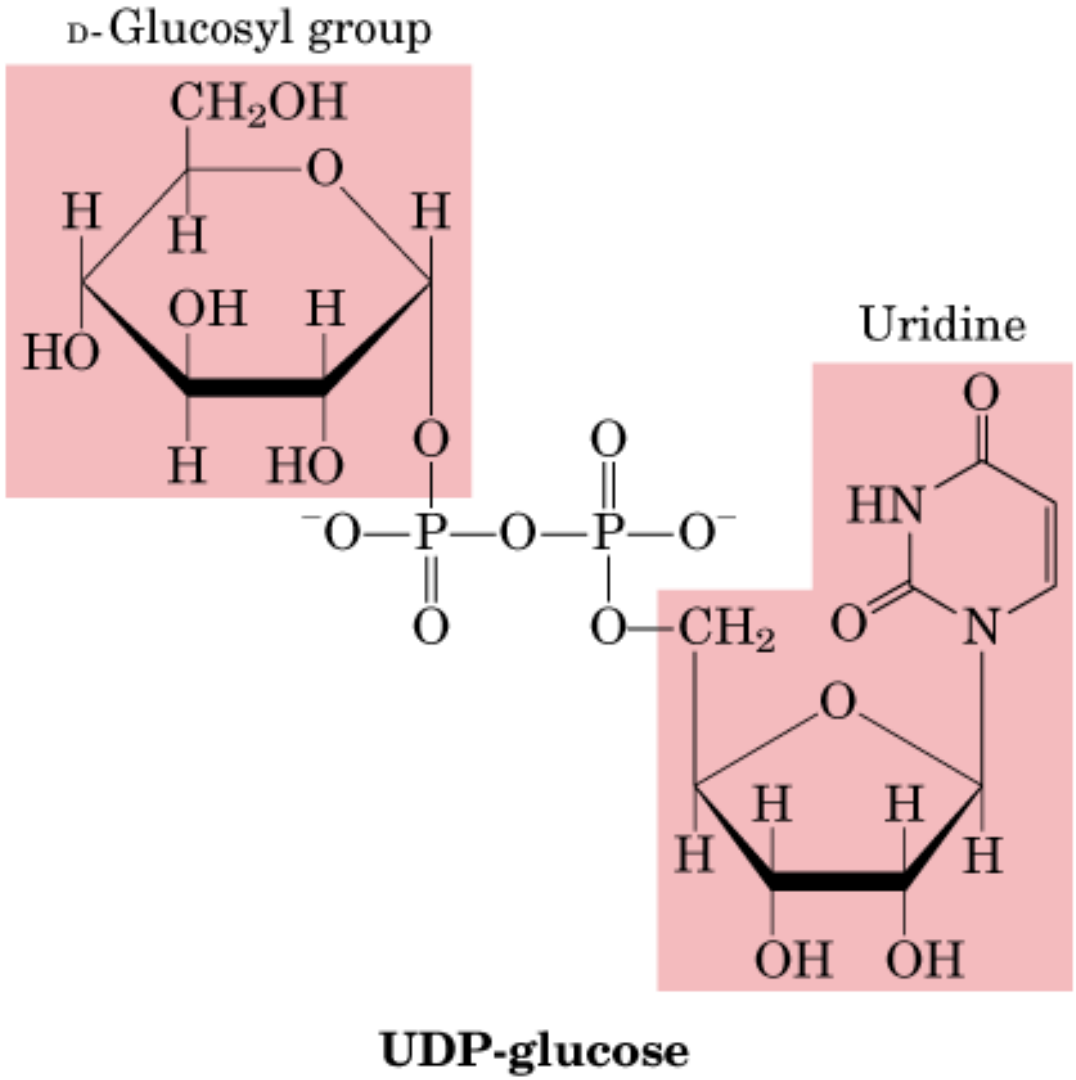
Salivary α-amylase (α- 淀粉酶) in the mouth hydrolyzes starch (glycogen) into short polysaccharides or oligosacchrides. Specific enzymes (e.g., lactase, sucrase, maltase,etc.) finally hydrolyze all disaccharides into monosaccharides.



3. Biosynthesis of glycogen, starch, and sucrose

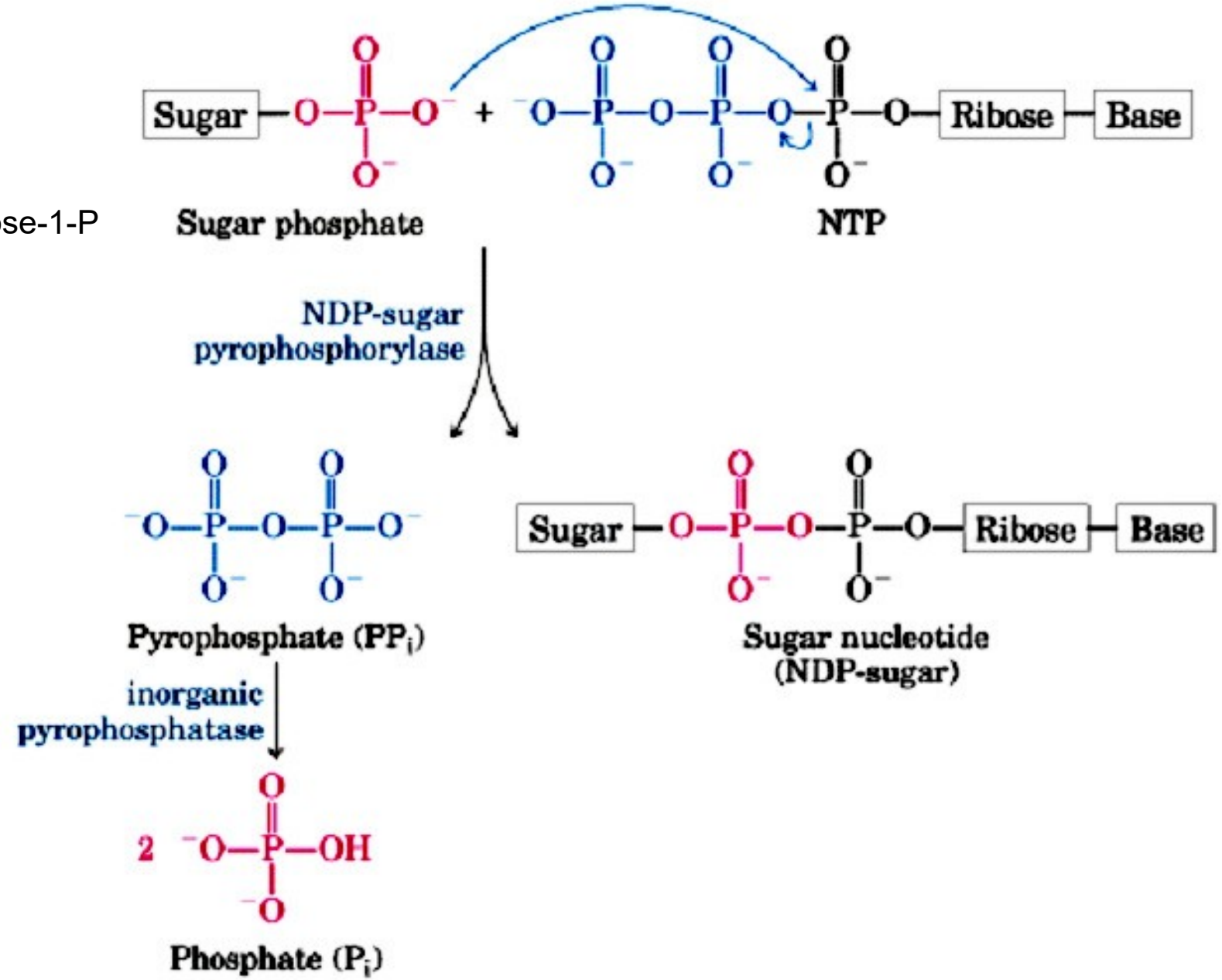
- Hexoses are converted to sugar nucleotides before being polymerized
- Hexose nucleotides are common precursors for carbohydrate transformation and polymerization!
- A hexose nucleotides is formed via a condensation reaction occurring between a NTP and a hexose 1-phosphate.





Sugar nucleotides were found to be the activated forms of sugars participating in biosynthesis.

Glucose-1-P


Sugar
Ribose
Base
Sugar phosphate
NTP
NDP-sugar pyrophosphorylase
Pyrophosphate (PP_i)
Sugar nucleotide (NDP-sugar)
inorganic pyrophosphatase
2
Phosphate (P_i)
Net reaction: Sugar phosphate + NTP ⟶ NDP-sugar + $2P_i$

- A sugar nucleotide is formed through a condensation reaction between a NTP and a sugar phosphate.

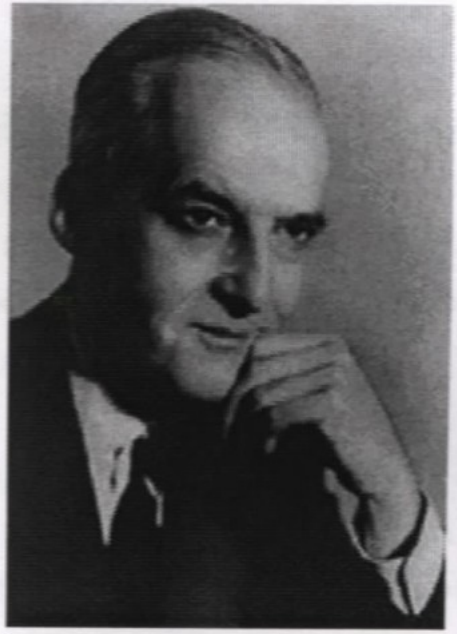
Glucose-1-P + UTP → UDP-glucose + PPi



The Nobel Prize in Chemistry 1970

"for his discovery of sugar nucleotides and their role in the biosynthesis of carbohydrates"

Presentation Speech



Luis F. Leloir

Argentina

Institute for Biochemical Research
Buenos Aires, Argentina

1906 - 1987

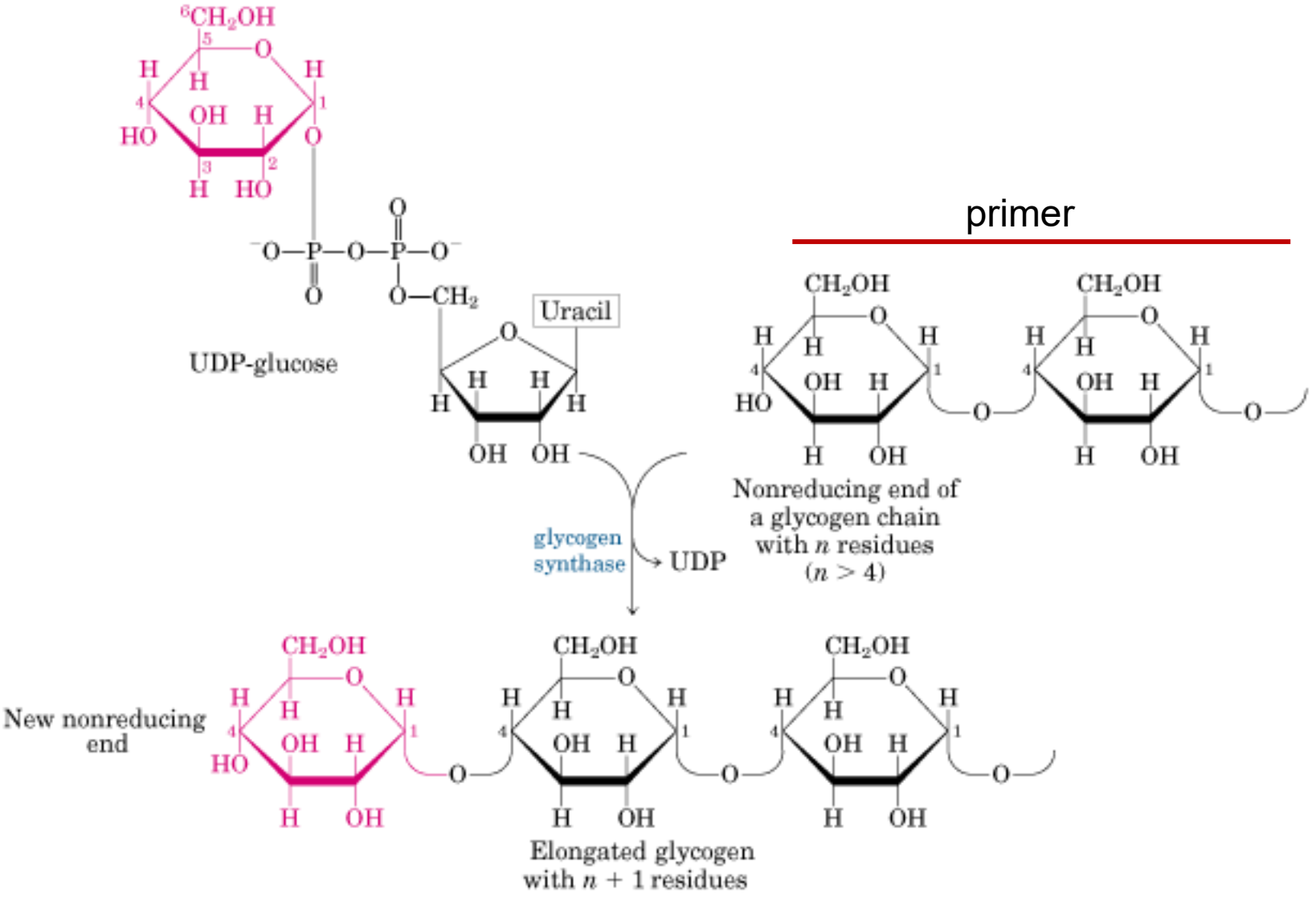
Biography

Glycogen is synthesized using UDP-glucose

The glucose residue of UDP-Glucose is transferred to the nonreducing end of a primer or glycogen branch (of at least 4 glucose residues) to make a new a-1,4 glycosidic bond in a reaction catalyzed by glycogen synthase.

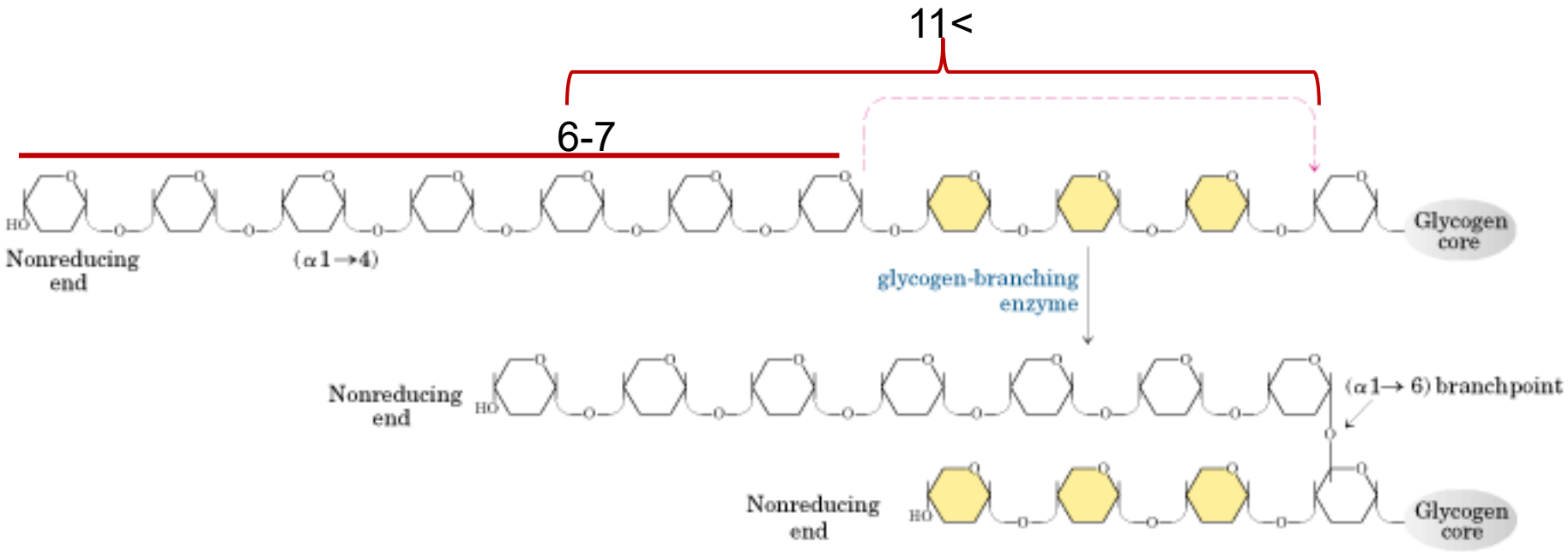


primer

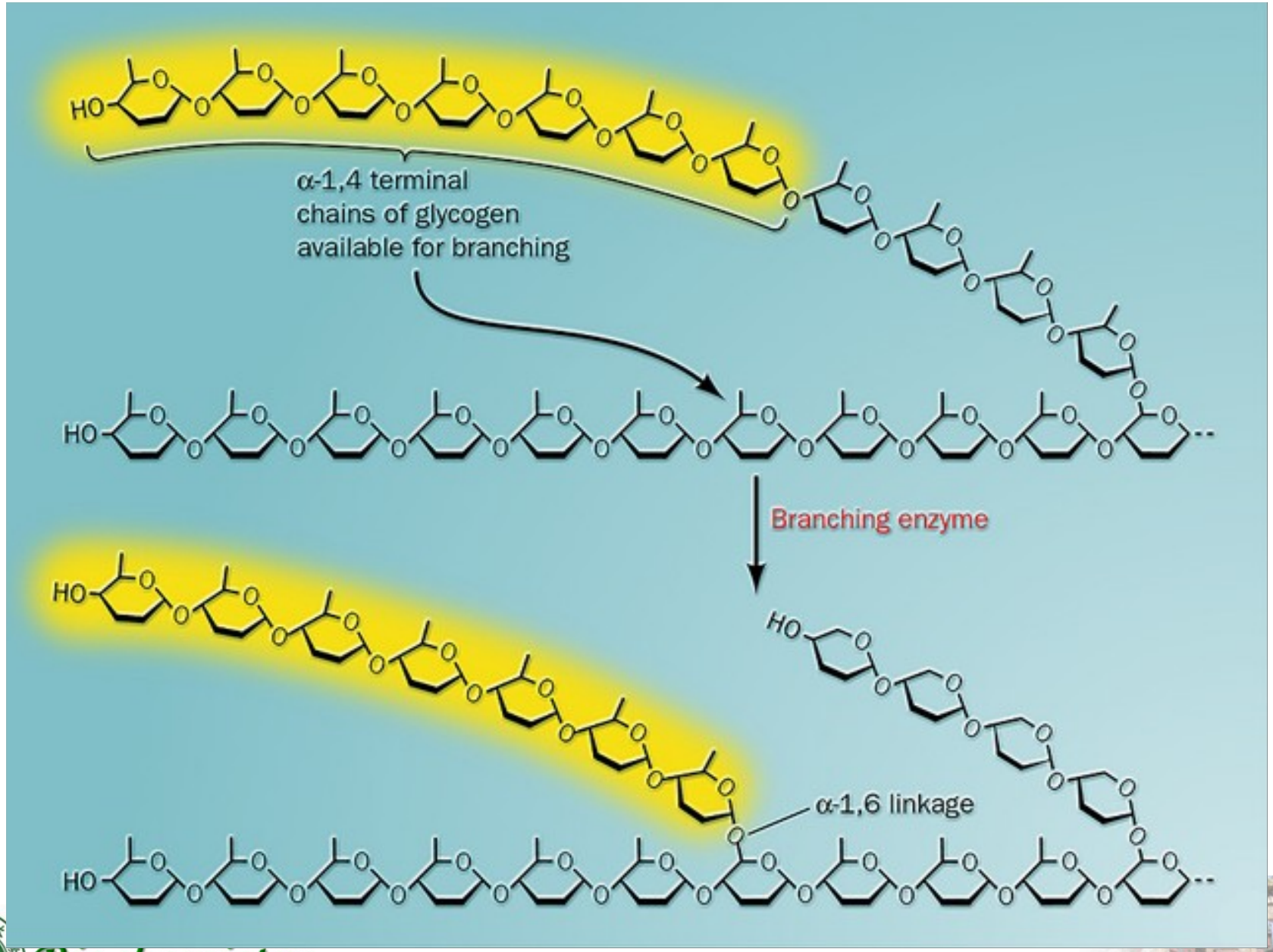

UDP-glucose
Uracil
glycogen synthase
UDP
Nonreducing end of a glycogen chain with n residues (n > 4)
New nonreducing end
Elongated glycogen with n + 1 residues

The formation of (a1→6) branches of glycogen is catalyzed by glycosyl-(4→6)-transferase: a terminal fragment of 6-7 residues is transferred from a branch having at least 11 residues to the C-6 hydroxyl group at a more interior position of the same or another glycogen chain.





A branching enzyme catalyzes the transferring of a short stretch of Glc residues from one nonreducing end to the interior of the glycogen to make an α1→6 linkage (thus a branch).


HO
α-1,4 terminal
chains of glycogen
available for branching
HO
Branching enzyme
HO
HO
α-1,6 linkage
HO

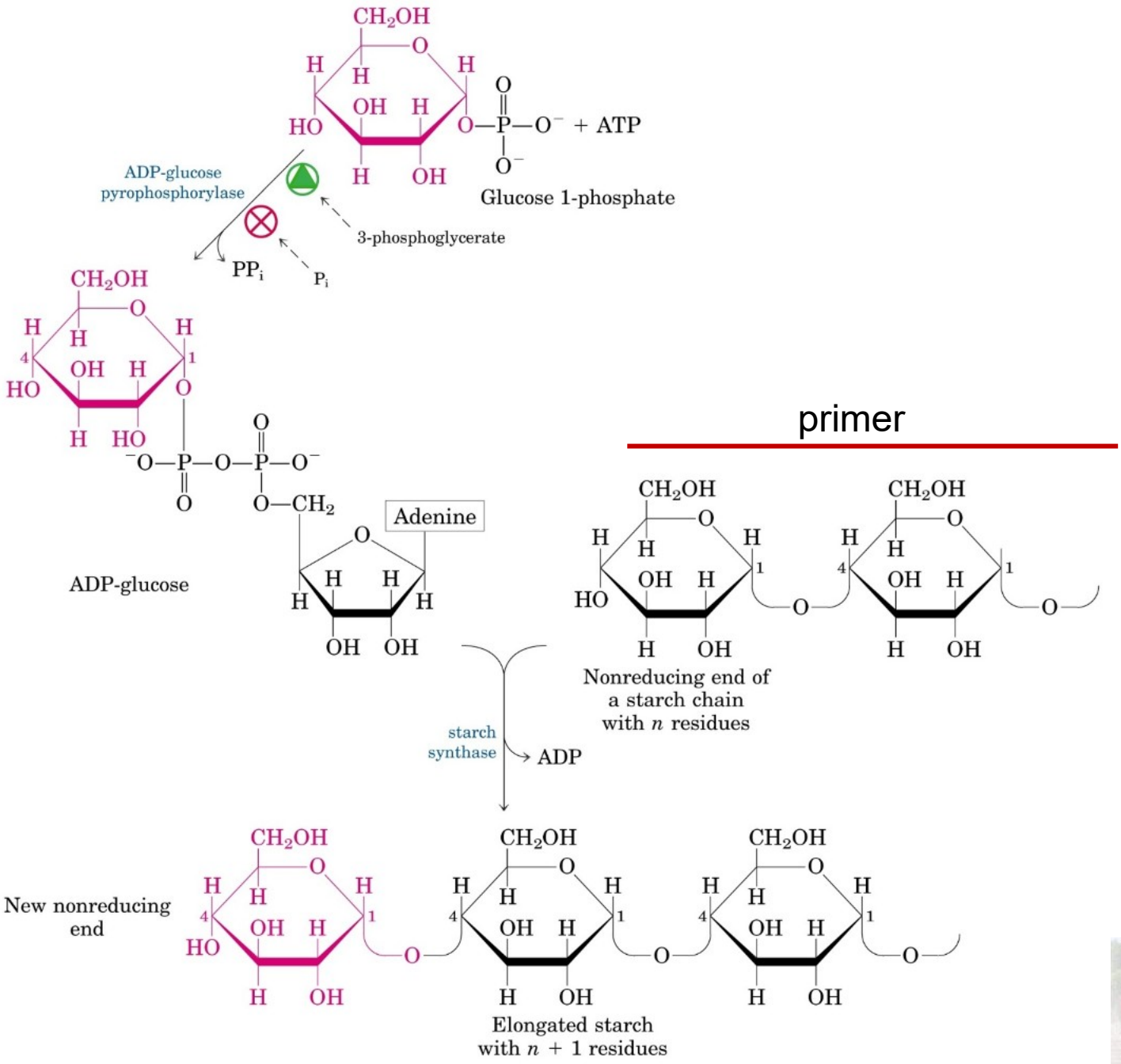
Starch synthesis in chloroplast stroma is similar to glycogen synthesis

But ADP-glucose is used as the precursor.

Starch synthase also transfers the glucose unit to the nonreducing end of a preexisting primer (form amylose 直链淀粉)

Branches in amylopectin (支链淀粉) are synthesized using a similar branching enzyme.

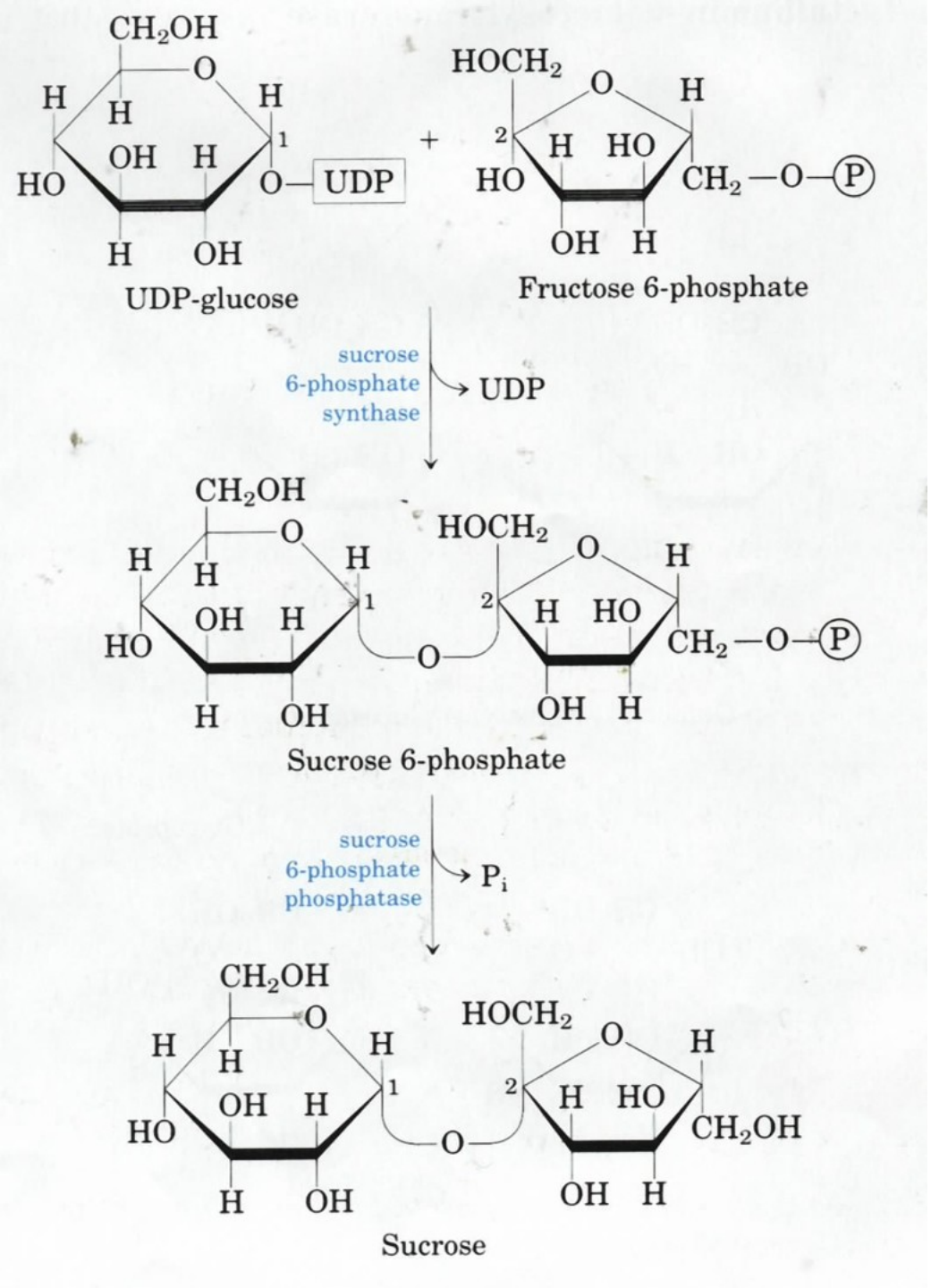



ADP-glucose pyrophosphorylase
3-phosphoglycerate
Glucose 1-phosphate
+ ATP
PPi
Pi
ADP-glucose
Adenine
primer
Nonreducing end of a starch chain with n residues
starch synthase
ADP
New nonreducing end
Elongated starch with n + 1 residues

Sucrose is synthesized from UDP-glucose and fructose 6-phosphate in the cytosol of plant cells

Sucrose 6-phosphate is first synthesized by the catalysis of sucrose 6-phosphate synthase.
The phosphate is then removed in a reaction catalyzed by sucrose 6-phosphate phosphatase.





Sucrose is synthesized from UDP-Glc and Fru 6-P



Carbohydrates can be synthesized from CO_2 in photosynthetic organisms





The Nobel Prize in Chemistry 1961

"for his research on the carbon dioxide assimilation in plants"

Presentation Speech



Melvin Calvin

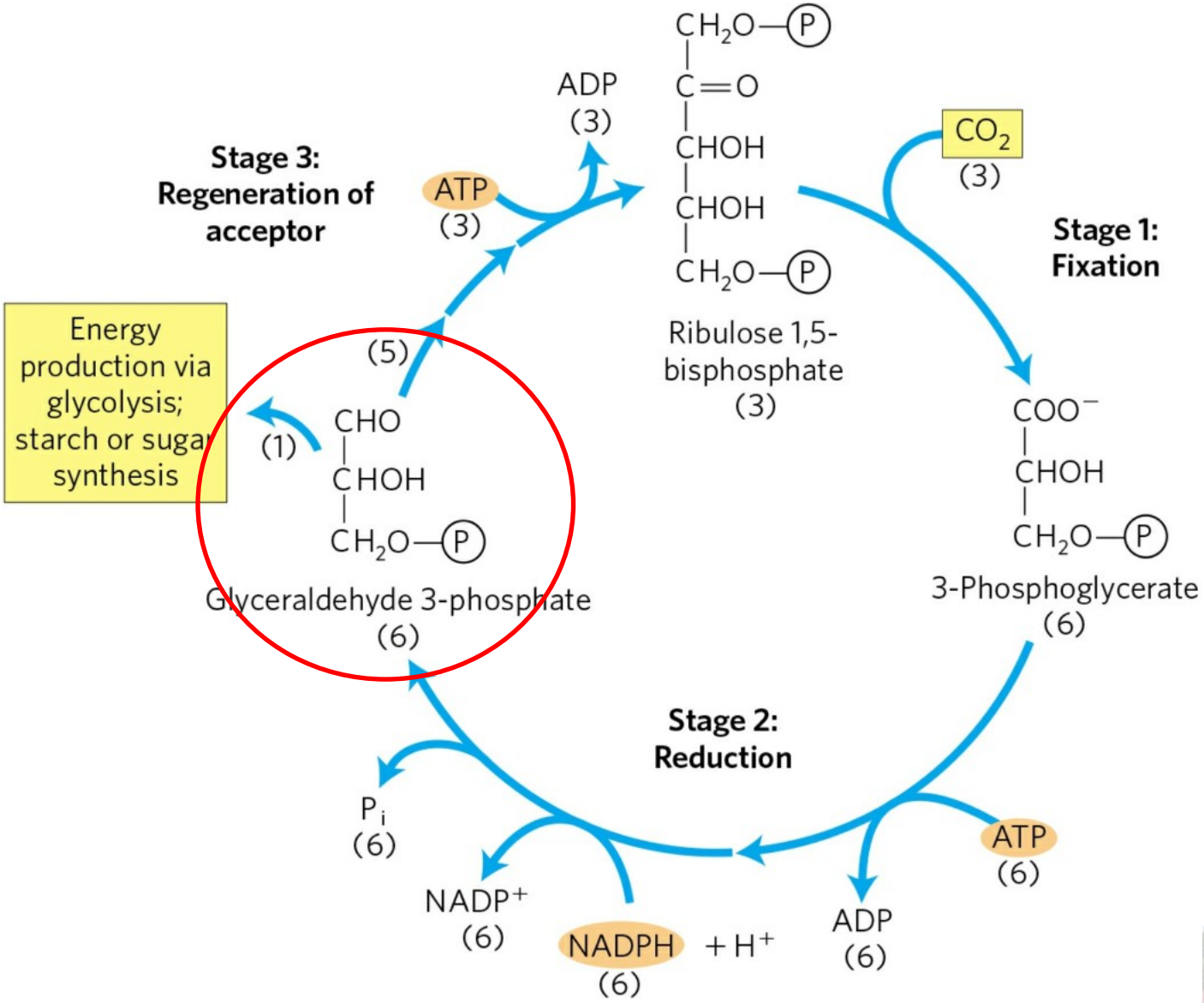
USA

University of California
Berkeley, CA, USA

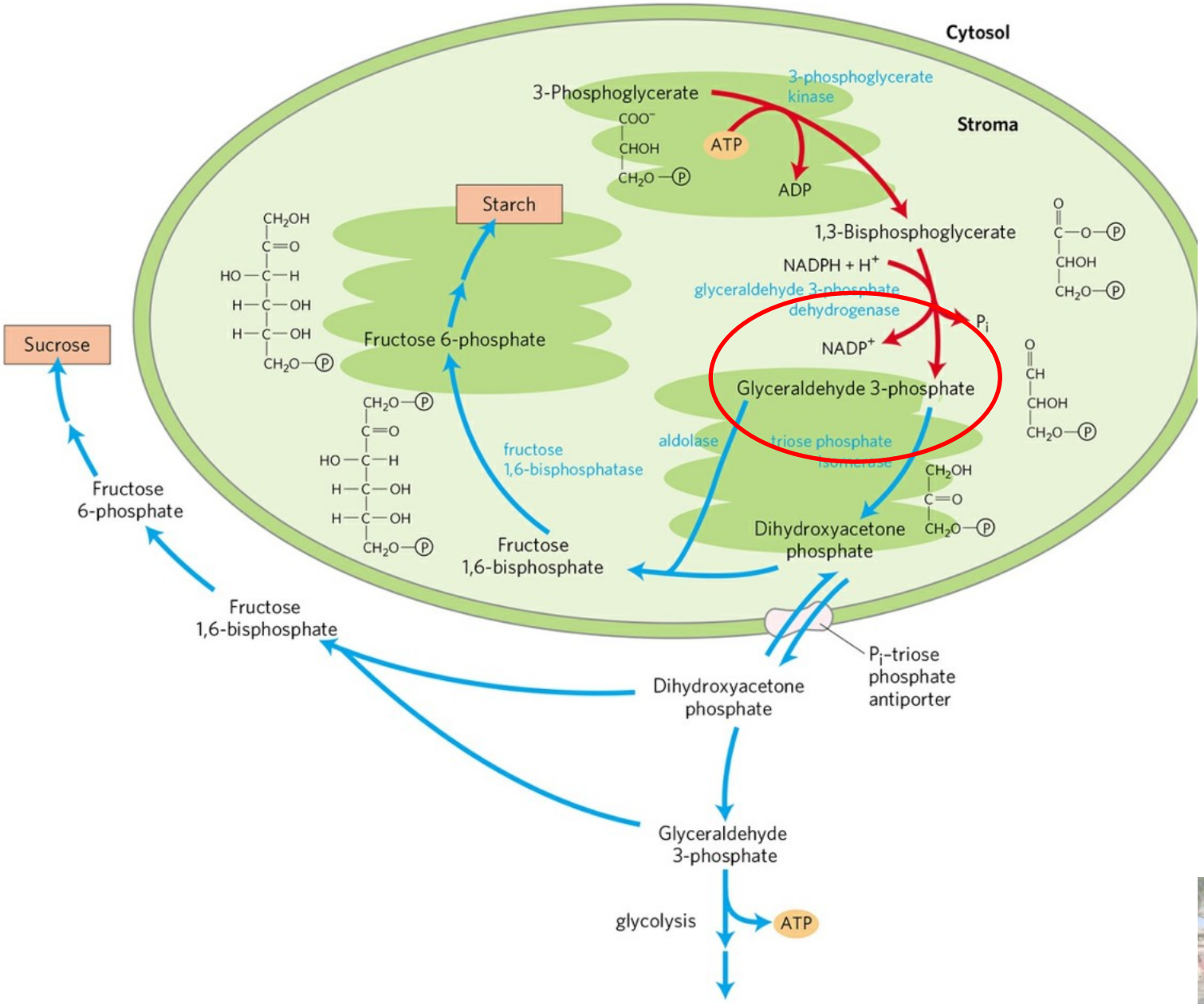
1911 - 1997

Biography

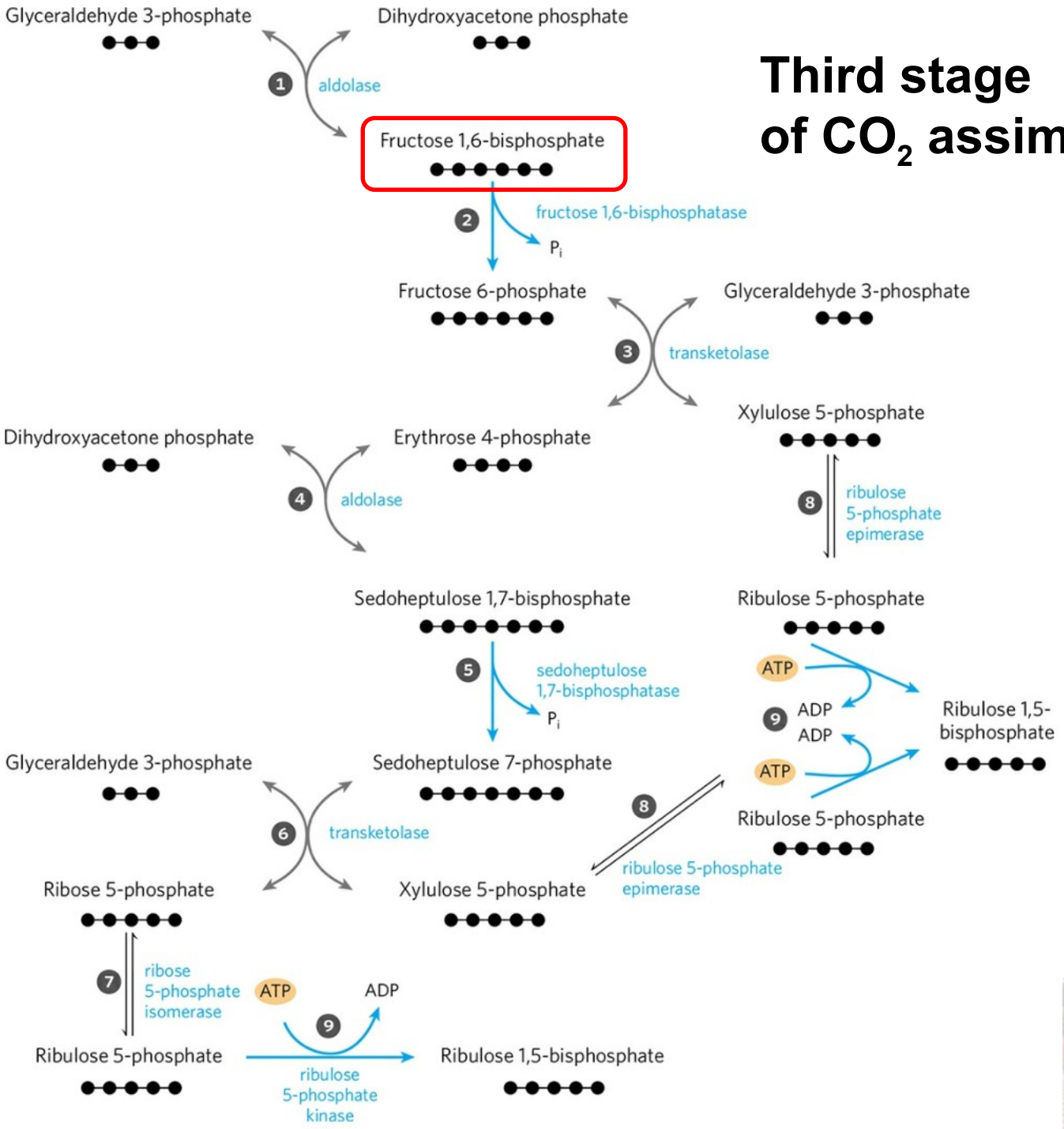
The three stages of CO2 assimilation in photosynthetic organisms



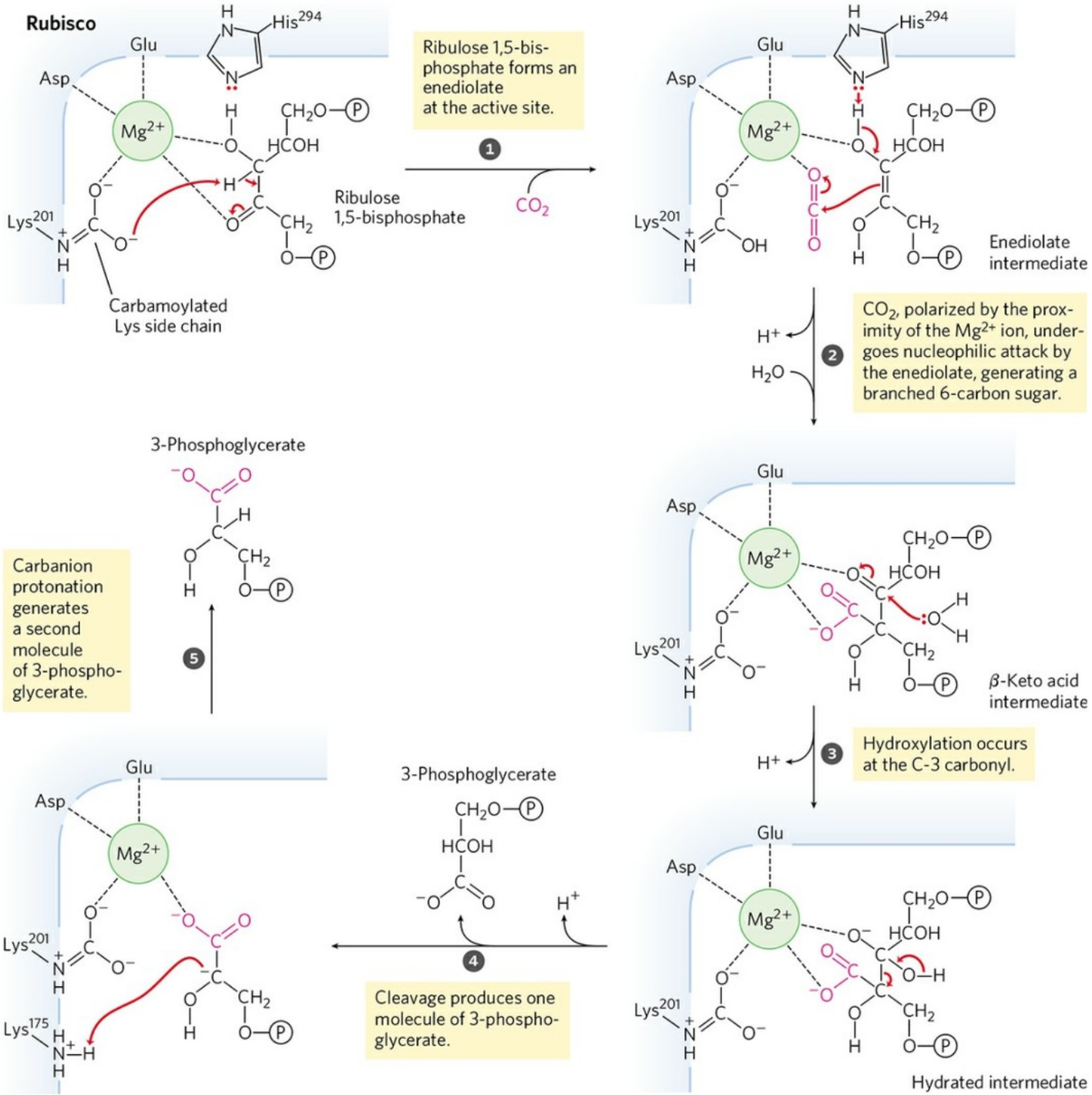
Second stage of CO_2 assimilation.

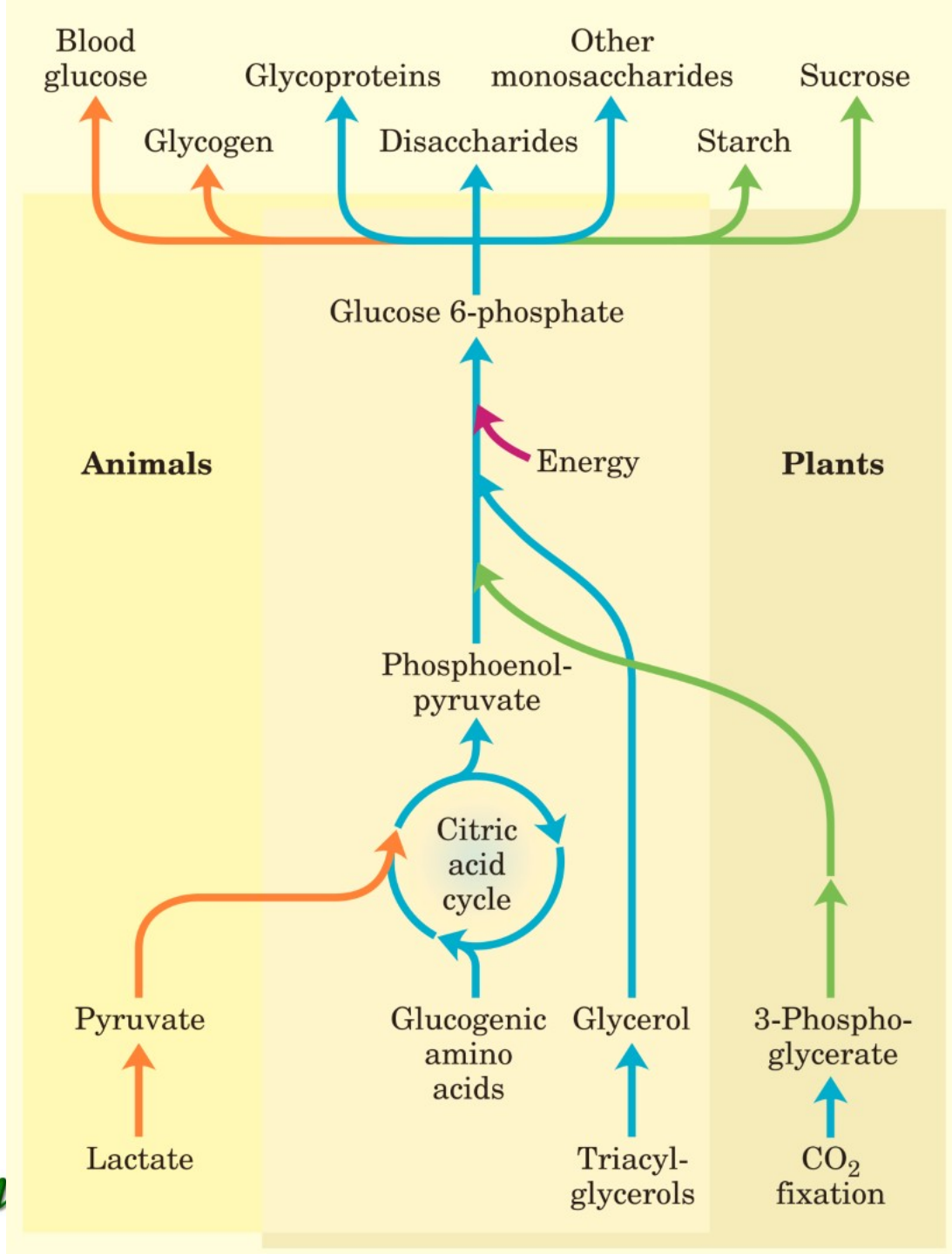


Third stage of CO_2 assimilation






Rubisco
Glu
His294
Asp
Mg2+
Lys201
Carbamoylated Lys side chain
Ribulose 1,5-bis-phosphate forms an enediolate at the active site.
1
CO2
Ribulose 1,5-bisphosphate
Enediolate intermediate
H+
H2O
2
CO2, polarized by the proximity of the Mg2+ ion, undergoes nucleophilic attack by the enediolate, generating a branched 6-carbon sugar.
β-Keto acid intermediate
3
Hydroxylation occurs at the C-3 carbonyl.
Hydrated intermediate
3-Phosphoglycerate
4
Cleavage produces one molecule of 3-phosphoglycerate.
Lys175
5
Carbanion protonation generates a second molecule of 3-phosphoglycerate.


Blood glucose
Glycoproteins
Other monosaccharides
Sucrose
Glycogen
Disaccharides
Starch
Glucose 6-phosphate
Animals
Energy
Plants
Phosphoenol-pyruvate
Citric acid cycle
Pyruvate
Glucogenic amino acids
Glycerol
3-Phospho-glycerate
Lactate
Triacyl-glycerols
CO_2 fixation

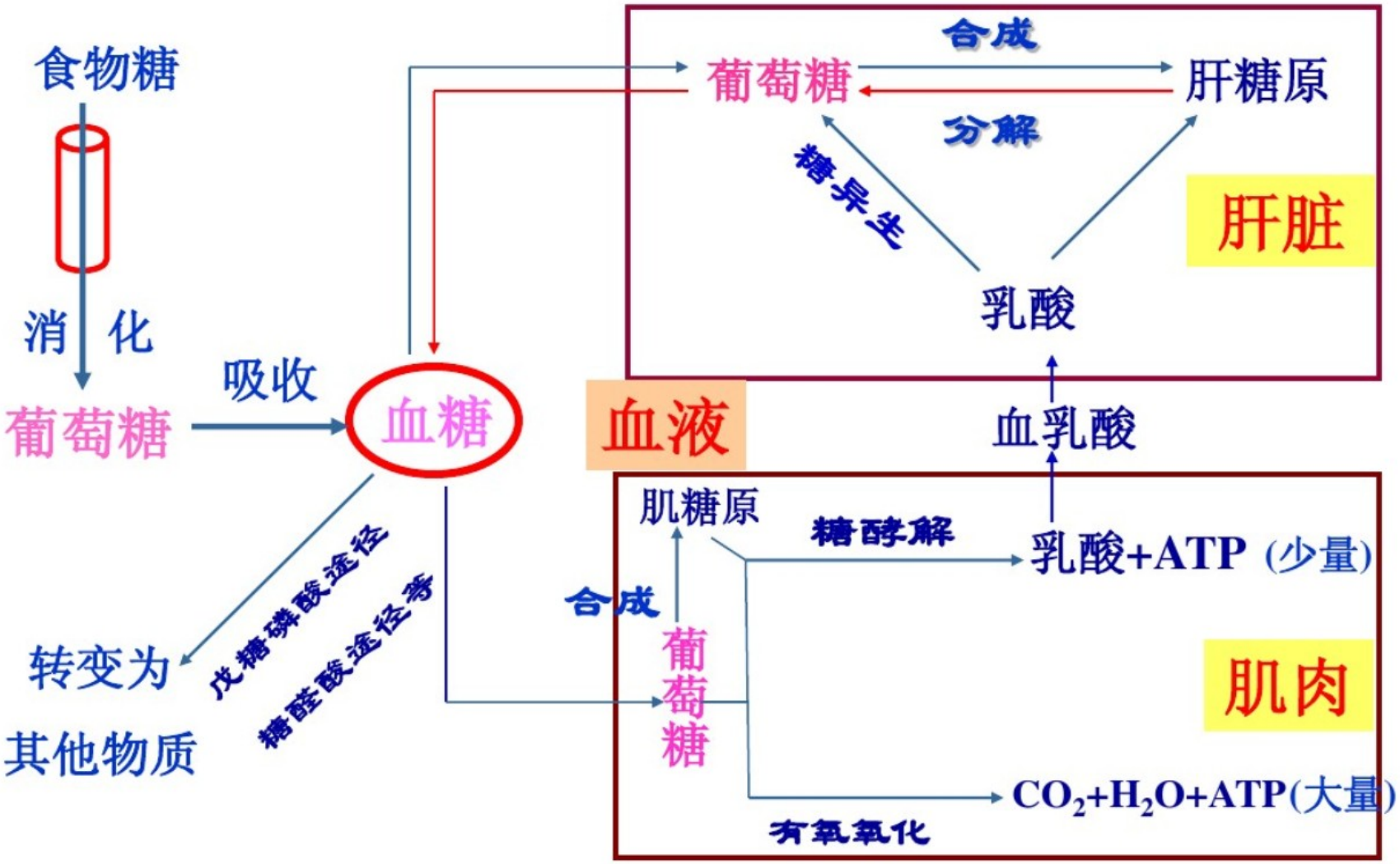
Biochem

END

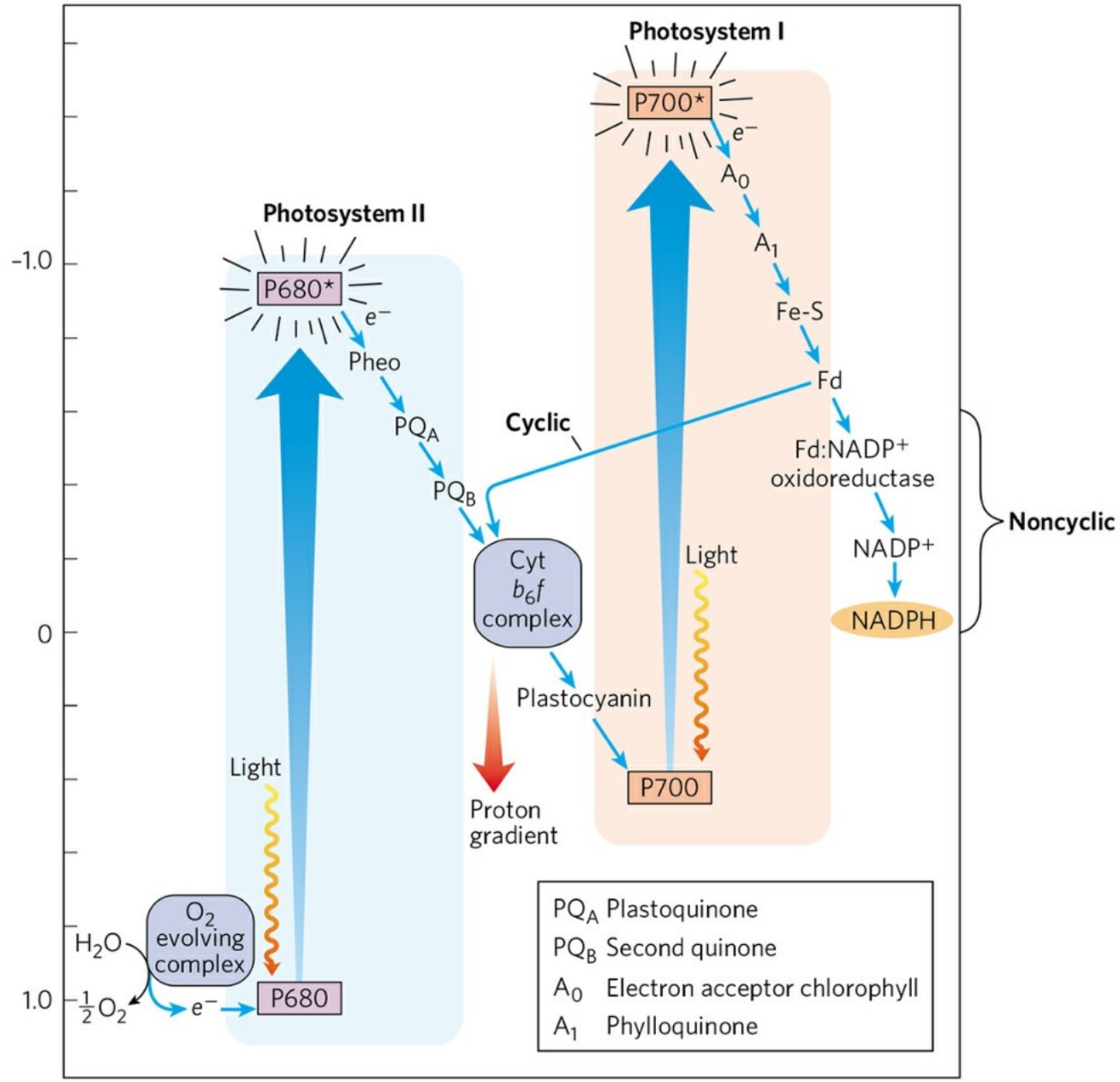
Thank you



高等动物糖代谢概况

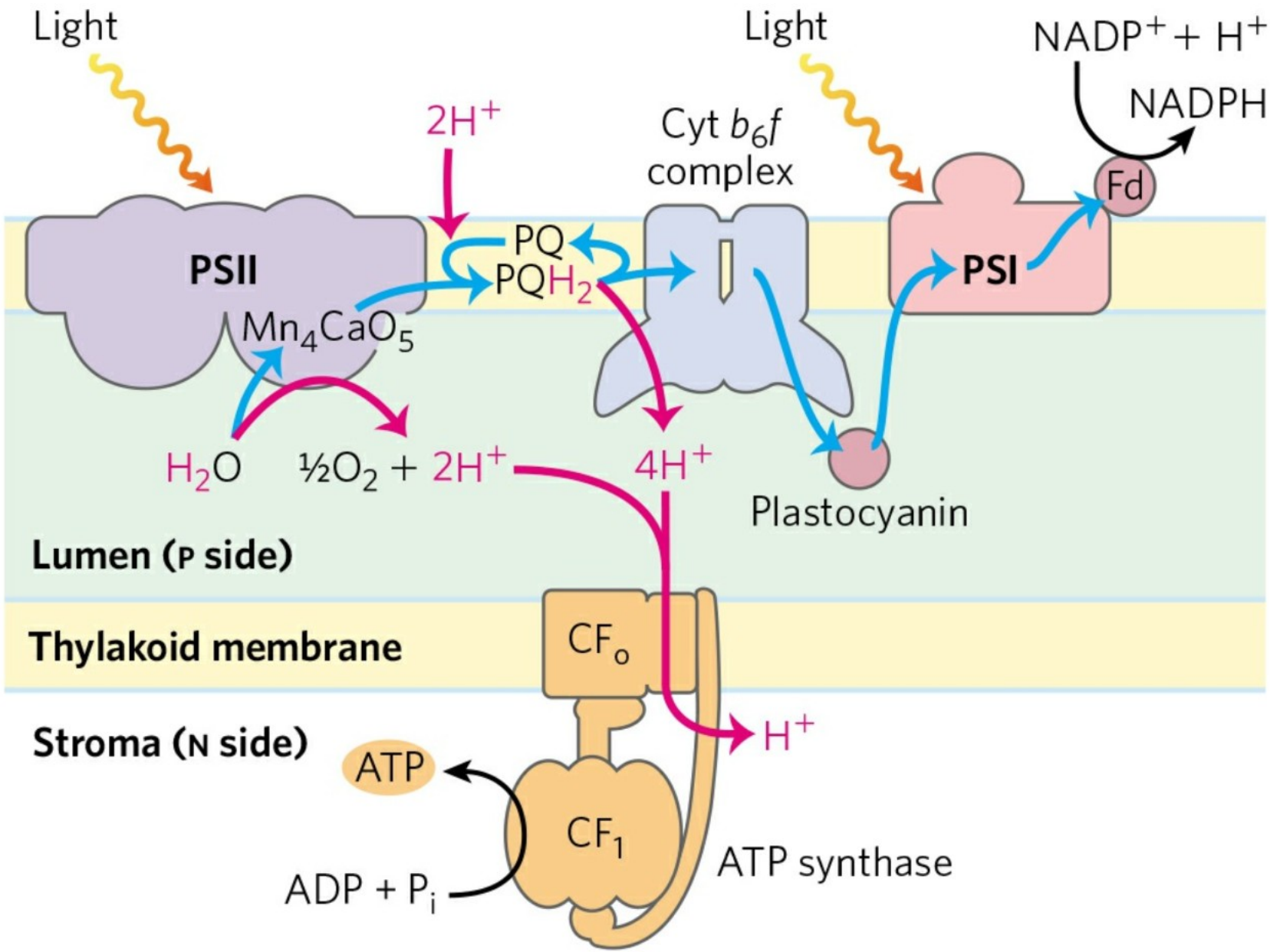

食物糖
消 化
葡萄糖
吸收
血糖
转变为
其他物质
戊糖磷酸途径
糖醛酸途径等
血液
合成
葡萄糖
肝糖原
分解
糖异生
乳酸
肝脏
血乳酸
肌糖原
糖酵解
乳酸+ATP (少量)
合成
葡萄糖
肌肉
CO_2+H_2O+ATP(大量)
有氧氧化

Integration of photosystems I and II in chloroplasts

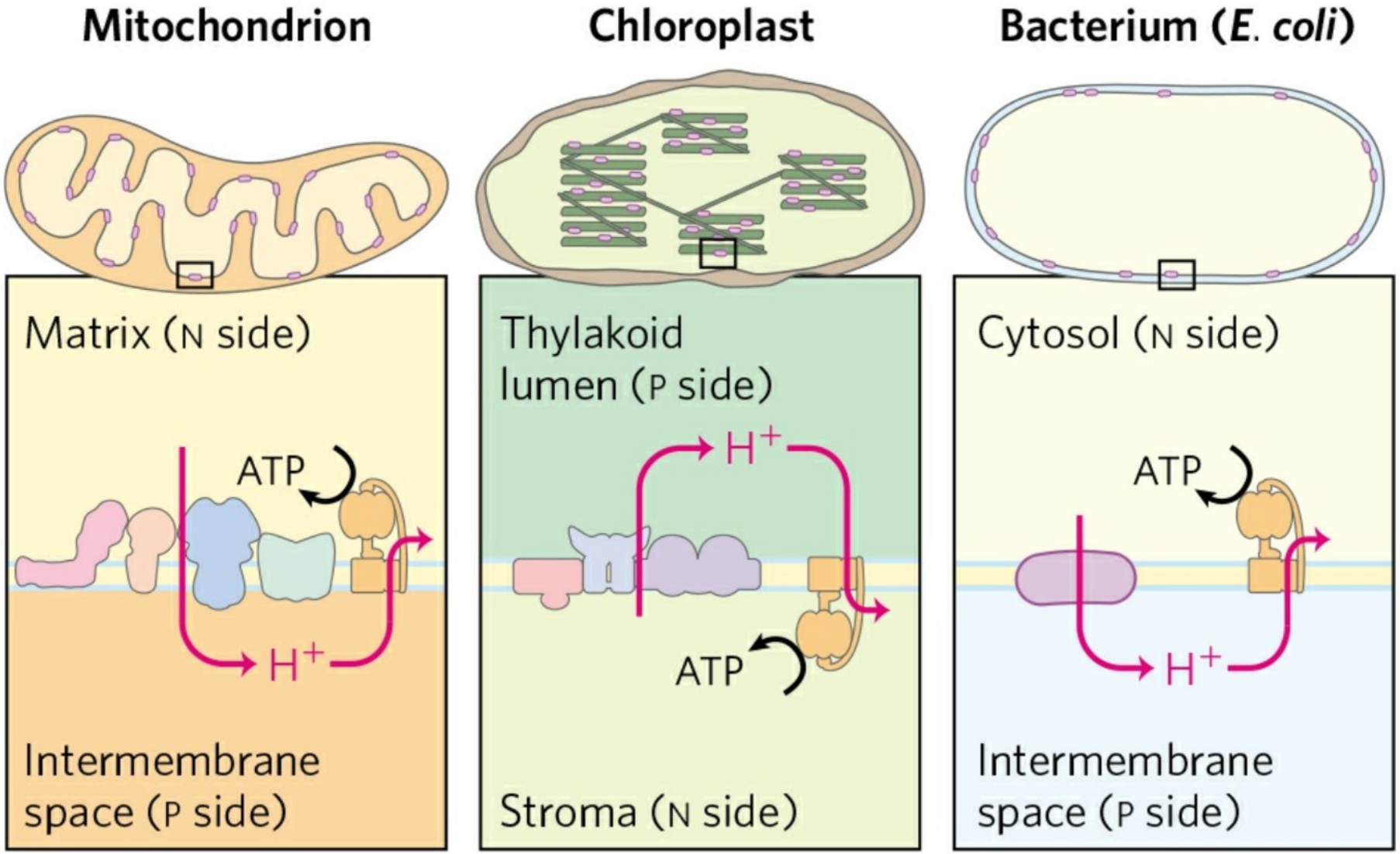


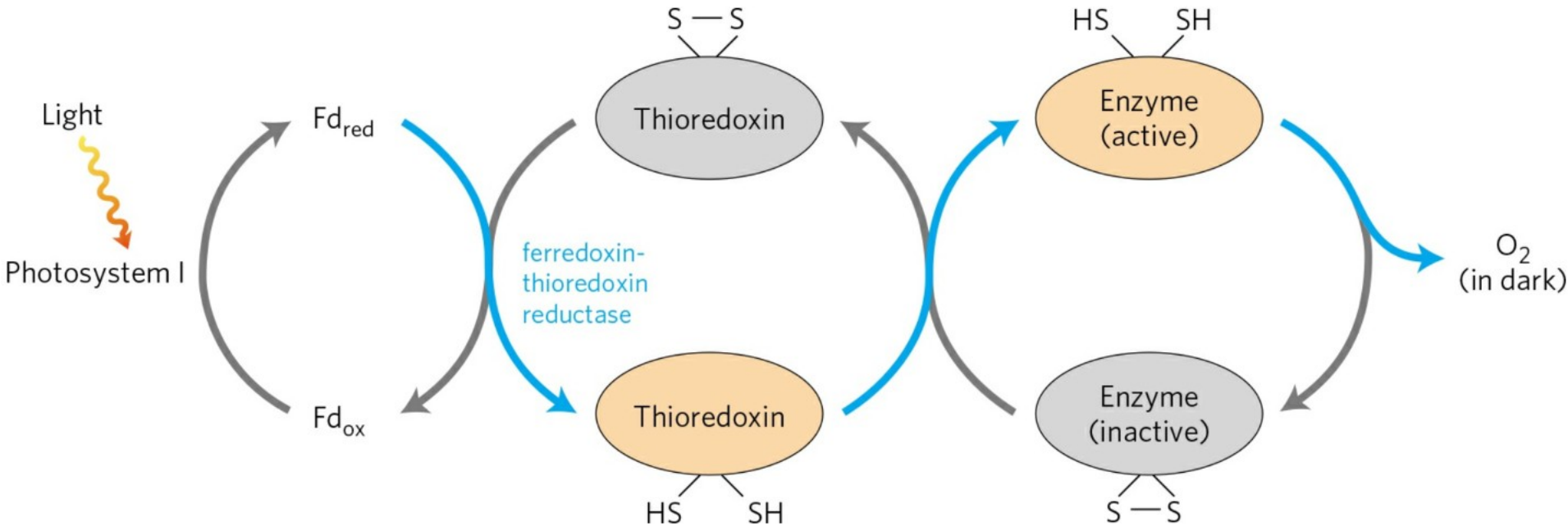


Proton and electron circuits during photophosphorylation.



Orientation of ATP synthase is fixed relative to the proton gradient

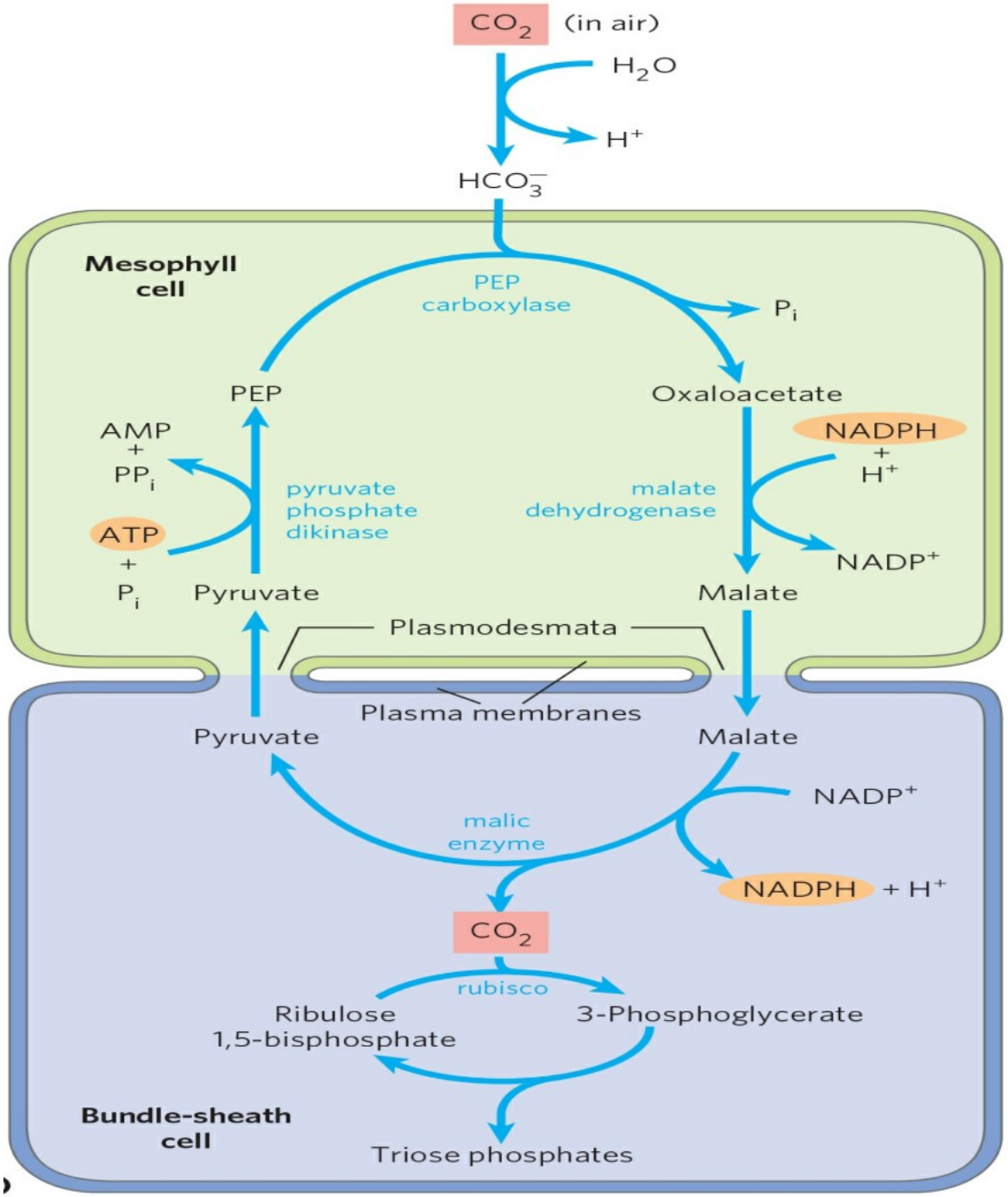




Light activation of several enzymes of the Calvin cycle



Carbon assimilation in C4 plants.



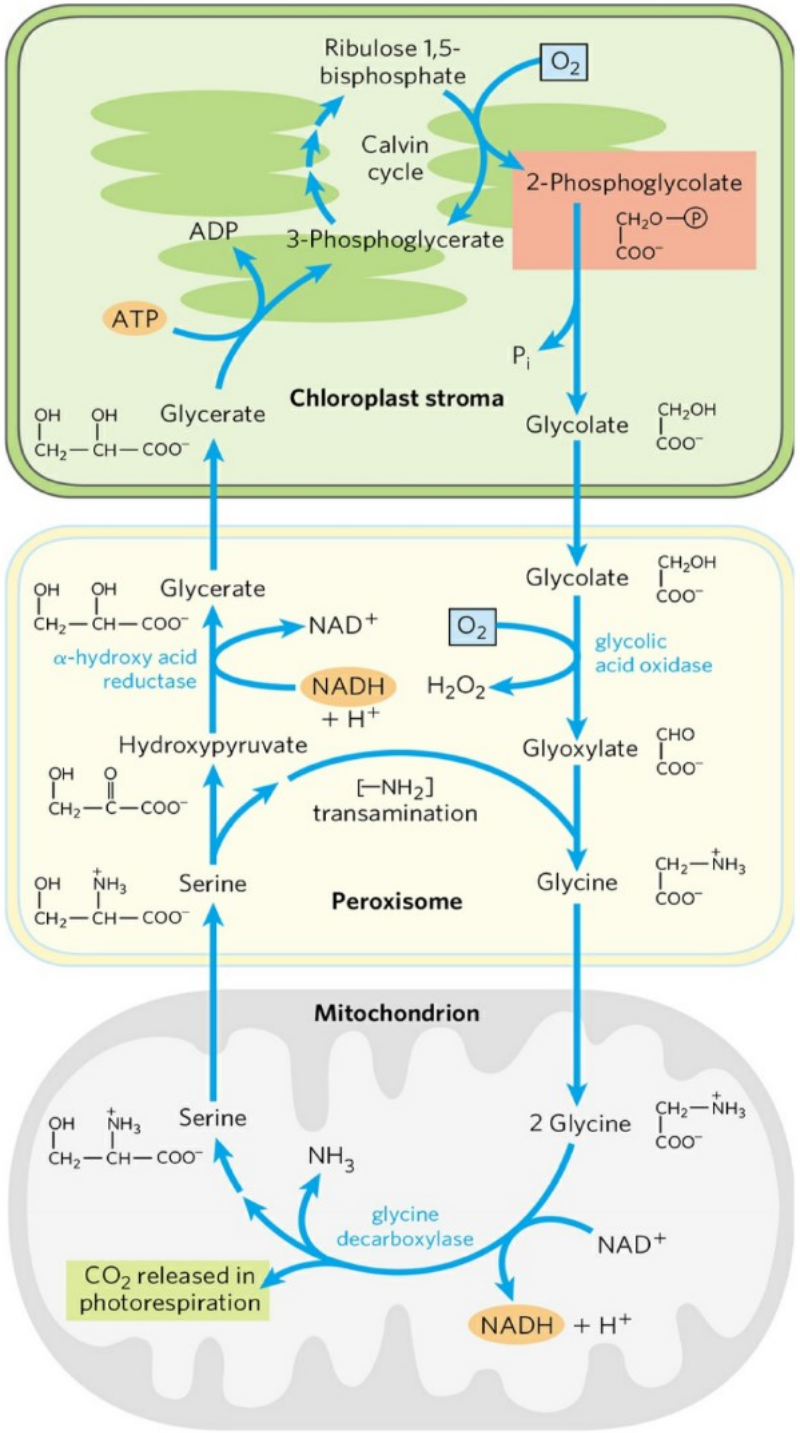
Biochemistry

Photorespiration Results from Rubisco's Oxygenase Activity

Rubisco is not absolutely specific for CO_2 as a substrate. Molecular oxygen (O_2) competes with CO_2 at the active site, and about once in every three or four turnovers, rubisco catalyzes the condensation of O_2 with ribulose 1,5-bisphosphate to form 3-phosphoglycerate and **2-phosphoglycolate** **(Fig. 20- 47)**, a metabolically useless product.



Glycolate pathway



Biochemistry